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Conception optimale de drones électriques : une approche
multidisciplinaire avec analyse des incertitudes, de la
tolérance aux pannes et des impacts environnementaux

Design optimization of unmanned aerial vehicles : a
multidisciplinary approach with uncertainty, fault-
tolerance, and environmental impact assessments

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Le roi des chamois était vêtu de vent. Dans la tempête, il se laissait envelopper par les rafales, c'était son manteau. Son pelage brillait, gonflé par l'explosion des éclairs, le roi fermait les yeux et se laissait étreindre par l'air déchaîné. Il était en sûreté là où toutes les autres créatures sentent une menace. Il était en alliance avec le vent, son coeur battait, léger, se chargeant de l'énergie lancée par le ciel sur la terre.

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Résumé

Les drones ont connu un développement intensif ces dernières années. En raison de leur rentabilité et de leur polyvalence, ces véhicules devraient gagner en popularité dans un large éventail d'applications, telles que la livraison de colis, la surveillance de lignes électriques et l'agriculture de précision. Parallèlement, le développement de nouvelles technologies et leur intégration sur diverses architectures de drones élargissent les possibilités de conception. Il en résulte un besoin accru pour des approches de conception holistiques permettant une meilleure intégration des technologies, un temps de développement plus court et une plus grande modularité. Cette thèse développe et met en application une méthodologie pour la conception de drones électriques multirotors, à voilure fixe et hybrides à décollage et atterrissage verticaux (VTOL). La plateforme ainsi développée permet le dimensionnement optimal d'un drone à partir de spécifications arbitraires sur la mission, les choix technologiques et l'architecture, en utilisant une approche multidisciplinaire. À partir d'un ensemble de modèles analytiques comprenant des lois d'échelle et des régressions, une méthodologie générique de dimensionnement est proposée. La méthodologie repose sur une formulation multidisciplinaire et optimale de conception (MDO) qui permet une convergence rapide vers la solution présentant les meilleures performances. En particulier, l'application de cette approche permet d'évaluer rapidement les effets d'une modification du cahier des charges. Dans un deuxième temps, les incertitudes relatives aux modèles et à la disponibilité des composants optimaux sur le marché sont évaluées. Pour atténuer les incertitudes critiques en matière de performances des drones, la méthodologie de dimensionnement est étendue pour permettre l'optimisation de la conception à partir de catalogues de composants existants en lieu et place de modèles physiques. Enfin, la thèse aborde deux aspects spécifiques de la conception des drones liés à la réglementation et aux enjeux sociétaux. D'une part, les récentes réglementations émises par l'Agence européenne de la sécurité aérienne (AESA) imposent un certain niveau de sûreté pour l'opération des drones. Dans cette optique, une approche est proposée pour évaluer la contrôlabilité de diverses architectures en cas de défaillance d'un rotor ou d'une surface de contrôle. L'évaluation est notamment reliée à la méthodologie de conception précédemment développée afin d'assurer un dimensionnement avec un certain niveau de tolérance aux pannes. D'autre part, l'acceptation sociétale des drones est étroitement liée aux préoccupations environnementales, incluant, entre autres, le changement climatique. Cette problématique est abordée par le développement et l'intégration

d'une discipline environnementale dans l'outil de conception. Cette approche permet d'évaluer la sensibilité des impacts environnementaux aux exigences de la mission et aux hypothèses technologiques, ainsi que de minimiser les impacts environnementaux dès les premières phases du processus de conception. La thèse contribue ainsi au développement d'un cadre de référence pour optimiser la conception des drones électriques avec une approche multidisciplinaire. De ce fait, les contributions de la thèse sont particulièrement adaptées pour la conception de futurs drones dont le déploiement sera soumis à des problématiques de marché, de sûreté et de contraintes environnementales.

Abstract

Unmanned aerial vehicles (UAVs) have undergone intensive development in recent years. Owing to their cost-effectiveness and versatility, UAVs are expected to gain popularity in a wide range of applications, such as parcel delivery, power line monitoring and precision farming. Concurrently, the development of new technologies and their integration into various drone concepts is expanding the range of design alternatives. This is driving the need for holistic design approaches with better technology integration, faster development time and greater modularity.

The thesis develops and implements a methodology for the early design of electric multirotor, fixed-wing and hybrid vertical take-off and landing (VTOL) UAVs. The framework enables the optimal sizing of a UAV from arbitrary specifications on the mission, technological choices and architecture, using a comprehensive multidisciplinary approach.

Starting from a set of analytical models including scaling laws and regressions, a generic sizing methodology is developed. The proposed methodology relies on an efficient multidisciplinary design and optimization (MDO) formulation, which enables fast convergence to the UAV candidate with the best performances. In particular, the application of this approach enables to rapidly assess the effects of changes in the requirements. Next, the uncertainties surrounding the models and the availability of optimal components on the market are assessed. To mitigate critical uncertainties in UAV performance, the sizing methodology is extended to allow the design to be optimized using catalogues of existing components instead of models. Finally, the thesis develops two specific aspects of UAV design related to regulatory and societal challenges. On the one hand, recent regulations issued by the European Union Aviation Safety Agency (EASA) impose a level of safety for certain categories of UAVs. To this end, an approach is proposed to assess the controllability of various architectures in the event of failures. The assessment is further linked to the design framework to achieve fault-tolerance sizing of the rotors and control surfaces. On the other hand, societal acceptance of UAVs is strongly related to environmental concerns, including but not limited to climate change. This challenge is addressed by developing and integrating an environmental discipline into the design framework. The novel approach enables to assess the sensitivity of environmental impacts to mission requirements and technological assumptions, as well as minimizing environmental burdens at the earliest design stages.

The thesis contributes to the development of a unified framework for optimizing the design of electric UAVs with a holistic approach. As such, it is relevant to future UAVs designed for applications subject to market, regulatory and environmental issues.

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Nomenclature

Abbreviations

(X)ACAI (Extended) Available Control Authority Index

BVLOS Beyond Visual Line Of Sight

CFRP Carbon Fiber Reinforced Plastic

CMA-ES Covariance Matrix Adaptation Evolution Strategy

DOC Degree Of Controllability

DOE Design Of Experiments

EASA European Union Aviation Safety Agency

ESC Electronic Speed Controller

FAA Federal Aviation Administration

FTC Fault Tolerant Control

FW Fixed-Wing

FW-VTOL Fixed-Wing Vertical Take-Off and Landing

G/LFP Graphite/Lithium Ferro Phosphate (or, Lithium Iron Phosphate) battery

G/NMC Graphite/Lithium Nickel Manganese Cobalt battery

IDF Individual Discipline Feasible

KKT Karush-Kuhn-Tucker

LCA Life Cycle Assessment

LCI Life Cycle Inventory

LCIA Life Cycle Impact Assessment

LTI	Linear Time-Invariant
MAC	Mean Aerodynamic Chord
MDA	Multidisciplinary Analysis
MDF	Multidisciplinary Design Feasible
MDO	Multidisciplinary Design Optimization
MP1	First Monotonicity Principle
MTOW	Maximum Take-Off Weight
NVH	Normalized Variable Hybrid
OSO	Operational Safety Objective
PCA	Principal Component Analysis
pdf	Probability Density Function
PEF	Product Environmental Footprint
SAIL	Specific Assurance and Integrity Level
SC	Special Condition
Si/NMC	Silicon/Lithium Nickel Manganese Cobalt battery
SLSQP	Sequential Least Squares Programming
SORA	Specific Operations Risk Assessment
SoSOS	Share of the Safe Operating Space
UAS	Unmanned Aerial System
UAV	Unmanned Aerial Vehicle
VTOL	Vertical Take-Off and Landing
XDSM	Extended Design Structure Matrix

Nomenclature for Part I

$\cdot^{FW}$	Fixed-wing propulsion system
$\cdot^{MR}$	Multirotor propulsion system

$\cdot_{climb}$	Climb flight scenario
$\cdot_{cruise}$	Cruise flight scenario
$\cdot_{hover}$	Hover flight scenario
$\cdot_{ref}$	Reference component
$\cdot_{stall}$	Stall flight scenario
$\cdot_{takeoff}$	Takeoff flight scenario
α	Angle of attack (rad)
β	Propeller pitch-to-diameter ratio
λ_f	Fuselage fineness ratio
$\Lambda_{LE_w}, \Lambda_{TE_w}$	Sweep angles of wing's leading and trailing edges (rad)
λ_w	Wing taper ratio
μ_{wires}	Wires linear mass (kg/m)
ρ_{air}	Air density
σ_{max}	Maximum allowable stress (Pa)
σ_{pro}	Propeller solidity
σ_{root}	Bending stress at arm/spar root (Pa)
θ_{FP}	Flight path angle (rad)
v_{bat}	Battery volume (m ³)
v_{ht}, v_{vt}	Volume coefficients of the horizontal and vertical tails
a_{to}	Takeoff acceleration (m/s ²)
AR	Aspect ratio
b_w	Wingspan (m)
C_f	Flat-plate skin friction coefficient
C_{bat}	Battery capacity (A.s)
C_{D_0}	Parasitic drag coefficient

c_{MAC}	Mean aerodynamic chord (m)
C_P	Propeller power coefficient
c_{root}, c_{tip}	Root and tip chords (m)
C_T	Propeller thrust coefficient
CL_{max}	Maximum lift coefficient of the wing
d_f	Fuselage diameter (m)
D_f	Drag (N)
d_{in}, d_{out}	Arm's inner and outer diameter (m)
d_{pro}	Propeller diameter (m)
e	Span efficiency factor
E_{bat}	Battery energy (Wh)
F	Thrust (N)
F/S	Thrust-to-weight ratio
g	Acceleration of gravity (m/s ²)
g^c	Consistency constraint for disciplinary couplings resolution
h_{spar}	Distance between two flanges (m)
I_{esc}	Electronic speed controller current (A)
I_{wires}	Wires rated current (A)
J	Advance ratio
K	Induced drag coefficient factor
k^c	Normalized variable for disciplinary couplings resolution
k^s	Sizing coefficient
K_V	Motor velocity constant (rad/V/s)
k_{arm}	Ratio of inner-to-outer diameters
K_{flange}	Ratio of flange thickness to flange length

l_f	Fuselage length (m)
l_{arm}	Arms length (m)
L_f	Downward aerodynamic force (N)
$m_.$	Component mass (kg)
m_{uav}	Mass of the UAV (kg)
n	Propeller rotational speed (rad/s)
N_{rotors}	Number of rotors
n_z	Vertical load factor
P_{bat}	Battery power (W)
P_{esc}	Electronic speed controller power (W)
q	Dynamic pressure (Pa)
R	Motor resistance (Ω)
Re	Reynolds number
S_{front}	Front surface of the multirotor frame (m ²)
S_{top}	Top surface of the multirotor frame (m ²)
S_w	Wing surface (m ²)
SM	Static margin
t/c	Thickness-to-chord ratio of airfoil
T_f	Motor friction torque (N.m)
T_{max}	Motor maximum torque (N.m)
T_{nom}	Motor nominal torque (N.m)
U_{bat}	Battery voltage (V)
U_{esc}	Electronic speed controller voltage (V)
V	Airspeed (m/s)
W	Weight of the UAV (N)

W/S	Wing loading (N/m ²)
x_{ac}	x-location of wing aerodynamic center (m)
x_{cg}	x-location of center of gravity (m)
x_{np}	x-location of neutral point (m)

Nomenclature for Part II

θ	Vector of model parameters
η	Absolute error
$\mathbf{x}$	Vector of model inputs
$\mathbf{y}$	Vector of model outputs
$\mathcal{N}$	Normal distribution
$\mathcal{U}$	Uniform distribution
μ	Mean
σ	Standard deviation
Var	Variance
Θ	Random variable for model parameter
$\tilde{\mu}_3$	Skewness
ε	Relative error
EE_i	Elementary effect of Morris
S_i	First-order Sobol sensitivity index for variable
S_{T_i}	Total-order Sobol sensitivity index
X	Random variable for model input
x^*, θ^*	Deterministic values

Nomenclature for Part III

'	Failure case
.0	Reference flight conditions

α	Required control authority parameter
δ	Control input vector
Δ_k	Vector of the effector magnitudes for the k -th set of hyper-plane segments
ξ_k	Axis of perpendicular distance between the k -th set of hyper-plane segments
$\mathbf{A}$	State matrix
$\mathbf{B}_f$	Control effectiveness matrix
$\mathbf{B}_{f,\text{lin}}$	Control effectiveness matrix of the linearized efforts
$\mathbf{B}_{f,l}$	Partition of the control effectiveness matrix relative to the linear control efforts
$\mathbf{B}_{f,\text{nl}}$	Partition of the control effectiveness matrix relative to the absolute control efforts
$\mathbf{B}$	Control matrix
$\mathbf{f}$	Rotor thrust vector
$\mathbf{H}$	Failure matrix
$\mathbf{K}$	Sizing factors vector
$\mathbf{u}$	Virtual control vector
$\mathbf{x}$	State vector
Ω	Virtual control vector constraint set
ϕ, θ, ψ	Roll, Pitch, Yaw angles (rad)
ρ	(Extended) available control authority index
ρ, η, ζ	Aileron, Elevator, Rudder deflection angles (rad)
τ	Propeller thrust setting
f	Thrust (N)
F_c	Center of symmetry of the virtual control constraint set
I_x, I_y, I_z	UAV moments of inertia (kg.m ²)
L, M, N	Roll, Pitch, Yaw control moments (N.m)
m_{uav}	UAV total mass (kg)

p, q, r Roll, Pitch, Yaw rotational speeds (rad/s)

u_b, v_b, w_b Longitudinal, Lateral, Vertical speeds (m/s)

X, Y, Z Longitudinal, Lateral, Vertical control forces (N)

Nomenclature for Part IV

η Component efficiency

B_j Background processes

C_D Airframe drag coefficient

$E_{mission}$ Energy from electrical grid consumed over a single mission (kWh)

m Component mass (kg)

$n_{cycles, battery}$ Number of charge-discharge cycles before replacing battery

n_{cycles} Number of times the mission is performed over the life cycle

P_i Parameters for the parametric LCA model

Chapter 1

Introduction

1.1 Background

1.1.1 The UAV market landscape and its challenges

In recent years, the use of unmanned aerial vehicles (UAVs), commonly known as drones, has seen a remarkable increase, reshaping various industries around the world. Initially developed for military purposes, UAV technology has rapidly evolved, becoming more accessible and finding applications in civilian and commercial domains.

The growing demand for UAVs is underscored by their unique ability to perform tasks autonomously or be controlled remotely, providing unique flexibility and maneuverability in various routine and specialized operations. From tasks such as precision agriculture, environmental monitoring and disaster management, to activities such as infrastructure inspection, filmmaking and e-commerce logistics, UAVs are finding new applications and changing the way industries operate [1–5]. In particular, recent advances in UAV technologies have allowed manufacturers to produce a wide range of models in different sizes, weights and shapes, capable of carrying different types of payloads. This diversity makes UAVs affordable for a wide range of applications, contributing to their widespread adoption.

Despite these advances, technological and regulatory challenges are expected to limit the growth of the UAV market. Technological advancements, particularly in battery technology, are central to overcoming limitations on flight duration and range. As drone manufacturers push the boundaries of innovation, they are simultaneously navigating the regulatory landscape that governs the use of UAVs. The Federal Aviation Administration (FAA), the European Union Aviation Safety Agency (EASA), and similar regulatory bodies play a critical role in ensuring the safe integration of UAVs into the airspace. Striking a balance between encouraging innovation and addressing safety and environmental concerns, these regulations influence the design and functionality of UAVs. Drone manufacturers such as AeroVironment, DJI Technology, Parrot, and Delair are at the forefront of technological innovation, working with regulators to shape standards that enhance safety, security, and environmental responsibility.

1.1.2 Categories of UAVs

To meet the diverse needs of industries and applications, designers have developed and tested various concepts of UAVs. Among these, multicopters, also known as multirotor UAVs, have gained popularity. Multirotor UAVs are rotorcraft composed of multiple propellers that provide both lift and thrust simultaneously. This design provides a higher degree of agility and precision compared to traditional helicopters, making multirotor UAVs ideal for tasks that require maneuverability and stability, such as aerial photography and inspection. They come in a variety of configurations¹, ranging from quadcopters with four rotors to hexacopters and octocopters with six and eight rotors, respectively. In addition, the rotors can be arranged coaxially so that a number of two propellers per arm are installed in a "push-pull" configuration.

Fixed-wing UAVs, on the other hand, use a more traditional aircraft design with wings and stabilizing surfaces. These UAVs rely on forward motion to generate lift, offering advantages in endurance and range. The adoption of fixed-wing UAVs is driven by the importance of endurance and range in applications that require extensive coverage, such as mapping vast agricultural fields or conducting long-range surveys.

The emerging concept of hybrid FW (fixed-wing) VTOL (vertical take-off and landing) UAVs combines the advantages of both multicopters and fixed-wing designs. These UAVs can take off and land vertically like a multicopter, but transition to horizontal flight using their lifting surfaces for increased speed and efficiency. The tilt-rotor configuration uses rotating rotors that tilt to switch between flight modes, while tilt-wing designs use wings that tilt for the same purpose. Tail-sitter UAVs position the entire airframe vertically during takeoff and landing, and transition to a horizontal position during flight. Lastly, quadplane configurations combine a quadcopter-style multicopter configuration with fixed wings. The innovative approach of hybrid FW-VTOL UAVs meets the needs of applications that require versatility in both urban and rural environments.



(a) DJI Matrice 600 Pro.
Source: DJI.



(b) Delair DT26.
Source: Delair.



(c) AeroVironment JUMP 20.
Source: AeroVironment.

Figure 1.1 Examples of multirotor (a), fixed-wing (b) and FW-VTOL (c) concepts.

¹For a given concept, the configuration refers to the physical layout of the UAV components.

Additional attributes that can be used to differentiate UAV categories include weight, size, and mission altitude. One possible classification is provided in Table 1.1, where the UAV class ranges from micro UAVs to medium- and high-altitude long-endurance (MALE/HALE) UAVs [6]. However, the classification of UAVs is not universal and depends on the context. For example, EASA provides several class labels for recreational and low-risk commercial activities based on the mass of the UAV (<250g, <900g, <4kg, <25kg) and operational limitations such as altitude and flight speed. In addition, UAVs weighing more than 25 kg fall into the so-called "specific" and "certified" categories.

Table 1.1 UAVs classification according to Sadraey [6].

UAV Class	Mass	Size	Operating Altitude	Range	Endurance
Micro	< 250 grams	≤ 10 cm	< 100 ft	0.1-0.5 km	≤ 1 hr
Mini	0.250-1 kg	10-30 cm	< 500 ft	0.5-1 km	≤ 1 hr
Very small	1-2.5 kg	30-50 cm	< 1 000 ft	1-5 km	1-3 hrs
Small	2.5-25 kg	0.5-2 m	1 000-5 000 ft	10-100 km	0.5-2 hrs
Medium	25-450 kg	5-10 m	10 000-15 000 ft	500-2 000 km	3-10 hrs
Large	2 500-13 500 kg	20-50 m	20 000-40 000 ft	1 000-5 000 km	10-20 hrs
MALE	450-4 500 kg	15-40 m	15 000-30 000 ft	20 000-40 000 km	20-40 hrs
HALE	> 2 000 kg	20-50 m	50 000-70 000 ft	20 000-40 000 km	30-50 hrs

UAVs can also be categorized based on the type of energy used for propulsion. Gasoline or kerosene-powered UAVs, equipped with piston engines or gas turbines, are prevalent in larger models and enable longer flight durations and increased payload capacities. Battery-powered UAVs, common in smaller designs, use electric motors powered by rechargeable lithium-ion batteries. Fuel cell-powered UAVs offer extended endurance by generating electricity through hydrogen fuel cells. Finally, UAVs equipped with hybrid propulsion systems combine multiple power sources to optimize performance, such as using photovoltaic cells to continuously recharge a battery.

1.1.3 Main components of UAVs

UAVs rely on a combination of essential components to ensure their operation, as illustrated in Figure 1.2. The primary component is the propulsion system. For electric UAVs, the propulsion chain typically consists of electric motors connected to propellers, one or more batteries, and electronic speed controllers (ESCs) to regulate the power supplied to each motor. In addition, a power distribution board (PDB) handles the transfer of power from the battery to the ESCs and motors. Propellers for UAVs are made of plastic, wood, or carbon fiber, the latter being the most expensive option. Although there are different types of electric motors, UAVs generally integrate brushless motors, which offer better efficiency and longer lifetime compared to brushed motors. Outrunner and inrunner configurations are two possible designs for brushless motors. Most solutions for multirotor and fixed-wing UAVs use an outrunner configuration, where the rotor rotates outside of the stator, providing higher torque than

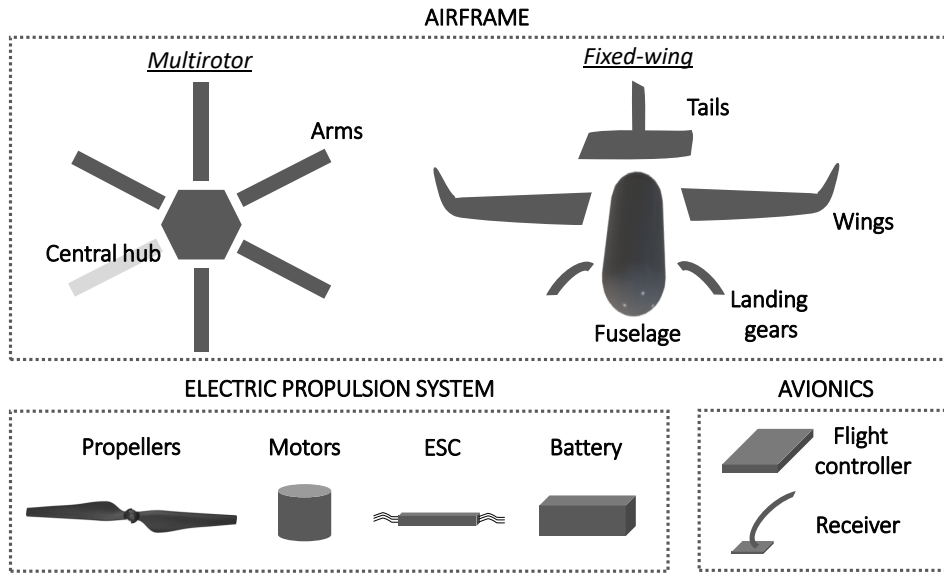


Figure 1.2 Schematic view of the main UAV components.

its inrunner counterpart. Batteries for UAVs use lithium-ion technology with either a liquid electrolyte or a polymer electrolyte. The presence of a polymer electrolyte (LiPo battery) provides increased current burst capability, but is prone to safety and durability issues. The cathode material consists of lithium, nickel, manganese and cobalt (NMC chemistry) or, less commonly, a mixture of lithium, iron and phosphate (LiFePO₄ or LFP chemistry). The latter offers significant advantages regarding lifetime at the expense of lower energy density.

The flight controller is another critical component that processes sensor data (e.g., accelerometers and gyroscopes) and adjusts motor speeds for stabilization and control. GNSS (Global Navigation Satellite Systems) receivers, such as GPS (Global Positioning System), enhance the UAV's navigation capabilities.

Another essential component is the airframe. The airframe provides the structure that houses and supports all the other components. For multirotor UAVs, the airframe typically consists of a central hub with arms, each supporting one (or two) motors and propellers. Fixed-wing UAVs, on the other hand, have airframes similar to traditional aircraft, with wings for lift and a tail assembly for stability. Control surfaces such as ailerons, elevators, and rudders provide flight control. In addition, a landing gear ensures a safe landing. Common airframe materials include carbon fiber, aluminum, and composite materials. Balsa wood and fiberglass are also commonly used for fixed-wing UAVs. The design of the airframe varies depending on the UAV's intended purpose and greatly influences the UAV's stability, maneuverability, and aerodynamic performance.

Finally, payload components are specific to the UAV's intended application. Typical payloads include cameras, sensors, lidar, and cargo payloads.

1.1.4 Design process of UAVs

UAV design is a decision-making process which typically involves multiple design criteria resulting in trade-offs. These criteria, including manufacturing cost, operating cost, UAV weight and size, maneuverability, and safety, guide the process by establishing priorities and design guidelines. For example, the decision to integrate either custom components (with superior performance) or off-the-shelf components (less expensive and readily available) may be driven by economic and development time factors. As the variety of design criteria indicates, a comprehensive design process is different from simply optimizing a configuration by minimizing its mass or improving its flight performance. The design process must also address concerns related to other phases of the UAV life cycle, including economic and, more recently, environmental considerations such as climate change.

The system design and development process for UAVs requires the integration of multiple disciplines throughout the three major phases of conceptual design, preliminary design, and detailed design [6] (Figure 1.3). As its name implies, the conceptual design phase involves

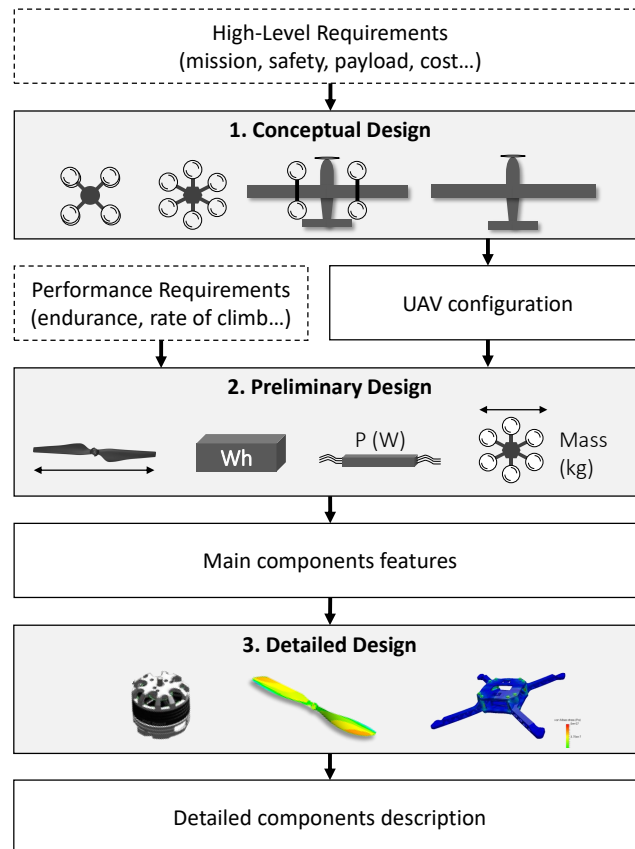


Figure 1.3 Main phases of the design process.

the design of the UAV at the concept level. The purpose is to identify the major components

(e.g., wings and propulsion system) that the UAV needs to meet the high-level requirements. The result of the conceptual design phase is a UAV configuration. The preliminary design phase begins with a set of performance requirements and returns an estimate of the major characteristics of the components for the selected configuration. For example, wing area, battery size, and motor power are determined. The phase can be further subdivided into the exploration of materials or technologies for the components, and the sizing of the components given this information. Preliminary design calculations are typically associated with low-fidelity models and include a degree of uncertainty. Finally, the detailed design refines and finalizes the calculations. The results of the detailed design are validated through a series of tests and validations, such as flight tests, before releasing the associated documentation (e.g., drawings) for production and assembly of the UAV.

1.2 Problem statement

The rapid development of UAVs for various applications underscores the critical need for design methodologies capable of capturing a range of decision criteria and requirements. This manuscript aims to develop a comprehensive framework that addresses multiple aspects of the conceptual and preliminary design of UAVs. Specifically, this research focuses on four major concerns: (1) the sizing of different UAV configurations, (2) the management of uncertainties in preliminary design, (3) the safety of UAVs through fault-tolerance analysis, and (4) the environmental assessment of UAVs. This holistic approach is essential to fostering the next generation of UAVs that are not only optimized for performance, but also exhibit enhanced safety and environmental sustainability with reduced development times.

1.2.1 Research questions and objectives

Each of these four aspects raises a number of questions, which are outlined here:

- RQ1.** *What methodologies can be employed for the preliminary design of UAVs?*
- RQ2.** *How can the sizing be optimized for various UAV configurations and applications?*
- RQ3.** *What are the primary sources of uncertainties in the preliminary design of UAVs, and how do they impact the results of the sizing process?*
- RQ4.** *What strategies can be developed to mitigate these uncertainties?*
- RQ5.** *What should be considered when evaluating the fault tolerance of different UAV configurations, and what methods can be applied?*
- RQ6.** *What metrics and methods can be used to establish a link between control analysis and UAV sizing?*

RQ7. *How can the environmental footprint of UAVs be systematically assessed in the design process?*

RQ8. *What are the main contributors to the environmental impacts of electric UAVs, and how can these impacts be minimized?*

The definition of the research questions clarifies the main objectives of the thesis. These objectives are:

O1. *To develop a methodology for sizing and optimizing different UAV configurations across a range of applications.*

O2. *To characterize, quantify, and effectively mitigate uncertainties in UAV design.*

O3. *To establish a methodology for the controllability assessment and fault-tolerant sizing of UAVs.*

O4. *To develop and integrate an environmental assessment model into a preliminary design framework in order to improve the sustainability of UAVs.*

1.2.2 Scope of the thesis

Although the thesis aims to encompass a variety of design problems, for example by covering most UAV concepts (multirotor, fixed-wing, and hybrid FW-VTOL UAVs), the research is carefully scoped to make a focused contribution to the broad field of UAVs. First, hybrid configurations with tilting rotors or tilting surfaces are excluded from the study. In particular, only the quad-plane configuration is covered for hybrid FW-VTOL UAVs. Second, the research focuses on UAVs ranging from a few hundred grams to a dozen kilograms, and is therefore of limited relevance to micro-UAVs and medium to large UAVs. Similarly, high-altitude applications are outside the scope. For the categories of UAVs defined above, the most common type of propulsion consists of an electric propulsion system with batteries, and is therefore the focus of the study. Another specificity of UAVs in this size range is the frequent use of off-the-shelf components instead of custom-made parts. This point is of particular importance when establishing a methodology for design optimization, as discussed further in the manuscript. All traditional engineering disciplines are covered by the thesis, including aerodynamics, structures, propulsion, weights and geometry, in addition to controllability and environmental sustainability. Finally, the design process is specifically targeted at the conceptual and preliminary design phases, excluding detailed design aspects. This deliberate scoping ensures a focused exploration of critical issues within well-defined boundaries.

1.3 Dissertation overview

In accordance with the four research objectives, the manuscript is divided into four parts. Each part is further subdivided into a chapter exploring the state of the art and a chapter describing the research contributions proposed by the thesis.

Part I covers the sizing and optimization of UAVs in the context of preliminary design. Chapter 2 provides an overview of the methodologies and tools for sizing UAVs. Chapter 3 develops and implements a multidisciplinary design optimization framework for sizing multirotor, fixed-wing and fixed-wing VTOL UAVs.

Part II focuses on the uncertainty quantification and mitigation in the context of preliminary design. Chapter 4 introduces the fundamentals of uncertainty analysis and its applications in the early design phases. Chapter 5 proposes a methodology for evaluating and mitigating uncertainty for systems that are ultimately assembled from off-the-shelf components. Particular emphasis is placed on the uncertainty arising from the limited availability of components on the market, as opposed to a mathematical resolution of the design process in a continuous domain.

Part III addresses the safety aspects of UAVs, in particular the fault tolerance property. Chapter 6 provides an overview of the regulatory framework and the concepts for assessing the controllability and fault tolerance of UAVs. Chapter 7 presents a methodology for assessing the controllability of various UAV configurations in the presence of faults, and for linking the assessment to the sizing process.

Part IV explores the environmental impacts of electric UAVs. Chapter 8 introduces the life cycle assessment (LCA) methodology and its applications to aerial vehicles. Chapter 9 develops an LCA model specific to UAVs. The model is implemented in the multidisciplinary design optimization framework to enable the systematic evaluation and minimization of the environmental footprint.

Finally, Chapter 10 summarizes the results and findings. Conclusions are made and perspectives are outlined.

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Part I

Sizing and optimization of UAVs

The first part of the manuscript focuses on the development and implementation of a comprehensive methodology for the sizing and optimization of UAVs. Chapter 2 provides a critical review of existing methodologies and tools, highlighting the need for an open-source, multidisciplinary optimization framework that can accommodate various UAV configurations. Chapter 3 introduces a methodology for sizing multirotor, fixed-wing, and fixed-wing VTOL UAVs. The methodology incorporates analytical models for disciplines such as geometry, structures, aerodynamics, and propulsion, and implements strategies for reducing the problem complexity. The presented framework serves as a foundation for preliminary design studies and allows for quick comparison of different UAV concepts.

Author's contributions

Conference papers

F. Pollet, S. Delbecq, M. Budinger, and J.-M. Moschetta. Design optimization of multirotor drones in forward flight. In *32nd Congress of the International Council of the Aeronautical Sciences*, Shanghai, China, September 2021. <https://hal.science/hal-03832135>

F. Pollet, S. Delbecq, M. Budinger, J.-M. Moschetta, and J. Liscouët. A common framework for the design optimization of fixed-wing, multicopter and VTOL UAV configurations. In *33rd Congress of the International Council of the Aeronautical Sciences*, Stockholm, Sweden, September 2022. <https://hal.archives-ouvertes.fr/hal-03832115>

Chapter 2

State of the art of methodologies and tools for sizing UAVs

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2.1 Introduction

The design of a system is essentially an inverse engineering problem. Starting from a set of requirements and specifications, such as the operational capabilities to be achieved by the system, the design process aims to find the appropriate characteristics of the system and its components. The concept is illustrated in Figure 2.1. Depending on the characteristics to be determined, the task may be referred to as a sizing problem. Sizing is a sub-process of design that focuses on finding the appropriate dimensions of the components once the system configuration and technological choices have been established. As such, in the context of UAVs, the sizing problem consists of deriving the characteristics of the components (e.g., propeller diameter, battery size) given a specific configuration (e.g., quadcopter, hexacopter) and a set of mission requirements and technological choices (e.g., materials, battery chemistry).

The problem is further referred to as optimal sizing, or more broadly, optimal design, when the goal is to find the "best" components rather than good enough ones.

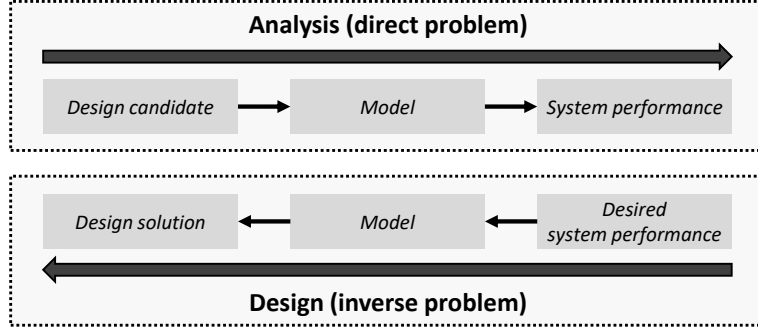


Figure 2.1 The inverse problem of system design.

Finding a solution to a design problem can be an arduous task. UAVs are complex systems in that they involve a number of components and disciplines. Solving an optimal design problem with respect to one discipline may lead to adverse effects in the other disciplines. For example, a UAV optimized for aerodynamic performance is more prone to structural damage: a high aspect ratio increases the aerodynamic efficiency of a wing, but compromises its structural strength. A well-designed UAV should consider the couplings between the disciplines to which it relates. A popular example is the detrimental "snowball effect" on the mass of any flying vehicle. Any increase in the mass of the system requires the installation of additional propulsion to counteract gravity, which in turn results in a mass increase. The snowball effect is particularly important for electric VTOL vehicles because the propulsion system, in particular the batteries, accounts for a large fraction of the total mass.

The purpose of this chapter is to introduce the means and tools for solving an optimal design problem when multiple disciplines are involved, and to review the existing methodologies for the preliminary sizing of UAVs.

2.2 Multidisciplinary design optimization

Multidisciplinary design optimization (MDO), also known as *multidisciplinary design analysis and optimization* (MDAO) is a key approach to the design process of complex engineering systems. It involves the simultaneous optimization of multiple disciplines within a system, aiming to find the "best" design that meets requirements and specifications. The history of MDO is intertwined with that of aircraft design [1]. Some of the earliest applications focused on coupling the aerodynamic and structural aspects of aircraft wings, considering them as interdependent disciplines and optimizing them together [2]. Before the recent emergence of MDO, the common approaches to design optimization were less systematic

and comprehensive. In the conventional design process, an initial design is proposed based on experience or generic principles. Its performance is evaluated and the designers decide whether it is sufficiently good based on the results. If not, modifications are made to the initial design. The process is repeated until a final satisfactory design is obtained. To speed up the search for a satisfactory solution, *designs of experiments* (DoE) can be used as a systematic approach to explore the design space, rather than relying on incremental changes. DoE aims to sample the design space to evaluate a wide range of design alternatives and understand the relationships between design variables and system performance. However, exploration of the entire solution space to exploit trade-offs and find a feasible and optimal design can be particularly challenging in complex systems.

The birth of computer science and algorithms enabled the adoption of numerical optimization to solve engineering problems. Design optimization enables a better end result to be achieved in a shorter development time. For systems involving multiple disciplines or components, the most common approach in the industry is to optimize each individual discipline sequentially, mimicking the organization of engineering groups. Even when interactions are accounted for, sequential optimization may converge to a suboptimal result. The advent of MDO represents a paradigm shift whose central idea is to find an optimal solution by simultaneously considering and optimizing a system as a whole with all relevant disciplines.

The following sections provide an overview of MDO, including the formulation of the optimization problem, the possible approaches to solving it, the optimization algorithms, and the available software resources.

2.2.1 Formulation of the optimization problem

The formulation of the optimization problem is a critical step that consists of translating the intent of the designer into a mathematical object that can be solved by an optimization algorithm. It involves defining the design variables that can be manipulated to generate design alternatives, the constraints to be met, and the objective function to be optimized. The choice of the design variables, constraints, and objective function in conjunction with the system description is up to the designer, and is guided by the specific goals and requirements of the design problem. The design variables are entries that are independently varied by the optimizer to represent different design solutions. Examples of design variables for a multirotor UAV are the length of the arms, the diameter of the propellers, and the size of the battery. Notably, different choices of variables can be used to describe the same object. For example, the cross-section of a hollow cylinder can be parameterized by its outer and inner diameters, or by its inner diameter and its ratio of inner-to-outer diameter. Finally, design variables can be either continuous or discrete, such as the choice of materials for the airframe. For each combination of design variables provided by the optimizer, the value of an objective function can be calculated. The objective function is a quantity that determines whether one design

is better than another. Typical objective functions for aerospace system design include mass (e.g., maximum takeoff weight), range, and aerodynamics (e.g., lift-to-drag ratio) [3]. In particular, the mass can be related to manufacturing and operating costs. The most common approach is to consider a single objective, but the combination of several objectives in the optimization problem is possible [4, 5]. Lastly, constraints are functions of the design variables whose values must be enforced to keep the solution within a feasible region. Constraints can be related to mission requirements (e.g., endurance), safety requirements (e.g., bending stress at the root of a wing), and system controls, among others.

Overall, the aim of the optimizer is to minimize (or maximize) the objective function by varying the design variables, while adhering to constraints. Mathematically, it can be formalized as follows:

$$\begin{aligned}
& \text{minimize} && f(\mathbf{x}) \\
& \text{with respect to} && x_{i,\min} \leq x_i \leq x_{i,\max}, \quad i = 1, \dots, n_x \\
& \text{subject to} && g_j(\mathbf{x}) \leq \mathbf{0} \quad j = 1, \dots, n_g \\
& && h_k(\mathbf{x}) = \mathbf{0} \quad k = 1, \dots, n_h
\end{aligned} \tag{2.1}$$

In this formulation, the objective function to be minimized is denoted as $f(\mathbf{x})$, where $\mathbf{x} = [x_1, \dots, x_{n_x}]$ represents a vector of design variables. These design variables, x_i , are subject to bounds specified as $x_{i,\min}$ and $x_{i,\max}$. The optimization problem also encompasses both inequality constraints and equality constraints. In practice, equality constraints can be replaced by two inequality constraints $h(\mathbf{x}) \leq \varepsilon$ and $-h(\mathbf{x}) \leq \varepsilon$, with ε sufficiently small and chosen with respect to the precision of the optimization. The bounds on the design variables are distinct from the constraints, as the optimizer has direct control over them, so are treated differently in the optimization process. Finally, a maximization objective can be expressed with the same convention by inverting the function ($\max[f(\mathbf{x})] = \min[1/f(\mathbf{x})]$) or changing its sign ($\max[f(\mathbf{x})] = -\min[-f(\mathbf{x})]$).

In practice, the calculation of the objective and constraint functions requires the solution of analyses that model the behavior of each component or discipline involved in the system. These analyses can take the form of explicit equations or a set of implicit models. In the latter case, the analysis can be solved using mathematical tools such as numerical solvers. Popular numerical solvers include linear fixed-point methods such as Jacobi or Gauss-Seidel, or the more sophisticated Newton-Raphson method. However, solving the equations explicitly, if feasible, will always result in shorter computation times.

To gain a comprehensive understanding of the optimization procedure, the *extended design structure matrix* (XDSM) can be employed as a visual representation [6]. The XDSM is a diagram that provides an illustration of data dependencies and process flow in MDO processes. Computational analyses are represented by rectangles. Data inputs and outputs are represented by parallelograms, and data exchange between computational processes is

highlighted by thick gray lines — vertical for inputs and horizontal for outputs. A rounded rectangle indicates an optimizer or solver driving the process. Finally, thin black lines enable visualization of the process flow, i.e. the sequence of executions. In an advanced version, the order in which the calculations are executed is defined by numbering. For clarity, the XDSMs provided in this manuscript are simplified by removing the numbering. In the problems presented, the sequence of executions can be deduced from the order of the analyses in the XDSM structure.

Figure 2.2 shows the resolution of a generic optimization problem. Starting from an initial value $\mathbf{x}^{(0)}$ of the design variables ¹, the optimizer runs the analyses for the corresponding design, and evaluates the objective and constraints. Based on the results, an updated set of design variables is computed. The sequence is repeated until the optimizer converges to an optimal solution $\mathbf{x}^*$. If feasible, the computations can be run in parallel, as indicated by the presence of two thin black lines for objective and constraint evaluation.

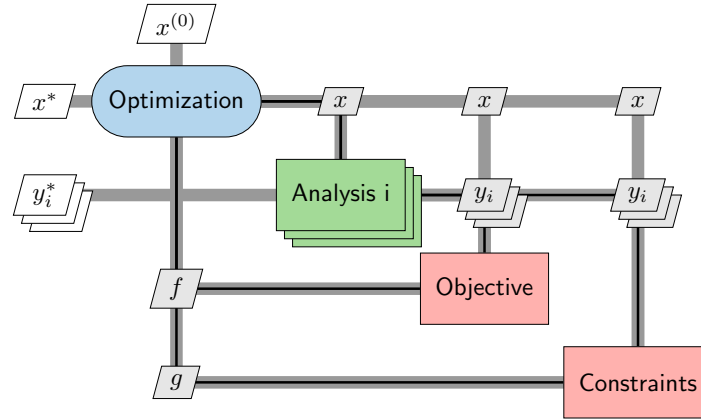


Figure 2.2 XDSM showing the optimization procedure.

2.2.2 Strategies for solving MDO problems

One of the fundamental challenges of MDO lies in the interdependencies between disciplines, with one analysis requiring the outputs of other analyses as inputs. This interdependence reflects the intricate relationships within the system, where changes in one discipline or component can impact and require updates in others. Figure 2.3 shows a generic system consisting of three components with coupled analyses. The coupling relationships can be categorized as either feedforward (analysis 1 $\rightarrow$ 2) or feedback (analysis 3 $\rightarrow$ 2) connections.

Some coupling relationships can be explicitly solved by sequentially computing the analyses, as is the case for the relationships 1 $\rightarrow$ 2 and 2 $\rightarrow$ 3 in Figure 2.3. This requires an appropriate ordering of the analyses. For example, switching the computational order

¹Note that some optimization algorithms do not require an initial guess of $\mathbf{x}$, as explained further below.

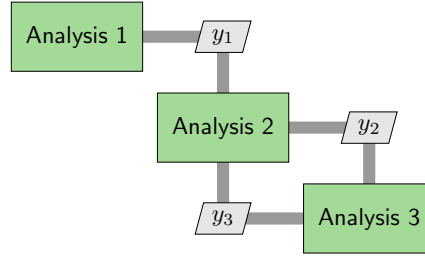


Figure 2.3 Dependencies for a generic three-component coupled system.

between analyses 1 and 2 would prevent relationship $1 \rightarrow 2$ from being resolved in a direct manner. Other coupling relationships cannot be resolved explicitly and require iterative techniques for convergence. This is the case for the *maximum takeoff weight* (MTOW) in aircraft design, whose calculation requires the aircraft geometry, propulsion system and fuel burn to be known. However, these analyses are derived from the MTOW [7], which leads to an algebraic loop. An illustration of an algebraic loop can be seen in Figure 2.3 between analyses 2 and 3. Overall, the process of resolving the coupled system is known as a *multidisciplinary analysis* (MDA). Different methods can be employed for the MDA. In the Gauss-Seidel method, the feedforward coupling relationships are explicitly solved by running the analyses in a sequence, whereas the remaining couplings are addressed through an iterative procedure. More specifically, each iteration starts with initial guesses y_i^t of the coupling variables and runs the analyses given these guesses. Once the analyses are solved, the values of the coupling variables are updated based on the actual solutions y_i returned by the computations. The process is iterated until it converges. Figure 2.4a shows an XDSM of the Gauss-Seidel method. The Jacobi method is similar, the only difference being that all coupling relationships are solved with the iterative procedure. That is, Jacobi does not use sequential computation to solve feedforward relations with the latest information available. This generally causes Jacobi to converge less quickly than Gauss-Seidel [8, 9]. However, since no analysis requires the prior computation of the others, Jacobi iterations can be performed in parallel, as shown in Figure 2.4b.

The MDA is only responsible for solving the coupling relationships and does not cover the optimization of the coupled system, i.e. the MDO. Different options exist for solving MDO problems, which are referred to as MDO architectures. Some of these architectures are described here. The *multidisciplinary design feasible* (MDF) architecture is the most intuitive approach in that it applies optimization directly to the MDA. In that sense, the multidisciplinary optimization is the same as a single-discipline optimization, except that the evaluation of the objective and/or constraints requires solving the coupled system. An XDSM of the MDF architecture is provided in Figure 2.5, where a Gauss-Seidel method is used for the MDA. Of course, other methods can be used to solve the coupled system, for example

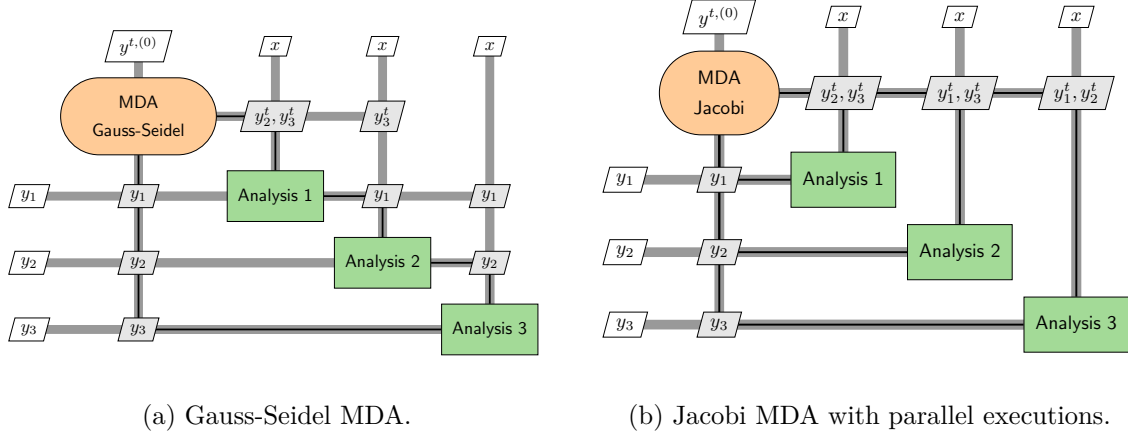


Figure 2.4 MDA procedures for solving a coupled system.

Jacobi. The main advantage of the MDF architecture is its simplicity of implementation and understanding. However, this approach requires solving an MDA at each iteration of the optimization process, which can be time-consuming and not always successful in converging.

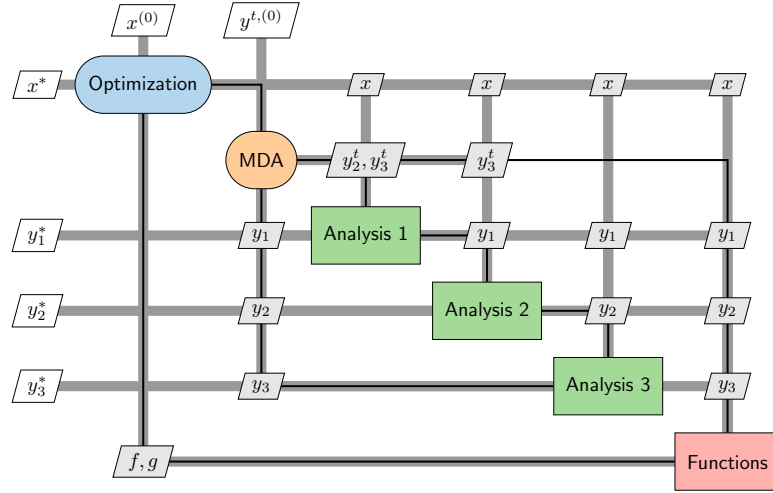


Figure 2.5 MDF architecture relying on a Gauss-Seidel MDA to solve the couplings at each optimization iteration.

To accelerate convergence to an optimal solution, the *individual discipline feasible* (IDF) architecture gives full control of the coupling variables to the optimizer. A dedicated MDA solver is no longer required. Instead, guesses of the coupling variables, y_i^t , also referred to as consistency variables, are introduced into the optimization problem, and constraints are added to ensure that the coupling variable guesses match the actual outputs of the analyses at the optimal solution: $h_i^c = y_i^t - y_i = 0$. Consequently, the optimization problem is reformulated

as follows²:

$$\begin{aligned}
 & \text{minimize} && f(\mathbf{x}, \mathbf{y}(\mathbf{x}, \mathbf{y}^t)) \\
 & \text{with respect to} && \mathbf{x}, \mathbf{y}^t \\
 & \text{subject to} && g(\mathbf{x}, \mathbf{y}(\mathbf{x}, \mathbf{y}^t)) \leq \mathbf{0} \\
 & && h^c(\mathbf{x}, \mathbf{y}(\mathbf{x}, \mathbf{y}^t)) = \mathbf{y}(\mathbf{x}, \mathbf{y}^t) - \mathbf{y}^t = \mathbf{0}
 \end{aligned} \tag{2.2}$$

The introduction of coupling variable copies $\mathbf{y}^t$ allows the analyses to be performed in parallel. Furthermore, the consistency of the coupled system is ensured only at the end of the optimization by solving the consistency constraints. This allows the optimization to take a more direct path to the optimal solution. However, if the process is interrupted prematurely, the multidisciplinary analysis will not be valid. The main drawback of the IDF architecture is the large number of variables to manipulate, which can be a challenge when many disciplines are involved. Figure 2.6 illustrates the IDF architecture.

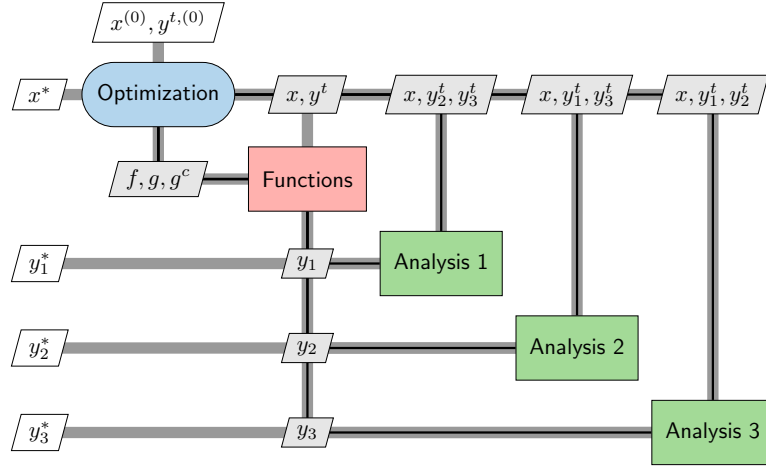


Figure 2.6 IDF architecture where the optimizer is responsible for solving the couplings. Each analysis can be executed in parallel.

Hybrid MDO architectures can be created by combining elements from MDF and IDF [10–13]. For example, some coupling relationships could be solved with a dedicated MDA solver, while the remaining ones are handled by the optimizer. Figure 2.7 shows an example of hybrid architecture that resembles the IDF, except that sequential computation is used to solve feedforward coupling relationships, while the remaining algebraic loops are handled by introducing coupling variable copies and consistency constraints.

Starting from a similar hybrid architecture, Delbecq [13, 14] proposed the *normalized variable hybrid* (NVH) formulation for solving design problems of actuation systems. First, the coupling variable copies are now expressed as multiplications of known quantities: $y_i^t = k_i \cdot y_0$. This approach is illustrated through the solving of an algebraic loop for an electromechanical

²Note, again, that equality constraints can be replaced by two inequalities.

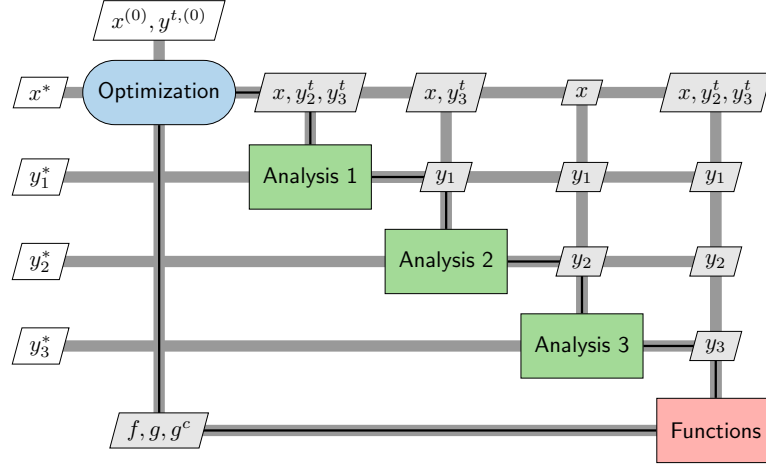


Figure 2.7 Hybrid architecture where the feedforward relationships are calculated in a sequence, while the feedback relationships are solved by the optimizer.

actuator [14, 15]. The electromagnetic torque T_{em} of the motor is a function of its inertia J , whose determination requires the electromagnetic torque to be known:

$$T_{em} = J \cdot A_{max} + T_{load} \quad (2.3a)$$

$$J = f(T_{em}), \quad (2.3b)$$

where A_{max} is the actuator acceleration rate and T_{load} is the external load to sustain, whose values are known. In the IDF architecture, the algebraic loop is solved by introducing copies of the motor inertia (J^t) and the electromagnetic torque (T_{em}^t) along with consistency constraints. A hybrid approach would be to keep only one coupling variable copy, J^t , to solve for the motor torque, and then compute the actual motor inertia directly from the current value of T_{em} . The NVH architecture proposes to obtain an initial guess of the motor torque by removing the unknown variable J from the torque equation and introducing a so-called sizing coefficient [15]:

$$T_{em}^t = k \cdot T_{load} \quad (2.4a)$$

$$J = f(T_{em}^t), \quad (2.4b)$$

where k is the sizing coefficient in the range $]0, +\infty[$ (in practice most likely in $[1, 10]$). The optimizer is responsible for varying the value of the sizing coefficient to ensure consistency between the torque guess T_{em}^t and its actual value T_{em} obtained from Equation 2.3a. In this formulation, the value taken by the coupling variable (here, k) is independent of the specification T_{load} . This is a major advantage over the IDF or hybrid architectures, where the converged values of T_{em}^t and J^t are specific to the specifications. The second modification

proposed by the NVH architecture is to replace the equality consistency constraint by a single inequality constraint. This is only possible for certain design problems where it is expected that optimization will inevitably lead to the inequality consistency constraint being active at the optimal point. An inequality constraint is active when $g(\mathbf{x}^*) = 0$. In the electromechanical actuator design problem, a mass minimization objective is chosen, such that the optimization leads to minimal motor torque. Indeed, the motor mass is proportional to its torque. As a consequence, the inequality $T_{em}^t \geq T_{em}$ is expected to be active at the optimum.

In the present manuscript, the NVH architecture applied to actuator systems design in [13] is formally described with a generic problem formulation:

$$\begin{aligned}
 & \text{minimize} && f(\mathbf{x}, \mathbf{y}(\mathbf{x}, \mathbf{k})) \\
 & \text{with respect to} && \mathbf{x}, \mathbf{k} \\
 & \text{subject to} && g(\mathbf{x}, \mathbf{y}(\mathbf{x}, \mathbf{k})) \leq 0 \\
 & && g^c(\mathbf{x}, \mathbf{y}(\mathbf{x}, \mathbf{k})) = \mathbf{y}(\mathbf{x}, \mathbf{y}^t) - \mathbf{y}^t(\mathbf{x}, \mathbf{k}) \leq 0
 \end{aligned} \tag{2.5}$$

The sign of the inequality consistency constraint may be reversed depending on the relationship between the coupling variable and the objective function. By reducing the number of variables compared to IDF and the number of constraints compared to other hybrid architectures, the NVH approach offers efficient computational cost [14]. Most importantly, it is robust to scale changes in the problem thanks to the sizing coefficients. The main drawback of the NVH architecture is that it requires a modification of the governing equations of the problem, making it more invasive than traditional MDO strategies.

Finally, other MDO architectures, generally referred to as distributed architectures, decompose the optimization problem into a set of smaller problems to optimize each discipline, and then coordinate them at the system level with a dedicated optimizer. Unlike the monolithic architectures described above, these distributed architectures mimic the structure of large engineering groups, which allows for better control of the design space to avoid unphysical solutions. However, they are less computationally efficient than monolithic architectures that use a single optimizer [16–18].

2.2.3 Optimization algorithms

Solving an MDO problem requires optimization algorithms to effectively guide the design toward an optimal solution. The choice of an appropriate algorithm is driven by its reliability and computational efficiency, both of which depend on the nature of the problem. The main decision criteria for selecting an optimization algorithm include the smoothness of the models (whether they are continuous and differentiable), the multimodality of the objective function (presence of a single global minimum or multiple local minima), the number of design variables and their nature (continuous vs. discrete), and, to a lesser extent, the presence of

optimization constraints.

Gradient-based optimization

Optimization algorithms are generally classified into two categories: gradient-based algorithms and gradient-free algorithms. Gradient-based algorithms use the gradient information of both the objective and the constraints with respect to the design variables to converge to the optimum more efficiently. A comparison between the cost of gradient-based algorithms and their gradient-free counterparts for a typical problem, the Rosenbrock function, is illustrated in Figure 2.8. Gradient-based algorithms require that the functions of the problem are sufficiently

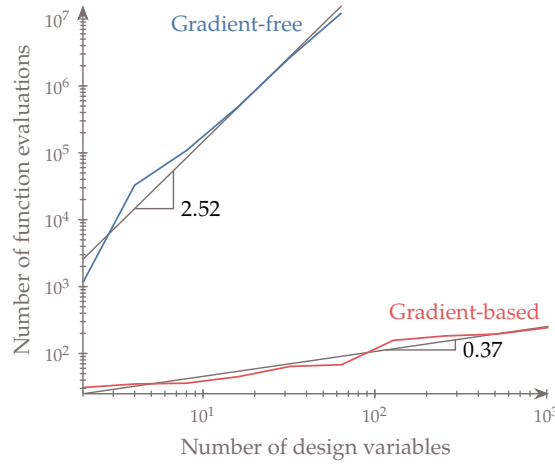


Figure 2.8 Cost increase of gradient-based versus gradient-free optimization algorithms with the number of design variables. Source: Martins *et al.* [8].

smooth. In theory, the differentiability property is required, but some discontinuities can be handled in practice as long as they do not occur near the optimum. All gradient algorithms operate by performing a sequence of steps leading from an initial point, $\mathbf{x}_0$, to an optimal point, $\mathbf{x}^*$, by moving along suitable directions in the design space. Different methods exist to find appropriate search directions at each iteration. The steepest descent approach is the most simple in that the search direction is defined as the opposite of the objective function's gradient, $-\nabla f$. Newton and Quasi-Newton methods like Broyden-Fletcher-Goldfarb-Shanno (BFGS) use second-order information to get better estimates for the search directions. The search direction $\mathbf{d}$ at each iteration is obtained by solving the linear system $H_f \mathbf{d} = -\nabla f$, where H_f is the Hessian matrix of the second-order partial derivatives. Compared to the steepest descent, Newton's methods converge to the optimum in fewer steps.

The sequence of steps taken iteratively in the design space terminates when optimality conditions are met. Sufficient optimality conditions for unconstrained optimization problems

are [19]:

$$\nabla f(\mathbf{x}^*) = \mathbf{0} \quad (2.6a)$$

$$H_f(\mathbf{x}^*) \text{ is positive definite.} \quad (2.6b)$$

The second condition ensures that $\mathbf{x}^*$ is a minimum rather than a saddle point or a maximum. Note that in most cases gradient-based algorithms cannot prove that the optimum found is a global optimum, but only a local one. If multimodality of the objective function is suspected, multi-start procedures, which consist of running the optimization for different start values $\mathbf{x}_0$, increase the likelihood of finding a global optimum.

For constrained optimization problems, the solution $\mathbf{x}^*$ may not correspond to an optimum point of the objective function, as shown in Figure 2.9. Null gradient and positive definiteness of the Hessian matrix are no longer sufficient conditions. Additional conditions are required to

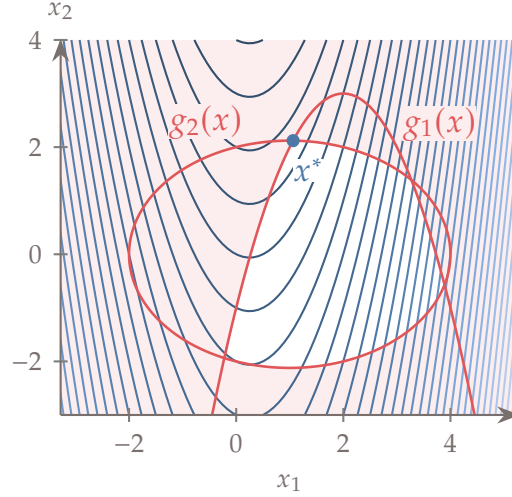


Figure 2.9 Solution for a constrained optimization problem. Blue lines represent the contours of the objective. Shaded red areas depict infeasible regions. Source: Martins *et al.* [8].

ensure that the solution lies within the feasible space and that any point that would be better in its vicinity does not belong to this feasible space. The Karush–Kuhn–Tucker theorem provides necessary conditions for the optimality of a solution $\mathbf{x}^*$ satisfying the constraints [19]:

$$\nabla f(\mathbf{x}^*) + \sum_{j=1}^{n_g} \sigma_j \nabla g_j(\mathbf{x}^*) + \sum_{k=1}^{n_h} \lambda_k \nabla h_k(\mathbf{x}^*) = \mathbf{0}, \quad \sigma_j, \lambda_k \geq 0 \quad (2.7a)$$

$$\sigma_j g_j(\mathbf{x}^*) = 0, \quad j = 1, \dots, n_g \quad (2.7b)$$

The first condition indicates that the gradient of the objective function is a linear combination of the gradients of the constraints that define the unfeasible space (Figure 2.10). In other

words, pursuing the descent direction $\nabla f(\mathbf{x}^*)$ would lead to entering the unfeasible space. The second condition allows to distinguish active constraints ($g_j(\mathbf{x}^*) = 0$) from inactive ones ($g_j(\mathbf{x}^*) < 0$), as only active constraints are involved to define the unfeasible space in the vicinity of $\mathbf{x}^*$. A constraint is inactive if $\sigma_j = 0$, such that inactive constraints are removed from the first condition equation. These necessary conditions become sufficient to define a

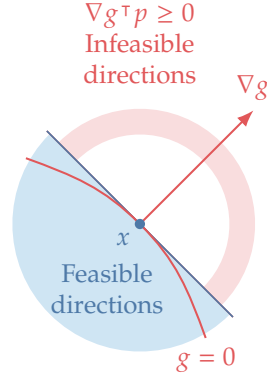


Figure 2.10 Illustration of the feasible and unfeasible directions expressed from the constraint gradients. Source: Martins *et al.* [8].

minimum solution if, in addition, the feasible directions around $\mathbf{x}^*$ have positive curvature, which can be verified from the Hessian matrices. A popular optimization method that handles constraints is the sequential quadratic programming (SQP) method. SQP can be seen as a generalization of Newton's methods that incorporates the constraint gradients into the computation of the search direction. Implementations of SQP include the open-source Python package SLSQP [20] and the `fmincon` function of Matlab.

The conditions to establish the optimality of a solution indicate the necessary computation of accurate derivatives of both the objective and the constraint functions. This information can be provided by the user in the form of analytical functions of the derivatives, or estimated by the algorithm through numerical calculation, for example with forward difference:

$$\frac{\partial f}{\partial x}(x) = \frac{f(x+h) - f(x)}{h} + O(h), \quad (2.8)$$

where h is a step size. Numerical computation of derivatives in place of analytical evaluation increases the overall computational time of the MDO [14]. When multiple models are involved in the coupled system, approaches such as MAUD assemble them to obtain the total system derivatives with minimal implementation effort [21].

Finally, scaling of the design variables, the objective function and the constraints is a critical issue in design optimization with gradient-based algorithms. Poor scaling can force the optimization algorithm to navigate a design space that is steep in one direction and shallow in the other, causing abrupt changes in search directions or premature convergence due to accuracy issues. Good practice is to scale the design variables, the objective function

and the constraints to be close to unity, depending on the problem. For this reason, the NVH architecture introduced earlier is particularly relevant because it tends to manipulate normalized variables.

Gradient-free optimization

When the models are not continuous and hardly differentiable, or the problem dimensionality is relatively small, gradient-free algorithms can be used. The rationale behind gradient-free methods is that only the function values are needed and no continuity assumption is made. Unlike gradient-based approaches, these methods do not always come with proofs of convergence to optimal solutions. Termination criteria for gradient-free algorithms usually involve a lack of improvement in the objective function or a maximum number of iterations [22]. A wide variety of gradient-free algorithms exist [23], and experimentation is often required to select the most appropriate one for the problem at hand. Algorithms may involve direct evaluations of the objective function (Nelder-Mead, genetic algorithms, particle swarm) or the construction and use of a surrogate model to guide the search process (COBYLA, EGO). In addition, they can be classified as deterministic or stochastic, depending on whether they rely on random features or not.

The Nelder-Mead simplex method [24] is a deterministic, direct search approach among the first gradient-free algorithms to be developed. The process starts with a set of points that form a simplex (i.e., a triangle in two dimensions, a tetrahedron in three dimensions). At each iteration, the algorithm modifies the simplex by replacing the vertex point with the worst objective value. It does so by applying transformations about the centroid of the simplex, such as reflection, expansion, and contraction. The method is simple to implement, however it shows limitations for high dimensional problems and is sensitive to initial conditions and scaling of the design variables. In its original implementation, the Nelder-Mead method is limited to unconstrained problems.

COBYLA [25] is another deterministic method. Its novelty is that it uses information about the constraints to achieve better efficiency. COBYLA involves forming linear approximations of the objective function and the constraints by interpolation at the vertices of a simplex. Similarly to the Nelder-Mead method, the algorithm iteratively proposes new vertices for this simplex until convergence. COBYLA was one of the first model-based approaches for solving constrained problems. In a general sense, model-based methods sample the search space (either locally or globally) to build a surrogate model of the objective function. Model-based methods are usually applied when the functions are costly to compute, as the surrogate model is easier to evaluate. One of the most widely used model-based algorithm is efficient global optimization (EGO) [26]. It employs a Bayesian strategy to model the objective as a random function. At each iteration, the additional sampling points are chosen to strike a balance between exploration (improving the accuracy of the surrogate model) and exploitation (finding the optimal solution). Recently, improved versions of EGO have enabled the handling

of constraints [27], better efficiency for high-dimensional problems [28–30], and optimization with mixed continuous, integer and categorical variables for aircraft design problems [31].

Finally, evolutionary algorithms are a broad class of stochastic methods that involve the nondeterministic generation of a large number of points at each iteration, often referred to as populations. They are particularly useful for discontinuous, noisy, multimodal functions, as the evaluation points are spread out in the design space. However, optimality is only judged statistically, so a large enough number of candidates should be evaluated to maximize the likelihood of approaching the actual optimum. Common evolutionary algorithms include genetic algorithms [32, 33], particle swarm optimization [34], simulated annealing [35], and CMA-ES [36–39]. CMA-ES, for Covariance Matrix Adaptation Evolution Strategy, is one of the most promising gradient-free algorithms with EGO and its variants, compared to which it demands higher computational budget but is more robust in finding the global optimum [40]. CMA-ES starts by sampling the search space with a multivariate normal distribution (maximum entropy principle), and iteratively updates the sampling points based on the latest information available. At each new sampling step, the multivariate normal distribution is updated to be centered on the favorite solution, and its covariance matrix, which represents the shape of the distribution, is adjusted to follow a natural gradient approximation of the objective obtained from the previous successful search steps. This allows the search path to accumulate information from successive generations to converge to the optimal solution faster, while the nondeterministic sampling prevents premature convergence. An illustration of CMA-ES is provided in Figure 2.11. Particular attention should be paid to the bounds on the design variables for optimization algorithms that sample the entire design space, such as evolutionary algorithms. In fact, the search tends to explore the entire design space specified by the variable bounds, so unnecessarily large bounds reduce the efficiency of the algorithms.

An additional reason for using gradient-free algorithms over their gradient-based counterparts is the presence of discrete design variables, since the concept of derivatives does not hold for such variables. Although some strategies exist to adapt gradient-based search to discrete variables [42], gradient-free algorithms are better suited to solve these types of problems.

2.2.4 Optimization software resources

Several proprietary and open-source platforms are available to implement and solve MDO problems [43]. Open-source frameworks include OpenMDAO [44], GEMSEO [45] and CoS-App [46], all of which are based on the Python programming language. They enable the multidisciplinary resolution of generic systems with diverse MDO architectures, and integrate algorithms from different libraries (SciPy [47], NLOPT [48], pyOptSparse [49]) for optimization and designs of experiments. In addition, specific tools such as FAST-OAD [7],

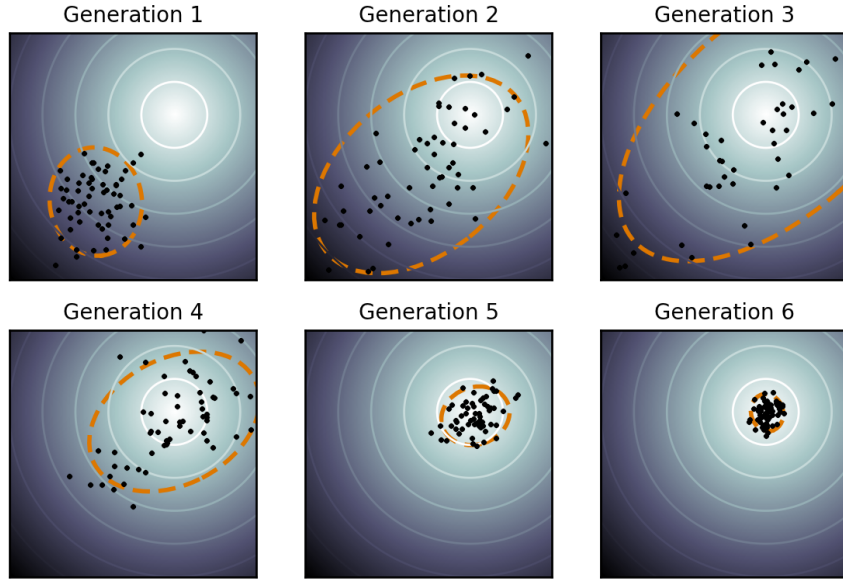


Figure 2.11 Illustration of the CMA-ES algorithm. Source: Wikipedia [41].

SUAVE [50–53], and OpenConcept [54] are tailored for overall aircraft design. Co-developed by ISAE-SUPAERO and ONERA, FAST-OAD enables aircraft design optimization and analysis. It provides model libraries for CS25 and CS23 aircraft and relies on the OpenMDAO framework to implement the MDO problems. It features pre- and post-processing utilities through Python notebooks [55] to easily define the problem to be solved, add or replace models that describe the system, and visualize the results. Similarly to FAST-OAD, OpenConcept enables the conceptual design of aircraft, with a focus on electric propulsion concepts. The tool is built on top of the OpenMDAO framework, and integrates OpenAeroStruct [56] for aerostructural aerodynamic models. SUAVE offers similar capabilities while addressing a broader range of aerospace vehicles, including distributed electric propulsion architectures [57], eVTOL aircraft [58] and UAVs [59]. In contrast to FAST-OAD and OpenConcept, SUAVE does not rely on a well-defined MDO approach. Therefore, the handling of multidisciplinary couplings is rather case-specific and requires additional implementation effort.

2.3 Model construction

Models are the building blocks of multidisciplinary design optimization. In fact, the models and their parameters define the possible problem formulations (design variables, constraints, and objective function), the multidisciplinary couplings to be solved, and the optimization algorithms that can be employed. For the early design stages, a good model captures the essential trade-offs to help the designer or the optimization algorithm make informed decisions with physical meaning. An optimizer will tend to exploit any weakness in the

models, potentially leading to unrealistic results. For this reason, one should start with simple models that provide a good understanding of the phenomena, and only then add complexity so that the possible weaknesses introduced by additional parameters and interactions are well identified. This section presents some techniques for the construction of simple models for use in design optimization.

2.3.1 Terminology

For the purpose of preliminary design, Budinger [60] proposes to distinguish the component parameters according to three categories:

- Integration parameters are used to evaluate the masses and main dimensions of the components.
- Simulation parameters (e.g. the resistance of an electric motor) are utilized by simulation models to perform the analysis of operating scenarios.
- Operating domain parameters (e.g. maximum torque of a motor) are used to represent the safe operating range of the components. They are associated with evaluation models, the goal of which is to check the ability of the components to safely fulfill their missions.

Describing a component requires determining the values of all three types of parameters, which can be a difficult process in the early stages of design when little knowledge is available. To reduce the complexity of the task, *estimation models* are used to link the integration, simulation, and operating domain parameters to a reduced set of primary characteristics called definition parameters. For example, most of the characteristics of an electric motor can be derived from its nominal torque and its torque/velocity constant. Estimation models are essential because they allow the full set of component parameters to be calculated prior to subsequent evaluation. The relationships between the parameters and models introduced here are summarized in Figure 2.12.

2.3.2 Modeling techniques

We now discuss a variety of approaches to support the modeling effort, with a particular focus on estimation models. Starting from a set of experimental observations, a simple way to build a model is to preserve the discrete structure of the collected data. For example, existing components with measured characteristics can be represented in the form of a database. The direct use of databases in the design process provides precise information on the parameters but leads to an explosion of combinatorial solutions and therefore significantly increases the computation time. On the other hand, mathematical models with algebraic forms are generally advantageous for optimization studies. For this reason, most design methodologies rely on continuous mathematical models rather than discrete data. Mathematical models

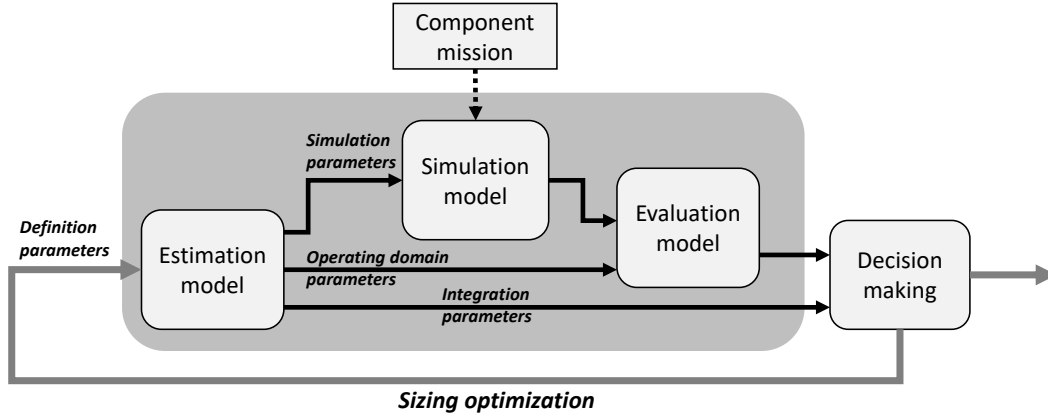


Figure 2.12 Types of parameters and models involved in the preliminary design of systems. Source: Budinger [60].

and the correlations they provide can be classified according to the degree of experimental data and physical relationships involved in their construction.

Empirical correlations

Empirical correlations are based entirely on statistical data from previous component or system designs. They provide rough estimates of the key parameters that drive the design, and are typically used in the early stages of conceptual design. Empirical methods have been widely used to derive weight estimation relationships for conventional aircraft [61–63]. Because they represent the data from which they are derived, empirical correlations have limited applicability. Using correlations originally developed for commercial aircraft could lead to over- or underestimation of the characteristics when applied to small UAVs. Several authors have proposed empirical correlations for different categories of UAVs [64–68]. Götten *et al.* [64] and Verstraete *et al.* [65] analyzed large databases of existing fixed-wing UAVs to derive correlations for initial design characteristics such as wing area and fuselage length as functions of total mass, as shown in Figure 2.13. Although the relationships are established across a wide range of UAV sizes and propulsion systems (e.g., electric and fuel-powered), they do not capture the specificities of the technologies used in the design (e.g., battery chemistry), nor do they describe novel architectures. Bershadsky *et al.* [68], Gur [69] and Alulema *et al.* [67] introduced correlations based on off-the-shelf components for the preliminary sizing of propulsion systems for UAVs. For example, the relation between the mass m_{bat} [kg] and energy [Wh] of a lithium-polymer battery may be given by [69]:

$$E_{bat} = 4.04m_{bat}^2 + 139m_{bat} + 0.0155 \quad (2.9)$$

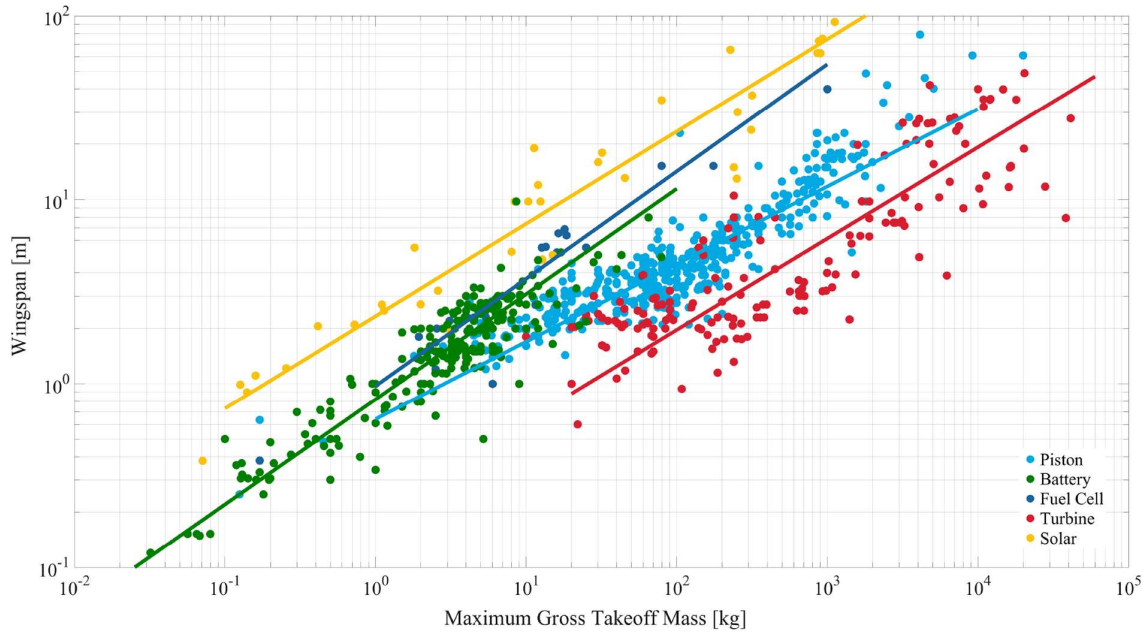


Figure 2.13 Wingspan as a function of maximum takeoff weight for various families of fixed-wing UAVs. Source: Verstraete *et al.* [65].

Empirical correlations can be obtained by curve fitting or more advanced methods. Curve fitting involves choosing a functional representation, typically a polynomial or power law, and adjusting its parameters to minimize the error between the data points and the functional curve. The best-fit problem is generally solved using the least squares method, which aims to find the parameters that minimize the sum of the squared deviations from the data points. More advanced methods, such as neural networks and kriging models, are able to fit observations without requiring a specific functional form of the model.

Because experimental correlations require exhaustive data collection of existing UAVs and components, the exploration of recent technological advances and concepts cannot be captured. This is an issue for the UAV design domain, where the integration of new technologies is constantly evolving.

Analytical and quasi-analytical models

Analytical and quasi-analytical models are based on physical phenomena. By improving the understanding of key physical and technological effects, such models are better suited for extrapolation than purely statistical methods. An example of an analytical model is the description of the mass of a component given its geometry and materials. The mass of a cube-shaped component with uniform composition is given by $m = \rho \cdot H \cdot h \cdot L$, where ρ is the material density and H , h , L are the component dimensions.

When an analytical expression is too complex (e.g., involves many unknown parameters), or the relationship between the physical parameters is not explicitly determined, dimensional analysis and Buckingham's theorem [70, 71] can be applied to obtain estimation models with a reduced number of inputs. The process, recalled in [60], starts by expressing a characteristic y of a component as a function f of its geometrical dimensions and physical or material properties:

$$y = f(\underbrace{L, d_1, d_2, \dots}_{\text{geometric}}, \underbrace{p_1, p_2, \dots}_{\text{physical}}) \quad (2.10)$$

$n \text{ parameters, } u \text{ physical units}$

The function f does not need to be explicitly stated. However, its variables must represent the main physical quantities involved in the problem. According to Buckingham's Theorem, Equation 2.10 can be rewritten with a reduced set of $n - u$ dimensionless parameters π_i , where n and u are respectively the number of parameters and physical units (e.g., time, mass, length) initially involved:

$$\underbrace{\pi_y = f'(\pi_{d_1}, \pi_{d_2}, \dots, \pi_{p_1}, \pi_{p_2}, \dots)}_{n - u \text{ parameters}}, \quad (2.11)$$

where

$$\pi_y = y \cdot L^{a_L} \prod_i p_i^{a_i} \quad (2.12a)$$

$$\pi_{d_i} = \frac{d_i}{L} \quad (2.12b)$$

$$\pi_{p_i} = L^{a_{p_i,0}} \prod_j p_j^{a_{p_i,j}} \quad (2.12c)$$

From this reduced form, an estimation model of the dimensionless number π_y can be derived, whose form depends on the properties of the dimensionless numbers π_{d_i} and π_{p_i} . In particular, several assumptions can be made. First, geometric similarity assumes that the geometric proportions are identical across a range of products. That is, a change in one dimension d_i is accompanied by the same relative change in the other dimensions d_j . For example, the length-to-diameter ratio of permanent-magnet motors in a given industrial range can be considered constant from one motor to another [60]. Second, material similarity assumes that the material properties are the same for all products in a range.

If both geometric and material similarity can be assumed, the dimensionless numbers π_{d_i} , π_{p_i} are considered constant. As a result, π_y is constant and a direct proportionality relationship can be established between the quantities involved in π_y :

$$\pi_y = y \cdot L^{a_L} \prod_i p_i^{a_i} = f'(\pi_{d_i}, \pi_{p_i}) = \text{Constant} \quad (2.13)$$

Again, if the material or physical properties p_i are assumed to be constant, the characteristic y can be estimated from the main dimension L of the component:

$$y \propto L^a \quad (2.14)$$

This relationship is referred to as a scaling law. It allows a component characteristic to be extrapolated from a single reference component whose parameters are known, with a typical power law:

$$\frac{y}{y_{ref}} = \left(\frac{L}{L_{ref}} \right)^a \iff y = k \cdot L^a = \frac{y_{ref}}{L_{ref}^a} \cdot L^a \quad (2.15)$$

The reference component is chosen to reflect a technology, a size range, or manufacturing choices. Application of Buckingham's theorem to brushless motors for UAVs leads to Equations 2.16a and 2.16b [72]:

$$J \propto L^{-0.5} \quad (2.16a)$$

$$\frac{T}{J} \propto L^4, \quad (2.16b)$$

where T is the electromagnetic torque of the motor, J the winding current density, and L the motor length. Combining both equations enables to estimate the motor mass m with respect to its torque:

$$m \propto L^3 \propto T^{3/3.5} \iff m = \frac{m_{ref}}{T_{ref}^{3/3.5}} \cdot T^{3/3.5} \quad (2.17)$$

Figure 2.14 shows the proposed scaling law fitted with three different reference motors, and its comparison with actual data.

In many cases, the components do not comply with the assumptions of geometric similarity or material similarity, and scaling laws cannot be derived. Instead, the function $\pi_y = f'(\pi_{d_i}, \pi_{p_i})$ must be approximated with data regressions. Compared to the empirical correlations introduced earlier, the use of a reduced number of dimensionless parameters to build a model has the advantage of requiring a significantly reduced amount of data, and increases the robustness of the regression [73]. In addition to traditional polynomial curve fitting and kriging techniques, the *Variable Power Law Metamodel* (VPLM) methodology [74] can be used to build a regression model using dimensional analysis.

The overall process for deriving estimation models based on dimensional analysis is summarized in Figure 2.15. Finally, it should be noted that the distinction between empirical correlations and regression models based on physical insights is not always clear. In fact, empirical correlations are often based on an implicit expert-based criterion for selecting the parameters of interest.

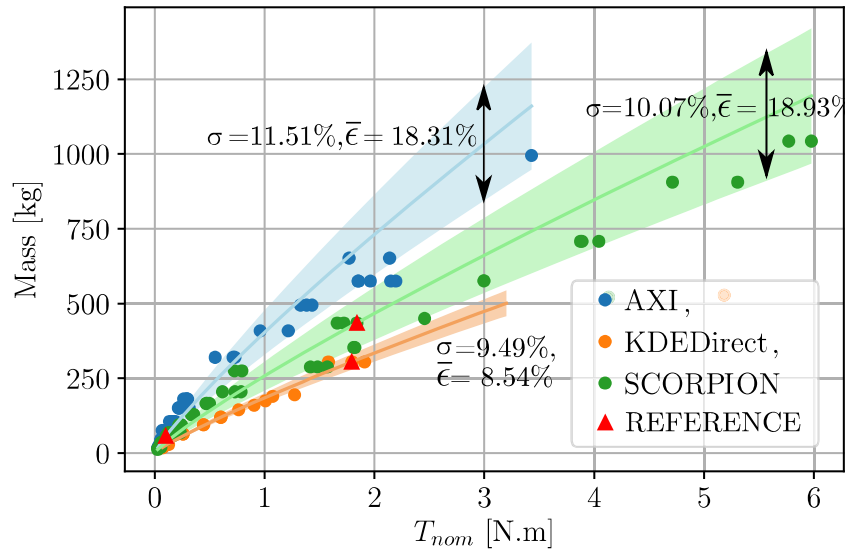


Figure 2.14 Mass as a function of nominal torque for different UAV motor families. Source: Budinger *et al.* [72].

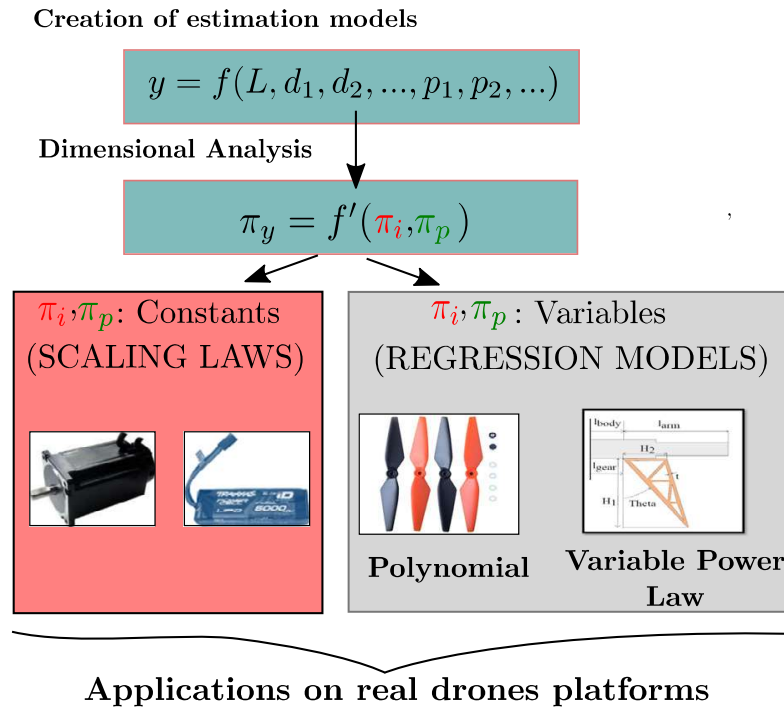


Figure 2.15 Application of dimensional analysis for reduction of model complexity. Source: Budinger *et al.* [72].

2.4 Existing tools for the preliminary sizing of UAVs

A variety of methodologies for sizing multirotor [68, 75–92], fixed-wing [69, 93–98] and hybrid FW-VTOL [59, 99–109] UAVs are proposed in the scientific literature. Historical approaches directly use databases of components to return an optimal combination for the propulsion system [80–82, 86–88, 90–92, 96]. Yet, to benefit from the high convergence performance made possible by the use of continuous models, most studies rely on empirical correlations and analytical models to estimate the characteristics of the components [59, 68, 69, 75–77, 84, 93, 95, 97, 99–107, 109, 110]. This tendency is illustrated in Figures 2.16 and 2.17, which show the occurrence of methodologies that rely on component databases, empirical correlations, and/or analytical models for the design process. Note that studies by the same authors have been grouped to avoid biased representations. The use of databases to select existing components, as opposed to their parameterization with empirical correlations and analytical methods, is specifically addressed in Part II of this manuscript. The present Part I focuses on continuous design problems as it offers better computational performance and allows the exploration of a wider design space.

Despite the wide variety of implementations, nearly three-quarters of the methodologies use numerical optimization capabilities to converge on an optimal candidate. However, as shown in Figure 2.16, few rely on a well-defined MDO implementation to ensure multidisciplinary optimality [76, 78, 79, 83, 97, 99, 101, 102, 104, 105]. The authors of the optimization framework CLOUDS [78, 79, 83] separate the estimation of the component characteristics from the performance analysis. From the design variables, the estimation module calculates the weights of the components, the geometry, and the simulation parameters. These parameters are then used by the analyzer to calculate the performance of the multirotor UAV during the mission. Finally, the constraints are checked and the design variables are updated using a genetic algorithm. The sequential approach allows the multidisciplinary problem to be decoupled so that no fixed-point iteration is required to determine the mass of the vehicle. An [99], Fayez [104], Nguyen [97], Tyan [101] and Zhang [102] adopt an MDF architecture to solve the multidisciplinary couplings with an iterative method. These studies use a gradient-based algorithm to solve the problem, except for Nguyen’s approach which uses a hybrid strategy that combines a genetic algorithm with a gradient-based method. Convergence times are seldom reported. The methodology developed by Nguyen converges to the optimal results in a hundred seconds. Delbecq *et al.* [76] have shown that the application of an efficient NVH strategy for sizing multirotor UAVs can reduce the resolution time by almost 4 compared to a standard MDO formulation. Therefore, the methodologies that rely on the MDF architecture are likely to be suboptimal in terms of numerical resources. Another advantage of the NVH MDO architecture proposed by Delbecq *et al.* is the use of normalized sizing coefficients as optimization variables whose bounds are independent of the specifications, enabling a generic

sizing code for a wide range of vehicle sizes. In fact, the results presented in [76] are validated on three multirotor UAVs of different scales, ranging from 1.5 kg to hundreds of kilograms.

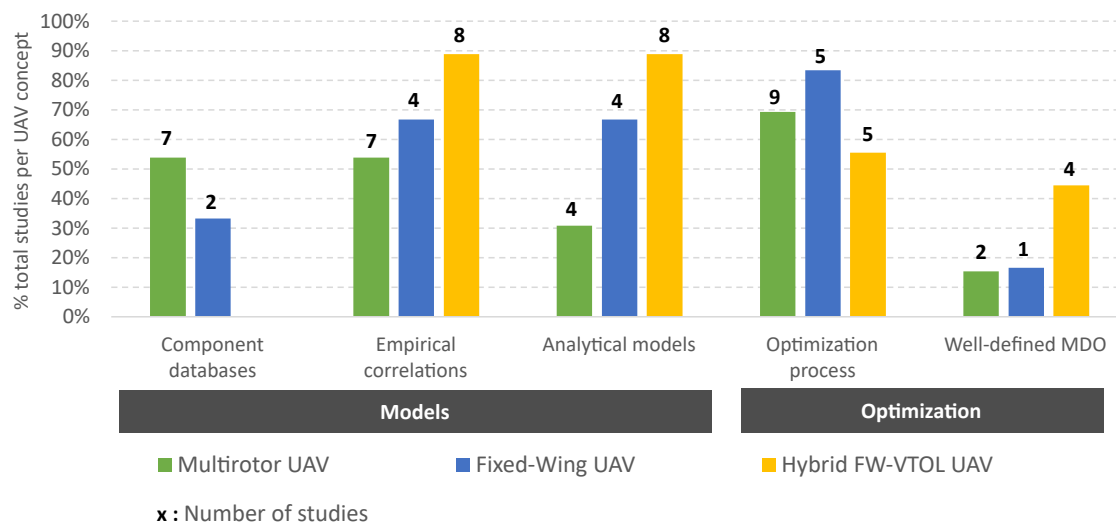


Figure 2.16 Occurrence of studies for the design of multirotor, fixed-wing and FW-VTOL UAVs, categorized according to the models involved and the optimization features.

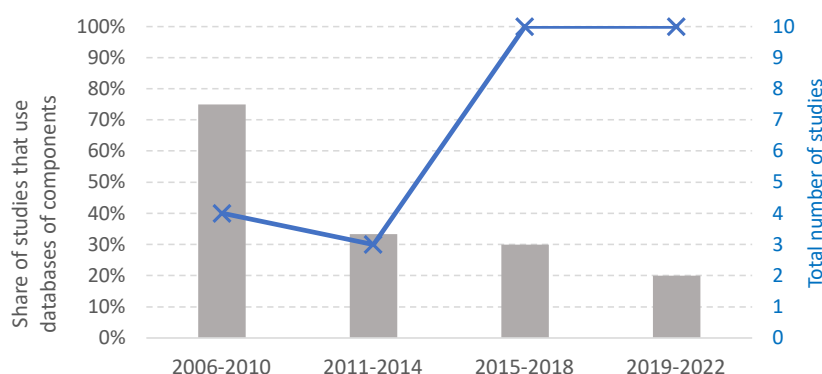


Figure 2.17 Occurrence of studies using component databases rather than analytical or empirical models. Evolution by time period.

The second important aspect worth exploring is the disciplinary coverage of the sizing methodologies. Most advanced MDO approaches focus on a limited number of design variables for sizing the vehicle. Zhang *et al.* [102] restrict the optimization to the propulsion system, the other components (e.g., wing geometry, rotors layout) being fixed. An *et al.* [99] and Tyan *et al.* [101] calculate the airframe mass for FW-VTOL UAVs as a fixed fraction of the total mass, which does not allow to optimize the structures and explore alternative materials. The most comprehensive methodologies include the framework CLOUDS [78, 79, 83] and the work of Fayez *et al.* [104, 105]. CLOUDS comprises the five main disciplines related to

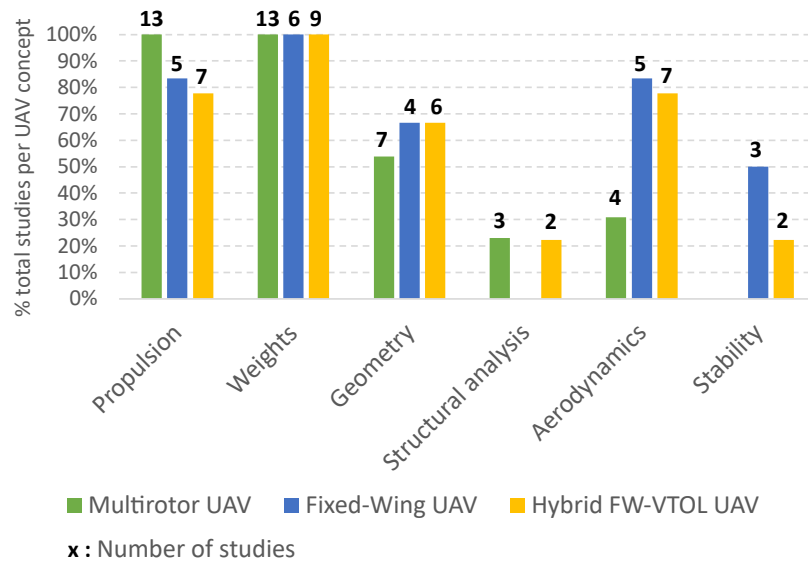
multirotor UAVs: propulsion, weight estimation, geometry, structural analysis, and airframe aerodynamics. The disciplines rely on a mix of analytical methods and empirical correlations for the estimation models. For example, the airframe drag model is obtained with a regression fitted to a CFD analysis and takes into account the geometric variations in the arms, center body, and landing gear. The CLOUDS methodology addresses multiple configurations of multirotor UAVs (e.g., quadcopters and hexacopters), but is not applicable to fixed-wing and FW-VTOL concepts. The sizing of different concepts of UAVs with a unified approach is addressed by Fayez *et al.*. The unified framework allows the sizing and comparison of multirotor and FW-VTOL UAVs for the same requirements. Particular effort is put into the modeling of the aerodynamics with the integration of a dedicated analysis tool, MachUp [111]. The authors point out that further optimization of the propulsion system should be addressed, as the selection of the best combination of components is not covered by the methodology. The disciplinary coverage of the methodologies reviewed in this section is summarized in Figure 2.18a.

Figure 2.18b provides an overview of the optimization objectives chosen in the studies. The weight function is the most commonly used objective for sizing UAVs. This is probably because this parameter is directly related to mission efficiency and vehicle cost [3].

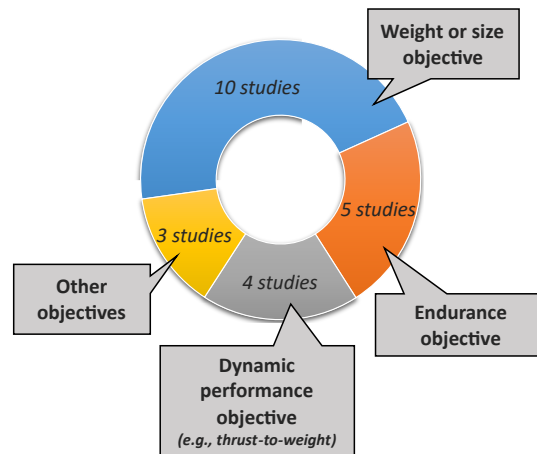
Finally, although some tools such as FlyEval [90–92] are available online, there does not seem to be any open source software for the preliminary design of UAVs at the moment. Such a tool would greatly benefit the scientific community by providing a framework to support the implementation of future applications and analyses.

2.5 Summary of the research gaps

To date, few methodologies address the sizing of multiple UAV concepts with a comprehensive discipline coverage, and even fewer leverage the capabilities of MDO. The most advanced methodologies are limited to a single concept (e.g., multirotor UAVs for CLOUDS [78, 79, 83] and Delbecqet *al.* [76]), optimize the vehicle with a limited number of design variables, and/or do not rely on an efficient MDO architecture to solve the problem. It also appears that no methodology is open source at the time of this writing, which precludes more in-depth analyses that could be conducted by the scientific community. Therefore, the literature lacks an open-source software tool for sizing and comparing multirotor, fixed-wing, and FW-VTOL UAVs within a unified and efficient MDO framework. In addition, the optimal design of small UAVs cannot be treated with the same models and tools as larger commercial aircraft. There are many differences related to the materials (e.g., wood and foam for fixed-wing drones), aerodynamics (low Reynolds numbers), variety of configurations, and rapidly evolving technologies (e.g., energy density of batteries). This requires analytical models that can capture such specificities, and a flexible framework to accommodate future evolutions of these models.



(a) Discipline coverage.



(b) Objective functions.

Figure 2.18 Occurrence of disciplines (left) and objective functions (right) in multirotor, fixed-wing and FW-VTOL UAV design studies.

Chapter 3

Efficient sizing and optimization of multirotor, fixed-wing and fixed-wing VTOL UAVs

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3.1 Introduction

The present chapter covers the development of a preliminary sizing methodology applied to multirotor, fixed-wing, and FW-VTOL UAVs. It is based on a comprehensive modeling of the main disciplines, including geometries, propulsion, structures, aerodynamics, and stability. In addition, the efficient formulation of the MDO problem allows sizing with short computation times. The implementation of the methodology in a software tool provides a means to size and compare different UAV concepts for similar specifications.

The work presented here reuses and improves the methodology proposed by Delebcq *et al.* [76] for sizing multirotor UAVs. The methodology of Delebcq *et al.* [76] relies on analytical models such as scaling laws and regression models, and implements specific resolution strategies to improve the computational efficiency of the algorithm. In particular, the NVH MDO architecture, which uses normalized coefficients as optimization variables whose bounds are independent of the specifications, allows the methodology to be applied in a generic manner to a wide range of UAV sizes. Building on the original methodology, the contributions of the present work are as follows:

- The flight scenarios, initially limited to takeoff, hover and vertical climb, are extended to include forward flight. In particular, the analytical model of the propeller is modified to account for non-axial flight conditions.
- The methodology is broadened to cover fixed-wing and FW-VTOL UAV concepts. The integration of new models (e.g., wings and stabilizing surfaces) introduces additional disciplinary couplings to be resolved.
- The methodology is implemented in a software tool that relies on advanced numerical capabilities to solve the problem and provides a graphical user interface.

The chapter starts by describing the scope of the methodology, including the architectures, configurations, and the main sizing assumptions. Section 3.3 introduces a set of models for the components and disciplines. The multidisciplinary design optimization and its resolution are presented in Section 3.4. Section 3.5 describes the implementation of the methodology into a software tool and the libraries supporting its development. Section 3.6 provides examples of sizing results obtained with the methodology. Finally, Section 3.7 concludes the chapter.

3.2 Scope of the methodology

The methodology focuses on specific configurations of multirotor, fixed-wing and FW-VTOL UAVs. Although the models are not restrictive for multirotor UAVs, configurations typically evaluated include quadcopters, hexacopters, and octocopters. In addition, each arm can support either a single propeller or two overlapping propellers in a push-pull configuration

(also known as coaxial). Many configurations for fixed-wing UAVs exist, but only the conventional aft-tail configuration with a single engine is addressed here. That is, the drone consists of wings, a fuselage, and an empennage mounted at its rear end. The propeller is located at the front of the fuselage. For FW-VTOL UAVs, the methodology covers the separate lift-thrust configuration, also known as quad-planes, where the forward flight uses a different propulsion system compared to the takeoff and hover flight. More precisely, four lifting propellers are attached to horizontal arms connected to the wings. Figure 3.1 provides examples of configurations covered by the methodology.

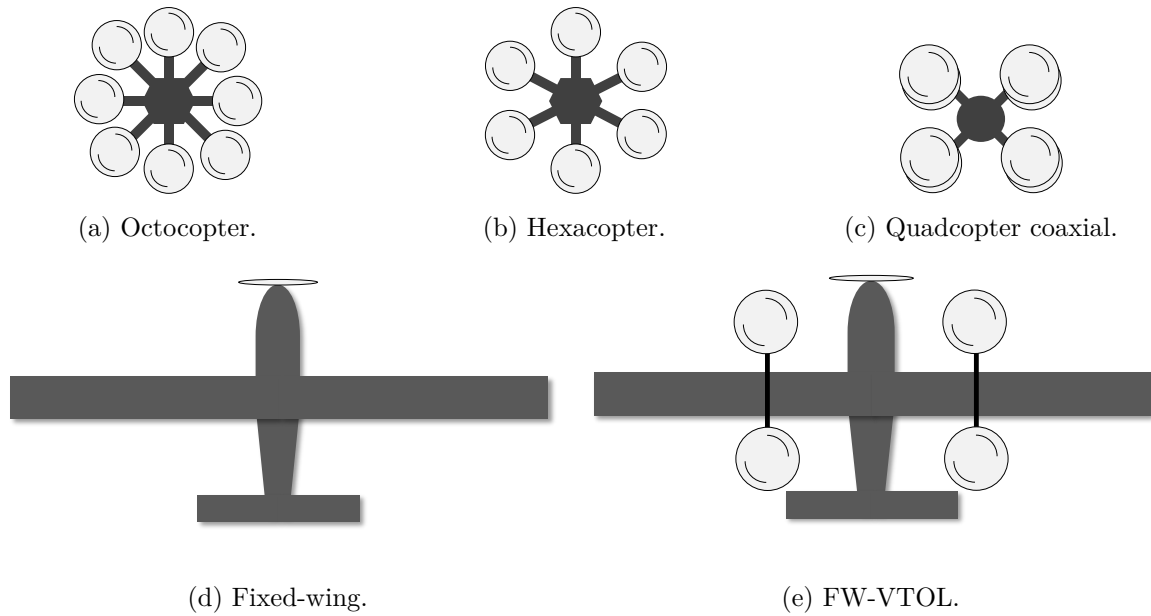


Figure 3.1 Examples of configurations covered by the methodology.

The modeling focuses on battery-powered UAVs with electric powertrains. Specifically, the propulsion chain consists of fixed-pitch propellers, permanent-magnet synchronous motors with an external (outrunner) rotor, electronic speed controllers (ESCs), batteries, and wires. Natural convection is assumed to be the only phenomenon involved in heat dissipation from the motors, ESC, and batteries. Therefore, thermal management systems are excluded from the study. This assumption can lead to suboptimal performance when designing larger UAVs, where the use of dedicated cooling systems can significantly improve power densities. In addition, avionics and sensors are not modeled, but a fixed mass and power consumption can be specified to account for these elements. In particular, the methodology may not be applicable to micro-UAVs where the mass fraction of the avionics and sensors cannot be neglected. Similarly, the landing gears are not included in the sizing and are assigned a constant mass. A conceptual view of the components modeled in the methodology is shown in Figure 3.2, for the FW-VTOL concept.

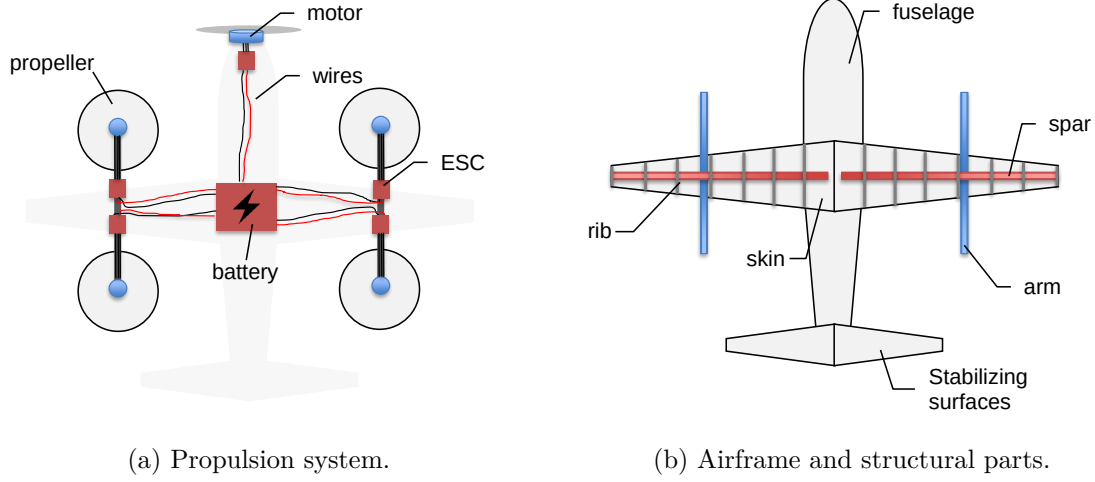


Figure 3.2 Components modeled in the methodology, for the FW-VTOL concept.

The flight scenarios addressed in this work consist of takeoff, climb (either vertical or with slope), cruise in forward flight and, if applicable, hover and low-speed flight (related to stall conditions). The definition of flight scenarios is crucial as it determines the sizing of the components through the so-called design drivers [60]. Specifically, the selection of the components is driven by both extreme performance criteria and endurance scenarios. For example, steady-state (endurance) scenarios like hovering flight and cruise flight are related to the maximum temperature the electric motors can withstand in continuous operation. The thermal operating limit for cylindrical brushless motors is determined by the nominal (rated) torque and the resistance through copper losses. The mechanical stress the arms must withstand is determined by the maximum transitory vertical acceleration, that is, the takeoff. The design drivers and sizing scenarios involved in the sizing of each component are provided in Table 3.1

3.3 Models for the preliminary design of UAVs

The methodology developed in this chapter proposes to parameterize the components with estimation models instead of relying on component databases, since the use of continuous mathematical models facilitates the implementation of computationally efficient solutions for sizing. A mixed approach using both estimation models and discrete sets of components is presented in Chapter 5. This section begins with the definition of the UAV simulation model that drives the selection of the components through the sizing flight scenarios. Component estimation models as well as structural and aerodynamic analysis equations are then presented.

In the following, superscript MR is used to describe the parameters related to the multirotor propulsion system or, for FW-VTOLs, the vertical takeoff and landing propulsion.

Table 3.1 Design drivers and sizing scenarios, based on [72, 76] for the propulsion system and extended to fixed-wing components.

Components	Design drivers	Parameters involved	Sizing scenarios
Propeller	Max. thrust	Diameter, Max. rotational speed	Extreme scenarios (Takeoff, Climb)
	Efficiency	Pitch	Endurance scenarios (Hover, Cruise)
Motor	Temperature rise	Nominal torque, Resistance	Endurance scenarios (Hover, Cruise)
	Max. voltage	Velocity (speed) constant, Resistance	Extreme scenarios (Takeoff, Climb)
ESC	Temperature rise	Max power, Max. current	Extreme scenarios (Takeoff, Climb)
Battery	Energy	Voltage, Capacity	Endurance scenarios (Hover, Cruise)
	Power	Voltage, Max. discharge rate	Extreme scenarios (Takeoff, Climb)
Wires	Temperature rise	Radius	Endurance scenarios (Hover, Cruise)
Arms	Max. stress	Dimensions	Extreme scenarios (Maximum takeoff thrust)
Wing spars	Max. stress	Dimensions	Extreme scenarios (Wind gust in cruise)
Wing	Max. lift	Area	Extreme scenarios (Stall)
	Fineness	Area, Aspect ratio	Endurance scenarios (Cruise)
Tails	Stability	Area, Tail arm	Disturbance scenarios (Wind disturbance in cruise)

Similarly, the superscript FW refers to the fixed-wing propulsion system. The subscripts $\cdot_{takeoff}$, $\cdot_{climb}$, $\cdot_{hover}$, $\cdot_{cruise}$ and $\cdot_{stall}$ indicate the relevant flight scenarios.

3.3.1 UAV simulation model

The goal of the UAV simulation model is to derive the main quantities for the selection of the vehicle's components, based on a few sizing scenarios. We focus here on the two most important parameters affecting UAV performance: the thrust-to-weight ratios F/W and, when applicable, the wing loading W/S .

Wing loading

For UAVs with lifting surfaces, the wing loading is the vehicle's total weight divided by its wing area. The wing loading must be optimized depending on the application. For example, low wing loadings (i.e. large wings) may require larger propulsion systems but offer lower stall speeds. In contrast, highly loaded wings are preferable for high speed performance. Raymer [61] provides methods to obtain estimates of optimal and/or acceptable wing loading

values under various performance conditions. This can be used to get an initial guess of the UAV sizing.

To maximize the range during cruise, in effect when flying at the speed for the maximum lift-to-drag ratio, the optimal wing loading must be such that:

$$(W/S)_{cruise} = q\sqrt{C_{D_0}/K} \quad (3.1a)$$

$$K = \frac{1}{\pi AR_w e} \quad (3.1b)$$

where $q = 0.5\rho_{air}V^2$ is the dynamic pressure at the selected airspeed V and air density ρ_{air} . AR_w is the wing's aspect ratio, C_{D_0} is the parasitic drag coefficient of the UAV, and K is the induced drag coefficient factor. The span efficiency factor e for straight wings can be calculated as follows [61]:

$$e = 1.78(1 - 0.045AR_w^{0.68}) - 0.64 \quad (3.2)$$

Ideally, this empirical span efficiency should include additional corrections to account for compressibility effects and wing sweep, among others. However, small UAVs typically fly at low speeds and have low-sweep wings, as this provides better performance under these conditions. Despite the lack of such corrections, and the fact that the empirical relationship was derived for larger commercial aircraft, Equation 3.2 is assumed to be valid for the scope of the study.

Optimizing the wing loading for cruise generally leads to small wings, and consequently a high stall speed. Although this issue is not critical for drones with lifting rotors, it can compromise the safety of a fixed-wing UAV. Given a required stall speed or, indirectly, an approach speed, the maximum wing loading not to be exceeded is expressed as [61]:

$$W/S \leq (W/S)_{stall} = qC_{L_{max}} \quad (3.3)$$

where $C_{L_{max}}$ is the maximum lift coefficient of the wing. The determination of the aerodynamics coefficients C_{D_0} and $C_{L_{max}}$ is covered in Section 3.3.5.

Thrust-to-weight ratios

Thrust-to-weight ratios are used to size the propulsion system for different flight scenarios. The calculation of the thrust to be provided by the rotors in each scenario is distinct for multirotor systems and fixed-wing UAVs.

For multirotor UAVs, the thrust calculation involves the equilibrium of forces acting on the vehicle, namely the weight (W), the aerodynamic forces (drag D_f and downward force L_f), and the thrust of the rotors (F_i). The parameters involved in the equilibrium equations are shown in Figure 3.3, where V_∞ is the free stream velocity and V_h the vertical speed, so

that θ_{FP} is the flight path angle of the drone [112]. Some assumptions are made here for simplicity. The rotors are assumed to be identical and symmetrical with respect to the center of gravity of the UAV. They are also assumed to be in the same plane, i.e. horizontal with respect to the UAV body frame. As a result, the tilting moment can be neglected and the rotors are treated as operating under the same conditions ($F_1 = F_2 = \dots = F$). Furthermore, the normal force generated by the rotors is neglected compared to the thrust since it is an order of magnitude lower [113]. The vertical and horizontal equilibrium equations are

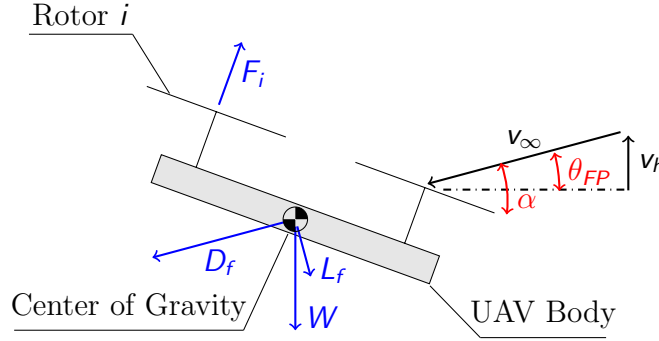


Figure 3.3 Quadcopter in climbing forward flight. Side view.

simplified to [112]:

$$N_{rotors}F \cos(\alpha - \theta_{FP}) = W + D_f \sin \theta_{FP} + L_f \cos \theta_{FP} \quad (3.4a)$$

$$N_{rotors}F \sin(\alpha - \theta_{FP}) = D_f \cos \theta_{FP} - L_f \sin \theta_{FP}, \quad (3.4b)$$

with N_{rotors} the number of rotors. Solving for the angle of attack α and the thrust F gives:

$$\tan(\alpha - \theta_{FP}) = \frac{D_f \cos \theta_{FP} - L_f \sin \theta_{FP}}{W + D_f \sin \theta_{FP} + L_f \cos \theta_{FP}} \quad (3.5a)$$

$$N_{rotors}F = \sqrt{(W + D_f \sin \theta_{FP} + L_f \cos \theta_{FP})^2 + (D_f \cos \theta_{FP} - L_f \sin \theta_{FP})^2} \quad (3.5b)$$

The calculation of the drag D_f and lift L_f of the frame is addressed further below. However, these quantities are functions of the angle of attack of the UAV, so that numerical solver (e.g., the `minimize` function from Scipy [47]) is required to converge Equation 3.5a. From the equilibrium equations, the thrust requirements for a multirotor UAV at takeoff, vertical

climb, hover and horizontal forward flight are obtained as follows:

$$N_{rotors} \cdot (F/W)_{hover}^{MR} = 1 \quad (3.6a)$$

$$N_{rotors} \cdot (F/W)_{takeoff}^{MR} = 1 + \frac{a_{to}}{g} \quad (3.6b)$$

$$N_{rotors} \cdot (F/W)_{climb}^{MR} = \sqrt{\left(1 + \frac{D_f}{m_{uav}g}\right)^2 + \left(\frac{L_f}{m_{uav}g}\right)^2} \quad (3.6c)$$

$$N_{rotors} \cdot (F/W)_{cruise}^{MR} = \sqrt{\left(1 + \frac{L_f}{m_{uav}g}\right)^2 + \left(\frac{D_f}{m_{uav}g}\right)^2} \quad (3.6d)$$

m_{uav} is the total mass of the UAV, g is the acceleration of gravity, and a_{to} is the acceleration requirement at takeoff.

When a UAV flies in a fixed-wing configuration, the thrust-to-weight ratios involve additional parameters related to the wing geometry [61]:

$$(F/W)_{cruise}^{FW} = \frac{qC_{D_0}}{W/S} + \frac{K}{q} \cdot W/S \quad (3.7a)$$

$$(F/W)_{climb}^{FW} = \frac{V_h}{V_{climb}} + \frac{q}{W/S} \cdot C_{D_0} + \frac{K}{q} \cdot W/S \quad (3.7b)$$

where W/S is the wing loading and V_h is the rate of climb. Note, N_{rotors} is equal to 1 here since only fixed-wing configurations with a single rotor are assessed. Takeoff is assumed to be performed via a catapult that launches the UAV at a speed ten percent greater than the stall speed requirement. Similarly to Equation 3.7b, the thrust-to-weight ratio for takeoff (or, initial climb) is expressed as:

$$(F/W)_{takeoff}^{FW} = \frac{V_h}{V_{takeoff}} + \frac{q}{W/S} \cdot C_{D_0} + \frac{K}{q} \cdot W/S \quad (3.8a)$$

$$V_{takeoff} = 1.1V_{stall} \quad (3.8b)$$

3.3.2 Estimation models for the electric propulsion system

A set of estimation models for the propulsion system, that is, models that return the main characteristics of the components from their so-called definition parameters, is provided by Budinger *et al.* in [72]. These estimation models take the form of scaling laws and regression models, which are recalled in Table 3.2. It is assumed that the models provided in [72] can be applied to both multirotor propulsion systems and fixed-wing propulsion systems with an appropriate change in the models' parameters (e.g., references for scaling laws) where required. In particular, the regression model for the performance of fixed-wing UAV propellers is obtained by applying the same methodology used for multirotor UAVs.

The models given by Budinger *et al.* for estimating the thrust and power coefficients of the propeller are only valid for static or axial flight conditions. However, there are flight scenarios where the airflow is not aligned with the propeller's axis of rotation, typically when

Table 3.2 Estimation models for the electric propulsion system [72].

Component	Parameter	Unit	Equation
Propeller	Definition parameters		
	Pitch-to-diameter ratio	-	β
	Diameter	m	d_{pro}
	Integration parameters		
	Mass	kg	$m_{pro} = m_{pro,ref} \cdot \left(\frac{d_{pro}}{d_{pro,ref}} \right)^3$
	Simulation parameters		
	Thrust coefficient	-	$C_T = 4.27 \times 10^{-2} + 1.44 \times 10^{-1} \cdot \beta$ (static flight) $C_T = 0.0279 - 0.0654 \cdot J + 0.1187 \cdot \beta + 0.2733 \cdot \beta^2 - 0.2885 \cdot \beta^3$ $+ 0.0210 \cdot J^3 - 0.2350 \cdot J^2 + 0.1868 \cdot \beta J^2$ (axial flight)
	Power coefficient	-	$C_P = -1.48 \times 10^{-3} + 9.72 \times 10^{-2} \cdot \beta$ (static flight)
			$C_P = 0.0181 - 0.0622 \cdot \beta + 0.0034 \cdot J + 0.3571 \cdot \beta^2$
			$-0.2377 \cdot \beta^3 + 0.0755 \cdot \beta J - 0.123 \cdot J^2$ (axial flight)
Motor	Definition parameters		
	Nominal (rated) torque	N.m	T_{nom}
	Velocity constant	rad/V/s	K_V
	Integration parameters		
	Mass	kg	$m_{mot} = m_{mot,ref} \cdot \left(\frac{T_{nom}}{T_{nom,ref}} \right)^{3/3.5}$
	Simulation parameters		
	Resistance	Ω	$R = R_{ref} \cdot \left(\frac{K_V}{K_{V,ref}} \right)^{-2} \cdot \left(\frac{T_{nom}}{T_{nom,ref}} \right)^{-5/3.5}$
	Friction torque	N.m	$T_f = T_{f,ref} \cdot \left(\frac{T_{nom}}{T_{nom,ref}} \right)^{0.69}$
	Operating domain parameters		
	Maximum torque	N.m	$T_{max} = T_{max,ref} \cdot \frac{T_{nom}}{T_{nom,ref}}$

Table 3.2 (continued).

Component	Parameter	Unit	Equation
ESC	Definition parameter		
	Nominal (rated) power	W	P_{esc}
	Nominal (rated) voltage	V	U_{esc}
	Integration parameters		
Battery	Mass	kg	$m_{esc} = m_{esc,ref} \cdot \frac{P_{esc}}{P_{esc,ref}}$
	Operating domain parameters		
	Nominal (rated) current	A	$I_{esc} = P_{esc}/U_{esc}$
	Definition parameters		
	Capacity	A.s	C_{bat}
	Voltage	V	U_{bat}
	Integration parameters		
	Mass	kg	$m_{bat} = m_{bat,ref} \cdot \frac{U_{bat}}{U_{bat,ref}} \cdot \frac{C_{bat}}{C_{bat,ref}}$
	Volume	m ³	$v_{bat} = v_{bat,ref} \cdot \frac{U_{bat}}{U_{bat,ref}} \cdot \frac{C_{bat}}{C_{bat,ref}}$
	Operating domain parameters		
Wires	Rated discharge current	A	$I_{max} = I_{max,ref} \cdot \frac{C_{bat}}{C_{bat,ref}}$
	Rated power	W	$P_{bat} = U_{bat} \cdot I_{max}$
	Energy	Wh	$E_{bat} = C_{bat} \cdot U_{bat}$
	Definition parameter		
	Nominal (rated) current	A	I_{wires}
	Integration parameters		
	Linear mass	kg/m	$\mu_{wires} = \mu_{wires,ref} \cdot \left(\frac{I_{wires}}{I_{wires,ref}} \right)^{4/3}$

a multirotor UAV is cruising in forward flight. Therefore, a modification of the regression models is proposed to estimate the thrust and power coefficients under nonaxial flight conditions.

Propeller coefficients in nonaxial conditions

The performance of a propeller can be expressed as a function of its thrust and power

coefficients — C_T and C_P respectively:

$$F = C_T \rho_{air} n^2 d_{pro}^4 \quad (3.9a)$$

$$P = C_P \rho_{air} n^3 d_{pro}^5 \quad (3.9b)$$

where ρ_{air} is the air density, n the rotational speed of the propeller and d_{pro} its diameter. The thrust and power coefficients in axial flight conditions (i.e., $\alpha = \frac{\pi}{2}$) are expressed by Budinger *et al.* as a function of the axial advance ratio $J_{axial} = \frac{V_\infty \sin \alpha}{n \cdot d_{pro}}$ and the pitch-to-diameter ratio β of the propeller. Specifically, the estimation models are obtained with regression techniques applied to a set of multirotor propellers from the manufacturer APC¹. A 3rd order polynomial model was retained by the authors for both thrust and power coefficients [72]:

$$\begin{aligned} C_T^{axial} = & 0.02791 - 0.06543 J_{axial} + 0.11867 \beta + 0.27334 \beta^2 - 0.28852 \beta^3 \\ & + 0.02104 J_{axial}^3 - 0.23504 J_{axial}^2 + 0.18677 \beta J_{axial}^2 \end{aligned} \quad (3.10a)$$

$$\begin{aligned} C_P^{axial} = & 0.01813 - 0.06218 \beta + 0.00343 J_{axial} + 0.35712 \beta^2 - 0.23774 \beta^3 \\ & + 0.07549 \beta J_{axial} - 0.1235 J_{axial}^2 \end{aligned} \quad (3.10b)$$

These aerodynamic coefficients are corrected to take into account non-zero incidence angles for the propeller. Leng *et al.* [113] propose an analytical model to express the thrust and power coefficients at non-zero incidence angle as ratios to their respective values in axisymmetric conditions. For a given blade geometry the authors suggest the following expressions:

$$C_{T,P}(\alpha, J) = \eta_{T,P}(\alpha, J) C_{T,P}^{axial}(J_{axial}) \quad (3.11a)$$

$$\eta_{T,P}(\alpha, J) = 1 + \frac{(J \cos \alpha / \pi \bar{r}')^2}{2(1 - J \sin \alpha / J_{0T,P}^{axial})} \delta(\alpha) \quad (3.11b)$$

Where:

- α is the angle of attack as defined in Figure 3.3.
- $J = \frac{V_\infty}{n \cdot d_{pro}}$ is the advance ratio.
- $J_{axial} = \frac{V_\infty \sin \alpha}{n \cdot d_{pro}}$ is the axial advance ratio.
- $J_{0T,P}^{axial}$ are the axial advance ratios where the thrust and power coefficients reach zero respectively. These ratios can be obtained by solving Equation 3.10 for $C_T^{axial} = 0$ and $C_P^{axial} = 0$.

¹<https://www.apcprop.com/technical-information/performance-data/>

- $\bar{r}'$ is the position of the representative section of the blade, in percentage radius (taken as 75%). For APC MR propellers, J_{0T}^{axial} and J_{0P}^{axial} follow the regression models:

$$J_{0T}^{axial} = 0.197 + 1.094 \cdot \beta \quad (3.12a)$$

$$J_{0P}^{axial} = 0.286 + 0.993 \cdot \beta \quad (3.12b)$$

- $\delta(\alpha)$ is a solidity term correction.

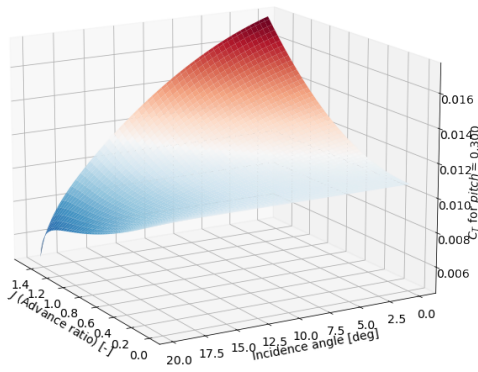
The solidity term correction is a function of its pitch angle p [rad] and solidity σ_{pro} [-] at the representative section of the blade $\bar{r}'$ [113]:

$$\delta(\alpha) = \frac{3}{2} \cos p \left[1 + \frac{\sigma_{pro}}{\tan p} \left(1 + \sqrt{1 + \frac{2 \tan p}{\sigma_{pro}}} (1 - \sin \alpha) \right) \right] \quad (3.13)$$

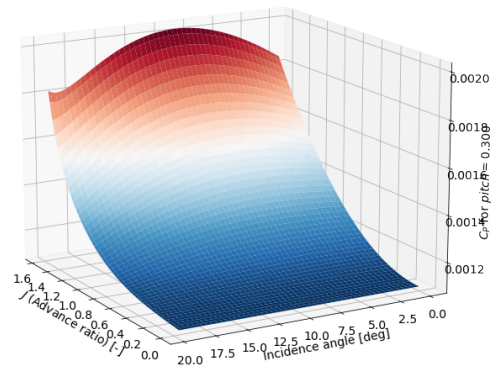
The solidity is obtained from the number of blades and the chord-to-radius ratio of the blades at the representative section: $\sigma_{pro} = \frac{N_{blades} c_{blade}}{\pi d_{pro}/2}$. Finally, given a pitch-to-diameter ratio $\beta = \frac{\text{pitch distance}}{\text{diameter}}$ and assuming the pitch distance remains constant along the blade, the pitch angle at the representative section $\bar{r}' = 75\%$ of the blade is obtained as:

$$p = \arctan \left(\frac{\text{pitch distance}}{\pi \bar{r}' d_{pro}} \right) = \arctan \left(\frac{\beta}{\pi \bar{r}'} \right) \quad (3.14)$$

Examples of estimated C_T and C_P for a propeller with a pitch-to-diameter ratio of $\beta = 0.3$ are shown in Figure 3.4. Appendix 3.A provides the full expressions obtained for both multirotor and fixed-wing propellers.



(a) Thrust coefficient C_T



(b) Power coefficient C_P

Figure 3.4 3D plots of estimated performance coefficients for a propeller with a pitch-to-diameter ratio $\beta = 0.3$.

3.3.3 Geometry

This section provides the main analytical models for the geometries of the different drone concepts.

Multirotor UAVs

The length of the arms supporting the rotors for multirotor drones is determined such that the propellers do not collide with each other. The evaluation of the top surface S_{top} and front surface S_{front} of the body frame (central hub) are obtained with scaling laws based on dimensional analysis. Specifically, it is assumed that the length ratios from one airframe to another are constant (geometric similarity), and that the total mass of the UAV is the main driver for these dimensions. Therefore, the following relationships are used to estimate the body frame surfaces:

$$\frac{S_{top}}{S_{top,ref}} = \left(\frac{m_{uav}}{m_{uav,ref}} \right)^{2/3} \quad (3.15a)$$

$$\frac{S_{front}}{S_{front,ref}} = \left(\frac{m_{uav}}{m_{uav,ref}} \right)^{2/3} \quad (3.15b)$$

where the subscript $.ref$ refers to the properties of a reference multirotor UAV. In the following Figure 3.5, surfaces data from several multirotor UAVs manufactured by DJI² are compared with their corresponding scaling laws. The reference UAVs used to calibrate the scaling laws are depicted in red. The deviations can be explained by the diversity of airframe shapes that can be found on the market and by the different missions for which the UAVs are designed. However, it is deemed acceptable for a preliminary model.

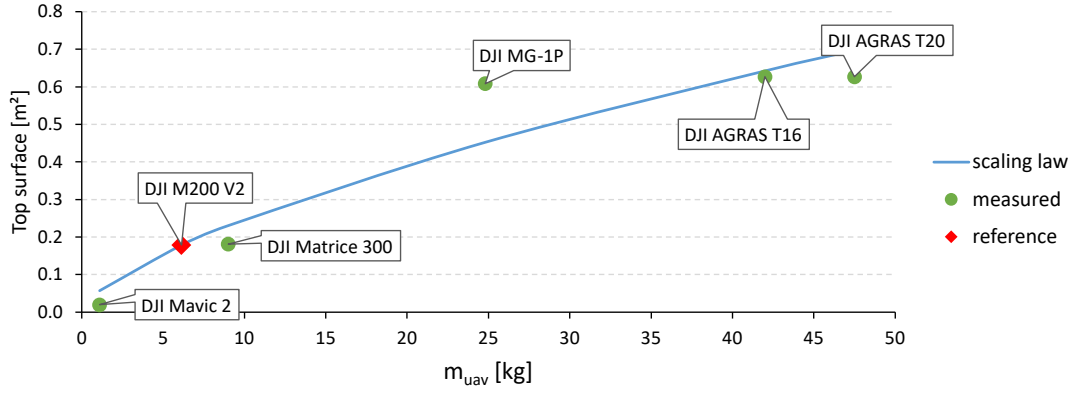
Fixed-wing

For the fixed-wing parts, the surface S_w of the wing is first derived from the total mass of the UAV and the previously selected wing loading. The wingspan b_w and chords lengths c at various positions y of the wing are derived from the dimensionless aspect ratio AR_w and taper ratio λ_w , assuming a trapezoidal wing planform:

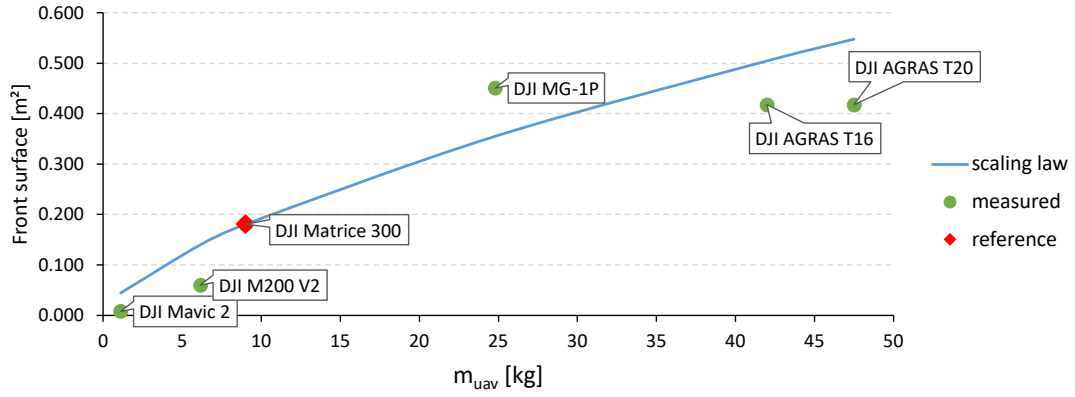
$$b_w = \sqrt{AR_w S_w} \quad (3.16a)$$

$$c(y) = \frac{2 S_w}{(1 + \lambda_w) b_w} \left[1 - \frac{1 - \lambda_w}{b_w} |2y| \right] \quad (3.16b)$$

²DJI: <https://www.dji.com/>



(a) Top surface variation with total UAV mass.



(b) Front surface variation with total UAV mass.



(c) DJI Mavic 2.



(d) DJI Matrice 300.



(e) DJI MG-1P.



(f) DJI Agras T20.

Figure 3.5 Body frame surfaces vs total UAV mass for several multirotor UAVs from DJI.

Specifically, the root and tip chords are defined as follows [63]:

$$c_{root} = \frac{2S_w}{(1 + \lambda_w)b_w} \quad (3.17a)$$

$$c_{tip} = \lambda_w c_{root} \quad (3.17b)$$

The wing's mean aerodynamic chord (MAC) is approximated by the mean geometric chord (MGC), which is a common assumption in aircraft design [63]:

$$c_{MAC_w} = \frac{2}{3} \frac{1 + \lambda_w + \lambda_w^2}{1 + \lambda_w} c_{root} \quad (3.18a)$$

$$y_{MAC_w} = \frac{b_w}{6} \frac{1 + 2\lambda_w}{1 + \lambda} \quad (3.18b)$$

The areas of the stabilizing surfaces are obtained from the horizontal tail volume coefficient v_{ht} and the vertical tail volume coefficient v_{vt} introduced by Raymer [61]. These non-dimensional ratios compare the product of the stabilizing surfaces to the wing geometry, and are strongly linked to the aircraft stability (see Section 3.3.6).

$$v_{ht} = \frac{l_t S_{ht}}{c_{MAC_w} S_w} \quad (3.19a)$$

$$v_{vt} = \frac{l_t S_{vt}}{b_w S_w} \quad (3.19b)$$

Here, l_t is the tail moment arm, that is the distance between the quarter of the wing's MAC and the quarter of the tail's MAC. Typical values for homebuilt aircraft are 0.50 for v_{ht} and 0.04 for v_{vt} [61], but other values may be applied depending on the application and stability requirements. The other geometrical parameters of the tails are derived similarly to the wing. However, lower aspect ratios will be preferred during optimization because the structural criterion is dominant over the aerodynamic criterion for these parts [62, 114].

Finally, the fuselage is made of a semi-spherical nose, a cylindrical mid-fuselage housing the batteries and the payload, and a conical rear-fuselage aft of the wing's trailing edge, as depicted in Figure 3.6. The total fuselage length is the sum of the wing position (with respect to the nose tip) and the tail moment arm. Its maximum diameter is derived from a fineness ratio provided by the designer. Appendix 3.B provides the geometrical relationships for the fuselage.

Fixed-wing VTOL

The VTOL propellers of a FW-VTOL drone must be located outside the high-turbulence area defined by the diameter of the fixed-wing propellers. Consequently, their distance from the axis of symmetry of the UAV must lie within the following range:

$$\left(\frac{d_{pro}^{FW}}{2} + \frac{d_{pro}^{MR}}{2} + \delta_{pro} \right) \leq y_{pro}^{MR} \leq \frac{b_w}{2} \quad (3.20)$$

where y_{pro}^{MR} are the locations of the propellers on the y-axis, d_{pro}^{FW} (d_{pro}^{MR}) is the diameter of the fixed-wing (multirotor) propellers, and δ_{pro} is the clearance distance. The location on the x-axis (i.e. along the fuselage and from the nose tip) is set such that the propellers do not collide or overlap with the wing. Therefore, the minimum length of the arms supporting

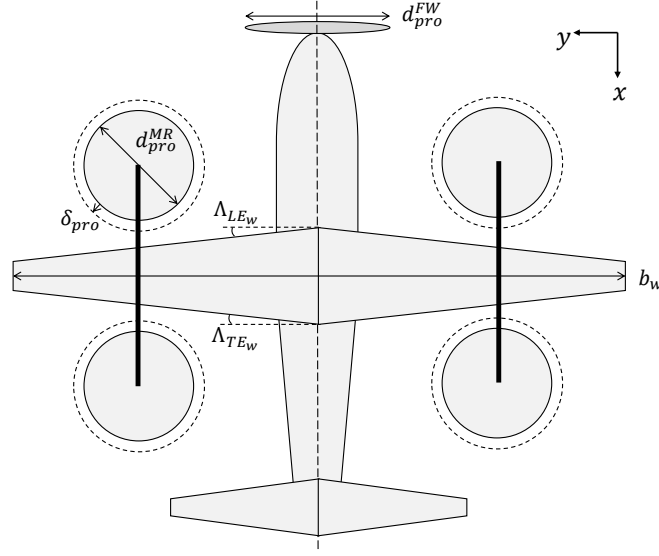


Figure 3.6 VTOL propellers location for FW-VTOL geometry

the VTOL propellers is:

$$l_{arm} \geq c_{root_w} + y_{pro}^{MR} (\tan \Lambda_{TE_w} - \tan \Lambda_{LE_w}) + (d_{pro}^{MR} + 2\delta_{pro}) \left(\frac{1}{\cos \Lambda_{TE_w}} + \frac{1}{\cos \Lambda_{LE_w}} \right) \quad (3.21)$$

where c_{root_w} is the wing root chord and Λ_{LE_w} (Λ_{TE_w}) is the sweep angle of the wing's leading edge (trailing edge). For a given sweep angle of the leading edge, the angle of the trailing edge is obtained from the aspect ratio AR_w and taper ratio λ_w of the wing [63]:

$$\tan \Lambda_{TE_w} = \tan \Lambda_{LE_w} - \frac{4}{AR_w} \left(\frac{1 - \lambda_w}{1 + \lambda_w} \right) \quad (3.22)$$

3.3.4 Structures

The primary structures, that is, the arms supporting the VTOL propellers and the spars ensuring the wing's integrity, are sized using a bending stress calculation based on the maximum static loads provided by the UAV simulation model. Dynamic loads (e.g., vibrations and fatigue) are not considered here. The arms are represented by cantilever beams of length l_{arm} with a circular hollow section of diameters d_{in} and d_{out} , at the extremity of which a maximum load F_{max} corresponding to the takeoff thrust is applied. The corresponding

bending stress at the root is expressed as:

$$\sigma_{root} = \frac{M_{root}}{I} \frac{d_{out}}{2} = \frac{32d_{out}M_{root}}{\pi(d_{out}^4 - d_{in}^4)} = \frac{32M_{root}}{\pi d_{out}^3(1 - k_{arm}^4)} \leq \sigma_{max} \quad (3.23a)$$

$$M_{root} = \frac{F_{takeoff}}{N_{rotors}^{MR}} l_{arm} \quad (3.23b)$$

where I is the moment of inertia of the beam, k_{arm} is the ratio of inner diameter to outer diameter, and N_{rotors}^{MR} is the number of VTOL propellers. The bending stress must not exceed the maximum allowable stress σ_{max} of the material chosen for the arms, obtained by applying a safety factor to the yield strength of the material.

An I-beam model is used for the wing spars, as I-shapes provide good structural properties at the cost of higher manufacturing complexity. However, other options such as hollow square beams could be implemented in future works. In a first approach, it is assumed that the bending moment is entirely reacted by the flanges of the I-beam. In addition, torsion and shear stresses are neglected for sizing the wing. Equation 3.23 is transformed into:

$$\sigma_{root} = \frac{M_{root} K_{flange} (1 + k_{spar})}{h_{spar}^3 k_{spar}^2 (1 + k_{spar}^2/3)} \leq \sigma_{max} \quad (3.24)$$

where K_{flange} is the ratio of flange thickness to flange length, generally in the order of 0.1 and taken constant here. k_{spar} is the flanges thickness divided by the distance h_{spar} between the two flanges. Two sizing cases are considered for the spars, as they must support the aerodynamic forces acting on the wing and, for FW-VTOL drones, the vertical takeoff thrust. The aerodynamic load on each half-wing is obtained from the equilibrium in cruise conditions and a load factor n_z . It is assumed to be applied at the wing's MAC location [61]. The thrust generated by the VTOL propellers at takeoff is transmitted to the spars via the arms. The point of application of this load is equal to the y-location of the arms, y_{pro}^{MR} . The resulting bending moments for each half-wing are:

$$M_{root}^{FW} = \frac{n_z m_{uav} g}{2} y_{MAC_w} \quad (3.25a)$$

$$M_{root}^{MR} = \frac{F_{takeoff}}{2} y_{pro}^{MR} \quad (3.25b)$$

Finally, the height of the spars must not exceed the minimum wing thickness:

$$h_{spar} \leq t_{wing,min} = c_{tip,w} \cdot \left(\frac{t}{c} \right), \quad (3.26)$$

with t/c the thickness-to-chord ratio of the selected airfoil.

The mass of the arms and the spars is obtained from their respective geometries and materials. Examples of materials are the 6082 aluminium alloy and carbon-fiber-reinforced polymers (CFRP). A conservative approach is adopted for sizing the ribs and the skin. The sizes and weights of these components are obtained from their materials properties (e.g. balsa

wood for the ribs) and the wing geometry, given a constant thickness for the ribs and an areal density for the skin. The spacing of the ribs is assumed to be one half of the mean aerodynamic chord [63].

No structural analysis is performed for the design of the tails and the fuselage. In particular, it is assumed that the tails are built from a full-height sandwich structure with areal density provided by the designer, and that no spar is required to support the structural loads. Similarly, the mass of the fuselage is obtained from its geometry and an assumed areal density representative of the construction. Stahl [100] provides a value of 1.05 kg.m^2 for the fuselage of a small UAV, considering a monolithic composite shell supported by an internal structure.

Finally, for multirotor UAVs, the total mass of the airframe (including arms and body) is obtained as a multiplication of the arms mass [76]:

$$m_{frame} = m_{frame,ref} \cdot \frac{m_{arms}}{m_{arms,ref}} \quad (3.27)$$

Further work could focus on developing a more detailed estimation model for the central hub and other structural parts of multirotor UAVs.

3.3.5 Aerodynamics

The calculations of the wing loading and thrust-to-weight ratios introduced in Section 3.3.1 require the determination of the aerodynamic forces applying to the UAV. The following equations distinguish between the aerodynamics of multirotor UAVs and those of fixed-wing and FW-VTOL UAVs.

Multirotor UAVs

For multirotor UAVs in climbing or cruising conditions, the airframe drag D_f and lift L_f are expressed as:

$$D_f = \frac{1}{2} C_D \rho_{air} V_{\infty}^2 S_{ref} \quad (3.28a)$$

$$L_f = \frac{1}{2} C_L \rho_{air} V_{\infty}^2 S_{ref} \quad (3.28b)$$

As depicted in Figure 3.3, L_f acts as a downward force when the drone moves forward — the aerodynamic surfaces on a multirotor UAVs are mostly disadvantageous. The effective area S_{ref} varies with the angle of attack and the surfaces facing the relative wind:

$$S_{ref} = S_{top} \sin \alpha + S_{front} \cos \alpha \quad (3.29)$$

The drag coefficient C_D and the lift coefficient C_L are highly dependent on the shape of the frame. Other parameters such as Reynold's number affect their value. Since the shape of the airframe is highly dependent on the application and payload, a generic model would

not be able to properly estimate the aerodynamic coefficients. Therefore, the drag and lift coefficients must be provided as constant values by the designer. In addition, the interactions between the rotors and the frame are neglected. Typical values for the drag coefficient of common geometries are given by Baker *et al.* [115]. Estimates of C_D are 0.47 for a sphere, 1.05 for a cube, and 2.05 for a long rectangular body with Reynolds numbers around 10^4 . In [116], the authors use high-fidelity CFD methods to analyze a quadcopter specialized for forward flight, the Hitec Endurance³. The results show that for the overall structure (including fuselage, landing gear, canards and arms) a drag coefficient of $C_D = 0.425$ and a beneficial lift of $C_L = 0.193$ are obtained. This provides a lower (upper) bound for the drag (lift) coefficient since the Hitec Endurance fairing is optimized to minimize drag and provide lift in forward flight mode, which is not the case for most multicopter designs. The sensitivity of the design results to the value of the drag coefficient is explored in Section 3.6.

The same approach is used for FW-VTOL UAVs in vertical takeoff conditions. That is, a constant drag coefficient value is used to calculate the drag that the VTOL propellers must compensate for. However, for fixed-wing and FW-VTOL UAVs in forward flight, a more detailed method is employed.

Fixed-wing and FW-VTOL

For fixed-wing and FW-VTOL UAVs in forward flight, the calculations of the wing loading and thrust-to-weight ratios require the value of the parasitic drag coefficient C_{D_0} and the maximum lift coefficient $C_{L_{max}}$. For wings with a fairly high aspect ratio, as is the case with most fixed-wing configurations, $C_{L_{max}}$ can be approximated as 90% of the maximum lift coefficient $C_{l_{max}}$ of the airfoil [61]. The airfoil optimization is not covered in this research, and a fixed value is considered for the maximum lift coefficient. The parasitic drag C_{D_0} is computed with a component buildup method [61], assuming no aerodynamic interaction between the different parts of the airframe, and with the propellers. It consists of estimating the parasitic drag of each part of the UAV with an equivalent flat-plate skin friction coefficient and a shape form factor:

$$C_{D_0} = \frac{1}{S_w} \left(\underbrace{\sum_{i=1}^{N_{surfaces}} C_{f,i} F F_i S_{wet,i}}_{\text{wings, fuselage, tails}} + \underbrace{C_{D_{pro}}^{MR} S_{pro}^{MR}}_{\text{stopped VTOL propellers}} \right) \quad (3.30)$$

The flat-plate skin friction coefficient C_f is calculated assuming a complete turbulent flow [117]:

$$C_f = \frac{0.455}{(\log_{10} Re)^{2.58}} \quad (3.31)$$

³Hitec Drone Services: <https://hiteccds.com/>

The Reynolds number is obtained from the flight conditions (altitude, temperature, and airspeed). The shape form factors FF are calculated as proposed by Raymer [61]:

$$FF_{fuselage} = 1 + \frac{60}{\lambda_f^3} + \frac{\lambda_f}{400} \quad (3.32a)$$

$$FF_{wings/tails} = 1 + \frac{0.6}{(x/c)_{max}} \left(\frac{t}{c}\right) + 100 \left(\frac{t}{c}\right)^4 \quad (3.32b)$$

where λ_f is the fuselage fineness ratio and t/c is the thickness-to-chord ratio of the wing's airfoil. The parameter $(x/c)_{max}$ is the chordwise location of the airfoil maximum thickness (0.3 for 4-digit NACA series) and Λ_m is the sweep at this location. Finally, the wetted areas of the wing and tails are taken as twice their areas, and the fuselage wetted area is calculated from its diameter d_f and length l_f [62]:

$$S_{wet fuselage} = \pi d_f l_f \left(1 - \frac{2}{\lambda_f}\right)^{2/3} \left(1 + \frac{1}{\lambda_f^2}\right) \quad (3.33)$$

The term $C_{D_{pro}}^{MR}$ in Equation 3.30 refers to the parasitic drag originating from the stopped propellers of the multirotor propulsion system, specifically for FW-VTOL configurations in forward flight. In feather position, when the blades are parallel to the airflow, Hoerner [118] provides a value of 0.1 for the drag coefficient of a single blade. This value is corrected with a solidity ratio to obtain the parasitic drag coefficient at the propeller scale:

$$C_{D_{pro}}^{MR} = 0.1 \sigma_{pro} = 0.1 \frac{N_{blades} c_{blade}}{\pi d_{pro}/2} \quad (3.34)$$

Based on the geometry of three-bladed APC propellers retrieved from UIUC data site [119], a mean value of 0.15 is adopted for the solidity ratio.

3.3.6 Stability

The longitudinal static stability of the fixed-wing and FW-VTOL concepts is assessed by the static margin of the UAV. The static margin is the relative distance between the center of gravity of the UAV and the neutral point. A center of gravity located ahead of the neutral point results in a positive static margin and a stable configuration. The center of gravity x_{cg} of the UAV is estimated from the individual masses and positions of the components. Among others, it is assumed that the batteries are centered at the wing position. The center of gravity of the lifting surfaces is approximated by the center of mass of the airfoil at the MAC position, which is located at 40% of the chord for most NACA airfoils.

The neutral point represents the most-aft position of the center of gravity before the UAV becomes unstable. At the neutral point, the pitching moment of the UAV is zero. Solving

the equation of the pitching moment for the neutral point yields [63]:

$$x_{np} = x_{ac} + c_{MAC_w} v_{ht} \frac{C_{L_{\alpha_{ht}}}}{C_{L_{\alpha_w}}} \left(1 - \frac{\partial \epsilon}{\partial \alpha} \right) \quad (3.35)$$

The contribution of the fuselage is neglected in Equation 3.35. The wing's aerodynamic center (ac) at low Mach numbers is located at the quarter-chord of the MAC [61]:

$$x_{ac} = x_{MAC/4} \quad (3.36)$$

The lift curve slopes C_{L_α} of the 3-dimensional wing and horizontal tail are calculated with the Prandtl's lifting-line theory:

$$C_{L_\alpha} = \frac{C_{l_\alpha}}{1 + \frac{C_{l_\alpha}}{\pi e AR}} = \frac{2\pi}{1 + \frac{2\pi}{\pi e AR}} \quad (3.37)$$

The ϵ term in Equation 3.35 is the downwash at the tail, originating from the trailing vortex behind the wing. The downwash slope with respect to the angle of attack is determined as follows:

$$\frac{\partial \epsilon}{\partial \alpha} = \frac{2C_{L_{\alpha_w}}}{\pi e AR_w} \quad (3.38)$$

Finally, the static margin resulting from the calculations of the center of gravity and the neutral point is obtained as:

$$SM = \frac{x_{np} - x_{cg}}{c_{MAC_w}} \quad (3.39)$$

3.3.7 Additional resources

Simulation models for calculating the performance of the sized components and the UAV in each flight scenario are provided in Appendix 3.C. In addition, reference values used for calibrating the estimation models (e.g., scaling laws parameters, material properties) are given in Appendix 3.D.

3.4 Sizing and optimization

An MDO approach is used to solve the design problem. Due to the large number of variables involved, the optimization process can be computationally expensive. However, the application of an efficient strategy as presented by Delbecq *et al.* [76] can significantly reduce the complexity of the problem. The methodology, originally applied to the sizing of multirotor UAVs, relies on an NVH architecture to solve the multidisciplinary couplings. In addition, the number of optimization constraints is reduced by 1) identifying and explicitly solving critical active constraints and 2) replacing the model-dependent design variables with

sizing coefficients that express inequality constraints. We propose here to extend the scope of the methodology developed in [76] to fixed-wing and FW-VTOL UAVs. However, for the sake of conciseness, only the design problem of FW-VTOL UAVs is presented in this section, since this concept encompasses the parameters of both multirotor and fixed-wing UAVs.

3.4.1 Initial problem formulation

The initial problem formulation for FW-VTOL UAVs is described in Table 3.3. It consists of a single objective function, a set of design variables, and several constraints, all of which are directly related to the quantities of the models introduced in the previous section.

The set of design variables includes wing loading, airframe geometric parameters (e.g., aspect ratio, wing position), structural parameters (e.g., arm and spar dimensions), and definition parameters of the propulsion system components (e.g., nominal motor torque). In particular, for the propulsion system, the choice of the design variables is supported by a correlation analysis. In fact, the design variables must have little or no dependence on each other, since the optimizer is free to choose their values independently. Methods to assist in this analysis include correlation matrices or principal component analysis (PCA). Figure 3.7 shows a PCA applied to a database of brushless motors from AXI⁴, Scorpion⁵, and KDE Direct⁶. The explained variance, that is, the proportion of variation in the database that can be attributed to each of the principal components (eigenvectors) generated by the PCA method, is shown in Figure 3.7a. The first two principal components are found to explain 75% of the variance within the database. An orthonormal basis is constructed from these two principal components, and the main physical parameters of the motor are projected into this basis, as depicted in Figure 3.7b. The direction of the vectors allows to find the correlations between the motor parameters. Perpendicular vectors indicate uncorrelated parameters. The results show that the nominal torque and the velocity constant are poorly correlated. In addition, the maximum torque and the resistance also appear to be uncorrelated. Therefore, a choice must be made to define which parameters will be set as design variables and which will be dependent variables. This choice is guided by the parameter categorization detailed by Budinger [60] and recalled in Chapter 2. The resistance is a simulation parameter that represents the non-ideal behavior of the motor as it relates to copper losses. The maximum torque is an operating domain parameter and specifies the safe operating range of the component. The nominal torque and the velocity constant are the so-called definition parameters that express the functional characteristics of electric motors. To better link the selection of a component to the system-level specifications, the design variables are chosen to be representative of functional parameters rather than imperfections. Consequently, for the electric motors, the nominal torque and the velocity constant are chosen as design variables,

⁴AXI Motors: <https://www.modelmotors.cz/>

⁵Scorpion: <https://www.scorpionsystem.com/>

⁶KDE Direct: <https://www.kdedirect.com/>

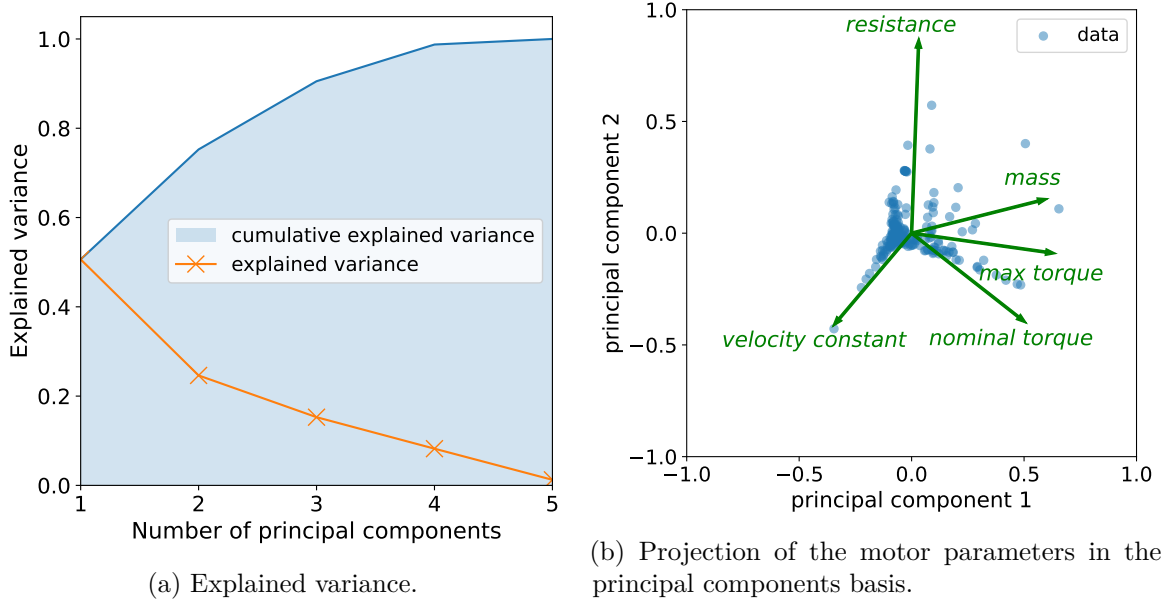


Figure 3.7 Principal component analysis applied to brushless motors.

and the remaining parameters are defined as dependent parameters to be obtained through the estimation models. Interestingly, the physical correlations provided by the estimation models (Table 3.2) are visible in the PCA: the resistance is a linear combination of the velocity constant and the nominal torque, while the maximum torque and the motor mass are mostly correlated with the nominal motor torque.

Optimization constraints include geometric considerations (e.g., spar height vs. wing thickness), operating range constraints (e.g., maximum motor torque), and system performance constraints (e.g., energy required to complete the mission).

Finally, several objective functions are suggested in this study: the flight endurance (or, equivalently, the range), the total mass, and the energy consumption. Additional optimization objectives are proposed in Chapter 9, along with an exploration of the tradeoffs that result from each objective.

3.4.2 Disciplinary couplings resolution

The multidisciplinary couplings are resolved by implementing a normalized variable hybrid (NVH) formulation.

The feedforward coupling relationships, which consist essentially of the mechanical and electrical quantities in the various flight scenarios, are explicitly resolved by computing the analyses in sequence. For example, the propeller speed in climbing flight is fed directly into the motor analysis so that the motor speed is equal to the propeller speed ($n_{mot,climb} = n_{pro,climb}$). Similarly, the battery current in hover condition is obtained directly from the motor current in the same flight scenario ($I_{bat,hover} = N_{rotors} I_{mot,hover}$).

Table 3.3 Initial optimization problem for FW-VTOL UAVs.

Design variables	Parameter	Description
W/S		Wing loading
AR_w, AR_{ht}, AR_{ot}		Aspect ratios of the wing and tails
λ_w		Taper ratio of the wing
$x_{MAC/4}$		Wing location w.r.t. nose tip
l_t		Tail moment arm
y_{pro}^{MR}		Location on y-axis of the VTOL propellers
l_{arm}		Arms length
d_{out}, h_{spar}		External dimensions of the arms and spars
k_{arm}, k_{spar}		Aspect ratios of the arms and spars
β		Propeller pitch-to-diameter ratio
d_{pro}		Propeller diameter
J_k		Propeller advance ratios for each flight scenario
T_{nom}		Motor nominal torque
K_V		Motor velocity constant
U_{bat}		Battery voltage
C_{bat}		Battery capacity
P_{esc}		ESC power
U_{esc}		ESC voltage

$\times 2$ (for fixed-wing propulsion and for VTOL propulsion)

Note 1: the flight speeds (e.g., V_{cruise}) can be set as design variables or fixed by the designer.
 Note 2: subscript $.k \in \{.takeoff, .climb, .hover, .cruise\}$ indicates the relevant flight scenarios.

The feedback couplings, that is the algebraic loops, are resolved by introducing normalized variables and consistency constraints. The first algebraic loop to be handled is the mass calculation. The propulsion systems and the structures are sized based on the thrust requirements, which require the total mass of the UAV to be known. However, the latter is calculated from the sum of the individual masses, resulting in a feedback loop. Following the NVH formulation, the mass loop is broken by setting an initial estimate for the total mass, which is assumed to be a multiple of the known payload mass [76]:

$$m_{uav,guess} = k_M^c \cdot m_{payload} \quad (3.40)$$

Table 3.3 (continued).

Constraints	Parameter	Description
	$W/S \leq (W/S)_{stall}$	Constraint on the stall speed
	$SM_{req} \leq SM$	Constraint on the static margin
	$v_{bat} + v_{pay} \leq v_f$	Constraint on the fuselage volume
	$g_1(y_{MR}^{pro}) \leq 0$	Constraint on the minimum y-location of VTOL propellers
	$g_2(y_{MR}^{pro}) \leq 0$	Constraint on the maximum y-location of VTOL propellers
	$g_3(l_{arm}) \leq 0$	Constraint on the minimum arms' length
	$\sigma_{root_arm} \leq \sigma_{max}$	Constraint on the arms' bending stress
	$h_{spar} \leq t_{wing}$	Constraint on the spars' height
	$\sigma_{root_spar}^{MR} \leq \sigma_{max}$	Constraint on the spars' bending stress (VTOL takeoff)
	$\sigma_{root_spar}^{FW} \leq \sigma_{max}$	Constraint on the spars' bending stress (wing load)
×2 (for fixed-wing propulsion and for VTOL propulsion)	$n_k \cdot d_{pro} \leq (n \cdot d_{pro})_{max}$	Constraints on the propeller tip speed
	$J_k n_k d_{pro} = V_k$	Constraints on the UAV flight speed
	$T_k \leq T_{nom}$	Constraints on the motor torque (steady state scenarios)
	$T_k \leq T_{max}$	Constraints on the motor torque (transient scenarios)
	$P_{bat,k} \leq P_{bat}$	Constraints on the battery power
	$U_{mot,k} \leq U_{bat}$	Constraints on the battery voltage
	$E_{mission} \leq E_{bat}$	Constraint on the battery energy
	$P_{esc,k} \leq P_{esc}$	Constraints on the ESC power
	$U_{esc,k} = U_{bat} \leq U_{esc}$	Constraints on the ESC voltage
	$m_{uav} \leq m_{uav,req}$	Constraint on the total mass not to be exceeded
Objectives	m_{uav} or $E_{mission}$ or t_{max} or R	Total mass of the UAV Energy consumption Flight endurance Flight range

Note 1: the flight speeds (e.g., V_{cruise}) can be set as design variables or fixed by the designer.

Note 2: subscript $\cdot_k \in \{\cdot_{takeoff}, \cdot_{climb}, \cdot_{hover}, \cdot_{cruise}\}$ indicates the relevant flight scenarios.

where k_M^c is a normalized variable in the range $[1, +\infty[$. The lower bound is equal to one because the total mass cannot be less than the payload mass alone. A consistency constraint is added to the optimization problem to ensure that the initial guess converges to the actual

mass after several iterations:

$$m_{uav,guess} \geq m_{uav} = \sum_i m_i, \quad (3.41)$$

with m_i the i -th component mass. A single inequality is used here instead of an equality constraint, which alleviates the numerical difficulties. This can be done because the optimization of the objective function inevitably causes the inequality to be active at the optimum. Indeed, a candidate design that is sized for a heavier mass than its actual mass will be suboptimal if efficiency or endurance criteria are targeted. However, special care should be taken when using objective functions other than those defined in this study, for example, when the goal is to maximize dynamic performance.

Another feedback loop is found when calculating the aerodynamic drag for fixed-wing and FW-VTOL UAVs. The parasitic drag coefficient is calculated from the geometry of the airframe and, for the FW-VTOL concept, the diameter of the VTOL propellers. However, the propulsion systems are sized from the thrust requirements, which in turn are derived from the parasitic drag coefficient (Equation 3.7). Therefore, an initial guess of the parasitic drag coefficient is obtained from a reference value, and a constraint is added to ensure that both the initial guess and the final value tend to a unique solution:

$$C_{D0,guess} = k_{C_{D0}}^c \cdot C_{D0,ref} \quad (3.42a)$$

$$C_{D0,guess} = C_{D0} \quad (3.42b)$$

In practice the equality constraint is treated as a pair of inequalities for better numerical handling. Note that the equality cannot be transformed into a single inequality constraint here. In fact, as can be seen in Equation 3.1a, overestimating the parasitic drag coefficient in the sizing process would result in a smaller wing and thus a lower mass penalty. However, it would worsen the aerodynamic performance and possibly require a larger (heavier) propulsion system. Therefore, there is no clear relationship between the parasitic drag estimate and the objective monotonicity, and the full equality constraint is kept.

Finally, an algebraic loop is found when computing the thrust-drag equilibrium in climbing flight, for FW-VTOL UAVs. The projected top area of the UAV is required for the drag calculation but is only accessible as an output of the geometry module as it requires the wing, tails and fuselage to be sized. For the FW-VTOL drones, an initial guess of the top area is obtained from the wing loading and the total mass of the UAV:

$$S_{top,guess} = k_S^c \cdot S_w = k_S^c \cdot \frac{m_{uav}g}{W/S} \quad (3.43a)$$

$$S_{top,guess} \geq S_{top}, \quad (3.43b)$$

with $k_S^c \geq 1$, since the projected top area is necessarily larger than the wing surface alone.

The consistency constraints and variables used for solving the disciplinary couplings are summarized in Table 3.4.

Table 3.4 Consistency constraints and normalized variables for solving feedback couplings.

	Parameter	Description
Design variables	k_M^c	Normalized variable for the total mass
	$k_{C_{D0}}^c$	Normalized variable for the parasitic drag coefficient
	k_S^c	Normalized variable for top (projected) area
Constraints	$g_1^c: m_{uav} \leq m_{uav,guess}$	Consistency constraint on the total mass
	$g_2^c: C_{D0,guess} = C_{D0}$	Consistency constraint on the parasitic drag coefficient
	$g_3^c: S_{top} \leq S_{top,guess}$	Consistency constraint on the top (projected) area

3.4.3 Problem reduction

The complexity of the optimization problem is reduced by applying the methodology proposed by Delbecq *et al.* in [76]. First, a monotonicity analysis is performed to transform the critical active inequalities into equalities. Then, a specific strategy is used to transform some of the remaining inequalities into sizing equations with a dedicated degree of freedom.

Monotonicity analysis

A design problem often consists of a large number of inequality constraints, which complicate the work of optimization algorithms. The monotonicity analysis enables to remove a number of inequality constraints by application of the First Monotonicity Principle (MP1). This principle, first introduced by Papalambros and Wilde [22], states that in a well-constrained minimization problem, every increasing variable (i.e., the objective function is increasing with the variable) is bounded below by at least one non-increasing active constraint (i.e., the constraint is not increasing with the variable). The principle also applies in the opposite case, when the objective function is decreasing instead of increasing. Mathematically, the MP1 can be expressed by the Karush-Kuhn-Tucker (KKT) conditions presented in Chapter 2 and recalled here:

$$\nabla f(\mathbf{x}^*) + \sum_{j=1}^{n_g} \sigma_j \nabla g_j(\mathbf{x}^*) = \mathbf{0}, \quad \sigma_j \geq 0 \quad (3.44a)$$

$$\sigma_j g_j(\mathbf{x}^*) = 0, \quad j = 1, \dots, n_g \quad (3.44b)$$

If the objective function is strictly increasing in a variable x_i , then $\left(\frac{\partial f}{\partial x_i}\right)^* > 0$ and there is at least one active constraint decreasing with x_i , i.e. $\sigma_j \left(\frac{\partial g_j}{\partial x_i}\right)^* < 0$, to satisfy the first KKT condition. The constraint activity is essential here to ensure $\sigma_j \neq 0$ as per the second KKT condition. When multiple constraints decrease with x_i , it is not possible to conclude about the activity of each of these constraints. In particular, several constraints may be active simultaneously. However, if a single constraint decreases, it is necessarily active and bounds

the objective from below. Such a constraint is said to be critical with respect to x_i , because if it were relaxed, the optimal value of the objective function would be different. Because of its properties, a critical constraint can be turned into an equality that restricts the value of x_i :

$$g_j(\mathbf{x}) \text{ critical w.r.t. } x_i \iff g_j(\mathbf{x}) = g_j(x_i, \mathbf{x}_{j \neq i}) = 0 \quad (3.45a)$$

$$\iff x_i = x_i(\mathbf{x}_{j \neq i}) \quad (3.45b)$$

If an explicit solution can be found from Equation 3.45, the variable x_i and the critical constraint g_j are eliminated from the optimization problem. Such a variable x_i is said to be *optimized out* [22] and leads to a reduction of the problem complexity.

In practice, the first step is to perform a monotonicity analysis of the objective and constraint functions with respect to the optimization variables. To keep track of the analysis, a monotonicity table [22] is constructed. A monotonicity table contains one column for each variable and one row for the objective function and each constraint. Monotonicity is indicated by a '+' for a strictly increasing function and a '-' for a strictly decreasing function. For a given variable, if the sign of a constraint derivative is the only opposite of the sign of the objective derivative, the constraint is critical and can be transformed into an equality. An example of monotonicity table is provided in Table 3.5. For conciseness, only the constraints and variables related to the geometry and structural analysis of the UAV are displayed.

Table 3.5 Monotonicity table to identify critical constraints. The expressions highlighted in red are critical with respect to some of the variables, because the sign of their derivative is the only opposite of the sign of the objective derivative.

	W/S	AR_w	λ_w	$x_{MAC/4}$	l_t	y_{pro}^{MR}	l_{arm}	d_{out}	k_{arm}	h_{spar}	k_{spar}	d_{pro}^{MR}	d_{pro}^{FW}
Objective $f : m_{uav}$	-		-	+			+	+	-	+	+	+	+
$g_1 : W/S \leq (W/S)_{stall}$	+												
$g_2 : v_{bat} + v_{pay} \leq v_f$													
$g_3 : g_3(y_{pro}^{MR}) \leq 0$	+	-				+							
$g_4 : g_4(y_{pro}^{MR}) \leq 0$						-						+	+
$g_5 : g_5(l_{arm}) \leq 0$	-	-	-			+	-					+	
$g_6 : \sigma_{rootarm} \leq \sigma_{max}$							+	-	+				
$g_7 : h_{spar} \leq t_{wing}$	+	+	-							+	+		
$g_8 : \sigma_{rootspar}^{MR} \leq \sigma_{max}$						+				-	-		
$g_9 : \sigma_{rootspar}^{FW} \leq \sigma_{max}$	-	+	+							-	-		

The following critical constraints are detected in the Monotonicity Table and solved explicitly:

g_5 (Eq. 3.21) critical w.r.t. l_{arm}

$$\longrightarrow l_{arm} = c_{root_w} + y_{pro}^{MR}(\tan \Lambda_{TE_w} - \tan \Lambda_{LE_w}) + (d_{pro}^{MR} + 2\delta_{pro}) \left(\frac{1}{\cos \Lambda_{TE_w}} + \frac{1}{\cos \Lambda_{LE_w}} \right) \quad (3.46)$$

g_6 (Eq. 3.23) critical w.r.t. d_{out}

$$\longrightarrow d_{out} = \left(\frac{32M_{root}}{\pi\sigma_{max}(1 - k_{arm}^4)} \right)^{1/3} \quad (3.47)$$

The two initial inequality constraints, g_5 and g_6 , are eliminated from the optimization problem, as well as the associated variables, l_{arm} and d_{out} . Finally, although the inequality constraint g_9 is critical with respect to the wing taper ratio, the equality cannot be explicitly solved for λ_w , so it cannot be optimized out.

Constraints transformation into sizing equations

The optimization problem can be further reduced by transforming some inequalities of the optimization problem into sizing equations with a degree of freedom that expresses the initial constraints. This special treatment of the optimization constraints was proposed by Reysset *et al.* [15] for the design of mechatronic systems. Based on physical considerations that translate optimality conditions, the authors first assume that all inequality constraints are potentially active, so are turned into equalities. For example, the motor torque constraint $T_{mot,k} \leq T_{nom}$ is likely to become an equality if a single flight scenario k is considered. Then, if the set of equations appears to be over-constrained, only a sufficient number of equalities are kept as explicit equations, and the excess inequalities are moved back to the optimization constraints. The original inequalities that were transformed into explicit equations are associated with sizing coefficients as degrees of freedom that enable the satisfaction of the design constraints. For the motor torque constraints, this strategy consists of transforming one of the inequality constraints into the following sizing equation:

$$T_{nom} = k_{mot}^s \cdot T_{mot,i} \quad (3.48)$$

The remaining inequalities $T_{mot,k \neq i} \leq T_{nom}$ are kept in the optimization constraints. The so-called *oversizing coefficient* k_{mot}^s lies in the range $[1, +\infty[$. The lower bound expresses the initial constraint since T_{nom} cannot be lower than $T_{mot,i}$. The unrestricted upper bound allows the remaining inequalities to be satisfied, for example, if the motor torque in a given flight scenario $k \neq i$ (e.g., cruise) is greater than that in scenario i (e.g., hover). The oversizing coefficient is set as a design variable in place of the nominal motor torque. Therefore, the strategy proposed by Reysset *et al.* [15] reduces the complexity of the problem in two ways:

1. It eliminates one or more constraints from the optimization problem,
2. And it replaces the initial design variables with sizing coefficients whose bounds are independent of the specifications.

A detailed application of the strategy for sizing multirotor UAVs can be found in [76]. In particular, the implementation of the strategy was shown to significantly reduce the number of iterations to convergence and the resolution time compared to the non-reduced problem. In the following, the methodology is specifically applied to the optimization problem of FW-VTOL UAVs.

For the FW-VTOL concept, the y-location of the VTOL propellers is defined from two inequalities (Equation 3.20) and one undetermined parameter, y_{pro}^{MR} . One of the inequalities is kept in the optimization constraints, and the remaining one is transformed into an equality by introducing a degree of freedom:

$$y_{pro}^{MR} = k_{y_{pro}}^s \left(\frac{d_{pro}^{FW}}{2} + \frac{d_{pro}^{MR}}{2} + \delta_{pro} \right) \quad (3.49a)$$

$$y_{pro}^{MR} \leq \frac{b_w}{2} \quad (3.49b)$$

where $k_{y_{pro}}^s$ is an oversizing coefficient within the range $[1, +\infty[$ and replaces the initial design variable y_{pro}^{MR} .

The structural sizing of the spars sets out two inequality constraints (Equations 3.24, 3.25). The resolution strategy involves introducing an oversizing coefficient to transform one of them into an equality:

$$\sigma_{root}^{FW} = k_{\sigma}^s \sigma_{max} \quad (3.50a)$$

$$\sigma_{root}^{MR} \leq \sigma_{max} \quad (3.50b)$$

Or, expressing Equation 3.50a in terms of the spars' height:

$$h_{spar} = k_{h_{spar}}^s \left(\frac{M_{root}^{FW} K_{flange} (1 + k_{spar})}{\sigma_{max} k_{spar}^2 (1 + k_{spar}^2/3)} \right)^{1/3} \quad (3.51a)$$

$$\sigma_{root}^{MR} \leq \sigma_{max} \quad (3.51b)$$

with $k_{h_{spar}}^s \in [1, +\infty[$ an oversizing coefficient that replaces the spars' height in the optimization variables.

Finally, a similar approach is adopted to replace the design variables of wing loading, tail moment arm, and wing position by sizing coefficients:

$$W/S = k_{WS}^s (W/S)_{stall} \quad (3.52a)$$

$$l_t = k_t^s b_w \quad (3.52b)$$

$$x_{MAC/4} = k_w^s b_w \quad (3.52c)$$

where $k_{WS}^s \in [0, 1]$ is an undersizing coefficient that expresses the initial constraint on the wing loading at stall speed. Unlike the other sizing coefficients, k_t^s and k_w^s are not derived from inequality constraints, but are useful to scale the physical parameters with respect to an initial guess.

3.4.4 Final problem formulation

The reduced optimization problem is summarized in Table 3.6, and an XDMS of the MDO architecture is provided in Figure 3.8. The problem reduction methodology reduced the number of design variables from 36 to 34 (by application of the MP1), and the number of constraints from 62 to 47. Specifically, 2 constraints were eliminated by application of the MP1 and 13 by introducing sizing coefficients to replace inequality constraints. In addition, the scaling of the variables allows for better handling by the optimization algorithms. The upper triangular structure of the XDMS clearly shows the absence of feedback relationships, thanks to the NVH formulation that delegates their resolution to the optimizer for better convergence.

The problem inputs consist mainly of the operating conditions for the sizing flight scenarios (e.g., flight speeds), the payload characteristics (e.g., mass and volume) and, for multirotor UAVs, the number of arms and rotors. In addition, the reference parameters for calibrating the scaling laws and analytical models can be modified by the designer to explore alternative technological assumptions. The "Performance" module allows the calculation of the UAV's performance for both the sizing mission and off-design missions specified by the designer. It essentially returns the energy consumption for the mission.

3.5 Implementation

The multidisciplinary models and the optimization problem are implemented in a dedicated software tool, FAST-UAV, which relies on the FAST-OAD open-source framework [7]. The object-oriented framework, written in Python and based on the OpenMDAO computing platform [44], allows easy switching between the models and adding or removing disciplines to match the desired drone concept. This capability is facilitated by the component-level description provided by the models. More precisely, the different (three) UAV concepts covered in this thesis share a significant number of models, and only the general structure of

Table 3.6 Final optimization problem for FW-VTOL UAVs.

Design variables	Parameter	Description
	k_{WS}^s	Undersizing coefficient on the wing loading
	AR_w, AR_{ht}, AR_{ot}	Aspect ratios of the wing and tails
	λ_w	Taper ratio of the wing
	k_w^s	Sizing coefficient on the wing position
	k_t^s	Sizing coefficient on the tail moment arm
	k_{ypro}^s	Oversizing coefficient on the y-location of VTOL propellers
	k_{hspar}^s	Oversizing coefficient on the spars' height
	k_{arm}, k_{spar}	Aspect ratios of the arms and spars
	β	Propeller pitch-to-diameter ratio
	k_{hdpro}^s	Undersizing coefficient on the propeller tip speed
	J_k	Propeller advance ratios for each flight scenario
	k_{mot}^s	Oversizing coefficient on the motor nominal torque
	k_{KV}^s	Sizing coefficient on the motor velocity constant
	k_{Ubat}^s	Oversizing coefficient on the battery voltage
	k_{Eibat}^s	Sizing coefficient on the battery energy
	k_{Pesc}^s	Oversizing coefficient on the ESC power
	k_{Uesc}^s	Oversizing coefficient on the ESC voltage
	k_M^c	Normalized variable for the total mass
	k_{CD0}^c	Normalized variable for the parasitic drag coefficient
	k_S^c	Normalized variable for top (projected) area
$\times 2$ (for fixed-wing propulsion and for VTOL propulsion)		
Disciplinary couplings resolution (NVH)		

Note: subscript $\cdot_k \in \{\cdot_{takeoff}, \cdot_{climb}, \cdot_{hover}, \cdot_{cruise}\}$ indicates the relevant flight scenarios.

the MDO needs to be modified. This MDO structure is simply declared via a configuration file (Figure 3.9), which does not require any programming knowledge on the user side. The configuration file also allows to specify the optimization objective and the optimization algorithm to be used. A high-level interface of FAST-UAV is accessible through Jupyter

Table 3.6 (continued).

	Parameter	Description
Constraints	$g_1 : SM_{req} \leq SM$	Constraint on the static margin
	$g_2 : v_{bat} + v_{pay} \leq v_f$	Constraint on the fuselage volume
	$g_3 : y_{pro}^{MR} \leq b_w/2$	Constraint on the maximum y-location of VTOL propellers
	$g_4 : h_{spar} \leq t_{wing}$	Constraint on the spars' height
	$g_5 : \sigma_{root,spar}^{MR} \leq \sigma_{max}$	Constraint on the spars' bending stress (VTOL takeoff)
	$g_6 : n_{k \neq i} \cdot d_{pro} \leq (n \cdot d_{pro})_{max}$	Constraints on the propeller tip speed
	$g_7 : J_k n_k d_{pro} = V_k$	Constraints on the UAV flight speed
	$g_8 : T_{k \neq i} \leq T_{nom}$	Constraints on the motor torque (steady state scenarios)
	$g_9 : T_k \leq T_{max}$	Constraints on the motor torque (transient scenarios)
	$g_{10} : P_{bat,k} \leq P_{bat}$	Constraints on the battery power
	$g_{11} : U_{mot,k \neq i} \leq U_{bat}$	Constraints on the battery voltage
	$g_{12} : P_{esc,k \neq i} \leq P_{esc}$	Constraints on the ESC power
	$g_{13} : E_{mission} \leq E_{bat}$	Constraint on the battery energy
	$g_{14} : m_{uav} \leq m_{uav,req}$	Constraint on the total mass not to be exceeded
Disciplinary couplings resolution (NVH)	$g_1^c : m_{uav} \leq m_{uav,guess}$	Consistency constraint on the total mass
	$g_2^c : C_{D_0,guess} = C_{D_0}$	Consistency constraint on the parasitic drag coefficient
	$g_3^c : S_{top} \leq S_{top,guess}$	Consistency constraint on the top (projected) area
	m_{uav} or $E_{mission}$ or t_{max} or R	Total mass of the UAV Energy consumption Flight endurance Flight range
Objectives		

Note 1: subscript $\cdot_k \in \{\cdot_{takeoff}, \cdot_{climb}, \cdot_{hover}, \cdot_{cruise}\}$ indicates the relevant flight scenarios.

Note 2: subscript $\cdot_i \in \{\cdot_{takeoff}, \cdot_{climb}, \cdot_{hover}, \cdot_{cruise}\}$ indicates the flight scenario for which an inequality constraint was transformed into a sizing equation.

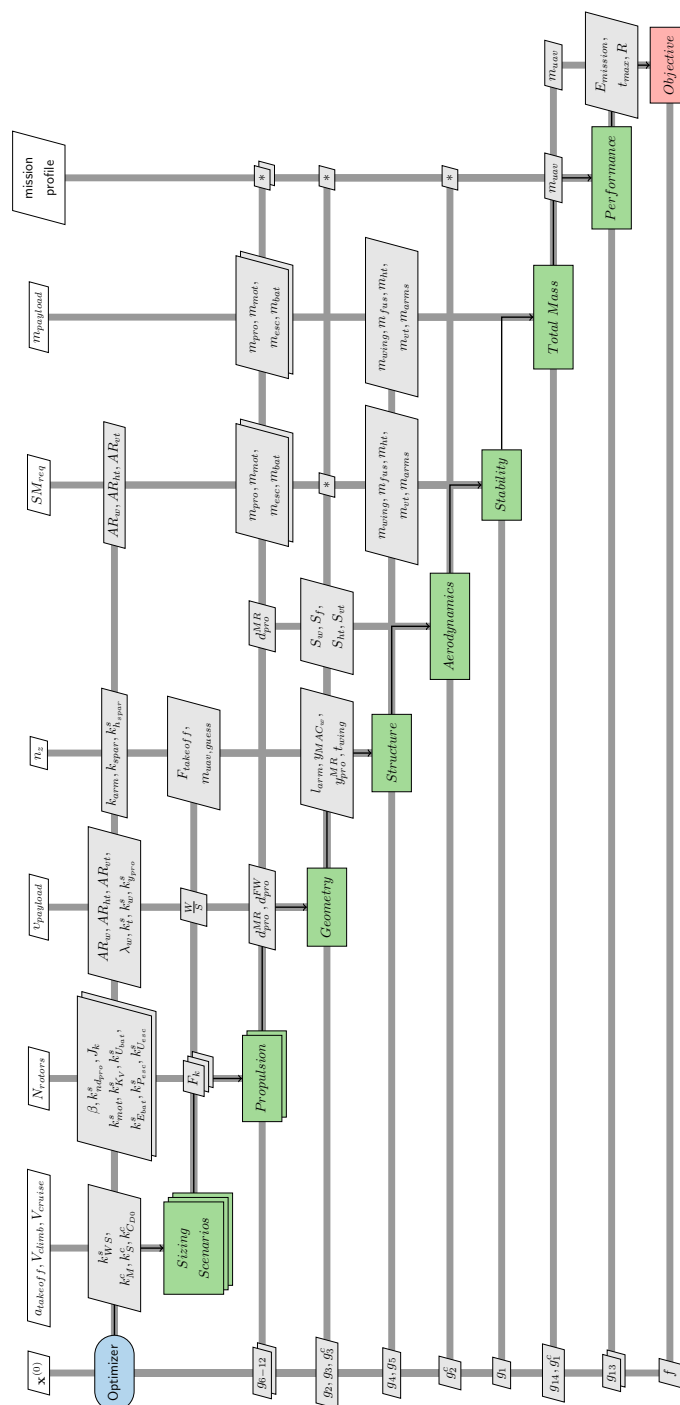


Figure 3.8 XDSM diagram for optimal sizing of FW-VTOL UAVs. Stacked boxes represent multiple instantiations of the analysis (e.g., two for the propulsion systems). The '*' symbol is used to indicate a large number of variables shared between analyses.

Notebooks [55]. It provides pre- and post-processing capabilities, including visualization of

the optimization problem with a N^2 diagram⁷, easy access to the input and output variables with interactive tables, and a variety of plots to display the geometry and mass breakdown of the sized UAVs (Figure 3.10). FAST-UAV is available open-source at the following GitHub repository: <https://github.com/SizingLab/FAST-UAV>.

```
# Definition of UAV model
model:
  scenarios:
    id: fastuav.scenarios.hybrid
  propulsion:
    id: fastuav.propulsion.hybrid
  geometry:
    id: fastuav.geometry.hybrid
  structures:
    id: fastuav.structures.hybrid
    spar_model: "I_beam"
  aerodynamics:
    id: fastuav.aerodynamics.hybrid
  stability:
    id: fastuav.stability.hybrid
  mtow:
    id: fastuav.mtow.hybrid
  performance:
    cruise_range:
      id: fastuav.performance.endurance.hybrid
    missions:
      id: fastuav.performance.mission
      file_path: ../../missions/missions_hybrid.yaml

# Definition of optimization algorithm
driver: om.ScipyOptimizeDriver(tol=1e-9, optimizer='SLSQP')

# Definition of the optimization objective
objective:
  # Mass minimization
  - name: data:weight:mtow

  # Endurance maximization
  #- name: data:performance:endurance:cruise
  # scaler: -1 # for maximization instead of minimization
```

(a) UAV concept definition.

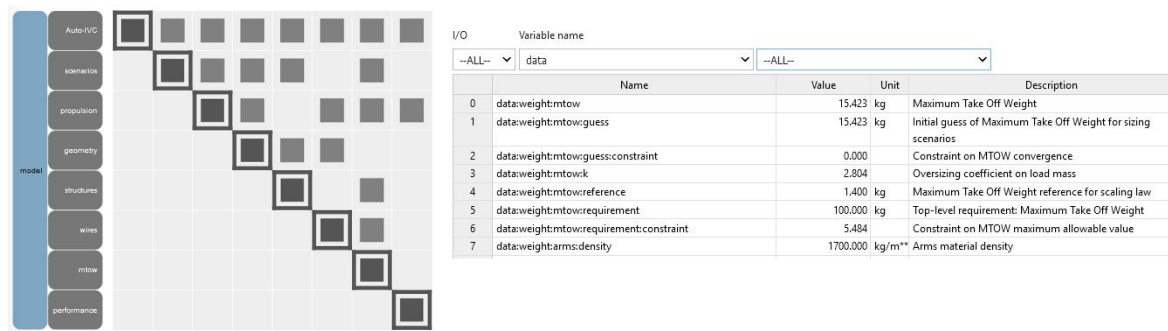
(b) Optimization algorithm and objective.

Figure 3.9 Example of declaration of the design problem in the configuration file.

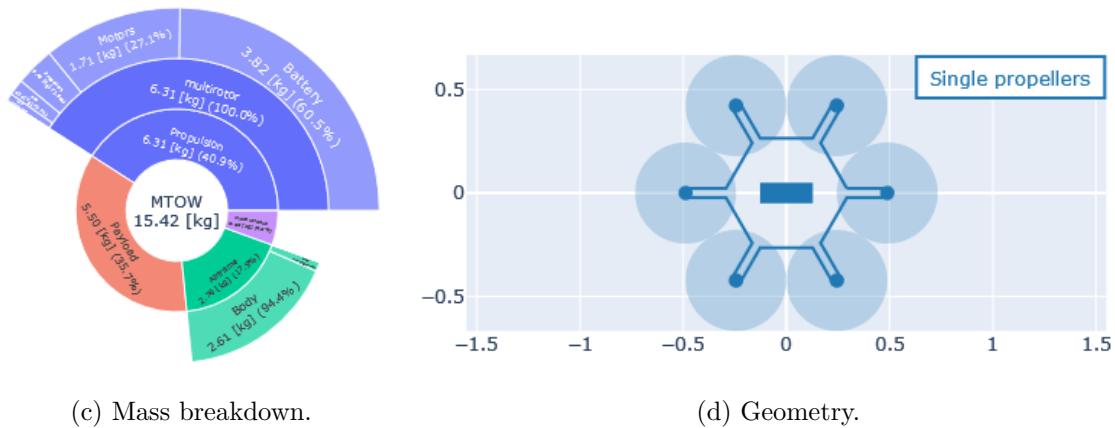
3.5.1 Optimization algorithm

The continuous nature of the models involved in the sizing process motivates the use of a gradient-based optimizer. In this study, the SLSQP implementation [20] is used. Where possible, the analytical expressions of the model derivatives are provided to the OpenMDAO computing platform. This greatly improves convergence by avoiding the numerical inaccuracies and computational time associated with a finite difference calculation. Importantly, the use of sizing coefficients as introduced in Section 3.4.3 provides a good estimate for the starting point of the gradient descent that is independent of the specifications. This is a significant improvement over non-normalized formulations. However, the gradient-based algorithm, like all other local search algorithms, is vulnerable to the multimodality of the problem. Specific strategies, such as multi-start methods, could be implemented in the future to increase the probability of convergence to a global optimum. The algorithm performance for typical optimization problems on a 1.8 GHz quad-core processor is shown in Table 3.7.

⁷ N^2 diagrams [120] are similar to XDSM, but focus on data connections rather than process flows.

(a) N^2 diagram.

(b) Variables viewer.



(c) Mass breakdown.

(d) Geometry.

Figure 3.10 Examples of plots and widgets in the FAST-UAV interface.

Table 3.7 Gradient-based algorithm performance for different concepts of UAVs.

	Multicopter	Fixed-Wing	FW-VTOL
Number of variables	12	21	34
Number of constraints	21	25	47
Number of function evaluations	101	139	129
Number of iterations	55	81	115
Resolution time (s)	4.1	14.2	21.8

3.6 Results and validation

A brief validation of the sizing tool and sensitivity studies are provided in this section. In addition, a case study is presented to illustrate the relevance of having a unified framework for sizing different concepts of UAVs. Additional results are provided throughout the manuscript.

3.6.1 Multirotor UAVs

Validation

A first validation of the sizing tool for multirotor UAVs is provided in [76]. However, the original publication do not consider forward flight conditions in the models and optimization. Therefore, the present study aims to extend the validation when considering the cruise flight segment.

The validation case focuses on the Parrot ANAFI USA⁸ shown in Figure 3.11. The quadrotor UAV has a total mass of 500 grams, and is equipped with a triple camera module which is assumed to weigh 250 grams. In addition, the airframe drag coefficient is assumed to be that of a cube, i.e. 1.0.



Figure 3.11 Parrot ANAFI USA. Source: Parrot.

The validation process consists of finding the optimal size and performance for a quadrotor UAV with similar mass and high-level requirements. Specifically, the optimization problem consists of maximizing the range of the UAV. The total mass of the UAV, the payload mass, and the takeoff acceleration are given as constraints. In addition to the design variables used to optimize the size of the components, the cruise speed is considered as an optimization variable. This allows the optimizer to find the optimal speed to maximize range.

Table 3.8 Specifications for the Parrot ANAFI USA optimization problem.

Requirements	UAV total mass:	500 grams
	Payload mass:	250 grams
	Payload consumption:	15 W
	Max. takeoff acceleration:	2.5 g
	Max. climb speed:	4 m/s
Design assumption	Airframe drag coefficient:	1.0
Objective function	Maximum flight range	

⁸Parrot ANAFI USA: <https://www.parrot.com/us/drones/anafi-usa>

Table 3.9 shows the sizing results for different components and flight parameters, and Table 3.10 lists the active constraints that have driven the results. The calculated characteristics are close to the manufacturer's data, with small deviations for the flight range and the optimum flight speed: -1.1% and +6.2% respectively. The constraints activity highlights that for this case study, the takeoff and hover scenarios are key to determining a feasible solution. In addition, the maximum range objective in cruise flight drives the sizing of the battery. Notably, three parameters remain unaffected by active constraints: the maximum speed of the propeller, the maximum torque of the motor, and the power of the battery. This is because the sizing of these components is driven primarily by steady-state (endurance) scenarios rather than extreme scenarios. For example, the substantial energy of the battery required to maximize the range inherently results in a high power rating, since for a given technology, power is proportional to energy. The resulting power is sufficient to withstand any of the flight scenarios. For the propeller, the tip speeds in the different flight scenarios were all found to be lower than the maximum speed allowed by the propeller sizing. This indicates that the propellers are actually oversized for the thrust they are required to provide. This allows the propellers to be more efficient in endurance scenarios (e.g. cruise), thereby reducing energy consumption and increasing the achievable range. This conclusion is specific to the objective of maximizing the range (or, equivalently, the efficiency of the UAV). In fact, a larger propeller diameter is more efficient, but adds weight, particularly due to the increased arms length required for sufficient clearance. As will be shown in Chapter 9, an energy efficient design is not necessarily lighter. Had the optimization objective been a mass minimization one, at least one constraint on the propeller tip speed in flight would have been active. Finally, the climb flight does not constrain the sizing here, but an increase in the climb speed requirement may have led to different conclusions. Overall, the sizing of the UAV is driven by a mix of flight scenarios, highlighting the need to consider all flight conditions to obtain a realistic design.

Table 3.9 Comparison of results for the Parrot ANAFI USA.

	Reference	Sizing	Relative difference [%]
Range [km]	17.4	17.2	-1.1
Optimal Velocity [m/s]	11.3	12.0	+6.2
Battery energy [Wh]	40.0	42.2	+5.5
Battery mass [g]	195	183	-6.2
Motor max. power [W]	46	49.8	+8.3
Top surface [m ²]	0.026	0.023	-11.5
Front surface [m ²]	0.009	0.010	+11.1

Table 3.10 Active constraints for the Parrot ANAFI USA design problem.

Component	Parameter	Active constraints
Propeller	Tip speed $n \cdot d_{pro}$	No active constraint
Motor	Nominal torque T_{nom}	$T_{nom} = T_{hover}$
	Maximum torque T_{max}	No active constraint
ESC	Voltage U_{esc}	$U_{esc} = U_{bat}$
	Power P_{esc}	$P_{esc} = P_{esc,takeoff}$
Battery	Voltage U_{bat}	$U_{bat} = U_{mot,takeoff}$
	Power P_{bat}	No active constraint
	Energy E_{bat}	$E_{bat} = E_{mission} = E_{max. range}$

Sensitivity study on the airframe aerodynamics

As explained in the modeling Section 3.3, the aerodynamic coefficients for multirotor UAVs lacks a parametric model, as these quantities depend on a number of criteria that are beyond the scope of the manuscript. In particular, the drag and lift coefficients are provided as constant values by the designer. However, depending on the values of the aerodynamic coefficients, there may be large variations in the performance of the UAV and thus in the sizing of the components. This effect is investigated here through a sensitivity study.

Specifically, the optimization problem consists of sizing an octocopter UAV for a 5 kg payload and a design range of 20 km. The optimization goal is to minimize the energy consumption over the 20 km mission, and the flight speed is set as an optimization variable. No constraint is set on the total mass of the UAV, which is an output of the sizing process. The optimization problem is run for multiple values of the aerodynamic drag coefficient, ranging from 0.4 to 1.2.

Table 3.11 Specifications for the optimization problem to investigate sensitivity to airframe aerodynamics.

Requirements	Number of arms:	8
	Number of propellers per arm:	1
	Payload mass:	5 kg
	Max. vertical acceleration:	2 g
	Design range:	20 km
Design assumption	Airframe drag coefficient:	0.4 - 1.2
Objective function	Minimum energy consumption	

The optimization results are provided in Figures 3.12 and 3.13. The mass breakdown Figure 3.12 shows the sensitivity of the vehicle mass to an increase in the value of the drag coefficient. The observed increase in total mass is primarily related to an increase in battery size, as the other components remain quite unchanged. In fact, the optimization of the flight speed (Figure 3.13) enables to keep the power requirements as low as possible when sizing the components, so that the propellers, motors, and ESCs are little affected by a change in

the drag coefficient. However, because energy consumption increases with drag, the battery must be oversized to provide enough energy to complete the mission.

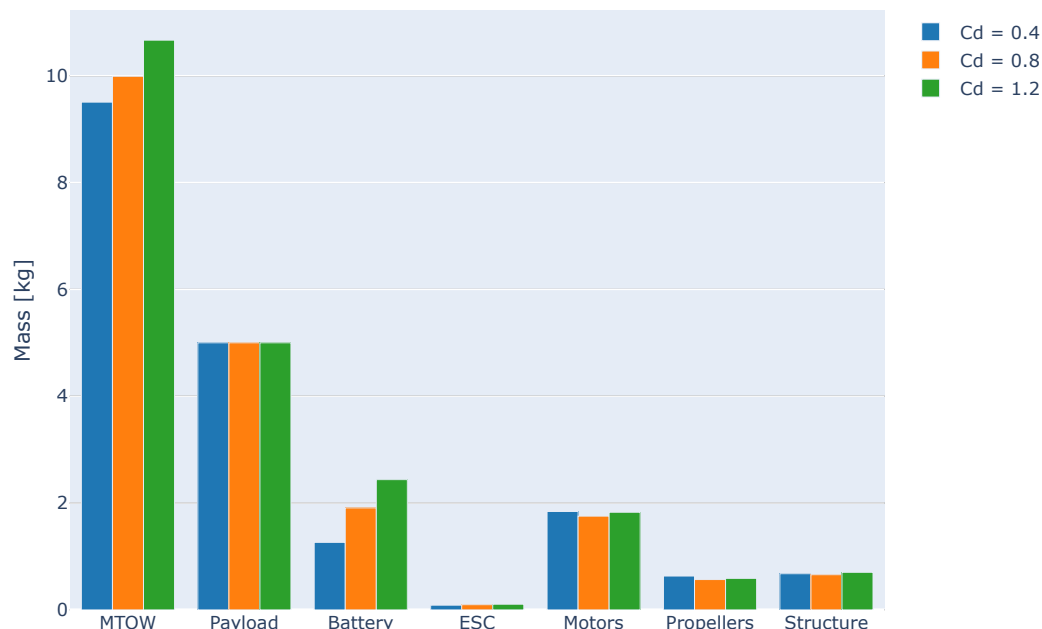


Figure 3.12 Evolution of the component's masses with respect to the airframe drag coefficient.

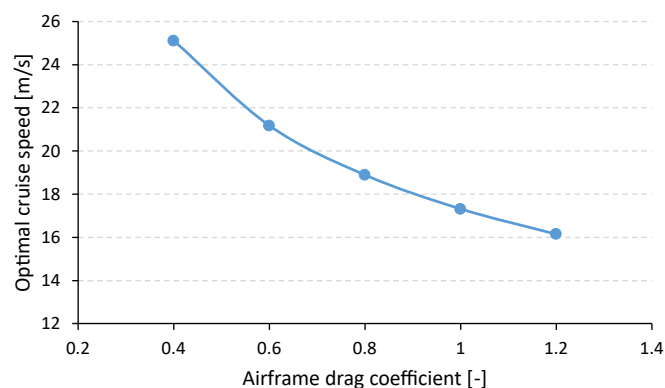


Figure 3.13 Evolution of the optimal flight speed with respect to the airframe drag coefficient.

Effect of increased range and payload on UAV size

The effect of range and payload specifications on the size of a hexacopter UAV is shown in Figure 3.14. For each payload mass and range requirement, the UAV is optimally sized with a mass minimization objective. The flight speed is kept constant. The snowball effect is clearly visible. Longer ranges require additional battery mass, which in turn increases the vehicle mass. For a given payload mass, there is a maximum range that can be achieved, depending

on the technology used in the design (e.g. battery chemistry and airframe materials). The increase in vehicle mass is all the more important as the range requirement approaches its maximum value. Finally, it is observed that smaller UAVs achieve better range capacity than heavier ones.

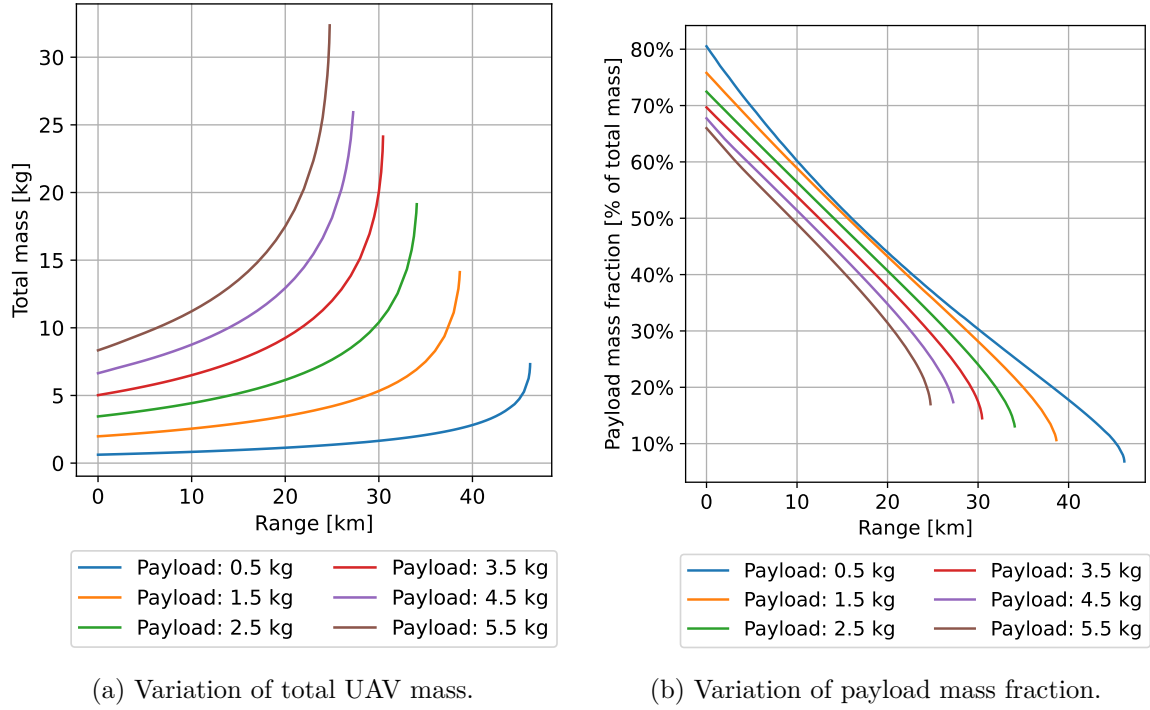


Figure 3.14 Variation of multirotor UAV design with range and payload requirements.

Effect of battery technology on UAV size

To investigate the importance of technological evolutions on UAV design, a design of experiments is carried out on the battery energy density and the flight endurance specification. All other specifications and technological assumptions are held constant. The results for a generic hexacopter UAV sized for a 2 kg payload mass are provided in Figure 3.15. The results quantify and demonstrate that the evolution of battery technology in recent years has enabled the emergence of multirotor UAVs with increased endurance.

3.6.2 Fixed-wing UAVs

The wide variety of materials and technologies used for fixed-wing and FW-VTOL UAVs and the lack of extensive data on existing vehicles makes it difficult to provide a comprehensive validation of these concepts. However, we propose here to validate the sizing of fixed-wing UAVs based on a high-level characteristic easily accessible from drone manufacturers: the product of payload mass and endurance as a function of the total mass. Examples of sizing

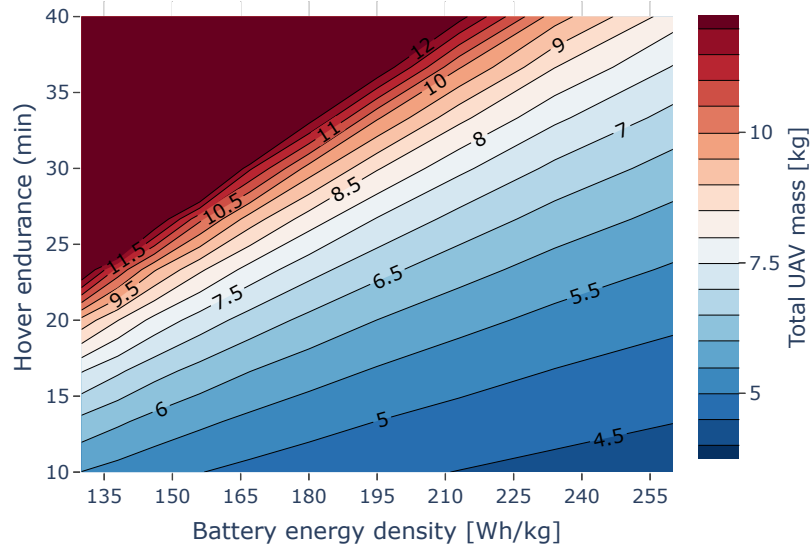


Figure 3.15 Evolution of UAV mass with respect to flight endurance specification and battery density assumption. The red area represent non-feasible designs.

results for five drones of different scales are depicted in Figure 3.16 and compared with existing vehicles⁹. The UAVs are sized for a constant share of payload mass, equal to 20% of the total mass. A fixed mass is added to account for the flight management system and the communication systems, which are not modeled in this manuscript. In addition, the same technological and material assumptions are applied to the five designs (see Appendix 3.D). Also provided in Figure 3.16 is the sizing correlation derived by Verstraete *et al.* [65] on the basis of 230 battery-powered fixed-wings introduced in the market before 2017.

The sizing results are consistent with the market trends, although they seem optimistic for small UAVs compared to Verstraete's regression. However, a great disparity is observed within the existing fixed-wings on the market. This is explained by the multitude of technological choices and materials available, in connection with economic factors. The low regression coefficient from Verstraete's correlation, equal to 0.827, highlights this phenomenon.

3.6.3 Comparative studies

The relevance of a common framework for sizing different concepts and configurations of UAVs is now illustrated with case studies.

Table 3.12 provides sizing examples for a multicopter, a fixed-wing and a FW-VTOL UAV given the same specifications and design constraints. In particular, the UAVs are sized to carry a 3 kg payload at a cruise speed of 16 m/s while maximizing cruise endurance. In addition, the total mass of the UAVs should not exceed 15 kg, as may be imposed by

⁹Delair DT26 Open-Payload, Delair DT18 AG, Lockheed Hawk III & IV, EOS Strix 300 & 400, Aeronautics Orbiter 2 & 3 AE, BlueBird SpyLite, AeroVironment Puma LE & AE, AeroVironment Raven B, AeroVironment Wasp AE.

Table 3.12 Sizing comparison of three UAV concepts for the same specifications.

		Hexacopter	FW-VTOL	Fixed-Wing
Inputs	Payload mass	3 kg	3 kg	3 kg
	Payload volume	0.015 m ³	0.015 m ³	0.015 m ³
	Total mass	15 kg	15 kg	15 kg
	Cruise speed	16 m.s ⁻¹	16 m.s ⁻¹	16 m.s ⁻¹
	Climb speed / Rate of climb	3 m.s ⁻¹	14 m.s ⁻¹ / 3 m.s ⁻¹	14 m.s ⁻¹ / 3 m.s ⁻¹
	Stall speed	-	13 m.s ⁻¹	12.5 m.s ⁻¹
	Vertical takeoff acceleration	0.8 g	0.3 g	-
	Materials	Carbon/fiberglass composites, balsa wood, foam		
Outputs	Max. cruise endurance	33 min	85 min	171 min
	Max. cruise distance	31.8 km	81.8 km	164.2 km
	Wing area	-	1.2 m ²	1.3 m ²
	Wing span	-	4.1	4.3
	Wing aspect ratio	-	14.1	14.1
	Taper ratio	-	1.0	1.0
	Fuselage length	-	1.6 m	1.7
	Footprint (length × width)	1.9 m ²	6.6 m ²	7.2 m ²
	Parasitic drag coefficient	0.5*	0.031	0.023
	Battery power in cruise	2067 W	211 W	178 W
	Battery energy	1428 Wh	374 Wh**	635 Wh
	FW propeller diameter	-	0.68 m	0.72 m
	MR propellers diameter	0.46 m	0.48 m	-
	Structures mass fraction	15%	44%	46%
	FW propulsion mass fraction	-	23%	35%
	MR propulsion mass fraction	65%	13%	-

* Drag coefficient for multirotor drones is a fixed input (see Section 3.3.5).

** Excluding VTOL battery (only the battery for fixed-wing propulsion).

when there is no severe limitation on the total mass, when the relative mass penalty of the VTOL system is low compared to cruise flight propulsion.

Table 3.13 Achievable flight endurance for various UAV configurations and total mass requirements, with a 1 kg payload specification.

Total mass	Quadcopter	Coaxial quadcopter	Hexacopter	Coaxial hexacopter	Octocopter	FW-VTOL	Fixed-wing
3 kg	33 min	38 min	31 min	37 min	29 min	37 min	70 min
5 kg	41 min	48 min	38 min	46 min	36 min	105 min	161 min
10 kg	44 min	52 min	41 min	50 min	38 min	152 min	229 min
15 kg	43 min	51 min	40 min	49 min	36 min	162 min	252 min
20 kg	42 min	50 min	38 min	48 min	34 min	163 min	263 min
40 kg	36 min	45 min	32 min	42 min	27 min	153 min	272 min

3.7 Conclusions and perspectives

Methodology and lessons learned

This chapter introduces a methodology for the optimal sizing of multirotor, fixed-wing and FW-VTOL UAVs. Analytical models for the different disciplines (geometry, structures, aero-

Table 3.14 Achievable flight endurance for various UAV configurations and total mass requirements, with a 5 kg payload specification.

Total mass	Quadcopter	Coaxial quadcopter	Hexacopter	Coaxial hexacopter	Octocopter	FW-VTOL	Fixed-wing
15 kg	23 min	29 min	21 min	28 min	18 min	15 min	62 min
20 kg	27 min	34 min	24 min	32 min	20 min	38 min	112 min
25 kg	29 min	36 min	25 min	34 min	21 min	58 min	152 min
30 kg	29 min	38 min	26 min	35 min	21 min	70 min	173 min
40 kg	29 min	39 min	25 min	36 min	21 min	82 min	197 min
50 kg	28 min	39 min	24 min	36 min	19 min	86 min	211 min

dynamics, propulsion and stability) are provided, and an optimization problem is formulated. The disciplinary couplings are resolved with a NVH formulation, and the complexity of the problem is reduced through the application of the first monotonicity principle and the introduction of sizing coefficients. These strategies enable to remove a number of constraints and design variables without affecting the results. Moreover, the formulation of the optimization problem introduces design variables whose bounds are independent of the application, making it appropriate to a wide range of specifications. The implementation of the methodology in an open-source software tool shows very low resolution times with the use of a gradient-based algorithm. As such, it is suitable for preliminary sizing. In addition, the component-level descriptions and object-oriented implementation make it simple to switch from one UAV concept to another.

The validity of the methodology is demonstrated for multirotor and fixed-wing UAVs by comparing the sizing results with existing drones. In particular, the results emphasize the need to consider different flight scenarios in order to obtain a realistic design. Additionally, the importance of airframe aerodynamics and battery technology is illustrated through sensitivity studies on multirotor designs. Finally, a first insight of the tool's ability to compare different concepts is provided through a case study. The hybrid FW-VTOL concept shows a significant reduction in flight endurance compared to its fixed-wing counterpart due to the mass penalty of the VTOL propulsion system. Therefore, this concept should only be dedicated to missions where vertical takeoff and landing or hovering is essential. Nevertheless, the FW-VTOL offers additional endurance compared to a hexacopter UAV with similar capabilities. The selection of a UAV concept, however, cannot be reduced to an endurance criterion and should also consider other aspects such as maneuverability and safety (including controllability and size).

Perspectives

The methodology could be further improved with the development of aerodynamic models to better describe the airframe performance of multirotor UAVs. In addition, the structural discipline could be refined, for example by modeling the landing gear of multirotor UAVs, or by considering wing torsion and empennage loads for fixed-wing configurations. Additional flight scenarios, such as maximum turn rate and runway takeoff with landing gear, could be

developed to further refine the methodology. Despite these limitations, the software tool already provides a solid framework for a variety of design studies and applications.

Appendix

Appendix 3.A Regression models for propellers

The performance data to fit the regression models is obtained from APC propellers manufacturer: <https://www.apcprop.com/technical-information/performance-data/>. Multirotor (VTOL) propellers are distinguished from fixed-wing UAV propellers.

3.A.1 Multirotor (VTOL) propellers

Based on "Multi Rotor (Electric Only)" APC propellers category.

Thrust coefficient

Static flight ($V = 0$) [72]:

$$C_T^{static}(\beta) = 4.27 \times 10^{-2} + 1.44 \times 10^{-1} \cdot \beta \quad (3.53)$$

Axial flight ($V > 0$, $\alpha = \pi/2$) [72]:

$$\begin{aligned} C_T^{axial}(\beta, J) = & 0.02791 - 0.06543J + 0.11867\beta + 0.27334\beta^2 - 0.28852\beta^3 \\ & + 0.02104J^3 - 0.23504J^2 + 0.18677\beta \cdot J^2 \end{aligned} \quad (3.54)$$

Nonaxial flight ($V > 0$, $\alpha \in [0, \pi/2]$):

$$C_T(\beta, \alpha, J) = \eta_T(\beta, \alpha, J) C_T^{axial}(\beta, J_{axial}) \quad (3.55)$$

$$\eta_T(\beta, \alpha, J) = 1 + \frac{(J \cos \alpha / \pi \bar{r}')^2}{2(1 - J \sin \alpha / J_{0T}^{axial})} \delta(\alpha) \quad (3.56)$$

$$J_{0T}^{axial} = 0.197 + 1.094 \cdot \beta \quad (3.57)$$

Power coefficient

Static flight ($V = 0$) [72]:

$$C_P^{static}(\beta) = -1.48 \times 10^{-3} + 9.72 \times 10^{-2} \cdot \beta \quad (3.58)$$

Axial flight ($V > 0$, $\alpha = \pi/2$) [72]:

$$C_P^{axial}(\beta, J) = 0.01813 - 0.06218\beta + 0.00343J + 0.35712\beta^2 - 0.23774\beta^3 + 0.07549\beta J - 0.1235J^2 \quad (3.59)$$

Nonaxial flight ($V > 0$, $\alpha \in [0, \pi/2]$):

$$C_P(\beta, \alpha, J) = \eta_P(\beta, \alpha, J)C_P^{axial}(\beta, J_{axial}) \quad (3.60)$$

$$\eta_P(\beta, \alpha, J) = 1 + \frac{(J \cos \alpha / \pi \bar{r}')^2}{2(1 - J \sin \alpha / J_{0P}^{axial})} \delta(\alpha) \quad (3.61)$$

$$J_{0P}^{axial} = 0.286 + 0.993 \cdot \beta \quad (3.62)$$

3.A.2 Fixed-wing propellers

Based on "Slow Fly (Electric Only)" APC propellers category.

Thrust coefficient

Axial flight ($V > 0$, $\alpha = \pi/2$):

$$C_T^{axial}(\beta, J) = 0.09613 - 0.26688J + 0.37102\beta \cdot J - 0.15240\beta \cdot J^2 \quad (3.63)$$

Nonaxial flight ($V > 0$, $\alpha \in [0, \pi/2]$):

$$C_T(\beta, \alpha, J) = \eta_T(\beta, \alpha, J)C_T^{axial}(\beta, J_{axial}) \quad (3.64)$$

$$\eta_T(\beta, \alpha, J) = 1 + \frac{(J \cos \alpha / \pi \bar{r}')^2}{2(1 - J \sin \alpha / J_{0T}^{axial})} \delta(\alpha) \quad (3.65)$$

$$J_{0T}^{axial} = 0.213 + \beta \quad (3.66)$$

Power coefficient

Axial flight ($V > 0$, $\alpha = \pi/2$) [72]:

$$C_P^{axial}(\beta, J) = 0.00440 + 0.08432\beta - 0.03854\beta^2 - 0.03864J - 0.08185J^3 + 0.12568\beta^2 \cdot J \quad (3.67)$$

Nonaxial flight ($V > 0$, $\alpha \in [0, \pi/2]$):

$$C_P(\beta, \alpha, J) = \eta_P(\beta, \alpha, J)C_P^{axial}(\beta, J_{axial}) \quad (3.68)$$

$$\eta_P(\beta, \alpha, J) = 1 + \frac{(J \cos \alpha / \pi \bar{r}')^2}{2(1 - J \sin \alpha / J_{0P}^{axial})} \delta(\alpha) \quad (3.69)$$

$$J_{0P}^{axial} = 0.207 + 1.036 \cdot \beta \quad (3.70)$$

Appendix 3.B Geometrical relationships for the fuselage

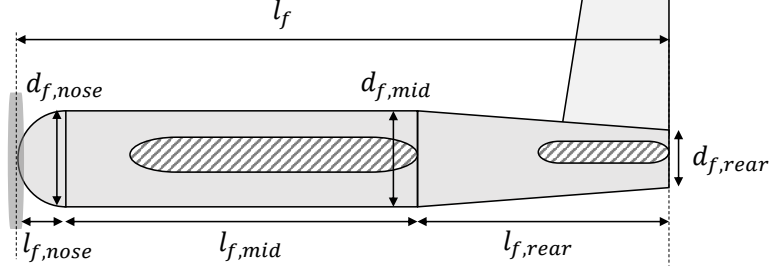


Figure 3.B.1 Fuselage geometry.

Fuselage length:

$$l_f = x_{TE_{ht}} = \underbrace{x_{MAC/4}}_{\text{wing w.r.t. nose tip}} + \underbrace{l_t}_{\text{tail moment arm}} + \underbrace{0.75 \times c_{MAC_{ht}}}_{\text{quarter-chord to TE}} \quad (3.71)$$

Mid-fuselage diameter:

$$d_{f,mid} = \frac{l_f}{\lambda_f} \quad (3.72)$$

Nose diameter and length:

$$l_{f,nose} = d_{f,nose} = d_{f,mid} = \frac{l_f}{\lambda_f} \quad (3.73)$$

Rear fuselage diameter:

$$d_{f,rear} = \kappa_f \cdot d_{f,mid} \quad (3.74)$$

Mid-fuselage and rear length:

$$l_{f,rear} = x_{TE_{ht}} - x_{TE_w} \quad (3.75)$$

$$l_{f,mid} = l_f - l_{f,rear} - l_{f,nose} \quad (3.76)$$

Surfaces:

$$S_{nose} = 2\pi \left(\frac{d_{f,nose}}{2} \right)^2 \quad (3.77)$$

$$S_{mid} = \pi \cdot d_{f,mid} \cdot l_{f,mid} \quad (3.78)$$

$$S_{rear} = \pi \left(\frac{d_{f,mid} + d_{f,rear}}{2} \right) l_{f,rear} + \pi \left(\frac{d_{f,rear}}{2} \right)^2 \quad (3.79)$$

Mid-fuselage volume (for housing payload and batteries):

$$v_f = \pi \left(\frac{d_{f,mid}}{2} \right)^2 l_{f,mid} \geq v_{bat} + v_{pay} \quad (3.80)$$

Appendix 3.C Simulation models for performance analysis

In the following, the subscript $\cdot_k \in \{\cdot_{takeoff}, \cdot_{climb}, \cdot_{hover}, \cdot_{cruise}\}$ indicates the relevant flight scenario.

3.C.1 Propeller [72]

Required thrust:

$$F_{pro,k} = (F/W)_k \cdot m_{uav} g \quad (3.81)$$

Propeller rotational speed:

$$\Omega_{pro,k} = \frac{2\pi}{60} n_{pro,k} = \frac{2\pi}{60} \sqrt{\frac{F_{pro,k}}{C_T \rho_{air} d_{pro}^4}} \quad (3.82)$$

Propeller power:

$$P_{pro,k} = C_P \rho_{air} n_{pro,k}^3 d_{pro}^5 \quad (3.83)$$

3.C.2 Motor [72]

Motor torque:

$$T_{mot,k} = \frac{P_{pro,k}}{\Omega_{pro,k}} \quad (3.84)$$

Motor current:

$$I_{mot,k} = K_V (T_{mot,k} + T_f) \quad (3.85)$$

Motor voltage:

$$U_{mot,k} = \frac{\Omega_{pro,k}}{K_V} + R I_{mot,k} \quad (3.86)$$

Motor power:

$$P_{mot,k} = U_{mot,k} I_{mot,k} \quad (3.87)$$

3.C.3 Battery [72]

Battery power:

$$P_{bat,k} = \frac{N_{rotors} P_{mot,k}}{\eta_{esc}} \quad (3.88)$$

Battery current:

$$I_{bat,k} = \frac{P_{bat,k}}{U_{bat}} \quad (3.89)$$

3.C.4 Electronic speed controller [72]

ESC apparent power:

$$P_{esc,k} = U_{bat} \cdot \frac{P_{mot,k}}{U_{mot,k}} \quad (3.90)$$

3.C.5 Structural analysis

Bending stress for circular hollow arms:

$$\sigma_{root} = \frac{M_{root}}{I} \frac{d_{out}}{2} = \frac{32d_{out}M_{root}}{\pi(d_{out}^4 - d_{in}^4)} = \frac{32M_{root}}{\pi d_{out}^3(1 - k_{arm}^4)} \quad (3.91)$$

$$M_{root} = \frac{F_k}{N_{rotors}^{MR}} l_{arm} \quad (3.92)$$

Bending stress for I-beam spars:

$$\sigma_{root} = \frac{M_{root} K_{flange}(1 + k_{spar})}{h_{spar}^3 k_{spar}^2 (1 + k_{spar}^2/3)} \quad (3.93)$$

$$M_{root}^{FW} = \frac{n_z m_{uav} g}{2} y_{MAC_w} \quad (3.94)$$

$$M_{root}^{MR} = \frac{F_k}{2} y_{pro}^{MR} \quad (3.95)$$

3.C.6 UAV performance

Mission energy consumption:

$$E_{mission} = \sum_k P_{bat,k} \cdot t_k \quad (3.96)$$

$$t_k = \frac{\text{distance}_k}{V_k} \quad (3.97)$$

Maximum hover/cruise endurance:

$$t_{max,k} = 0.8 \frac{C_{bat}}{I_{bat,k}} \quad (0.8 \text{ factor for minimum battery reserve}) \quad (3.98)$$

Range:

$$R = t_{max,cruise} \cdot V_{cruise} \quad (3.99)$$

Appendix 3.D Reference values

The values provided in the following Table 3.D.1 are based on:

1. Propeller: APC Propellers 11 x 4.5 (<https://www.apcprop.com>)
2. Motor: AXI 5325/16 GOLD LINE (<https://www.modelmotors.cz>)
3. ESC: Turnigy K-Force 70A-HV (<https://turnigy.com>)
4. Battery: DJI Matrice 30 Series TB30 (<https://www.dji.com/>)
5. Wires: PVC-insulated single wiring cables (https://www.engineeringtoolbox.com/wire-gauges-d_419.html)
6. Multicopter frames: MK Quadro (<https://wiki.mikrokoetter.de/en/EasyQuadro>), Concordia University Student project ("*Capstone Aerospace Engineering Design Project*").
7. Frame materials for fixed-wing UAVs: [100, 121]

Table 3.D.1 Reference values for the models.

Component	Parameter	Value	Unit	Description
Propeller	$m_{pro,ref}$	0.015	kg	Reference propeller mass
	$d_{pro,ref}$	0.28	m	Reference propeller diameter
	$(n \cdot d_{pro})_{max}$	44.45	m/s	Maximum tip speed from manufacturer [76]
	σ_{pro}	0.15	-	Propeller solidity
Motor	$m_{mot,ref}$	0.575	kg	Reference motor mass
	KV_{ref}	33.3	rad/V/s	Reference velocity constant
	$T_{nom,ref}$	2.32	N.m	Reference nominal torque
	$T_{max,ref}$	2.82	N.m	Reference maximum torque
	$T_{f,ref}$	0.03	N.m	Reference friction torque
	R_{ref}	0.03	Ω	Reference resistance
	$m_{esc,ref}$	0.115	kg	Reference ESC mass
ESC	$P_{esc,ref}$	3108	W	Reference ESC power
	$U_{esc,ref}$	44.4	V	Reference ESC voltage
	η_{esc}	0.95	-	ESC efficiency
	$m_{bat,ref}$	0.68	kg	Reference battery mass
Battery	$U_{bat,ref}$	22.8	V	Reference battery voltage
	$C_{bat,ref}$	20,520	A.s	Reference battery capacity
	$I_{max,ref}$	200	A	Reference discharge current
	$v_{bat,ref}$	500	cm ³	Reference battery volume
Wires	$\mu_{wires,ref}$	0.191	kg/m	Reference linear mass
	$I_{wires,ref}$	120	A	Reference wires rating
Frame aerodynamics	C_D^{MR}	0.5	-	Drag coefficient (for multirotor UAVs)
		1.0	-	(for FW-VTOL UAVs in vertical climb)
	C_L^{MR}	0	-	Lift coefficient (for multirotor UAVs)
	$C_{D_{pro}}^{MR}$	0.015	-	Drag coefficient for stopped VTOL propellers
	$C_{L_{max}}^{FW}$	1.2	-	Maximum lift coefficient (for fixed-wings)

Table 3.D.1 (continued).

Component	Parameter	Value	Unit	Description
Frame geometry	$S_{top,ref}^{MR}$	0.02	m ²	Reference top surface (for multirotor UAVs)
	$S_{front,ref}^{MR}$	0.045	m ²	Reference front surface (for multirotor UAVs)
	$m_{uav,ref}^{MR}$	1.4	kg	Reference UAV mass (for multirotor UAVs)
	v_{ht}	0.55	-	Horizontal tail volume coefficient
	v_{vt}	0.035	-	Vertical tail volume coefficient
	δ_{pro}	0.1	m	VTOL propellers clearance from wing
	ΛLE_w	0.0	rad	Sweep of wing's leading edge
	t/c	0.15	-	Thickness-to-chord ratio (wing and tails)
	$(x/c)_{max}$	0.3	-	Chordwise location of maximum airfoil thickness
	λ_f	5.0	-	Fuselage fineness ratio
	κ_f	0.2	-	Fuselage tail cone ratio
	σ_{max}	$280 \times 10^6/4$	N/m ²	Maximum allowable stress for composite (divided by factor 2 for dynamics and 2 for stress concentration)
	$\rho_{composite}$	1700	kg/m ³	Composite material density for arms and spars
	K_{flange}	0.1	-	Flanges thickness-to-length ratio
Frame structures and mass	t_{rib}	0.0035	m	Ribs thickness
	ρ_{balsa}	160	kg/m ³	Balsa wood density for the ribs
	ρ_{skin}	0.2	kg/m ²	Areal density for wing skin
	ρ_{fus}	2.0	kg/m ²	Areal density for fuselage
	$\rho_{ht,vt}$	2.0	kg/m ²	Areal density for tails
	$m_{arms,ref}^{MR}$	0.14 (very small UAV)	kg	Reference arms mass (for multirotor UAVs)
		1.4 (small UAV)	kg	
	$m_{frame,ref}^{MR}$	0.49 (very small UAV)	kg	Reference frame mass (for multirotor UAVs)
		23.7 (small UAV)	kg	

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Part II

Uncertainty quantification and mitigation

The second part of the manuscript addresses the critical aspect of uncertainty in UAV design and proposes methods for its quantification and mitigation. Chapter 4 conducts a thorough review of uncertainty analysis in preliminary design, revealing gaps in handling uncertainties associated with off-the-shelf components availability. Chapter 5 characterizes uncertainties related to both design models and component availability, and proposes a methodology to mitigate these uncertainties in the design optimization process. Specifically, the sizing methodology developed in the first part of the thesis is extended to allow the integration of off-the-shelf component databases instead of analytical models. The presented approach demonstrates its effectiveness through a case study of a multirotor UAV, demonstrating a significant reduction in uncertainty regarding both flight endurance and mass.

Author's contributions

Journal articles

F. Pollet, M. Budinger, S. Delbecq, J.-M. Moschetta, and J. Liscouët. Quantifying and mitigating uncertainties in design optimization including off-the-shelf components: Application to an electric multirotor UAV. *Aerospace Science and Technology*, vol. 136, p. 108179, May 2023, doi: 10.1016/j.ast.2023.108179.

Conference presentations

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Chapter 4

State of the art of uncertainty analysis in the context of design optimization

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4.1 Introduction and context

The design and manufacture of a product is a process involving a series of decisions and actions. Any poor decision can result in non-compliance with requirements and the need to make corrective changes, the cost of which increases with late identification and implementation [1]. Identifying and managing the sources of uncertainty throughout the project is key to making informed decisions to reduce technical and cost risks. The necessity to deal with uncertainty is highlighted by the presence of verification and validation phases in the V-model, used to manage and guide the product development process and shown in Figure 4.1.1. The verification and validation phases in the V-model enable the evaluation of functions and performances and confront them to requirements and specifications. They are intended to detect any non-conformity resulting from uncertainties introduced during the design,

manufacturing, and integration phases. The impossibility of ensuring a property implies additional iterations in the development cycle until all requirements are met. Although very few approaches in the literature explicitly define the sources of uncertainty arising throughout the V-model [2], it seems appropriate to offer a brief description of these and to contextualize the research conducted in this manuscript.

The development process in the V-model starts with the descending phase in which the system is gradually defined. First, a set of requirements is declined into technical specifications with the use of functional models. The translation of functional requirements into specifications reveals a first source of uncertainty due to the ambiguity and difficulty in interpreting user expectations.

The technical specifications are then fed into the preliminary design process. This stage involves the use of models to evaluate a number of candidate solutions in line with the specifications. On the one hand, the choice of a design solution at this stage is often a mere estimation of the solution that will ultimately be chosen. The description of a design variable depends on a level of knowledge that increases as the project progresses, and it is not always possible to implement a given choice in practice. For example, the choice of a component may be restricted by its supply on the market, or by manufacturing constraints. On the other hand, the use of models and their implementation lead to results that are only an approximation of physical reality, for example, because assumptions have been made to reduce their complexity or because phenomenological knowledge is incomplete.

The design choices and the results obtained from the models are then passed on, with their uncertainties, to the subsequent stages of detailed design and prototype production. In addition to the uncertainties associated with detailed design modeling, the manufacturing of a component is subject to dispersions and inaccuracies, as it is impossible in practice to repeatedly produce the theoretical dimensions of a product.

Finally, the ascending part of the V-model is dedicated to the testing of each development phase. It encompasses the verification of the components and the integrated system in accordance with the design process and the validation of the functionalities with respect to the requirements. More specifically, the verification steps aim at identifying any non-conformity arising from uncertainties in the design and production processes, while the validation ensures that the customer's requirements are met, despite the uncertainties involved in translating them into system functions and then into technical specifications.

The list of uncertainties arising during the product development process and provided here is far from exhaustive. For example, there are a number of uncertainties relating to project planning and management, which involve human behavior and economic aspects. The conditions in which the industrialized product will operate are another source of uncertainty. Although such uncertainty emerges downstream of the verification and validation steps, the designer can address them in the early stages of development by considering robustness and reliability properties.

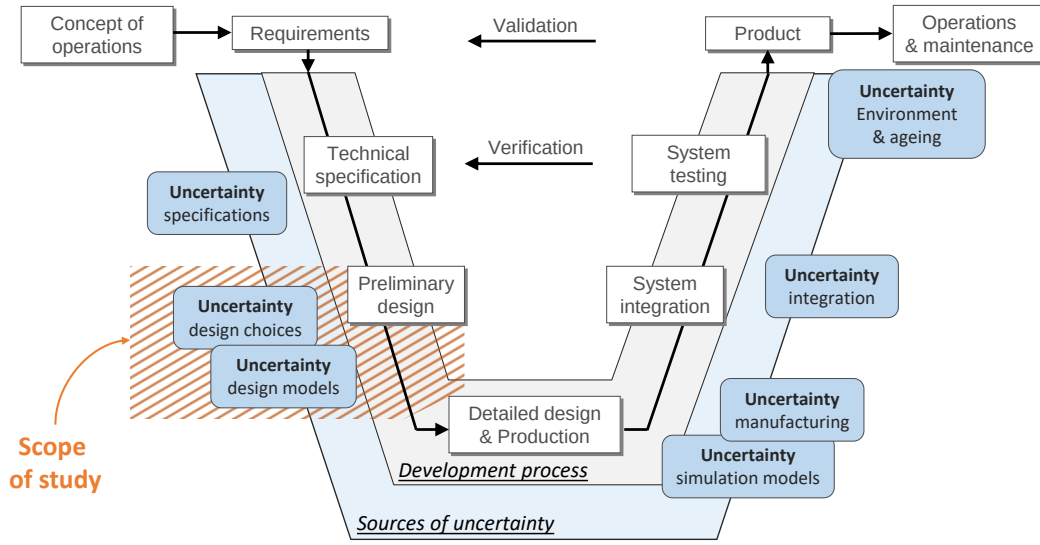


Figure 4.1.1 The V-model and sources of uncertainties throughout the development process (non-exhaustive list).

As summarized in Figure 4.1.1, a large number of uncertainties can arise during the development process of an electric UAV. In particular, the time gap between decisions made in the early design phases and the completion of the development process is a challenging aspect. In fact, the consequences of decisions become more pronounced over time as the effects of those decisions are magnified throughout the development. As emphasized by Ullman [3], 75% of manufacturing costs are committed before the end of the early design phases. This justifies the particular interest that the designer must have in addressing uncertainty at this stage of product development, and is therefore the focus of Part II of the manuscript.

4.2 Uncertainty analysis

Uncertainty analysis, or uncertainty assessment, involves *"how to identify, classify, characterize, quantify, and combine uncertainties within an application, with the expressed goal of understanding how to manage uncertainties"* [4]. As expressed by De Rocquigny, the goals of uncertainty analysis typically fall into four categories which are not mutually exclusive [5]:

- *Understand* the influence of uncertainties and rank their importance in order to guide the decision-making process.
- *Accredit* and validate a model, and better calibrate its settings if necessary.
- *Select* or optimize a design by comparing its relative performance in an uncertain context.

- *Comply* with a decision criterion, e.g. by demonstrating a low probability of exceeding a threshold in safety or reliability.

The achievement of these objectives with a systematic approach has been covered by many authors [2, 5–7]. In what follows, the methodological framework introduced by De Rocquigny [5] to characterize the uncertainties, propagate them through the system model and analyze their relative importance is described. It consists of a 4-step approach, as depicted in Figure 4.2.1.

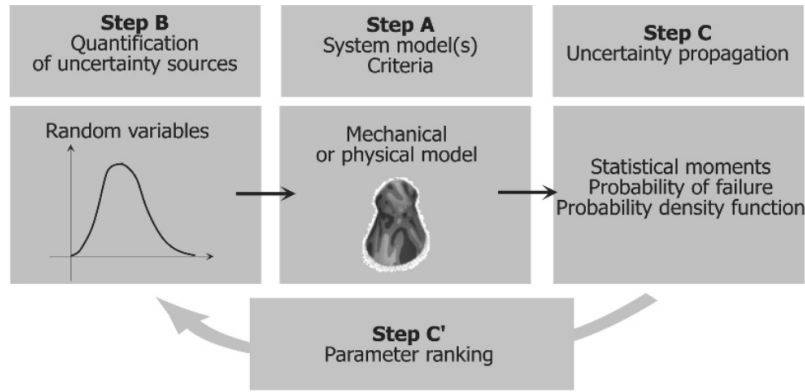


Figure 4.2.1 Methodological principles for the treatment of uncertainty, as proposed by De Rocquigny [5]. Source: Lemaire [2].

- **Step A.** Uncertainty assessment first requires a *pre-existing model* describing the system. Although the model may be of varying nature and complexity, it can be viewed conceptually as a function linking inputs to output variables of interest. The outputs generally represent physical variables related to performances such as the flight endurance of a UAV. They may be associated with decision criteria about their representative values (also referred to as quantities of interest): mean, variance, probability of exceeding a threshold, etc. The model inputs, in the context of preliminary design, typically correspond to design variables (e.g. propeller diameter) or technological assumptions (e.g. material properties). These inputs can be fixed or uncertain, depending on the scope of the uncertainty assessment and the level of control over them. The parameters of the model itself may be considered uncertain, for example, to reflect the goodness of fit of the model to measured data. Model parameters may be considered as inputs to the model or as distinct from those inputs. This inclusion depends on the authors [2, 5, 6, 8]. Overall, the system can be represented formally by the equation

$$\mathbf{y} = F(\mathbf{x}, \boldsymbol{\theta}), \quad (4.1)$$

where F is the system model linking a set of inputs $\mathbf{x}$ to a set of outputs $\mathbf{y}$ through equations involving specific model parameters $\boldsymbol{\theta}$.

- **Step B.** The second step, referred to as *uncertainty modeling* (or *uncertainty characterization*), consists of quantifying the sources of uncertainty. This involves estimating a measure of uncertainty in the inputs $\mathbf{x}$ and the model parameters $\boldsymbol{\theta}$. The probabilistic framework is generally adopted at this stage, and the measure of uncertainty will then be a set of probability densities with or without assumptions about the independence of the variables. Identifying the distributions of the random variables requires collecting as much information as possible, either by direct measurement, physical argument or expert judgment.
- **Step C.** The third step consists of translating the uncertainty in the inputs and the model parameters into uncertainty in the outputs of the system model. The *uncertainty propagation* returns estimates of the quantities of interest of the outputs, for instance, their full probability density functions (pdf) or their first statistical moments. The propagation of uncertainties is based on various methods, such as Monte Carlo sampling, depending on the quantities to be estimated (e.g., full pdf or first statistical moments) and the subsequent analysis step.
- **Step C'.** In the last step, the relative importance of the various sources of uncertainty is analyzed. *Sensitivity analysis* or *importance ranking* consists of examining the uncertainty propagation results to study how the variation in the output can be apportioned to the uncertain variables x and θ . By identifying the critical uncertainties, the designer can further refine the uncertainty analysis, for instance by collecting more data on the key parameters, or taking actions on the system to reduce uncertainty.

The following sections provide further information on steps B, C and C'.

4.2.1 Uncertainty characterization

This section starts with a discussion of the classical distinctions made in the literature about the nature of uncertainty, before describing the methods for modeling uncertainty.

Classifications

Definitions and classifications of uncertainty vary widely depending on the discipline and context. The most general classification found in the literature distinguishes between aleatory (irreducible) and epistemic (reducible) sources of uncertainty [2, 5, 6, 8]. Aleatory uncertainty concerns unpredictable variations inherent to physical systems and their environments, and as such, cannot be controlled or anticipated. The amplitude of the gusts that an aircraft will encounter in its lifetime is an example of such uncertainty. On the other hand, epistemic

uncertainty arises from a lack of information in the system, the level of which depends on the maturity of physical knowledge and, in the context of engineering, on the development stage of the project. Epistemic uncertainty can be reduced by collecting additional data, by improving the accuracy of the collected data through measurement, or by resolving design choices (i.e., by fixing a design parameter among a range of possibilities). However, the boundary between reducible and irreducible uncertainty is sometimes unclear in practice [5]. For example, the uncertainty of a measurement, although it may seem reducible, cannot be entirely resolved because of technical, industrial, or even economic constraints (e.g., costly measurements).

A finer distinction in the context of design problems would be the one proposed by DeLaurentis [9, 10] and illustrated in Figure 4.2.2. Input uncertainty occurs when the requirements that define a design problem lack precision or clarity. Model parameter uncertainty arises from the inherent limitations of mathematical models in accurately representing physical systems due to incomplete knowledge. Measurement uncertainty occurs when the desired response cannot be observed directly from the mathematical model and requires inference (e.g. in control systems). Finally, operational and environmental uncertainties arise from unanticipated or uncontrollable external disturbances.

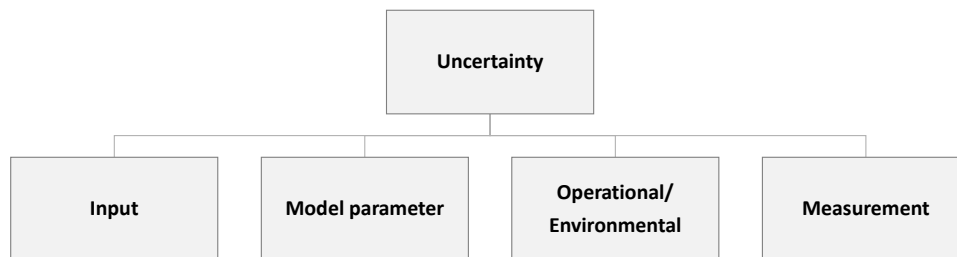


Figure 4.2.2 Uncertainty classification proposed by DeLaurentis for aerospace design problems [9, 10].

Thunnissen [8] proposes a further detailed classification. Input uncertainty is divided, among others, into requirement uncertainty and design uncertainty, the latter relating to unresolved choices among a set of design alternatives. Model uncertainty is divided into phenomenological uncertainty due to as-yet-unknown physical processes or events, approximation errors, numerical errors, and programming errors. Approximation errors are due to model deviations, which decrease with the level of fidelity and goodness of calibration. Numerical and programming errors arise from finite computational precision and implementation mistakes, respectively.

Modeling

Once the sources of uncertainty have been identified, they need to be associated with

uncertainty models built on the basis of available information. This information typically includes direct observations (statistical data) and expert judgment. The models can be refined with physical arguments such as positivity (e.g., the physical dimensions of an object cannot be negative). Although there are a variety of mathematical paradigms to represent uncertainty (e.g. possibility theory, Dempster-Shafer theory) [11], the probabilistic framework has benefited from significant theoretical developments. In addition, probabilistic approaches are often the most appropriate for use with numerical methods for uncertainty propagation and sensitivity analysis [2]. In the probabilistic theory, the uncertain variables are modeled as random variables. These statistical estimators are characterized by a probability distribution which must be identified as accurately as possible. In addition to graphical analyses (empirical vs. theoretical cumulative distribution function, Q-Q plots), some statistical tests, such as the chi-square test, can be used to identify the goodness of fit of distributions to the observed data. The robustness of the estimation can be measured by confidence intervals, which decrease proportionally to the square root of the sample size. Thus, the quality of the uncertainty model increases with the amount of information. In a sense, the confidence in the uncertainty model gives rise to a second level of uncertainty, a sort of "uncertainty about uncertainty", which is the subject of more elaborate studies, notably addressed by the Bayesian framework [5, 11].

Without getting into the details of all existing probability distributions, two principles often guide the choice of these. The central limit theorem will favor the normal distribution as the sample size increases. Conversely, when information is scarce, the *maximum entropy principle* [12] provides a means of translating engineering judgment into a model without adding additional, unverified information. Suppose an uncertain variable is only known to be bounded by a and b . In that case, the maximum of entropy states that a uniform distribution is a reasonable choice, as no value is favored. Alternatively, if a value $c \in [a, b]$ is expected to be the most likely, a triangular distribution could be used.

Finally, an important issue when modeling multiple uncertain variables is the question of dependencies. Although the assumption of independence of the input variables allows for a simplified treatment, neglecting the dependencies can significantly distort the results of the study. In addition to physical arguments, the hypothesis of independence can be supported (or rather rejected) by correlation studies such as Pearson's correlation or Spearman's correlation, bearing in mind that zero correlation is not always sufficient to conclude that two variables are independent. When independence cannot be assumed, multivariate statistical tools are needed to model dependence. This includes for instance the use of copulas [13].

4.2.2 Uncertainty propagation and analysis

Determining the probabilistic outputs $\mathbf{y}$ of a physical model comes down to solving

$$\mathbf{y} = F(\mathbf{x}, \boldsymbol{\theta}), \quad (4.2)$$

where:

- $\mathbf{x}$ is the vector of input variables that are either deterministic or probabilistic, that is, represented by random variables $\mathbf{X}$.
- F is a function representing the behavior of the system.
- $\boldsymbol{\theta}$ is the vector of parameters used in the model and applied to the inputs. Similarly to the input variables, the model parameters are deterministic or uncertain (random variables $\boldsymbol{\Theta}$).

The problem of uncertainty propagation consists of determining the uncertainties in $\mathbf{y}$, knowing those in $\mathbf{x}$ and $\boldsymbol{\theta}$. Then the importance ranking step, or more broadly the sensitivity analysis, aims to compare the relative influence of the sources of uncertainty.

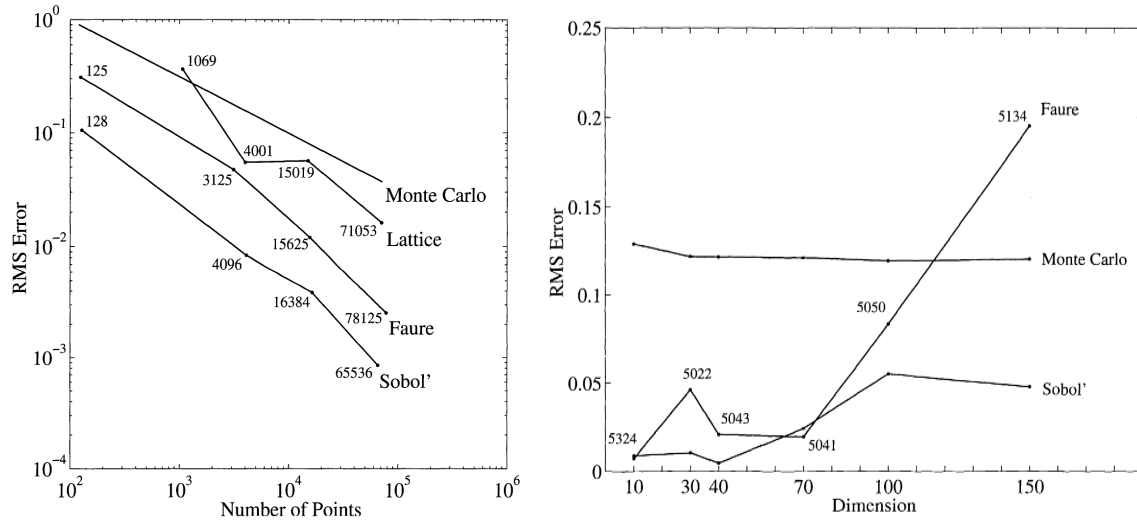
Uncertainty propagation

It is not always necessary to characterize the full probability distribution of the outputs, and some studies may focus on a few quantities of interest, such as the first statistical moments (mean, variance and skewness). In these particular cases, referred to as *dispersion analyses*, the Taylor approximation can be used for the propagation of uncertainty. The following formula provides the Taylor quadratic approximation for estimating the output variance of a function $y = F(\mathbf{x})$ [5]:

$$\text{Var}(Y) \approx \sum_i \left(\frac{\partial F}{\partial x_i} \right)^2 \text{Var}(X_i) + \sum_{i \neq j} \left(\frac{\partial F}{\partial x_i} \right) \left(\frac{\partial F}{\partial x_j} \right) \text{Cov}(X_i, X_j) \quad (4.3)$$

In other cases when the full distribution is needed (*distribution analysis*), simulation techniques such as the well-known Monte Carlo method can be used. Simulation methods repeat the deterministic calculation $\mathbf{y}_k = F(\mathbf{x}_k, \boldsymbol{\theta}_k)$ for a high number of realizations k of the random variables $\mathbf{X}$ and $\boldsymbol{\Theta}$. From the samples, it is possible to generate an estimate of the density function of the output. The sampling of the input space, that is the generation of the random realizations $\mathbf{x}_k, \boldsymbol{\theta}_k$, is achieved with different methods: Simple Random Sampling (SRS), Latin Hypercube Sampling (LHS) [14], and Quasi-Monte Carlo Sampling (e.g., Sobol' sequences [15]), among others. Apart from the ease of implementation, which favors SRS, the choice of an appropriate sampling method depends on the specific problem and its dimensionality [16]. However, in the absence of knowledge of the model function, Quasi-Monte Carlo methods appear to perform better as fewer samples are required to obtain an accurate estimate of the output properties [16, 17]. For SRS and LHS, the error in the output estimate decreases with the square root of the number of samples but is independent of the input space dimension. In comparison, Quasi-Monte Carlo has a convergence rate of $O((\log N)^m/N)$, where N is the number of samples and m is the problem dimension. Figure 4.2.3 illustrates the variation of the root mean square error (RMSE) in the output

estimate with increasing number of samples (4.2.3a) and increasing dimensions (4.2.3b), for different sampling methods [18].



(a) RMSE with increasing number of samples.

(b) RMSE with increasing problem dimension.

Figure 4.2.3 Example of root mean square error of the output estimate for different numbers of samples (4.2.3a) and problem dimensions (4.2.3b). Lattice, Faure, and Sobol' are Quasi-Monte Carlo methods. The numbers displayed on the plots indicate the number of samples used in the simulations. Source: Glasserman [18].

Using Monte Carlo to estimate the output distribution requires a large number of system model runs to obtain an accurate result, typically several thousand. Time or resource constraints can therefore become a concern if the model is computationally intensive. For the reliability assessment problem class, which typically consists of calculating the probability of failure of a system rather than determining the full distribution of uncertainty, other numerical methods offer advantages in terms of computational efficiency. In *reliability analysis*, the probability that a system's performance exceeds or falls below a threshold can be obtained using the First- and Second-Order Reliability Methods (FORM/SORM). FORM and SORM methods involve finding the point of maximum probability failure by solving an optimization problem. Then, the failure boundary, that is the frontier between the acceptable domain and the failure domain, is estimated using a linear (FORM) or quadratic (SORM) approximation around this point. Finally, the probability of failure is obtained by integrating the subspace of the probability density function defined by the approximated failure boundary.

However, when a full understanding of the output distribution is required, simulation techniques like Monte Carlo are still needed. If no alternative is available, calculation time can be reduced by using metamodels (or surrogate models) to approximate the pre-existing system model, thereby reducing its complexity. Two popular metamodeling techniques are

Polynomial Chaos Expansions (PCE) [19] and Kriging [20]. The main methods for uncertainty propagation described in this section are summarized in Table 4.2.1.

Table 4.2.1 Main methods for uncertainty propagation.

	Quantities of interest	Common methods
Dispersion analysis	Statistical moments (mean, variance, skewness, and high-order)	Taylor approximation, Monte Carlo simulation
Distribution analysis	Full distribution (pdf or cdf)	Monte Carlo simulation
Reliability analysis	Probability of exceeding a threshold (failure probability)	FORM/SORM, Monte Carlo simulation

Sensitivity analysis

Sensitivity analysis aims to identify the variables that, combined with their probabilistic descriptions, most influence the output, thereby helping to determine the focus for future uncertainty mitigation.

If a Taylor approximation is used to propagate the uncertainty, the relative importance of the sources of uncertainty can be estimated by comparing their respective contributions in Equation 4.3. More advanced methods analyze the samples returned by Monte Carlo simulations to decompose the variance of the output, among which the Sobol' method [21, 22] is the most commonly used. Based on Monte-Carlo simulations, this method decomposes the variance of the output into terms of increasing dimensions, called sensitivity indices, that are attributed to the uncertain variables and their distributions. The first-order indices measure the effect of varying a given input alone within its domain of definition, averaged over the variations of the other inputs [22]:

$$S_i = \frac{\text{Var}_{X_i}(E_{X_{-i}}(Y|X_i))}{\text{Var}(Y)}. \quad (4.4)$$

Here, $E_{X_{-i}}(Y|X_i)$ denotes the average value of the output y when all inputs except x_i vary, and $\text{Var}_{X_i}(E_{X_{-i}}(Y|X_i))$ is the variance of the previous quantity when x_i varies. Note that for ease of reading, the uncertain model parameters θ_i are considered blindly in the x_i notation. The second-order indices describe the coupled effect of two variables on the output, and so on for higher-order indices. Finally, total-order effects [23] capture the contribution of an input to output variance by taking into account its interactions of any order with all other variables [22]:

$$S_{T_i} = 1 - \frac{\text{Var}_{X_{-i}}(E_{X_i}(Y|X_{-i}))}{\text{Var}(Y)}. \quad (4.5)$$

The Sobol' indices in Equations 4.4 and 4.5 are proportional to the variances of the inputs, hence taking into account their respective probability distributions. Finally, it should be noted that Sobol' indices presented here have been developed to deal with independent inputs.

In the case of dependent variables, generalized methods such as the ones proposed by Saltelli and Tarantola [24], Kucherenko *et al.* [25] and Owen *et al.* [26, 27] can be employed.

Although variance-based methods provide accurate results, they require a large number of simulations (i.e., model evaluations) to collect enough information for calculating sensitivity indices. For m uncertain inputs, the Sobol' method requires $N \times (m + 2)$ model evaluations to obtain the first and total-order indices, where N is the number of samples needed to calculate the indices of a given x_i with sufficient accuracy (typically 500 or higher) [28]. More specifically, $m \times N$ simulations are necessary to compute Y when isolating the i -th input for $i = [1, \dots, m]$, and $2N$ simulations are needed for computing Y when all inputs vary (that is, to obtain $\text{Var}(Y)$). If the second-order indices are to be calculated, the number of model evaluations increases to $N \times (2m + 2)$. To save time, the least-influential parameters can be removed beforehand with a screening study. Screening techniques such as the method of Morris [29] allow obtaining an estimate of the relative importance of uncertainty sources prior to a more detailed sensitivity analysis. The original implementation of Morris method considers a set of m uncertain parameters mapped into the unit hypercube, that is, each parameter x_i varies in the range $[0, 1]$. The uncertain parameters are varied one at a time, and the output variations are observed. The elementary effect of a parameter x_i is defined as

$$EE_i = \frac{f(x_1, x_2, \dots, x_i + \Delta, \dots, x_m) - f(x_1, x_2, \dots, x_i, \dots, x_m)}{\Delta}, \quad (4.6)$$

where $\Delta \in]0, 1]$ is the experiment step. More precisely, a Morris experiment starts by selecting a first sampling point randomly in the input space: $\mathbf{x}^{(1)} = (x_1^{(1)}, \dots, x_m^{(1)})$. The second sampling point, $\mathbf{x}^{(2)}$, is obtained by increasing or decreasing one component of $\mathbf{x}^{(1)}$ by Δ , such that $\mathbf{x}^{(2)} = (x_1^{(1)}, \dots, x_i^{(1)} \pm \Delta, \dots, x_m^{(1)})$. Then, a third sampling point $\mathbf{x}^{(3)}$ is generated by increasing (or decreasing) the value of another parameter $x_j \neq x_i$ by Δ . The process is repeated until all inputs have been changed. The succession of sampling points $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(m+1)}$ defines what is called a trajectory, and enables the calculation of the elementary effects EE_i . Examples of trajectories are provided in Figure 4.2.4.

The set of elementary effects is recalculated for r different trajectories in the input space. Then, for each parameter, the mean μ_i^* of the absolute elementary effects and their standard deviation σ_i are obtained. A high value for the mean effect μ_i^* indicates a strong influence on the output's magnitude. The standard deviation σ_i characterizes both non-linear effects and interactions with other sources of uncertainty. If the input parameters do not lie in the range $[0, 1]$ as expected in the original method, a scaling factor is applied to take into account the actual distribution of each parameter. Consequently, as in Sobol' method, the results are biased towards parameters with large uncertainties.

The Morris method requires $r \times (m + 1)$ model evaluations, where the number r of different trajectories typically ranges from 4 to 10 [31]. Thus, this method allows ranking the sources of uncertainty with a low computational cost. Differentiation between influential and irrelevant

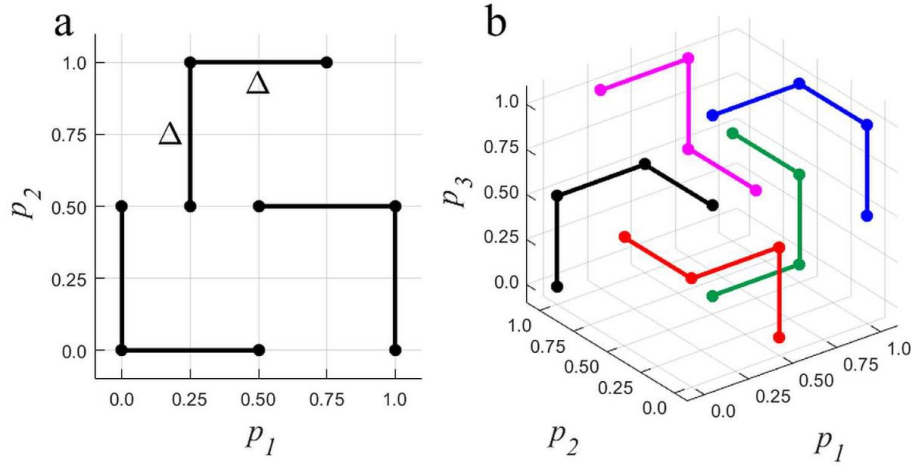


Figure 4.2.4 Examples of trajectories when $m = 2$ (a) and $m = 3$ (b). Source: Likhachev [30].

parameters based on the results from Morris method is an important aspect. According to Menberg *et al.* [32], the choice of a quantitative criterion for separating the parameters is rarely discussed in the literature and seems rather case-specific. It depends in particular on the computational resources available to the designer for the subsequent analyses and the desired level of representativeness of the final results. Finally, it should be noted that, as with any screening method, Morris' results should be interpreted qualitatively rather than quantitatively. While it can rank the uncertain parameters in order of importance, it cannot quantify the amount of output variance accounted for by each parameter (contrary to the Sobol' method).

4.2.3 Feedback process

Once the uncertainty in the model output and the contributions from the different sources are known, there is an essential feedback process to be undertaken by the decision-maker. Actions typically fall into three categories: refining the uncertainty study, reducing the uncertainty in the model output, and adjusting the design. The first type of action relates to the iterative nature of the uncertainty assessment, which might involve a simple uncertainty modeling at the beginning before a more in-depth study of the most influential variables identified in the sensitivity analysis step. The second type of action might involve the collection of additional data and better modeling (e.g., model calibration or replacement) to reduce the uncertainty in the model output. The third type of action aims to improve the design to enhance its performance in spite of uncertainties. The latter process falls into the realm of robustness and reliability and is typically addressed by specific design methodologies such as uncertainty-based MDO (UMDO) [33–37] to obtain a reliable (RBMDO) [38–40] and/or a robust (RMDO) [37, 40–44] design candidate.

To conclude this section, a visual summary of the techniques and methods involved in the uncertainty analysis is proposed in Figure 4.2.5.

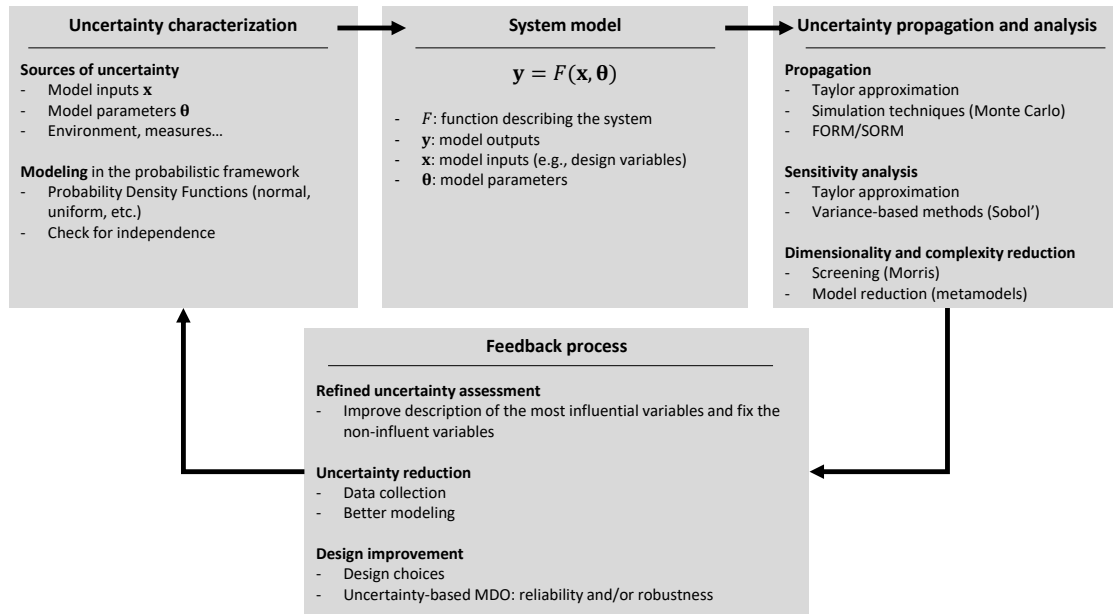


Figure 4.2.5 Techniques and methods for uncertainty analysis.

4.3 Uncertainty management in the design of aerospace systems

This section focuses on the treatment of uncertainty in the early design phases of aerospace systems. First, an overview of existing research is given. Then, a focus is placed on the optimization strategies that allow the replacement of continuous mathematical models by the discrete selection of off-the-shelf components, thereby reducing the sources of uncertainty.

4.3.1 Literature review

The analysis of uncertainties and their impact in the early design phases of aerospace systems have been addressed in previous studies [40, 45–53]. Smith and Mahadevan [50] were among the first to integrate reliability analysis methods in the conceptual design of aerospace systems. The authors successfully optimized the geometry of a launch vehicle with a constraint on the probability of exceeding a pitching moment coefficient, using the FORM method. However, little attention is given to the quantification of sources of uncertainty. Uncertainties in the

design variables and the models are estimated from expert judgment. In [40], the authors present a procedure for reliable and robust optimization of an aircraft in the early design phases. Uncertainties in the disciplinary models and the design variables are described by normal laws using historical data. The parameters of the normal distribution used to quantify model uncertainty are adjusted according to the point at which the model is evaluated. Uncertainties in design variables are roughly quantified on the basis of historical data representing changes in their values between the preliminary design phase and the final design. The design optimization problem considers both robustness and reliability constraints, leading to an increase in aircraft weight but better reliability and a less sensitive optimum. Daskilewicz *et al.* [47] investigate the effects of uncertainty on performance objectives in the design optimization of a light jet aircraft. Uncertainties in aerodynamics, propulsion and weight metrics are considered, but are modeled with arbitrary distribution laws and no attempt is made to quantify the actual uncertainties. Neufeld *et al.* [49] use historical data to quantify the uncertainty introduced by the use of low-fidelity models in aircraft design. The design optimization is solved with a reliability-based MDO strategy. As in [40], a tradeoff is found between the performance of the design and its robustness, that is, the probability that all requirements will be met in spite of uncertainties. More recently, Lim *et al.* [48] investigated the uncertainty propagation in flight performance of multirotor UAVs. Input uncertainties due to manufacturing tolerances and imperfections in the main components of the propulsion chain are modeled with normal distributions whose parameters are obtained by expert judgment. It is shown that due to uncertainties in battery capacity, rotor diameter, motor speed constant and no-load current, there is a 19% risk that the quadrotor UAV under study will not complete the mission for which it was designed. The authors conclude that increasing the nominal battery capacity by at least 3% compared to the initial value ensures a much higher probability of completing the flight mission despite the uncertainties.

The literature survey shows that many authors have taken an interest in quantifying and managing uncertainty in the design process of aerospace systems. However, the current review reveals two areas of potential research gaps. Firstly, no study seems to address the uncertainties associated with the lack of knowledge about the actual availability of the components specified during the preliminary design phase. Yet this is a recurring problem in the design of systems involving low- to mid-grade mechatronic components, such as UAVs. Because these components are often purchased from the market rather than custom-made, the designer's specifications cannot always be guaranteed. Secondly, the management of uncertainties, i.e. the feedback process, is generally addressed by robust or reliable optimization methods. These approaches, which are very valuable for ensuring a level of reliability and robustness, nevertheless demonstrate a necessary compromise between system performance and the certainty that all the requirements will be met by the final design. Complementary actions to mitigate the sources of uncertainty, rather than the ultimate impacts of uncertainty, can be implemented in the early design phases. Replacing low-fidelity

models with higher-fidelity ones, as suggested by multi-fidelity tools [54–58], is one possibility. However, multi-fidelity approaches only address model uncertainties and in no way reduce uncertainty about the inputs.

In summary, the trend towards the increasing integration of pre-defined commercial components, whose range is limited, introduces a new source of uncertainty into the design process. Indeed, the choices made by the designer in the early design phases are ultimately constrained to a finite space of possibilities that may not coincide. This is all the more true when the design variables used in MDO tools to describe the components are defined in the continuous domain, as opposed to the discrete number of components available on the market. To address this particular source of uncertainty, one approach is to restrict the selection of components to a predefined set of alternatives available in manufacturers' catalogs, as early as the preliminary design phase. However, since component catalogs offer discrete choices (as opposed to continuous design models), specialized optimization tools are needed to solve the design problem efficiently. The following section explores this issue in more detail.

4.3.2 Transition from continuous domain optimization to mixed continuous and discrete optimization

Reducing uncertainty arising from a limited choice of components requires transitioning from a pure continuous-domain optimization to a mixed continuous and discrete optimization, enabling both custom definition and selection of off-the-shelf components from a database. In this manuscript, the term *custom* refers to the parameterization of hypothetical components with continuous mathematical models, as opposed to *off-the-shelf* which describes the discrete selection of existing components in a database. In the literature, various methodologies allow the exploration of databases to find the best combination of off-the-shelf components [59–67].

The most straightforward method consists of an exhaustive evaluation of all combinations of components from a database. Although some authors have proposed filtering operations to speed up the process [59], this technique remains computationally very costly because of the combinatorial explosion.

Other approaches rely on advanced algorithms to solve the discrete optimization problem [60–64, 67]. Magnussen [60, 61] and Pawlus [62] use a mixed-integer linear program solver to optimally select components for a multirotor UAV and an industrial electric drivetrain, respectively. Each element of the discrete set of components is assigned a boolean decision variable whose value equals one if the element is selected and zero otherwise. Consequently, the dimension of the design variables vector is equal to the number of elements in the database (up to 3091 variables in Magnussen's case study), making it impractical for complex problems.

Ng [63] and Arellano-Quintana [64] rely on genetic algorithms to find the best combination of components for multirotor UAVs. Discrete integer design variables representing the indices of the selected components are introduced to solve the optimization problem. Using a more sophisticated approach, Saves *et al.* [67] efficiently solves an aircraft design problem

containing categorical variables to represent different tail geometries and engine positions. The method relies on Bayesian optimization [68], a powerful strategy for solving problems with complex objective functions. In its original implementations, Bayesian optimization is not suitable for non-continuous design variables. However, some authors have extended it to discrete integer and categorical variables by continuously relaxing the design variables [69, 70]. Relaxation of the categorical variables increases the number of hyperparameters involved in the optimization process, making it impractical for high-dimensional problems (e.g. large databases to be explored). Saves [67] further improves the Bayesian optimization strategy by providing an adaptive procedure to reduce the number of hyperparameters. Application to an aircraft design problem shows a significant improvement over genetic algorithms.

In the approaches proposed by Ng [63], Arellano-Quintana [64] and Saves [67], the bounds on the discrete design variables are defined according to the number of components stored in the database. The design space for m design variables is defined as $\Omega = \{1, \dots, N_1\} \times \dots \times \{1, \dots, N_m\}$, where $N_1, \dots, N_m$ are their respective number of choices offered by the database. As a result, the range of the discrete design variables varies when an element is added or removed from the components database. In the context of this manuscript, the variability of the bounds of the design variables contradicts the efficient MDO formulation implemented in Chapter 3, which interest was precisely to make the formulation of the problem independent of the specifications. Adopting these approaches would require a significant effort to reformulate the problem and the equations involved in the sizing procedure, complicating the transition from continuous domain optimization to mixed continuous and discrete optimization.

A final alternative approach is to solve the problem in the continuous domain and use the solution as a reference to find the closest components in a database [65, 66]. Due to the offline discretization process, this two-step method breaks the inter-dependencies between the components, and no feedback on the design is performed to verify the constraint satisfaction and optimality of the discrete solution.

4.4 Summary of the research gaps

The literature review revealed a significant research gap regarding the management of uncertainty in the design process of systems that will eventually be assembled from off-the-shelf components. Firstly, none of the research studied has quantified the uncertainty associated with the conjunction of pre-design tools, which generally explore a continuous design space, and future constraints (e.g. during the production phase) whereby choices may be limited to a discrete set of components. Secondly, to reduce this source of uncertainty, the exploration of the design space in the preliminary design phase needs to be constrained in order to converge exclusively on solutions containing available components. In other words, it requires transitioning from a continuous-domain optimization to a discrete one (or, possibly, a mixed continuous and discrete optimization). Despite the many approaches to

mixed-variable optimization, none of them seem to allow a continuous mathematical model to be replaced by a database of off-the-shelf components without a major reformulation of the design problem. Furthermore, the relevance of continuous mathematical models for the sizing of custom components, as proposed in Chapter 3, cannot be disregarded. It is therefore necessary to develop an approach that allows the designer to decide whether to use custom-made or off-the-shelf components, depending on the context, without having to formulate a separate MDO problem for each design case.

Chapter 5

Uncertainty quantification and mitigation in the sizing of systems including off-the-shelf components

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5.1 Introduction

A vast majority of the design optimization frameworks that have emerged in recent years, such as the one described in Chapter 3, typically parameterize the key components of the system with mathematical models and solve the optimization problem in the continuous domain [71–76]. This strategy yields critical issues in the context of preliminary design.

Firstly, the models employed in the preliminary design phase are often associated with large uncertainty because of their low fidelity. Secondly, designers are increasingly focusing on integrating off-the-shelf components instead of custom-made parts to improve development times and production costs. A priori, this approach, which constrains the selection to a finite number of components, is inconsistent with a continuous-domain optimization. As pointed out in Chapter 4, this inconsistency leads to some uncertainty about the actual characteristics of discretely available components as opposed to the custom definition provided by the sizing process in the continuous domain.

The present chapter aims to introduce a methodology for evaluating and mitigating uncertainties in the context of design optimization, with a focus on the uncertainty driven by the discrete availability of components optimally sized in a continuous domain.

The methodological framework is described in Section 5.2. The major sources of uncertainty in the early design phases are characterized and quantified in Section 5.3. Methods for propagating uncertainties through a system's model and for ranking their importance are proposed in Section 5.4, and applied through a case study of a multirotor UAV. Section 5.5 introduces a methodology for the optimal sizing of systems including both custom and off-the-shelf components, and compares three designs of multirotor UAVs with different levels of uncertainty. Finally, Section 5.6 concludes the chapter.

5.2 Methodological framework for uncertainty assessment and mitigation

This section presents the underlying framework of the uncertainty study, building on the generic methodology described in Chapter 4. The proposed methodological framework for quantitative uncertainty assessment and mitigation consists of four steps:

1. Identify and model the system. The model generally consists of one or several deterministic functions $\mathbf{y} = F(\mathbf{x}, \boldsymbol{\theta})$ linking a set of inputs $\mathbf{x}$ to a set of outputs $\mathbf{y}$ through equations involving specific model parameters $\boldsymbol{\theta}$. In this study, the model introduced in Chapter 3 for describing multirotor UAV designs is considered.
2. Quantify and characterize the sources of uncertainty. Section 5.3 covers the probabilistic description of uncertainties arising from incomplete knowledge, including model accuracy and uncertain values for the model inputs.
3. Propagate uncertainties through the system model to get a measure of uncertainty in the output variables of interest. In this study, the Monte Carlo method is performed to obtain the probabilistic distributions of the outputs. A screening method is applied as a preliminary analysis for reducing the number of uncertain variables prior to the

Monte Carlo simulations. This approach is described and applied to a case study in Section 5.4.

4. Perform a sensitivity analysis to assess the relative importance of uncertainties. Section 5.4 develops and applies an analytical method, as well as the more advanced method of Sobol' for importance ranking.
5. Mitigate uncertainties. In this chapter, the focus is on removing important sources of uncertainty arising from the unknown availability of the components defined in a continuous design space. In contrast, selecting a component directly from a database provides better knowledge of its parameters. For this purpose, Section 5.5 introduces a versatile methodology for the optimal sizing of systems including custom and/or off-the-shelf components.

The five steps described in this process can be repeated until the level of uncertainty is deemed acceptable.

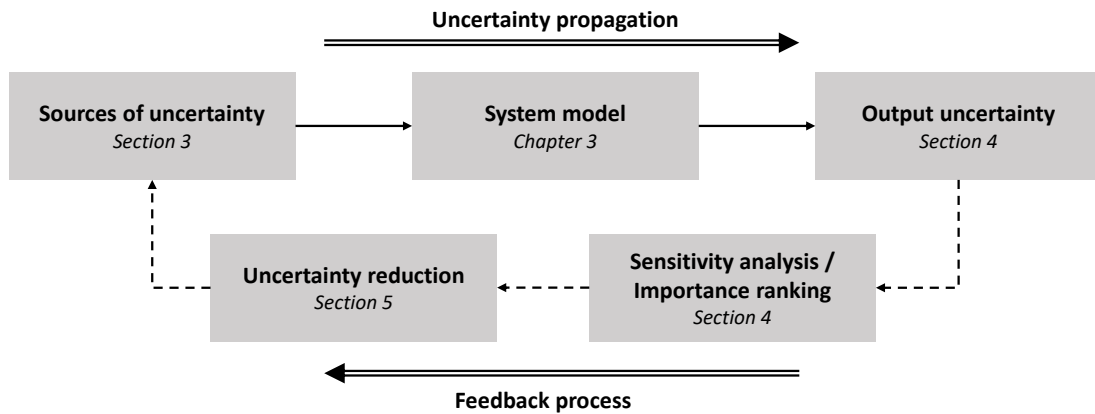


Figure 5.2.1 Methodological framework for uncertainty assessment. Adapted from [2, 5].

5.3 Sources of uncertainty

Building on the uncertainty classification provided in Chapter 4, the present study addresses two specific sources of uncertainty commonly encountered during the design process:

- *Model uncertainty* relates to the accuracy with which a model reflects reality. Model uncertainty includes, among others, the inadequate structure of the model, low numerical resolution, and erroneous fitting of the model parameters. In this study, all sources of model uncertainty are considered without distinction in the model parameters.
- *Input uncertainty* is associated with the input design variables of a model. In the context of preliminary design, estimation models are used to estimate the functional parameters

of the components from a few definition parameters, as introduced in Chapters 2 and 3. Input uncertainty relates to the level of information available on the input design variables (or, definition parameters), which varies according to the design stage. It is reduced to zero after all design variables have been determined, that is when the development process is finalized.

Model uncertainty acts on the model parameters θ , while input uncertainty affects the input variables $\mathbf{x}$. Consequently, the uncertainties on the outputs $\mathbf{y} = F(\mathbf{x}, \theta)$ arise from both model uncertainty and input uncertainty.

In the following sections, the model uncertainties and the input uncertainties associated with the multirotor UAV system described in Chapter 3 are characterized. The focus is on the estimation models used to derive the properties of the key components of the electric propulsion system. The main scaling laws and regression models for sizing the propellers, motors, batteries, and electronic speed controllers (ESCs) are recalled in Table 5.3.1. The continuous models presented here are calibrated on the average properties of commercially available components.

Although there are uncertainties in all design disciplines, only the propulsion part of the design is addressed in this study. Specifically, insufficient data on the airframe aerodynamics (e.g., drag) prevents quantifying the associated uncertainties. Hence, the analysis is restricted to hover flights to exclude the aerodynamic uncertainties from the problem. However, future work should consider such uncertainty, as the sensitivity of the design to the aerodynamic performance of the airframe (Chapter 3, Section 3.6) could significantly change the conclusions of the study.

Table 5.3.1 Estimation models for the preliminary design of multirotor UAVs [77].

	Output variable y	Model equation	Model parameters θ	Input variables x
Propeller	Mass M [kg]	$M = \theta_1 D^3$	$\theta_1 = \frac{M_{ref}}{D_{ref}^3}$	Diameter D [m]
	Thrust coefficient C_T [-]	$C_T = \theta_2 + \theta_3 \beta$	$\theta_2 = 4.27 \times 10^{-2}$ $\theta_3 = 1.44 \times 10^{-1}$	Pitch-to-diameter β [-]
	Power coefficient C_P [-]	$C_P = \theta_4 + \theta_5 \beta$	$\theta_4 = -1.48 \times 10^{-3}$ $\theta_5 = 9.72 \times 10^{-2}$	Pitch-to-diameter β [-]
Motor	Mass M [kg]	$M = \theta_6 T_{nom}$	$\theta_6 = \frac{M_{ref}}{T_{nom,ref}}$	Nominal torque T_{nom} [N.m]
	Resistance R [Ω]	$R = \theta_7 K_V^{-2} T_{nom}^{-5/3.5}$	$\theta_7 = \frac{R_{ref}}{K_{V,ref}^{-2} T_{nom,ref}^{-5/3.5}}$	Velocity constant K_V [rad/V/s]
				Nominal torque T_{nom} [N.m]
	Friction torque T_f [N.m]	$T_f = \theta_8 T_{nom}^{0.69}$	$\theta_8 = \frac{T_{f,ref}}{T_{nom,ref}^{0.69}}$	Nominal torque T_{nom} [N.m]
ESC	Mass M [kg]	$M = \theta_9 P$	$\theta_9 = \frac{M_{ref}}{P_{ref}}$	Power P [W]
Battery	Mass M [kg]	$M = \theta_{10} E$	$\theta_{10} = \frac{M_{ref}}{E_{ref}}$	Energy E [Wh]

Subscript $_{ref}$ refers to reference components used for fitting the models.

5.3.1 Model uncertainty

The scaling laws and regression models introduced by Budinger [77] require a limited number of input variables and, as such, are well suited for preliminary design. However, several

simplifying assumptions had to be made to obtain the models. These assumptions, detailed in Chapter 2, are briefly recalled here. Geometric similarity assumes constant length ratios between the sized components and the reference components used for fitting the models. Material similarity assumes similar material properties across the size range. Finally, the uniqueness of the design driver states that only one dominant physical phenomenon drives the evolution of the characteristic to be estimated. When these assumptions are not satisfied, the accuracy of the model deteriorates, and model uncertainty arises. The estimated value of the output, $\hat{y}$, deviates from the true value y :

$$y = (1 + \varepsilon_\theta)\hat{y} = (1 + \varepsilon_\theta)f(x, \theta^*), \quad (5.1)$$

where ε_θ is the relative error and $f(x, \theta^*)$ is the model equation whose parameter θ^* is obtained with appropriate fitting methods (e.g., error minimization). Since the models presented in Table 5.3.1 are linear with respect to their parameters θ_i^* , it follows that

$$y = f(x, (1 + \varepsilon_\theta)\theta^*) = f(x, \theta), \quad (5.2)$$

with $\theta = (1 + \varepsilon_\theta)\theta^*$. The inaccuracies of the models are described by deviations in the model parameters, that is, model uncertainties. According to [77], the model uncertainties of scaling laws and regressions can be modeled with independent normal distributions:

$$\varepsilon_\theta \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2) \quad (5.3)$$

$$\Leftrightarrow \theta \sim \Theta = \mathcal{N}\left((1 + \mu_\theta)\theta^*, \sigma_\theta^2\theta^{*2}\right) \quad (5.4)$$

The mean μ_θ and standard deviation σ_θ of the normal distributions are mainly influenced by the choice of the reference components for fitting the equations and the form of the equations since uncertainty increases with the exponent of the scaling law [77]. Figure 5.3.1 shows the distribution of the relative error for the estimated battery mass, and its comparison to a normal distribution.

The choice of a distribution law to approximate the model uncertainty depends on the specific model under study. The actual error distribution may be skewed or have heavier tails than a normal distribution, which may lead to a poor estimation of the model uncertainty if an inappropriate distribution law is employed. As explained in Chapter 4, if sufficient data is available, graphical analyses or statistical tests can be used to determine the goodness of fit of the distribution to the measurements. In [77], the authors use a normal probability plot to assess the adequacy of the observed data to a normal distribution. If a limited amount of data is available, it may not be possible to assess the adequacy of a specific distribution law. In such situations, a uniform distribution is most appropriate for modeling the uncertainty, as it is the distribution with maximum information entropy [12, 78]. The parameters of the uniform distribution are obtained from an observed or assumed bounded interval on the

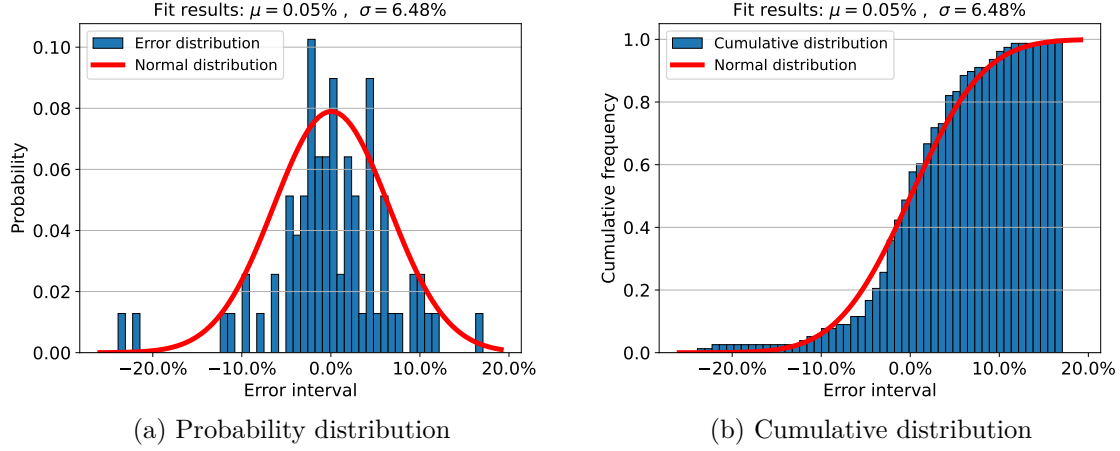


Figure 5.3.1 Relative error distribution for the estimated battery mass [77].

model error:

$$\theta \sim \Theta = \mathcal{U}(\theta^* + \varepsilon_{\theta, \min}, \theta^* + \varepsilon_{\theta, \max}) \quad (5.5)$$

5.3.2 Input uncertainty

Uncertainty about the design variables of the components (i.e., the inputs of the design models) may arise from various sources. In particular, the impossibility of guaranteeing a component's availability on the market is a major source of uncertainty at the early stages of the design process. Indeed, manufacturers limit their production to specific ranges of components for economic reasons. Thus, during the design process, a gap exists between the characteristics of a hypothetical component provided by the continuous-domain optimization and the characteristics of the closest off-the-shelf component available. If the catalogs of components are not known a priori by the designer, for instance, at the conceptual design stage, the latter faces uncertainty on the values of the input variables to be used for the models (e.g., propeller diameter). This section describes the input uncertainty associated with an incomplete market offer.

In [79], Pahl *et al.* suggest that manufacturers follow a size range approach when designing their products by limiting their production to a few selected sizes in order to reduce costs. A standardized methodology is often adopted by manufacturers to determine the best distribution of components, i.e. choosing an optimal size range while best meeting consumer demand. Starting from an initial design, new designs are derived, which are arranged in a mathematical sequence. While an arithmetic progression on a design variable's value is straightforward, Pahl *et al.* advise choosing these quantities according to a geometric progression within a range. Whether an arithmetic progression or geometric progression approach is adopted depends on the market.

The arithmetic progression approach consists of deriving the new designs with a constant increment on a parameter describing the component (e.g., propeller diameter). For the n -th component in the size range, the value x_n of its parameter is obtained from an initial design x_0 as follows:

$$x_n = x_0 + nd, \quad (5.6)$$

with d the increment step. When a customer selects a component from a catalog, he chooses the component whose characteristics are nearest (or nearest upper, or nearest lower) to his specifications. The selection process is represented in Figures 5.3.2 and 5.3.3, where a continuous set of input values is mapped to a countable set, with a finite number of elements built on an arithmetic progression. The associated absolute error between the continuous variable x^* and the discrete variable x is plotted in Figure 5.3.3.

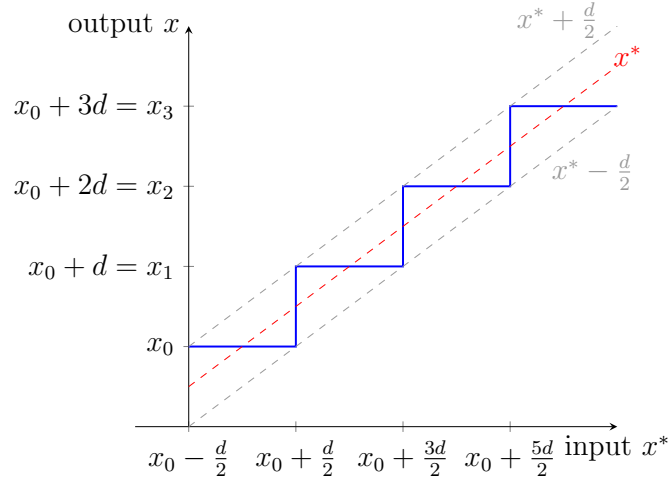


Figure 5.3.2 Arithmetic progression mapping.

The absolute error has a uniform distribution bounded by half the increment step:

$$\eta_x = x - x^* \sim \mathcal{U}\left(-\frac{d}{2}, \frac{d}{2}\right) \quad (5.7)$$

$$\Leftrightarrow x \sim X = \mathcal{U}\left(x^* - \frac{d}{2}, x^* + \frac{d}{2}\right) \quad (5.8)$$

Hence, a presumed denser market offer yields less uncertainty. The APC multirotor propellers¹ provide an example of arithmetic scaling on the diameter. The size range covers diameters from 0.153 to 0.406 meters, with an average difference d of 0.0254 meters, that is, exactly one inch. Figure 5.3.4 depicts the absolute error distribution of the diameter and its comparison with a uniform distribution law.

¹APC propellers for multirotor UAVs: <https://www.apcprop.com/product-category/multi-copters-drones/>

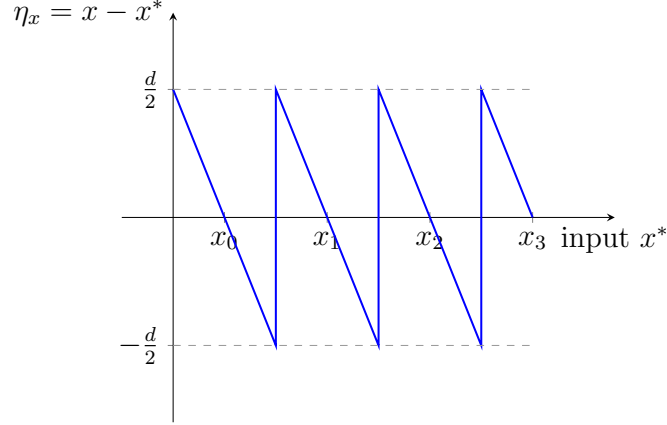


Figure 5.3.3 Absolute error of the arithmetic progression mapping.

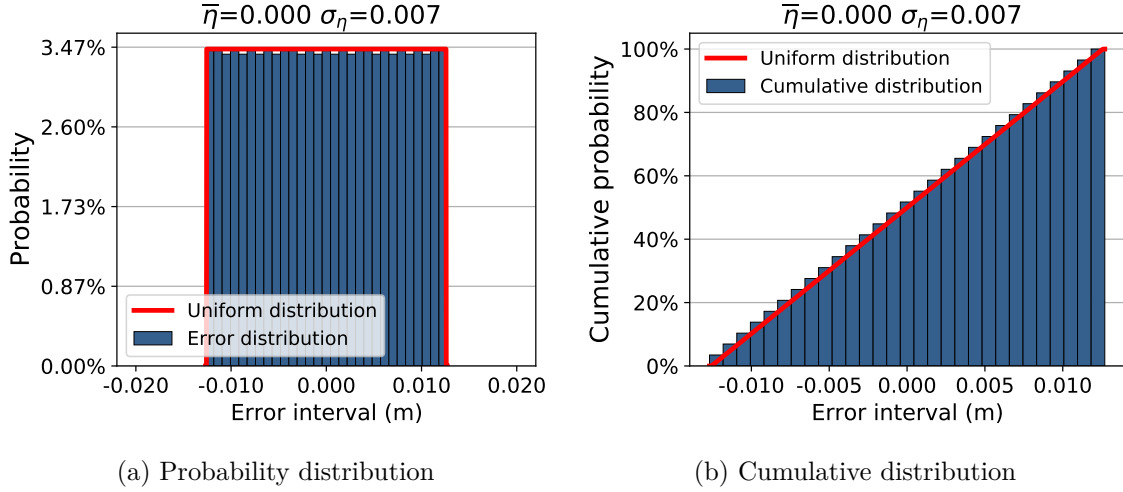


Figure 5.3.4 Absolute error distribution of the propeller diameter and comparison to a uniform distribution.

For other components available on the market, a geometric progression might be more representative [79]. It consists of multiplying the value of a parameter describing the component by a constant ratio $r > 1$ in order to derive the new designs:

$$x_n = x_{n-1}r = x_{n-2}r^2 = \dots = x_0r^n, \quad (5.9)$$

The mapping for a size range built on a geometric progression is represented in Figures 5.3.5 and 5.3.6. The magnitude of the absolute error η_x between the continuous variable and the real variable is significantly correlated to the magnitude of the input. A probabilistic description of the absolute error would hide this behavior and is therefore not desirable. Instead, a probabilistic model of the relative error ε_x is adopted, as its magnitude is not

correlated with the input (Figure 5.3.6). The relative error distribution follows a uniform law:

$$\varepsilon_x = \frac{x - x^*}{x^*} \sim \mathcal{U}(-\delta, \delta) \quad (5.10)$$

$$\Leftrightarrow X = \mathcal{U}(x^*(1 - \delta), x^*(1 + \delta)), \quad (5.11)$$

with $\delta = \frac{r-1}{r+1}$. Here again, a high number of components available within a size range (i.e., $r \rightarrow 1$) reduces the variable's uncertainty.

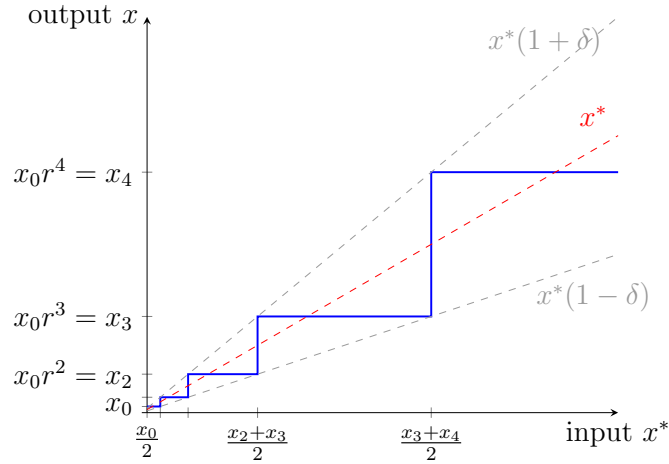


Figure 5.3.5 Geometric progression mapping.

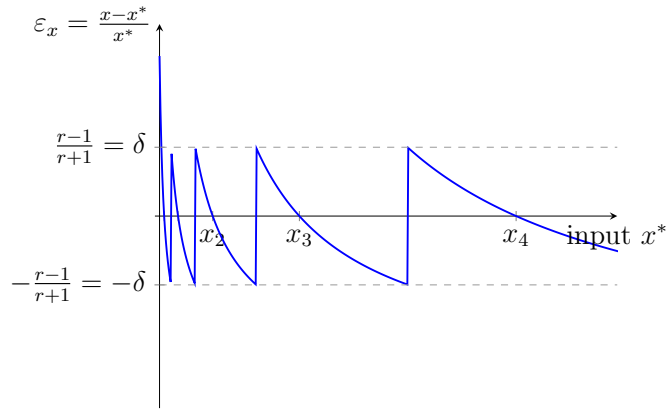


Figure 5.3.6 Relative error of the geometric progression mapping.

A quasi-geometric scaling of the nominal motor torque is observed on a family of 15 ABB products² over the range 1-200 N.m. As shown in Figure 5.3.7, the relative error falls between -17% and 17% with a uniform probability distribution.

²ABB Motors: <https://new.abb.com/motors-generators/iec-low-voltage-motors>

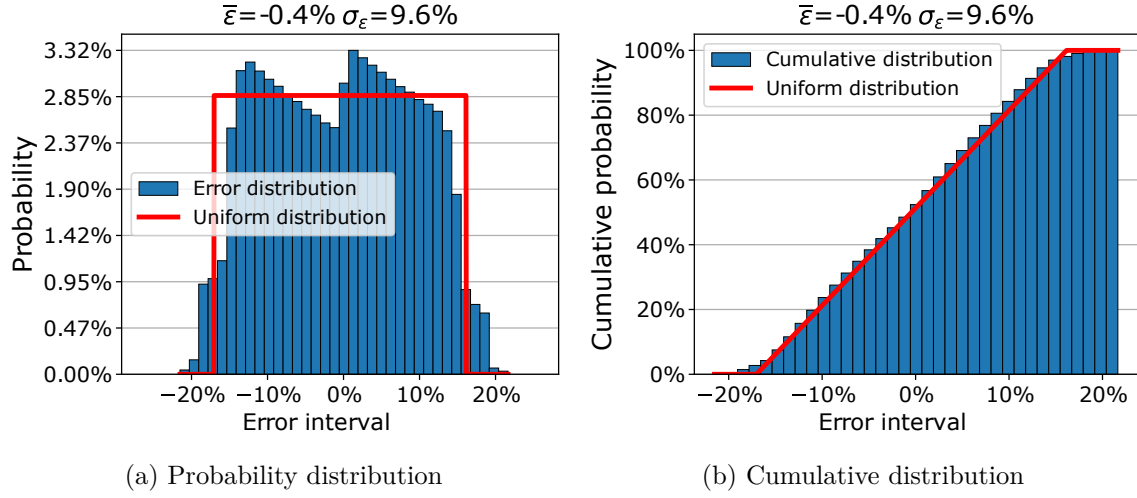


Figure 5.3.7 Relative error distribution of the nominal motor torque and comparison to a uniform distribution.

The choice between an arithmetic or geometric approach for the mapping can be made from the study of a database representative of the market offer if available to the designer. Typically, a progression is considered arithmetic if the mean statistical difference observed between two terms (d_1) is close to the theoretical difference obtained from the size range bounds (d_2):

$$d_1 = \frac{1}{N_c - 1} \sum_{i=1}^{N_c-1} (x_{i+1} - x_i) \quad (5.12)$$

$$d_2 = \frac{\max(x_i) - \min(x_i)}{N_c - 1}, \quad (5.13)$$

where N_c is the number of distinct components in the database. A progression is considered as geometric if the mean statistical ratio observed between two terms (r_1) is close to the theoretical ratio obtained from the size range bounds (r_2):

$$r_1 = 10^{\frac{1}{N_c-1} \sum_{i=1}^{N_c-1} \log\left(\frac{x_{i+1}}{x_i}\right)} \quad (5.14)$$

$$r_2 = \left(\frac{\max(x_i)}{\min(x_i)}\right)^{\frac{1}{N_c-1}} \quad (5.15)$$

When no database is available to the designer, estimating the most appropriate mapping method is more difficult. As described by Pahl [79], the mapping and selection of the step sizes are strongly related to the market situation and production costs. Such data would require manufacturer-specific knowledge and is beyond the scope of the thesis.

Finally, when multiple design variables are involved in the selection of a component, the random variables describing the uncertainties are assumed to be mutually independent. This

assumption is supported by the fact that the design variables involved in the optimization process are uncorrelated by nature, as described in Chapter 3, Section 3.4. However, it should be noted that the uncorrelation property is only a necessary condition for independence. Therefore, future work should focus on analyzing the interactions between the design variables describing a component in more detail and, if necessary, using advanced methods to model dependence.

5.4 Uncertainty propagation and sensitivity analysis

This section first introduces the methods used for uncertainty propagation and sensitivity analysis. The methodology is then applied to the flight performance of a multirotor UAV.

5.4.1 Uncertainty propagation

The previous sections have introduced the characterization and quantification of two sources of uncertainty in preliminary design. The uncertain input variables $\mathbf{x}$ and model parameters $\boldsymbol{\theta}$ can be modeled as random variables denoted $\mathbf{X}$ and $\boldsymbol{\Theta}$, respectively. Given the uncertainties on the set of input variables and model parameters describing the design of a system, the next step is to obtain a characterization of the outputs' distributions:

$$\mathbf{Y} = F(\mathbf{X}, \boldsymbol{\Theta}) \quad (5.16)$$

This is achieved by propagating the uncertainties through the system model F using the Monte Carlo method.

An important point is the definition of the overall system model F on which the uncertainties are propagated. It is recalled that, in the context of this manuscript, each component of the UAV is described by a set estimation models and simulation models. Estimation models (Table 5.3.1) allow the characteristics of the components to be derived from a few parameters. Simulation models calculate the performance of the components and the system given the characteristics of the components and the operating conditions. Simulation models for a multirotor UAV in hovering conditions are recalled in Appendix 5.A. In most studies, uncertainty propagation is performed after the design has been optimized for the requirements. In this approach, the uncertainties are propagated through each component analysis independently. This allows the evaluation of uncertainties in a 3-dimensional output space:

1. **Uncertainty about the compatibility and integrity of the components.** For example, due to uncertainties, the motor rating may not be sufficient to handle the propeller's needs without overheating. In the context of product development, this uncertainty expresses the risk of not passing the verification stage of system integration.

2. **Uncertainty about the performance of the system.** This uncertainty implies the risk of not achieving the technical specifications set prior to the design stage ("System testing" in the V-model), for example hovering endurance.
3. **Uncertainty about the dimensions and weight of the system.** In the product development process, this translates into uncertainty about manufacturing costs, among others.

The conventional approach for uncertainty propagation ensures a comprehensive analysis. However, it combines multiple verification activities that occur at different stages of the product development process, making the analysis hazardous. For this reason, we propose here to limit the scope to the last two points, i.e. the uncertainties about the system performance and its dimensions. In other words, it comes down to assessing the remaining uncertainties once the correct system integration has been verified, possibly after some iterations in the product development process. To this end, the model F on which the uncertainties are propagated must contain the part of the design process responsible for solving the compatibility and integrity constraints.

In practice, for the sizing methodology presented in Chapter 3, the proposed approach involves solving the constraints on the components' integrity (e.g., $T_{nom} \geq T_{mot,hover}$) for each simulation of the Monte Carlo. The other constraints (e.g., $E_{bat} \geq E_{mission}$) are left unresolved. For the particular case covered in this chapter, where only hover flight is considered, the component integrity constraints are explicitly enforced by the sizing coefficients, so no optimization is required for this purpose. Finally, the disciplinary couplings (e.g., the mass loop) are resolved for each Monte Carlo simulation to ensure consistency.

5.4.2 Sensitivity analysis

A first approach for ranking uncertainties consists of analyzing the form of the equations involved in the modeling. As a preliminary analysis, the relative contributions of input uncertainty and model uncertainty in each elementary model f_i are assessed.

Let suppose a model f with uncertain input $x = (1 + \varepsilon_x)x^*$ and uncertain model parameter $\theta = (1 + \varepsilon_\theta)\theta^*$. A first-order approximation with Taylor series expansion at (x^*, θ^*) for small relative deviations ε_x and ε_θ provides:

$$\begin{aligned}
 y &= f(x, \theta) \\
 &= f(x^* + x^*\varepsilon_x, \theta^* + \theta^*\varepsilon_\theta) \\
 &= f(x^*, \theta^*) + x^*\varepsilon_x \left. \frac{\partial f}{\partial x} \right|_{x^*, \theta^*} + \theta^*\varepsilon_\theta \left. \frac{\partial f}{\partial \theta} \right|_{x^*, \theta^*} + o\left(\sqrt{(x^*\varepsilon_x)^2 + (\theta^*\varepsilon_\theta)^2}\right)
 \end{aligned} \tag{5.17}$$

This allows for determining, with a first-order approximation, the variations of the output caused by input uncertainty and those caused by model uncertainty. The variances of their distributions are compared to assess the relative effects of the different sources of uncertainty.

The variations of y due to input uncertainty are considered equal to those due to model uncertainty if

$$\begin{aligned} x^* \varepsilon_x \frac{\partial f}{\partial x} \Big|_{x^*, \theta^*} &= \theta^* \varepsilon_\theta \frac{\partial f}{\partial \theta} \Big|_{x^*, \theta^*} \\ \iff (x - x^*) \frac{\partial f}{\partial x} \Big|_{x^*, \theta^*} &= (\theta - \theta^*) \frac{\partial f}{\partial \theta} \Big|_{x^*, \theta^*} \\ \iff (x - x^*)^2 \frac{\partial f}{\partial x} \Big|_{x^*, \theta^*}^2 &= (\theta - \theta^*)^2 \frac{\partial f}{\partial \theta} \Big|_{x^*, \theta^*}^2 \end{aligned} \quad (5.18)$$

Such that, for N realizations of the random variables:

$$\begin{aligned} \frac{1}{N} \sum_{i=1}^N (x_i - x^*)^2 \frac{\partial f}{\partial x} \Big|_{x^*, \theta^*}^2 &= \frac{1}{N} \sum_{i=1}^N (\theta_i - \theta^*)^2 \frac{\partial f}{\partial \theta} \Big|_{x^*, \theta^*}^2 \\ \iff \text{Var}(X) \frac{\partial f}{\partial x} \Big|_{x^*, \theta^*}^2 &= \text{Var}(\Theta) \frac{\partial f}{\partial \theta} \Big|_{x^*, \theta^*}^2 \\ \iff x^{*2} \text{Var}(\epsilon_x) \frac{\partial f}{\partial x} \Big|_{x^*, \theta^*}^2 &= \theta^{*2} \text{Var}(\epsilon_\theta) \frac{\partial f}{\partial \theta} \Big|_{x^*, \theta^*}^2 \\ \iff \left| x^* \frac{\partial f}{\partial x} \Big|_{x^*, \theta^*} \right| \sigma_x &= \left| \theta^* \frac{\partial f}{\partial \theta} \Big|_{x^*, \theta^*} \right| \sigma_\theta, \end{aligned} \quad (5.19)$$

where σ_x (resp., σ_θ) is the standard deviation of the relative error ε_x (resp., ε_θ). For the input uncertainties with uniform distributions introduced in Section 5.3, the standard deviation σ_x is linked to the increment step d or constant ratio r of the components' database:

$$\begin{cases} \sigma_x = \frac{d}{2\sqrt{3}} \frac{1}{|x^*|} & \text{for an arithmetic database} \\ \sigma_x = \frac{2\delta}{2\sqrt{3}} = \frac{1}{\sqrt{3}} \frac{r-1}{r+1} & \text{for a geometric database} \end{cases} \quad (5.20)$$

Equations 5.19 and 5.20 enable finding the assumption on database density (i.e., d or r) such that the two sources of uncertainty have similar effects:

$$\text{Eq. 5.19} \iff \begin{cases} d = 2\sqrt{3} \left| \theta^* \frac{\frac{\partial f}{\partial \theta}}{\frac{\partial f}{\partial x}} \Big|_{x^*, \theta^*} \right| \sigma_\theta & \text{for an arithmetic database} \\ r = \frac{\left| x^* \frac{\partial f}{\partial x} \Big|_{x^*, \theta^*} \right| + \left| \theta^* \frac{\partial f}{\partial \theta} \Big|_{x^*, \theta^*} \right| \sigma_\theta \sqrt{3}}{\left| x^* \frac{\partial f}{\partial x} \Big|_{x^*, \theta^*} \right| - \left| \theta^* \frac{\partial f}{\partial \theta} \Big|_{x^*, \theta^*} \right| \sigma_\theta \sqrt{3}} & \text{for a geometric database} \end{cases} \quad (5.21)$$

For power laws (e.g., scaling laws) in the form $f(x, \theta) = \theta x^a$, with a the power law exponent, criterion 5.21 simplifies to:

$$\text{Eq. 5.19} \iff |a| \sigma_x = \sigma_\theta \iff \begin{cases} d = \frac{2\sqrt{3}|x^*|\sigma_\theta}{|a|} & \text{for an arithmetic database} \\ r = \frac{|a| + \sigma_\theta \sqrt{3}}{|a| - \sigma_\theta \sqrt{3}} & \text{for a geometric database} \end{cases} \quad (5.22)$$

This approach provides a rough estimate for ranking uncertainties when a single model is involved in the propagation. For the estimation models introduced in Section 5.3 Table 5.3.1,

the outputs' variations due to model uncertainty are presented in Table 5.4.1. Estimates on the databases' densities such that the variations caused by input uncertainty match those caused by model uncertainty are presented in Table 5.4.2. For the motors database, a constant ratio $r = 1.50$ between two consecutive values of the nominal torque leads to equal effects of model and input uncertainty in the mass calculation. Considering a range within 1 - 10 N.m, there should be at least $\log_r(10/1) \approx 6$ products in the motor database for the input uncertainty to be secondary.

Table 5.4.1 Outputs' variations caused by model uncertainty for the UAV models.

Output		Model uncertainty deviation σ_θ
Motor	Mass M	11.5%
	Resistance R	13.2%
	Friction torque T_f	33.4%
Battery	Mass M	7.1%
ESC	Mass M	25.4%
Propeller	Mass M	10.0%
	Thrust coefficient C_T	8.8%
	Power coefficient C_P	7.5%

Table 5.4.2 Assumptions on databases' densities to match the variations caused by input uncertainty with those caused by model uncertainty.

Output		Input	Database progression such that $ a \sigma_x = \sigma_\theta$	
Motor	Mass M	Nominal torque T_{nom}	Geometric	$r = 1.50$
	Resistance R	Velocity constant K_V	Geometric	$r = 1.26$
		Nominal torque T_{nom}	Geometric	$r = 1.38$
	Friction torque T_f	Nominal torque T_{nom}	Geometric	$r = 11.38$
Battery	Mass M	Energy E	Arithmetic	$d = 179.5$ Wh
ESC	Mass M	Power P	Arithmetic	$d = 782.7$ W
Propeller	Mass M	Diameter D	Arithmetic	$d = 0.056$ m
	Thrust coefficient C_T	Pitch-to-diameter β	Arithmetic	$d = 0.091$
	Power coefficient C_P	Pitch-to-diameter β	Arithmetic	$d = 0.078$

To refine the results obtained with the Taylor approximation, the Sobol' method [21, 22] introduced in Chapter 4 is adopted on the full-order model F to derive the global impact of uncertain variables, taking into account the interactions between them. Due to the relatively high dimensionality of the case study, the screening method of Morris is applied to select a reduced set of variables. Here, the variables retained for the Sobol' sensitivity study are selected such that their cumulative mean effects estimated with the method of Morris account for 90% of the total variation in the output.

5.4.3 Case study: performance uncertainty of a multirotor UAV

This section applies the methodology and tools for uncertainty assessment to study the flight performance of a multirotor UAV. The case study considers the specifications of the DJI M600 Pro hexacopter drone³. With a maximum payload of 5.5 kg, this UAV is designed for a wide range of industrial applications. The main specifications of the DJI M600 Pro at maximum takeoff weight are provided by the manufacturer and summarized in Table 5.4.3.

Table 5.4.3 Main specifications for the DJI M600 Pro at maximum takeoff weight. Source: DJI.

Specification	Value
Number of propellers:	6
Payload:	5.5 kg
Max thrust-to-weight ratio:	2
Hover endurance:	18 min

The steps for the case study are illustrated in Figure 5.4.1. From the technical specifications in Table 5.4.3, a deterministic design of the UAV is obtained using the methodology introduced in Chapter 3. The results obtained for the deterministic design are provided in Appendix 5.B. In a second step, the uncertainty analysis is performed. The uncertainties are propagated through the same set of equations that allowed to obtain the deterministic design, except that the system performance constraints are removed from the optimization problem. Specifically, each simulation for the uncertainty assessment consists in introducing some deviations (ε_x , η_x , ε_θ or η_θ) in the models inputs and parameters, then re-run the relaxed design problem. The result is an updated design whose performance (and dimensions) differ from the optimal deterministic one because of uncertainties. The uncertainty analysis is performed in three phases. First, the Morris screening method is used to select a reduced number of uncertain variables that most influence the output performance. Then, the Monte Carlo method is used to obtain the output distribution. Finally, the Sobol' indices are calculated to analyze the contributions to the output variance.

Probabilistic model and implementation

The study focuses on 15 uncertain variables of the propulsion system and their impact on hover flight endurance. The models for estimating the hovering time are recalled in Appendix 5.A. The assumptions on the uncertain variables are detailed in Appendix 5.C. The probabilistic distributions are derived according to the methods introduced in Section 5.3. In particular, the uncertainties on the components' availability are estimated from catalogs of commercially available products⁴. For the purpose of reproducibility, a Jupyter Notebook [80]

³DJI M600 Pro: <https://www.dji.com/matrice600-pro>

⁴Manufacturers' catalogs: APC Propellers for multirotors (46 products), AXI Model Motors (52 products), ProLiteX batteries (78 products), YGE ESCs (17 products)

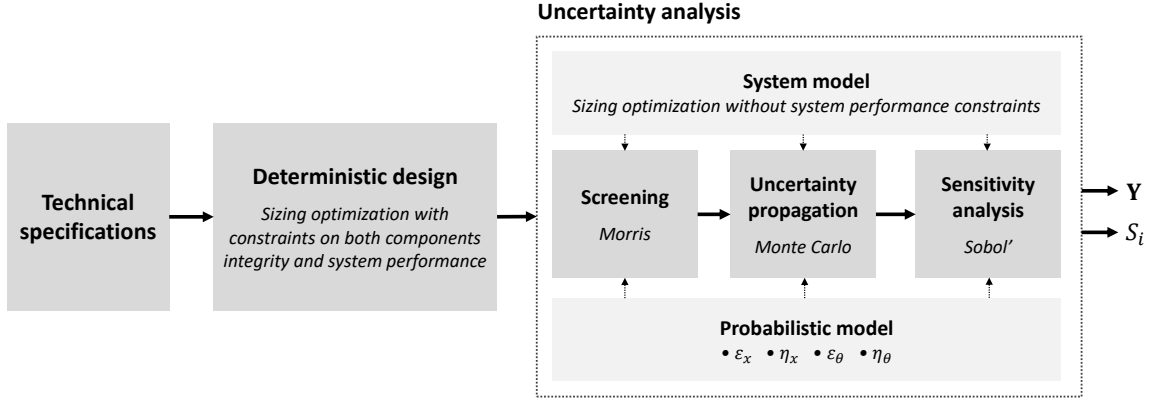


Figure 5.4.1 Overall process for the case study.

for estimating input uncertainties from databases is provided in Appendix 5.D. In addition to input uncertainty and model uncertainty, some parameters obtained from experts' judgments without specific models (e.g., the efficiency of the speed controller) are considered to be distributed uniformly within a range of values. Indeed, the uniform distribution is the best adapted to describe limited information (principle of maximum entropy) [78]. Finally, the distributions are assumed to be mutually independent.

The statistical methods described in Section 5.4 are implemented using Python libraries. In particular, the Monte Carlo simulations and sensitivity analyses are performed using OpenTURNS [81] and SALib [82], respectively.

Results and discussion

As a preliminary analysis, the relative importance of model uncertainty and input uncertainty for each individual model is assessed from the form of the equations with the criteria derived in Equation 5.22. The calculations are made from the model uncertainties provided in [77] and the assumptions on the databases presented in Appendix 5.C. The results are shown in Table 5.4.4. For each model of the propulsion system, the uncertainty arising from model discrepancy has more effect on the output than the uncertainty on the inputs. Thus, model uncertainties are expected to outweigh input uncertainties. However, this analysis does not allow for ranking the final effects of the sources of uncertainty on the hover flight endurance, as the scope is limited to each individual equation.

Screening

The screening method of Morris is applied to select a reduced set of uncertain variables that most influence hover endurance. The method is implemented with 10 different trajectories, such that 160 calls to the complete UAV model are required. Figure 5.4.2 shows the mean effects μ^* and standard deviations σ for each uncertain variable. The effects are scaled, that is,

Table 5.4.4 Deviations caused by model uncertainty and input uncertainty within each model.

	Output	Model uncertainty deviation σ_θ	Input uncertainty deviation $a \sigma_x$	
Motor	Mass M	11.5%	Nominal torque T_{nom}	3.6%
	Resistance R	13.2%	Velocity constant K_V	4.8%
Battery	Friction torque T_f	33.4%	Nominal torque T_{nom}	5.2%
	Mass M	7.1%	Nominal torque T_{nom}	2.5%
ESC	Mass M	25.4%	Energy E	0.4%
Propeller	Mass M	10.0%	Power P	10.2%
	Thrust coefficient C_T	8.8%	Diameter D	4.6%
	Power coefficient C_P	7.5%	Pitch-to-diameter β	1.9%
			Pitch-to-diameter β	1.9%

the probability distributions of the uncertain variables are taken into account when evaluating their influence on the output. The most influential variables are located in the upper-right quadrant of the σ versus μ^* plot. It is observed that 3 variables are highly important to explain the hover endurance: the propeller thrust coefficient C_T and power coefficient C_P obtained from a regression model, and the ESC efficiency. The model uncertainty on the battery mass, obtained from an assumed energy density, also has a significant impact on flight endurance. Furthermore, the model uncertainties of the motor resistance and motor mass have a great influence. The most influential uncertainties originate from discrepancies in the models rather than their inputs, which is consistent with the preliminary study presented in Table 5.4.4. This outcome reflects the high number of components available in the UAV market. Although some of these parameters are directly related to hover endurance (e.g., battery energy), their low uncertainty results in a negligible absolute influence on the output.

The 7 most influential variables in the μ^* plot represent 91% of the total mean effect, so the remaining variables are discarded in subsequent studies.

Output distribution and sensitivity analysis

Propagation of uncertainty using Monte Carlo experiments is performed on the 7 most influential variables identified with the method of Morris. This allows obtaining a distribution of the hover endurance and computing the sensitivity indices using Sobol' method. To achieve high precision in the output distribution and sensitivity indices, a high number of model evaluations is required. As explained in Chapter 4 the total number of model evaluations depends on the number m of uncertain variables (here, $m = 7$) and the sample size N required to accurately estimate the output distribution and the sensitivity indices for each given variable. Following recommendations of Owen [83] for improving the accuracy of the sensitivity indices, the number of samples is chosen to be a power of two. Figure 5.4.3 shows convergence of hover endurance mean and standard deviation for increasing sample sizes. The quantities quickly stabilize around their final values, such that 64 samples would be

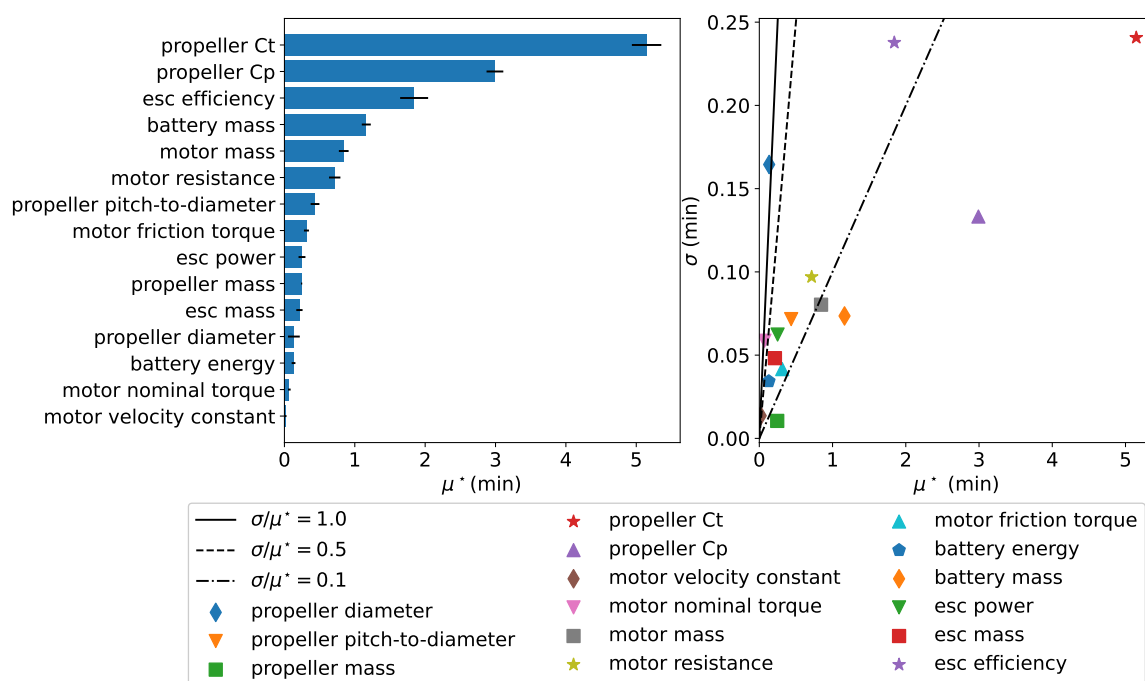


Figure 5.4.2 Screening results using Morris method showing the impact of uncertainty in each variable on the hover endurance.

sufficient to accurately estimate the mean and standard deviation of the output distribution. However, accurate calculation of the sensitivity indices requires a higher number of simulations. Figure 5.4.4 shows convergence of the total-order index and the margin of error corresponding to a 95% confidence level for the most influential variable, that is, the propeller thrust coefficient. For a 10% margin of error criterion on the total-order index, 1024 samples are needed. Overall, the sampling process being repeated for calculating the sensitivity indices of each of the 7 variables and the output distribution, 16384 model evaluations are required. Hence, the following results are obtained with 16384 model evaluations (1024 samples).

The distribution of the flight endurance is shown in Figure 5.4.5. The observed distribution fits a normal law with a mean value $\mu = 18.1$ min and standard deviation value $\sigma = 3.2$ min.

The patterns and trends of the Monte Carlo simulation can be visualized with the parallel coordinate plots in Figure 5.4.6. Parallel coordinate plots display multivariate data on a series of parallel axes, where each axis corresponds to a different variable. In its interactive version, the plot can be filtered to display a reduced set of simulations. For example, it is apparent that the combined overestimation and underestimation of the propeller thrust and power coefficients gives a high probability of achieving an endurance of less than 18 minutes. However, the identification of trends for the other sources of uncertainty is more difficult and requires advanced analysis methods.

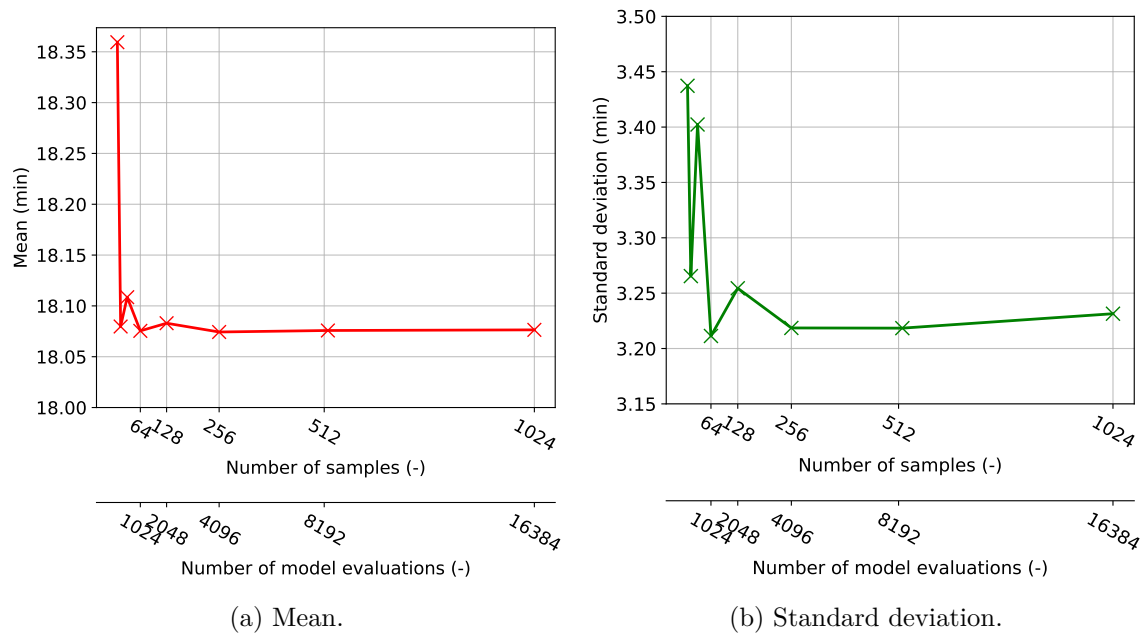


Figure 5.4.3 Convergence of the mean (left) and standard deviation (right) of hover endurance with increasing sample sizes.

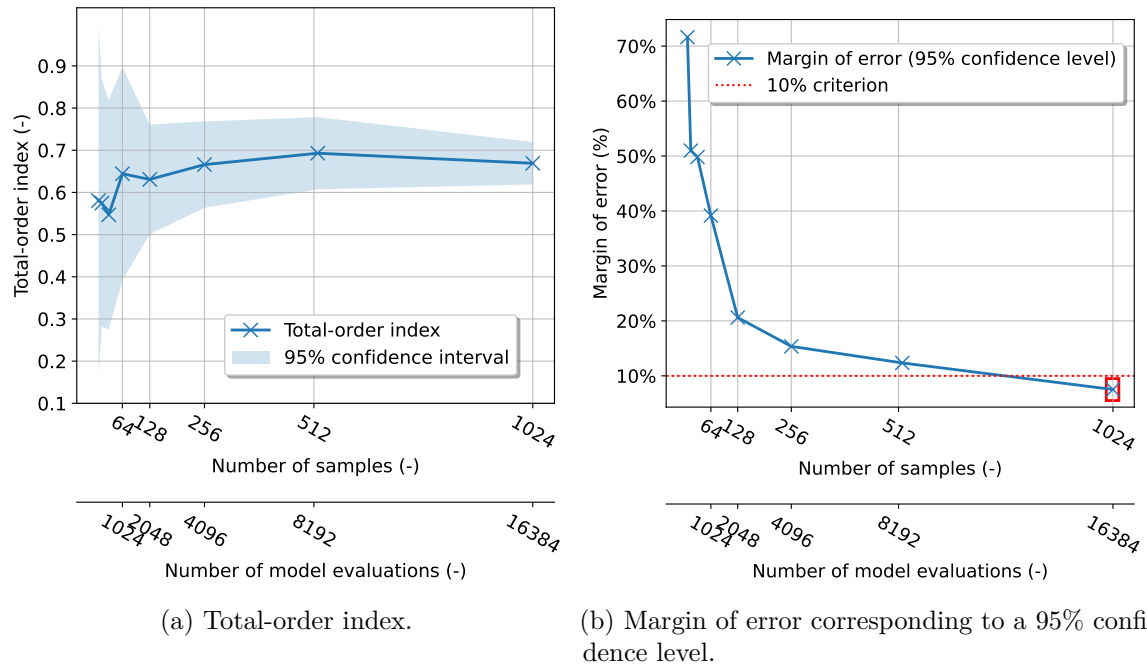


Figure 5.4.4 Convergence of the total-order index (left) and its margin of error corresponding to a 95% confidence level (right) for the most influential variable (propeller C_t).

The total- and first-order contributions to the output variance obtained with the method of Sobol' are given in Figure 5.4.7. The second and higher-order contributions are close to

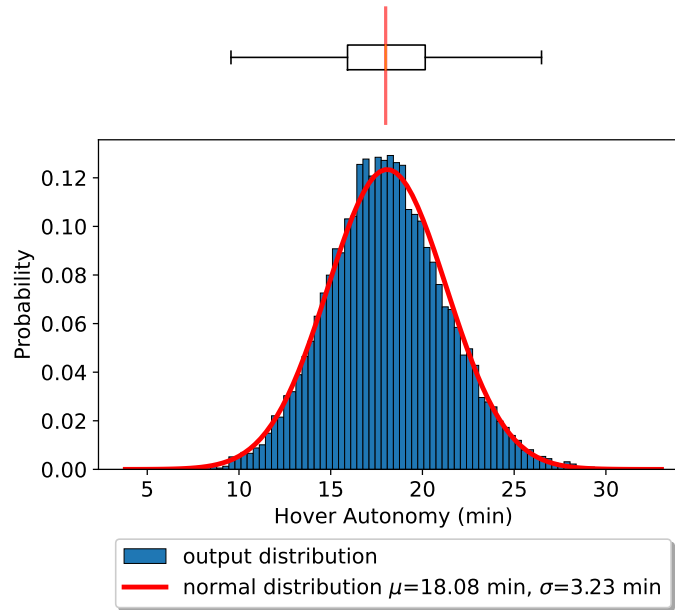
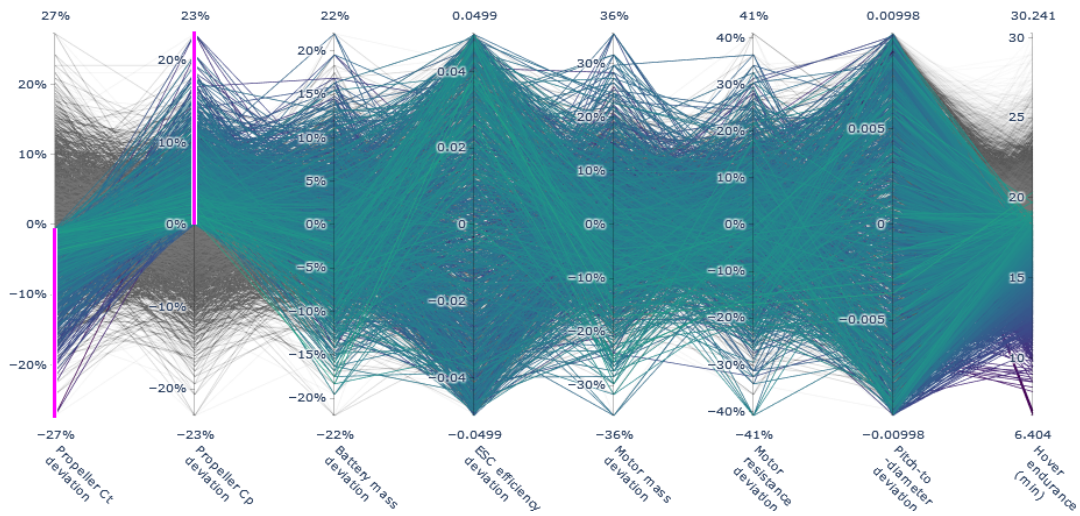
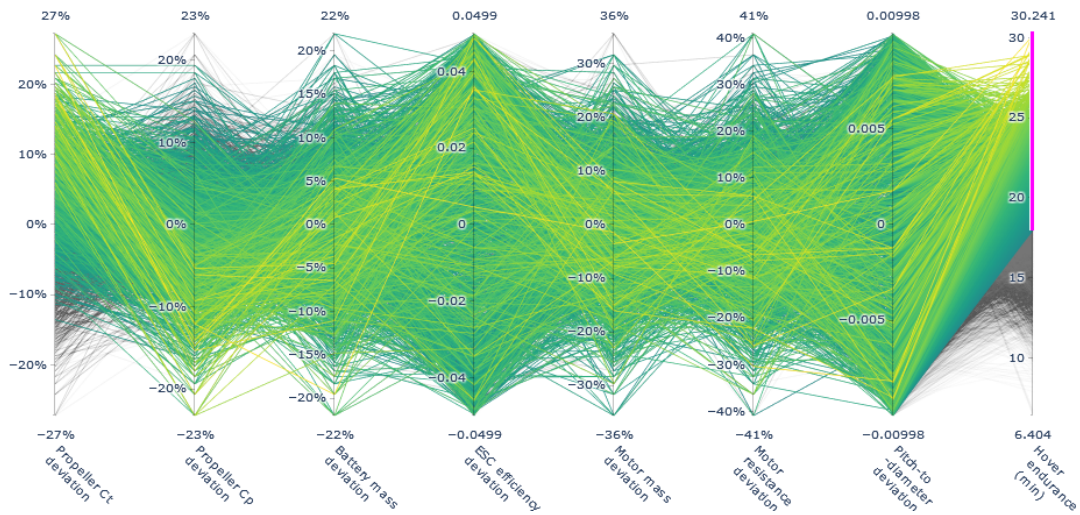


Figure 5.4.5 Distribution of the estimated hover endurance.

zero, highlighting the negligible interactions between the pairs of variables involved in the analysis. The standard deviations found in the Morris analysis are, therefore, more related to non-linearities in the models than interaction effects. A major conclusion of the sensitivity analysis is that uncertainty on the aerodynamic performance of the propellers significantly impacts flight endurance. This is because the mechanical power is obtained directly through the propeller performance, whose models come with great uncertainty. The variation in the component masses (e.g., battery mass) also affects the mechanical power, but the small variations of the estimated masses with respect to the total mass of the UAV limit their influence. The electrical power, which defines the flight endurance together with the battery energy, is calculated from the mechanical power, and as such inherits the uncertainties on the aerodynamic performance of the propellers. In addition, electrical power is influenced by the efficiency of the electrical components (e.g., motor resistance and ESC efficiency). The uncertainty on ESC efficiency is ranked 4 in the sensitivity analysis, highlighting the need to develop a specific model instead of an expertise-based value to reduce uncertainty. Nevertheless, the impact of the electrical parameters uncertainties remains secondary. Finally, the sensitivity to battery energy is negligible because of the large number of UAV batteries available on the market, hence the low uncertainty on this design variable.



(a) Plot filtered to show the hover endurance achieved when the propeller thrust and power coefficients are lower and higher respectively than their estimated values.



(b) Plot filtered to show simulations achieving a hover endurance greater than 18 minutes.

Figure 5.4.6 Parallel coordinate plots of the Monte Carlo simulation, where each polyline represents a single simulation run. The first seven axes are the independent uncertain variables. The last column is the output (hover endurance).

5.5 Hybrid design optimization for uncertainty reduction

The previous sections have shown how uncertainty in various variables affects the knowledge of the system's outputs of interest. In particular, there is a significant risk of overestimating the performance of the system. The actual design may, therefore, not be able to meet the performance requirements specified by the customer. The sensitivity analysis offers the possibility to identify the most influential sources of uncertainty so that actions can be

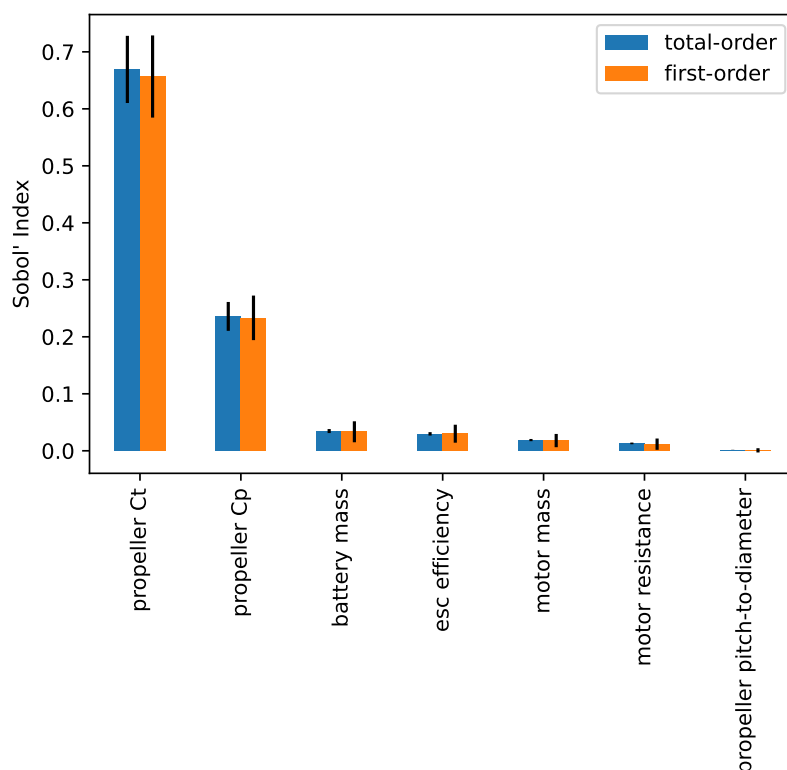


Figure 5.4.7 First- and total-order Sobol' indices showing the impact of uncertainty in each variable on the hover endurance.

taken to mitigate them. Among the possible actions, the designer can opt for increasing the fidelity of the models in order to improve their representativeness. Another approach is to restrict the design space to a set of existing components listed in a database instead of parameterizing components with continuous models. Using off-the-shelf components whose parameters are listed in datasheets provides a deterministic description (Figure 5.5.2) in place of the probabilistic one associated with modeling of custom components (Figure 5.5.1). For example, the aerodynamic model of the propeller, which has large uncertainties, can be replaced by the actual values of the thrust and power coefficients measured experimentally by the manufacturer. Apart from uncertainty reduction, restricting the design to a fixed set of components reduces costs and improves development times, making it popular among designers in a variety of technical fields. However, it is rare that a system can be designed solely from off-the-shelf components. For example, an aircraft designer will not find a ready-made airframe on the market. Designers may also want to explore future technologies for which databases are not yet available, hence requiring analytical models. Therefore, there is a need for a design optimization methodology allowing sizing systems that combine both custom and off-the-shelf components. Such a *hybrid* approach benefits from the detailed

description provided by datasheets and the ability of continuous analytical models to explore a greater design space.

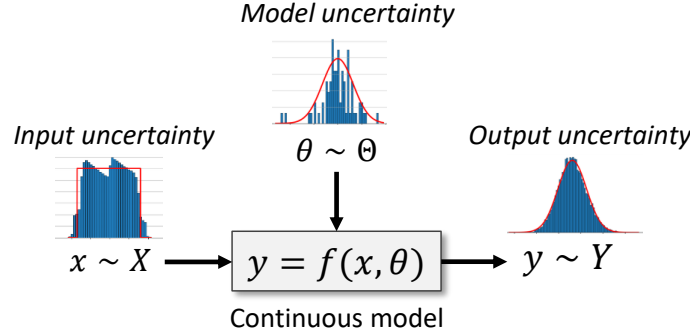


Figure 5.5.1 Custom component described with models.

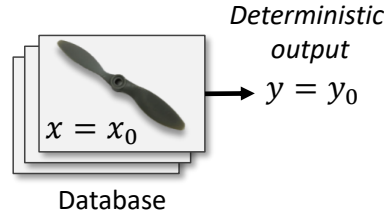


Figure 5.5.2 Existing component described with deterministic data.

5.5.1 Proposed methodology

Figures 5.5.3, 5.5.4a and 5.5.4b provide a generic overview of the proposed hybrid design optimization with an extended design structure matrix (XDSM) diagram [84].

As described in Chapter 2, the design optimization problem is expressed as follows:

$$\begin{aligned} & \underset{\mathbf{x}^*}{\text{minimize}} && f(\mathbf{x}^*) \\ & \text{subject to} && g_j(\mathbf{x}^*) \leq 0, \quad j = 1, \dots, m. \end{aligned} \tag{5.23}$$

where f is the objective function to be minimized (or maximized) by solving for the set of continuous design variables $\mathbf{x}^*$, and $g_j(\mathbf{x}^*) \leq 0$ are inequality constraints. Note that the superscript $*$ here refers to the continuous nature of the variables and is not to be confused with the optimal value of the variables (unlike the notation provided in Chapter 2). The hybrid design optimization addresses the design problem by solving the individual component analyses (depicted by green boxes in Figure 5.5.3) with either continuous models or discrete off-the-shelf selections without the need to reformulate the optimization problem. In particular, the continuous nature of the design variables is preserved⁵.

⁵For this reason, the term "hybrid" is used here instead of "mixed", as the latter most commonly refers to optimization problems with both continuous and discrete variables.

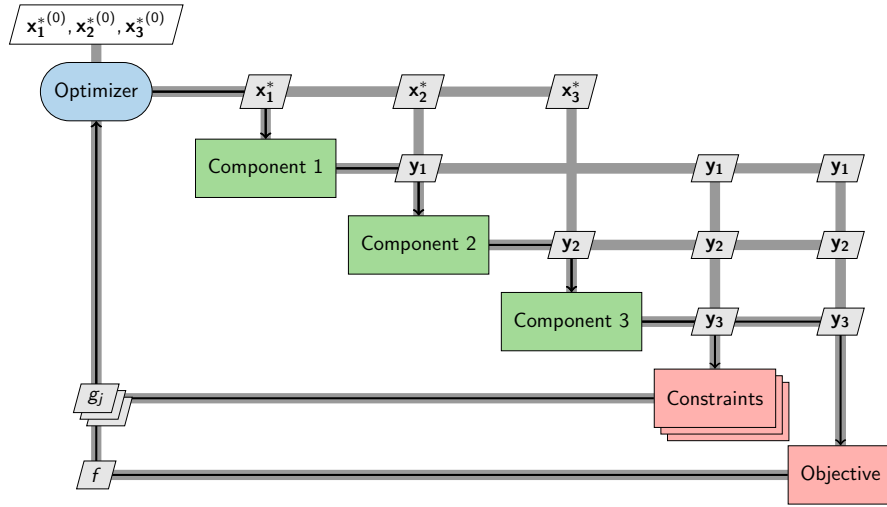


Figure 5.5.3 XDSM for the hybrid design optimization.

When no database is available to the designer, an analytical model is used to return an estimate of the characteristics of the component based on the value of its design variables. This sizing process is illustrated in Figure 5.5.4a, where the continuous design variables $\mathbf{x}_i^*$ and model parameters θ_i^* may contain deviations from physical reality, as explained in Section 5.3. The illustration is simplified for clarity. For more complex problems, the outputs of the upstream component analyses can be used as inputs to the estimation model, adding to the propagation of uncertainty.

When a database with off-the-shelf components is available, the sizing of a component is a two-step process, illustrated in Figure 5.5.4b. First, the candidate component that most closely matches the values of the continuous design variables is selected. This translates the continuous design variables $\mathbf{x}^*$ into discrete variables $\mathbf{x}$ describing an existing component. The dependent parameters of the corresponding component are then extracted from its datasheet instead of being estimated with models. This approach allows for removing both input uncertainty and model uncertainty within the scope of this chapter.

The hybrid design optimization offers several advantages:

- Depending on the available data, the designer can define which components are selected from a database and which are parameterized with design models by replacing the corresponding component analyses. A hybrid design combining both off-the-shelf components and hypothetical components with average properties is therefore obtained.
- The formulation of the optimization problem remains the same as in a continuous design approach, with a reduced number of design variables. The choice regarding which method is used to size the components does not change the problem formulation, as the interfaces between the optimizer and the components are preserved. In addition,

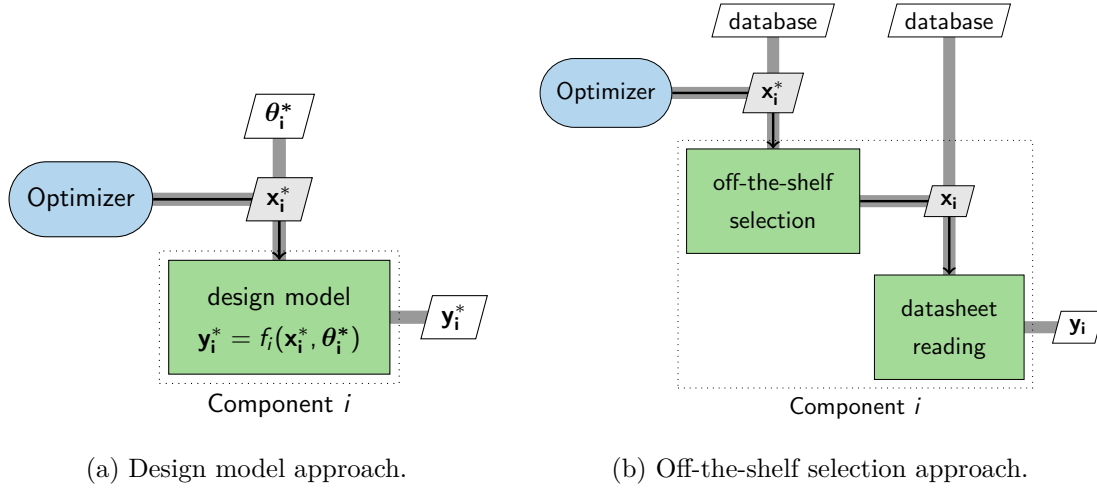


Figure 5.5.4 Sizing process at component level.

the bounds on the design variables are independent of the number of components stored in the databases.

- The nested analyses inherent to an MDO approach ensure that an optimal and consistent design is obtained. This is a significant advance compared to approaches where an optimization in the continuous space is followed by an offline selection of off-the-shelf components, which no longer guarantees optimality and compliance with optimization constraints as well as consistency of the multidisciplinary couplings.

The hybrid design optimization methodology proposed here offers a single, unique framework for both custom and off-the-shelf solutions. It can be derived from most continuous design methodologies in a minimally invasive way and requires no reformulation effort of the optimization problem. However, its success relies on two key factors: the adoption of an effective selection procedure, and the use of an optimization algorithm adapted to the discontinuities introduced by the databases.

5.5.2 Good practices for an efficient off-the-shelf selection

Selecting a component from a database can be done in many different ways, but not all will lead to the same result. It is, therefore, necessary to adopt a methodological approach to perform this task.

First, appropriate variables must be chosen to differentiate the many products in the database. Efficient discrimination requires the use of variables that are not, or only weakly, correlated. This property was ensured by a correlation analysis when choosing the design variables for the optimization problem, in Chapter 3, Section 3.4. Therefore, the variables used to select the components in the databases are the same as the design variables defined in the optimization problem (e.g., nominal torque and velocity constant for the motors).

Secondly, for off-the-shelf components, cleaning the database is a critical, if not mandatory, part of the methodological approach. Filtering the database reduces its dimension and thereby speeds up the selection process. It also allows removing components that have no interest in the application, in effect, components that do not correspond to an optimal solution of the design problem. Such restriction of the search space leads to better convergence properties. A Pareto analysis enables to narrow the database to a reduced set of efficient candidates. Figure 5.5.5 shows a torque versus weight plot for brushless motors. Pareto-efficient solutions with a high torque-to-weight ratio are highlighted in red. The other components are discarded, as they exhibit higher mass penalties for similar mechanical performance. Particular attention must be given to this step, as excessive filtering can lead to sparse data coverage and compromise the search for an optimal combination of components at the system level.

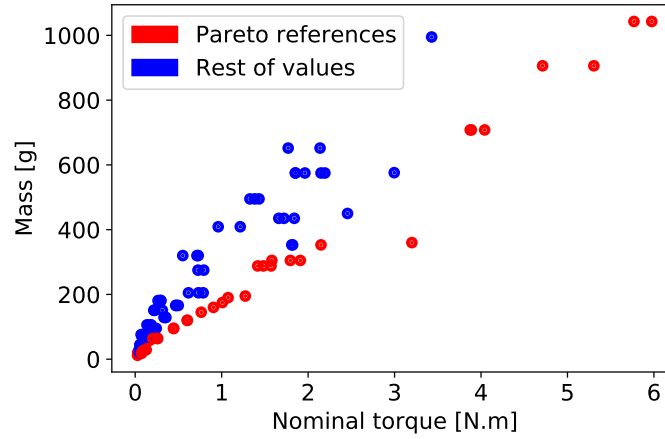


Figure 5.5.5 Pareto front for UAV motors. Manufacturers: AXI Model Motors, Scorpion Motors, KDE Direct.

Finally, a selection technique must be chosen. The problem of finding the candidate component in a database that is closest to given values of the design variables consists of a nearest neighbor search (NNS). It requires defining a distance metric based on which candidate solutions will be compared. The brute force method computes the distance from the given point to every other point in the database, and as such has a running time complexity of $\mathcal{O}(D \times N)$, where D is the number of design variables and N is the number of candidate components in the database. The search process can be sped up using space-partitioning methods such as ball tree or k-d tree, with a running time cost of approximately $\mathcal{O}(D \times \log(N))$ [85]. Lastly, normalization of the metric space is recommended. However, it can be desirable to bias the search in favor of one design variable by weighting the corresponding metric axis. In the present work, the ball tree nearest-neighbor algorithm from the Python library Scikit Learn [85] is employed with equal weighting for the variables

selection. In addition, for some of the variables, a filtering routine of the database is applied to enforce the selection of the closest upper value rather than the nearest one. This allows the hierarchical relationships provided by the original sizing procedure to be preserved. For example, in the continuous sizing procedure from Chapter 3, the nominal torque of the motor is chosen to be greater than the torque required by the propeller in hover conditions ($T_{nom} = k_{mot}^s \cdot T_{mot,hover}$, $k_{mot}^s \geq 1$), thus ensuring that the hover performance constraint is satisfied by explicit resolution. To enforce this relationship in the discrete selection process, candidate components whose nominal torque is less than the continuous value are automatically discarded.

5.5.3 Optimization algorithm

The design process presented in Section 5.5 contains discontinuities introduced by the discretization of the components' design space in case off-the-shelf components are integrated. Therefore, the optimization problem is poorly suited to gradient-based optimizers. Indeed, gradient-based algorithms such as the one used in Chapter 3 require the functions to be sufficiently smooth. Occasional discontinuities are tolerated in practice, but not if the discontinuity is near the optimum, as it may be the case when optimizing with off-the-shelf components. For their part, some derivative-free optimization methods are more robust against ruggedness and discontinuities. Among them, the covariance matrix adaptation evolution strategy (CMA-ES) [86–89] has proven to be successful in many applications with complex functions and requires minimal tuning. A major advantage of CMA-ES is its ability to deal with the stepwise and discontinuous nature of the objective function. Indeed, the projection of continuous variables into a discrete set of elements, as proposed by the present methodology, can lead to the repeated evaluation of the same combination of components if the variations applied by the optimizer to the continuous design variables are too small. This behavior can be avoided by a careful balance between exploration and exploitation, as provided by the evolutionary nature of CMA-ES and its step size control. Compared to other state-of-the-art algorithms such as EGO [90], CMA-ES requires a higher computational budget, but the UAV design models are fast to evaluate, so this is no limiting factor in the present work. The Python implementation of CMA-ES from the *pycma* library [91] was used for the following case study. In particular, the optimization constraints were handled through an artificial penalty on the objective function. The performance of the CMA-ES algorithm and its comparison with a gradient-based algorithm is discussed in Section 5.5.5.

5.5.4 Case study: design optimization of a multirotor UAV

The hybrid design optimization is performed on the hexacopter drone introduced in Section 5.4.3, with the specifications provided in Table 5.4.3. The optimization problem consists

of minimizing the total mass of the UAV while ensuring a hover endurance greater than or equal to 18 min. Three different designs are optimized and compared:

1. A *custom* design obtained with the continuous-domain MDO framework introduced in Chapter 3. As a reminder, the models employed in the continuous methodology return estimates for component characteristics that are representative of the average properties of commercially available components.
2. A *hybrid* design whose components are parameterized with estimation models, apart from the propeller which is directly selected off-the-shelf to minimize the main source of uncertainty.
3. An *off-the-shelf* design in which all components of the propulsion system are selected directly from a database.

The off-the-shelf components used in the design optimization are taken from manufacturers' catalogs available online⁶.

Deterministic results

The results from the design optimizations are considered as deterministic references, whose main parameters are provided in Table 5.5.1 and Figure 5.5.6. All three deterministic designs meet the 18-minute hover endurance requirement, which confirms the successful convergence of the optimization process. However, differences are observed in the mass minimization objective. Specifically, the custom design is heavier than its off-the-shelf counterpart, which might be counter-intuitive. Indeed, a discrete solution is expected to be suboptimal compared to a continuous solution. The current finding indicates that there are components in the catalogs that provide better performance compared to the component descriptions provided by the continuous models. Indeed, the models employed in this study are only reflecting the average properties of commercially available components. More specifically, the propeller parameterized with models offers a slightly higher efficiency (higher C_T/C_P ratio), which results in lower energy consumption for the same thrust. This is reflected in the battery energy and mass, which is minimal for the custom design: the battery size can be reduced for the same hover endurance requirement of 18 minutes. On the other hand, the off-the-shelf propellers are smaller, which allows for shorter arms, hence reducing the airframe's weight. The motors selected off-the-shelf are also significantly lighter, as depicted in Figure 5.5.6.

Uncertainty analysis

An uncertainty analysis is carried out around the deterministic designs to quantify the uncertainty in the flight performance and UAV's mass, applying the methodology introduced

⁶Manufacturers' catalogs: APC Propellers, AXI Model Motors, Scorpion Motors, KDE Direct, Gens ACE, Gens Tattu, Kokam, Thunder Power Rampage, Thunder Power ProLiteX, YGE, Turnigy, KOSMIK ESCs

Table 5.5.1 Deterministic results of the different UAV design optimizations.

	1. Custom design	2. Hybrid design	3. Off-the-shelf design
Propeller	Design model $D = 0.489 \text{ m}$, $\beta = 0.300$ $C_T = 0.0859$, $C_P = 0.0277$	APC 18x5.5MR $D = 0.457 \text{ m}$, $\beta = 0.306$ $C_T = 0.0839$, $C_P = 0.0284$	APC 18x5.5MR $D = 0.457 \text{ m}$, $\beta = 0.306$ $C_T = 0.0839$, $C_P = 0.0284$
Motor	Design model $T_{nom} = 1.025 \text{ N.m}$ $K_V = 20.78 \text{ rad/V/s}$	Design model $T_{nom} = 1.012 \text{ N.m}$ $K_V = 22.90 \text{ rad/V/s}$	KDE 4215XF-465 $T_{nom} = 1.273 \text{ N.m}$ $K_V = 48.7 \text{ rad/V/s}$
ESC	Design model $P = 890 \text{ W}$, $U = 34 \text{ V}$	Design model $P = 992 \text{ W}$, $U = 34.8 \text{ V}$	Jeti Spin 48 Opto $P = 1776 \text{ W}$, $U = 37.0 \text{ V}$
Battery	Design model $E = 730 \text{ Wh}$, $U = 34 \text{ V}$	Design model $E = 819 \text{ Wh}$, $U = 34.8 \text{ V}$	36Ah 6S 25C LiPo* $E = 799 \text{ Wh}$, $U = 22.2 \text{ V}$
Total mass (objective)	15.42 kg	15.49 kg	15.06 kg
Hover endurance (requirement)	18.00 min	18.00 min	18.07 min

* 6-P Thunder Power RC TP6000-6SPX25

Note: "Design model" refers to hypothetical components sized with the continuous estimation models presented in Chapter 3 and recalled Table 5.3.1.

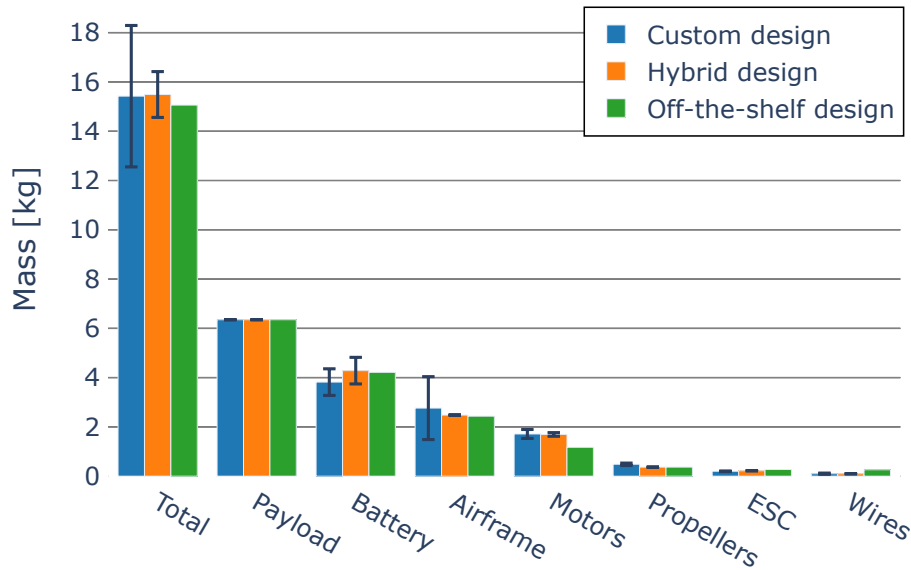


Figure 5.5.6 Mass breakdown for the different UAV designs. The black vertical segments represent the 95% confidence intervals obtained with the uncertainty analysis.

in Section 5.4. The results from the uncertainty analysis are provided in Table 5.5.2 and Figure 5.5.8. The 95% confidence intervals obtained for each component mass are also shown as vertical segments in Figure 5.5.6. In addition, the geometries of each UAV design in Figure 5.5.7 are overlayed to illustrate the mean, upper, and lower bounds of the 95% confidence intervals. Interestingly, the use of an off-the-shelf propeller for the hybrid and off-the-shelf designs freezes the diameter and thus the arms length (but not necessarily the

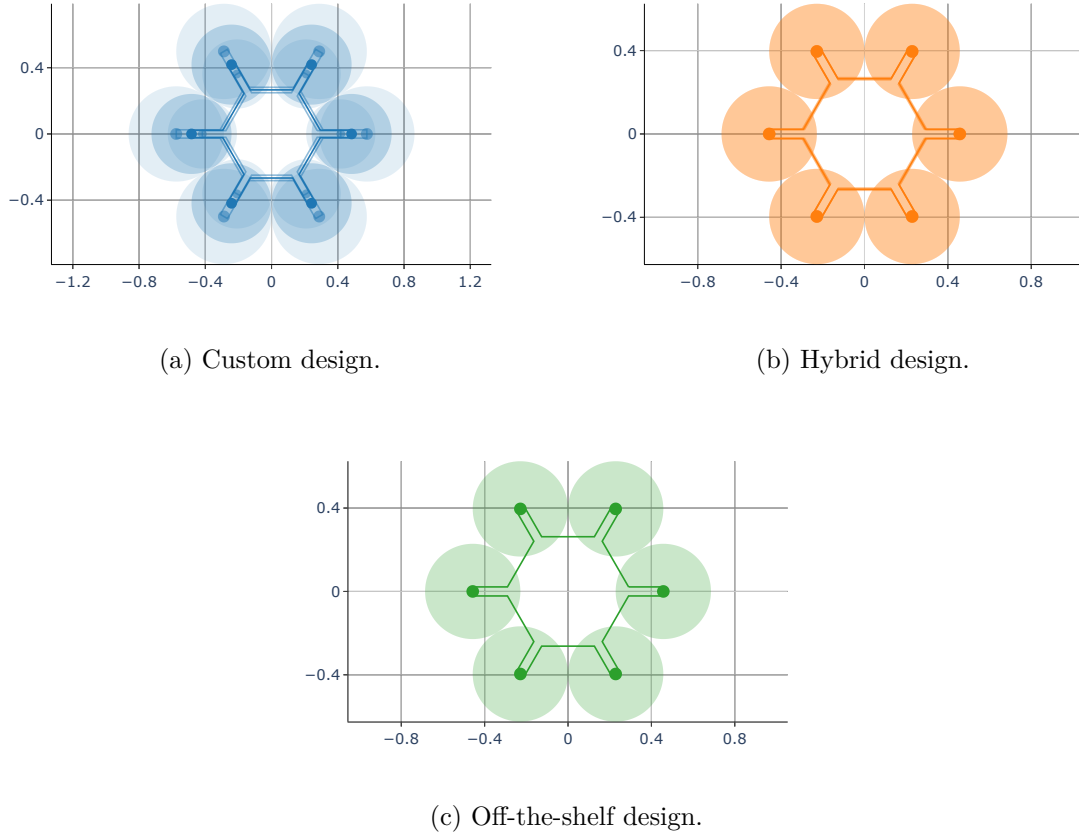


Figure 5.5.7 Geometries for the different UAV designs. For each case, the geometries are overlaid to illustrate the mean, upper, and lower bounds of the 95% confidence intervals obtained from the uncertainty analysis.

mass of the other components). This explains the constant dimensions observed for these designs as opposed to the ambiguous dimensions obtained for the custom design.

Due to input and model uncertainties, the custom design exhibits a standard deviation of 3.2 minutes for hover endurance. This translates to a 31% risk of having a flight of less than below 16.5 minutes instead of the required 18 minutes, which would require additional design feedback in the development process. The probability decreases to 7% for the hybrid design and is null for the off-the-shelf design (taking into account only the sources of uncertainty covered by this study). The hybrid design offers the advantage of combining low uncertainty with the convenience of using continuous models when databases are not available for all components.

The skewness⁷ of the mass distribution observed for the custom design shows non-linearities in the continuous models. Specifically, the mass skewness is due to the airframe mass which increases with longer arms. The arms length is sized from the diameter of the propeller, and the latter is derived from the propeller performance (uncertain variables C_t and C_p) in a non-linear way.

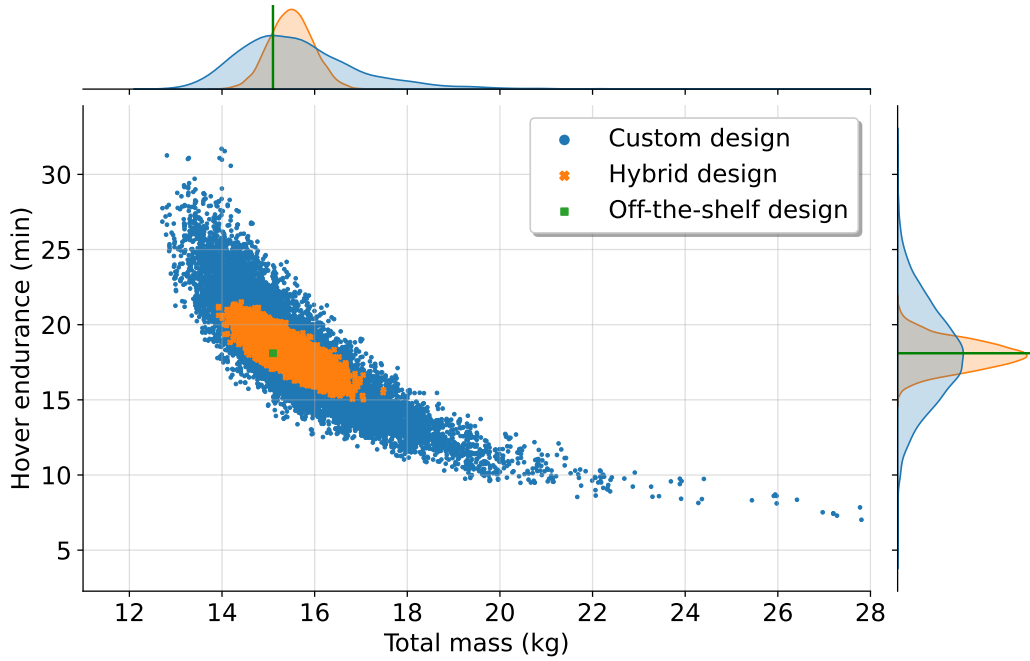


Figure 5.5.8 Distributions of the hover endurance and total mass estimations resulting from Monte Carlo experiments on the custom, hybrid and off-the-shelf designs.

Table 5.5.2 Mean μ , standard deviation σ and skewness $\tilde{\mu}_3$ of the estimated mass and hover endurance distributions for the different UAV designs.

	1. Custom design	2. Hybrid design	3. Off-the-shelf design
Total mass	$\mu = 15.66$ kg $\sigma = 1.43$ kg $\tilde{\mu}_3 = 2.74$	$\mu = 15.49$ kg $\sigma = 0.47$ kg $\tilde{\mu}_3 = 0.04$	$\mu = 15.06$ kg $\sigma = 0.0$ kg $\tilde{\mu}_3 = 0.0$
Hover endurance	$\mu = 18.08$ min $\sigma = 3.23$ min $\tilde{\mu}_3 = 0.14$	$\mu = 18.02$ min $\sigma = 0.98$ min $\tilde{\mu}_3 = 0.17$	$\mu = 18.07$ min $\sigma = 0.0$ min $\tilde{\mu}_3 = 0.0$

5.5.5 Numerical performance

The following Table 5.5.3 shows the performance of a gradient-based algorithm (SLSQP) and the CMA-ES algorithm for solving the "custom", "hybrid" and "off-the-shelf" optimization

⁷Skewness is a measure of the asymmetry of a distribution. It is the third standardized moment (dimensionless) and is calculated as: $\tilde{\mu}_3 = \mu_3/\sigma^3 = E[(X - \mu)^3]/(E[(X - \mu)^2])^{3/2}$.

problems. For the off-the-shelf design and, to a lesser extent, the hybrid design, the optimization had to be run several times because a feasible solution could not always be found. On average, when all components have to be selected off-the-shelf, 1 out of 5 optimization runs converges to a feasible and optimal solution. The reported computation times correspond to the optimizations that returned the best feasible solution. The lack of robustness of the optimization highlights the complexity of the MDO problem. In addition to the set of best practices provided in Section 5.5.2, a general guideline for numerical performance is to limit the number of components that are to be selected off-the-shelf rather than estimated with analytical models. In this regard, the uncertainty analysis performed beforehand is critical in determining which components must be selected off-the-shelf to significantly reduce uncertainties and which can still be modeled analytically. A possible approach is to start the design process with a "custom design" problem, then evaluate the uncertainties and re-run the optimization by replacing the most critical component (here, the propeller) with an off-the-shelf selection. The process is repeated until the output uncertainty is deemed acceptable, possibly before having to replace all analytical models with databases of components. Finally, the use of advanced implementations of CMA-ES with better constraint handling, as proposed in [92–94], is expected to improve the convergence properties since the problem is highly constrained (21 constraints).

Table 5.5.3 Performance of the SLSQP and CMA-ES algorithms on the multirotor UAV design problem.

	1. Custom design		2. Hybrid design		3. Off-the-shelf design	
	Resolution time	Best feasible solution	Resolution time	Best feasible solution	Resolution time	Best feasible solution
SLSQP	1.9s	15.42 kg	13.2s	15.50 kg	No feasible solution found	
CMA-ES	2min 14s	15.42 kg	2min 17s	15.49 kg	2min 44s	15.06 kg

5.6 Conclusions and perspectives

Methodology and lessons learned

The present chapter introduces a new approach for quantifying and mitigating uncertainties in design optimization by addressing uncertainties related to both design models and off-the-shelf component availability. The contributions of the present methodology are twofold: (1) it formalizes the uncertain availability of components in the market, and (2) it enables the design optimization of systems with both custom and off-the-shelf components.

More specifically, the main sources of uncertainty in the early design phases have been characterized. The achievements in this respect are recalled here:

- Input uncertainty, which refers to the unknown availability of a component in the market, is characterized by the distribution of the design variables' values within a size range. This distribution follows either an arithmetic or a geometric progression.

- Input uncertainty was compared to model uncertainty arising from an inadequate structure of the design models or erroneous fitting of their parameters. An analytical criterion on the distribution of components in the market was developed to evaluate the relative importance of input and model uncertainty with a first-order approximation.
- A more advanced analysis relying on numerical methods (Monte Carlo, Morris and Sobol') was performed to propagate uncertainties through the system models and to rank the effects of different uncertain variables on the output variation.
- Application to an electric multirotor UAV showed that uncertainty in the aerodynamic model of the propeller is critical regarding hover flight endurance. In addition, the large number of components available in the UAV market makes input uncertainties of secondary importance.

Next, a methodology for uncertainty reduction in multidisciplinary design optimization was proposed. In this methodology, uncertainty mitigation is achieved by replacing the design models with datasheets describing a specific component selected off-the-shelf. More precisely:

- The methodology enables the design optimization of a system whose components are either parameterized with continuous models or selected off-the-shelf. This is achieved by replacing the individual component analyses in the MDO process rather than replacing the continuous optimization variables with discrete ones.
- Compared to state-of-the-art approaches, the design methodology presented in this chapter allows easy switching between custom and off-the-shelf components without having to reformulate the optimization problem. In addition, the sizing process returns a truly feasible and optimal design rather than mapping the analytical solution to the most proximate set of components.
- Best practices for improving the performance of the design optimization were provided. In particular, the exploration of component databases can be made more efficient by applying a Pareto filter. In addition, the use of an advanced optimization algorithm such as CMA-ES is essential.
- The relevance of the proposed approach was demonstrated with a multirotor UAV case study. Three designs with different degrees of custom versus off-the-shelf components were optimized, and uncertainty analyses performed around the deterministic solutions. The results showed a significant reduction in performance uncertainty of 69% when the most critical component, namely the propeller, is selected off-the-shelf.

Perspectives

Further research should address the following limitations and perspectives of the present

work. First, the uncertainties modeled in Section 5.3 were assumed to be mutually independent, which may not always be verified in practice. The use of generalized sensitivity analysis methods capable of accounting for dependencies between variables would address this limitation. The modeling of dependencies may require the identification of a primary design variable for each component, whose uncertainty distribution follows an arithmetic or geometric progression as proposed in Section 5.3, and a set of secondary design variables whose uncertainties are weakly correlated to those of the primary variable. Second, with regard to the design optimization approach, the influence of the choice of the variables used to select the components in the databases could be investigated. For example, the prioritization of one variable over another in the discrete selection process should be investigated by analyzing the sensitivity of the final objective to each variable. Third, a benchmark could be performed to compare the proposed optimization approach with mixed discrete and continuous variables optimization. This would however require a reformulation effort of the optimization problem. Finally, the case study would benefit from an extension to dynamic flight scenarios. This requires the quantification of uncertainties in the airframe aerodynamic model, and the extension of propeller performance uncertainties to cover non-axial conditions in forward flight [95].

Appendix

Appendix 5.A Flight performance model (from Chapter 3)

Thrust per propeller in hover: $F_{pro,hover} = \frac{m_{uav}g}{N_{rotors}}$

Propeller speed: $\Omega_{pro} = \frac{2\pi}{60}n_{pro} = \frac{2\pi}{60}\sqrt{\frac{F_{pro}}{C_T\rho_{air}d_{pro}^4}}$

Propeller power: $P_{pro} = C_P\rho_{air}n_{pro}^3d_{pro}^5$

Motor torque: $T_{mot} = \frac{P_{pro}}{\Omega_{pro}}$

Motor current: $I_{mot} = K_V(T_{mot} + T_f)$

Motor voltage: $U_{mot} = \frac{\Omega_{pro}}{K_V} + RI_{mot}$

Motor power: $P_{mot} = U_{mot}I_{mot}$

Battery power: $P_{bat} = \frac{N_{rotors}P_{mot}}{\eta_{esc}}$

Hover endurance: $t_{hover} = 0.8\frac{E_{bat}}{P_{bat,hover}}$

Appendix 5.B Results for the deterministic (custom) design

The results for the deterministic design obtained with the continuous-domain methodology from Chapter 3 are presented in Table 5.B.1. Also provided are the reference values from DJI for the M600 UAV, for comparison.

Table 5.B.1 Results for the deterministic design obtained with the continuous-domain methodology, and comparison with the DJI M600 Pro.

	Reference (DJI)	Sizing results	Relative error (%)
Total mass (kg)	15.5	15.4	- 0.50
Batteries mass (kg)	4.08	3.82	- 6.43
Motors mass (kg)	1.38	1.71	+ 24.12
Propeller diameter (m)	0.533	0.489	- 8.31
Battery energy (Wh)	780	729	- 6.43
Motor max. power (W)	1000	890	- 11.04

Appendix 5.C Uncertain variables for the case study

Table 5.C.1 Assumptions on uncertain variables.

Name	Type of uncertainty	Nominal value	Error distribution
Propeller diameter (m)	Input	0.489	$\mathcal{U}(-0.013, 0.013)$
Propeller pitch-to-diameter ratio (-)	Input	0.30	$\mathcal{U}(-0.01, 0.01)$
Propeller mass (kg)	Model	0.080	$\mathcal{N}(0, 10.0\%)$
Propeller coefficient C_T (-)	Model	0.0859	$\mathcal{N}(0, 8.8\%)$
Propeller coefficient C_P (-)	Model	0.0277	$\mathcal{N}(0, 7.5\%)$
Motor velocity constant (rad/V/s)	Input	20.78	$\mathcal{U}(-3.5\%, 3.5\%)$
Motor nominal torque (N.m)	Input	1.025	$\mathcal{U}(-4.9\%, 4.9\%)$
Motor mass (kg)	Model	0.285	$\mathcal{N}(0, 11.5\%)$
Motor resistance (V/A)	Model	0.248	$\mathcal{N}(0, 13.2\%)$
Motor friction torque (N.m)	Model	0.0149	$\mathcal{N}(0, 33.4\%)$
Battery energy (Wh)	Input	730	$\mathcal{U}(-5, 5)$
Battery mass (kg)	Model	3.82	$\mathcal{N}(0, 7.1\%)$
ESC power (W)	Input	890	$\mathcal{U}(-294, 294)$
ESC mass (kg)	Model	0.0329	$\mathcal{N}(0, 25.4\%)$
ESC efficiency (-)	Expert judgment	0.95	$\mathcal{U}(-0.05, 0.05)$

Values are based on manufacturers' catalogs (APC Propellers, AXI Model Motors, YGE, Thunder Power ProLiteX), and Budinger *et al.* [77].

Appendix 5.D Numerical Resources

The Jupyter Notebook and the databases used to obtain the results on input uncertainty in Section 5.4, Figures 5.3.4 and 5.3.7 are available at the following link: https://github.com/SizingLab/input_uncertainty_quantification.

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Part III

Fault-tolerant design for UAV safety

Part III of the manuscript focuses on improving the safety of UAVs by assessing their controllability under fault conditions and sizing the lift-thrust control system accordingly. Chapter 6 reviews the regulatory framework and existing research, identifying gaps in controllability assessment and fault-tolerant sizing for multirotor, fixed-wing, and hybrid FW-VTOL UAVs. Chapter 7 presents a novel methodology for the controllability assessment and fault-tolerant sizing of UAVs, addressing the challenges posed by faults in rotors and control surfaces. A new metric for assessing the control authority of UAV designs in the presence of faults is proposed, and sizing factors are introduced to bridge the gap between controllability analysis and sizing of the effectors. Case studies demonstrate the effectiveness of the proposed approach and show its potential for safety-critical applications.

Author's contributions

Journal articles

J. Liscouet, F. Pollet, J. Jézégou, M. Budinger, S. Delbecq, and J.-M. Moschetta. A methodology to integrate reliability into the conceptual design of safety-critical multirotor unmanned aerial vehicles. *Aerospace Science and Technology*, p. 107681, Jun. 2022, doi: 10.1016/j.ast.2022.107681.

F. Pollet, J. Liscouet, M. Budinger, S. Delbecq, and J.-M. Moschetta. Controllability assessment and fault-tolerant sizing of UAVs under effector failures. *AIAA Journal of Guidance Control and Dynamics*, Under review.

Chapter 6

State of the art of safety-critical UAVs

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6.1 Introduction and context

Unmanned aerial vehicles are capable of performing a wide range of tasks in complex environments, including the delivery of medical equipment, aerial inspections and parcel delivery. Such operations, usually beyond visual line of sight (BVLOS) and over people in densely populated areas, are considered safety-critical. Indeed, technical failures leading to uncontrolled UAV behavior can result in catastrophic collisions with humans, other aircraft or infrastructure. Practical examples of safety-critical applications include a joint project between Swiss Post and drone manufacturer Matternet to deliver medical samples across various Swiss cities since 2017. Interestingly, the Swiss Post withdrew from the project in 2022 because they could not be profitable while complying with current regulations [1]. Specifically, two flight operation incidents occurred in 2019 in Zurich [2]. In the second incident, an emergency procedure led to the deployment of the drone's parachute, but the connection cord tore and the drone hit the ground in a forest near a group of children. According to the final report of the Swiss Transportation Investigation Safety Board (STSB), the parachute was triggered automatically after the UAV entered an oscillating state, most likely due to an external factor such as a bird strike or pre-existing propeller damage [3]. It is concluded that

for a quadcopter concept such as the Matternet M2 drone, the failure of a propulsion unit will inevitably lead to a crash, or at least the activation of the automatic flight termination system¹. The STSB also indicates that with sufficient sizing, a hexacopter can still land more safely in the event of failed propellers. The Matternet incident constitutes a major concern for European citizens. Indeed, a recent EASA survey showed that safety is the main concern of European citizens when it comes to delivery drones, including emergency use cases [4]. This underscores the need for safety-driven UAV design, as opposed to classical energy efficiency or mass considerations. Unlocking the potential of UAVs in safety-critical applications necessitates comprehensive investigations, comparative assessments of diverse design concepts, and the optimization of UAV sizing to meet safety criteria.

6.2 Regulatory framework

Determining the safety and reliability requirements for the design process requires an analysis of the regulations that apply to Unmanned Aerial Systems (UAS). Here, the term UAS includes the aircraft (UAV) itself and the entire system that supports and controls the UAV. Among the agencies responsible for civil aviation safety, the EASA has published a comprehensive regulatory framework for certification specific to UAS. Specifically, Regulations (EU) 2019/947 [5] and (EU) 2019/945 [6] define the applicable safety requirements for the operation, design and manufacture of UAS. These regulations detail the high-level safety objectives set out in EASA's basic regulation (EU) 2018/1139 [7] through an operation-centered, risk-based approach that enforces safety requirements proportionate to the assessed level of operational risk. The main steps of this process are described here.

First, define the category of operation as either 'Open', 'Specific' or 'Certified' based on the UAS concept of operations. The 'Open' category is the main reference for the majority of recreational and low-risk commercial drone activities. Other operations, for example, if flying above 120m or beyond visual line of sight, are considered in the 'Specific' or 'Certified' categories. In addition, operations involving large UAVs (>3m) or the transport of dangerous goods or people are considered in the 'Certified' category. As an illustrative case, the operation of a drone in an urban environment with a total weight of more than 4 kg will fall under the 'Specific' or 'Certified' categories.

The next steps vary according to the category of operation. For UAS operations in the 'Open' category, the UAV is expected to be labeled with a class identification ranging from C0 to C4. The class identification determines the rules to be applied with regard to the operation and the operator/pilot. For example, a UAV weighing between 250 and 900 grams may be operated under a C1 Class Identification, provided the pilot is over 16 years of age and no flight over uninvolved persons is expected. For the 'Certified' category, detailed guidelines

¹A Flight Termination System (FTS) is a system that terminates the flight when triggered. Its purpose is to ensure that an out-of-control UAV does not enter adjacent areas with an undefined trajectory.

have yet to be published by EASA in 2023. Finally, for applications in the 'Specific' category, the Specific Operations Risk Assessment (SORA) provides a methodology to assess the intrinsic risks of the intended operation. The SORA combines ground and air risk analyses related to the operation into a Specific Assurance and Integrity Level (SAIL) ranging from I to VI, which determines the subsequent actions to be taken. The ground and air risk analyses typically involve the dimensions of the UAV and the operational scenarios, for example, if BVLOS over a populated area or in the vicinity of other aircraft is considered. The SAIL then leads to the identification of Operational Safety Objectives (OSOs) to be achieved with a certain level of robustness, i.e. a level of integrity (safety gain) and a level of assurance (proof that the safety gain can be obtained). The level of robustness to be demonstrated tends to increase as the SAIL increases. An extract of the OSOs is shown in Figure 6.2.1, together with the level of robustness to be demonstrated depending on the SAIL.

OSO number (in line with Annex E)		SAIL					
		I	II	III	IV	V	VI
	Technical issue with the UAS						
OSO#01	Ensure the UAS operator is competent and/or proven	O	L	M	H	H	H
OSO#02	UAS manufactured by competent and/or proven entity	O	O	L	M	H	H
OSO#03	UAS maintained by competent and/or proven entity	L	L	M	M	H	H
OSO#04	UAS developed to authority recognised design standards ¹	O	O	L	L	M	H
OSO#05	UAS is designed considering system safety and reliability	O	O	L	M	H	H

Figure 6.2.1 Examples of OSOs and their level of robustness depending on the SAIL [8]. 'L' stands for low robustness, 'M' for medium robustness, and 'H' for high robustness. 'O' means the OSO is optional.

For low-risk operations (SAIL I and II), the UAV may be operated with a class identification label (as for the 'Open' category) instead of demonstrating compliance with the technical OSOs. For medium-risk (SAIL III and IV) and high-risk (SAIL V and VI) operations, the airworthiness objectives of the Special Conditions SC Light-UAS - Medium Risk [9] and SC Light-UAS - High Risk [10] may be used to demonstrate compliance with the design-related OSOs. Within the subset of SC Light-UAS requirements, "Light-UAS.2510-Equipment, Systems and Installation" provides the applicable system safety requirements. Equipment and systems, including the rotors and the control surfaces, must be designed so that a single failure will not result in a catastrophic failure condition. Additionally, for the high-risk requirement, each catastrophic condition resulting from multiple failures must be extremely improbable. A visual summary of the EASA regulatory framework and how it relates to the safety requirements of the design is shown in Figure 6.2.2.

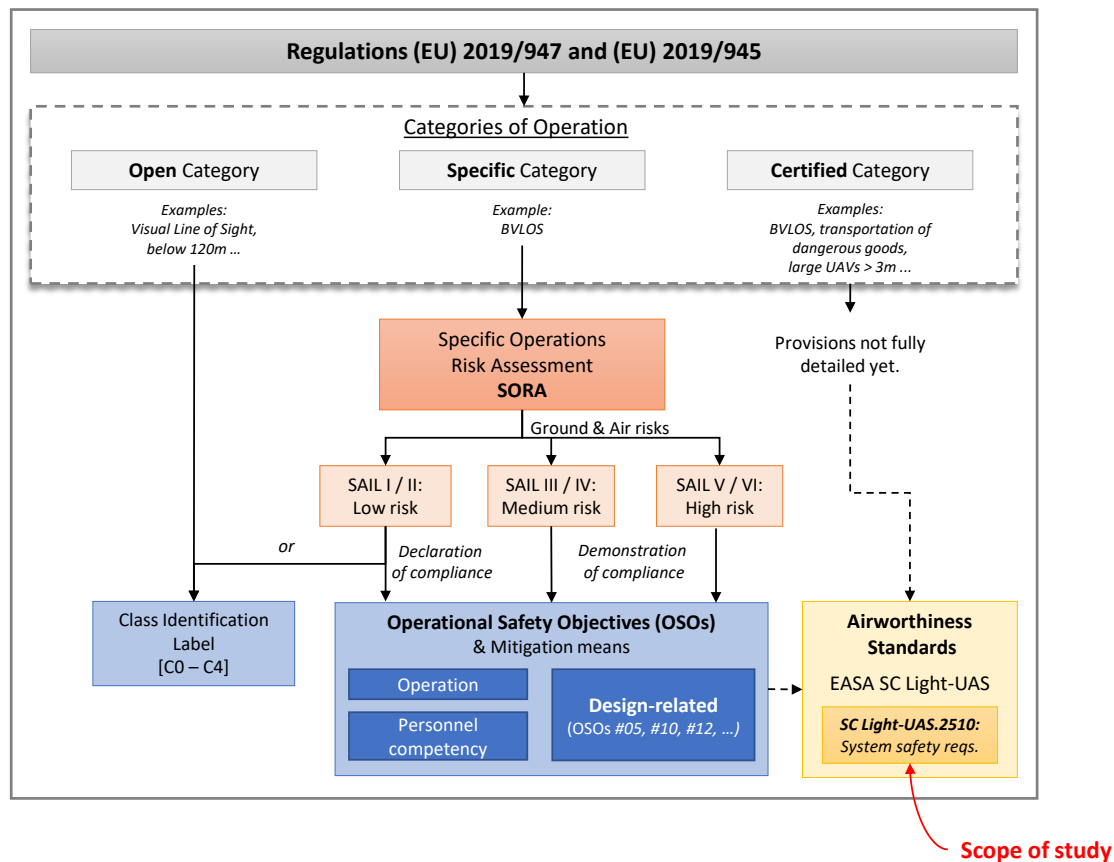


Figure 6.2.2 EASA Regulatory Framework for Unmanned Aerial Systems.

Means to demonstrate compliance with SC-Light-UAS.2510 Medium Risk have recently been proposed by EASA [11]. It is stated that catastrophic failure conditions should be understood as those failure conditions leading to the loss of control of operation. Thus, compliance with the safety requirements implies that *"failure conditions leading to loss of control of the operation are not probable"* and *"should not result from a single failure"*. Note that the activation of a flight termination system or a parachute implies that control of the operation is already considered to have been lost. These features are considered emergency systems intended to reduce the risk after the control of the UAV has been lost, and therefore cannot be used to demonstrate compliance with SC-Light-UAS.2510.

Therefore, the regulation on UAS for medium- and high-risk operations suggests, among other things, that UAV manufacturers must be able to:

1. Assess the controllability of a given UAV configuration in the event of a failure,

2. Specify emergency ratings for the lift-thrust-control system (i.e., rotors and control surfaces), such as the maximum thrust to be provided by each rotor in critical scenarios, and
3. Size the lift-thrust-control system accordingly.

6.3 Safety-critical design of UAVs

6.3.1 General definitions

Before going into a review of the literature on the subject, it is worth defining the relevant notions. In a general sense, a *fault* is an event that changes in an undesired way the characteristics of a system or component, eventually leading to degraded performance or even loss of system function. Faults can be classified as system faults, sensor faults and actuator faults. System faults change the dynamic properties of the system, for example due to an alteration in the aerodynamics of the airframe. Sensor faults prevent the controller from correctly assessing the state of the system. Finally, actuator faults cause the control of the system to be interrupted or modified. If, in addition, the faulty component is unable to perform its function and must be shut down, it is said to be in a *failure* state.

A system is *fault-tolerant* if a component fault or failure does not cause a failure at the system level, i.e. the system is still able to accomplish its function [12]. Fault tolerance should be distinguished from robust design. Robust design must deal with uncertainties, such as those introduced in Part II, that cause the real operating conditions or the system behavior to differ from those assumed in the design. Fault tolerance aims to deal with well-defined conditions resulting from faults that are expected to occur. Various strategies, including sliding mode control, adaptive control, and robust control, can achieve fault tolerance with or without explicit fault diagnosis [13]. The success of these strategies presupposes the existence of a solution to the control problem. In this sense, the possibility of finding a controller that satisfies the control objectives for the faulty system is a property called *reconfigurability* [14]. In other words, in a reconfigurable system, the original control objectives specified for the fault-free system can still be achieved by shutting down the faulty components and using only the healthy ones. Inadequate reconfigurability, for example, insufficient redundancy, prevents effective fault tolerance, regardless of the control strategy [15]. Reconfigurability can be evaluated through the remaining *controllability* and *observability* of the faulty system [15].

Safety is a broader concept that describes the absence of danger for the system and its environment. As shown in Figure 6.3.1, the performance of a system can be categorized into four distinct regions. Faults place the system in a state of degraded performance. In such a situation, the fault-tolerant control should take corrective action to prevent further degradation towards the unacceptable performance region. Beyond the region of unacceptable performance is a region of danger. If the system reaches this area, a safety system (e.g.,

flight termination system) must be invoked to interrupt the operation. As noted above, flight termination systems cannot be used to demonstrate compliance with the requirement SC-Light-UAS.2510. Thus, the present manuscript focuses on fault tolerance rather than safety.

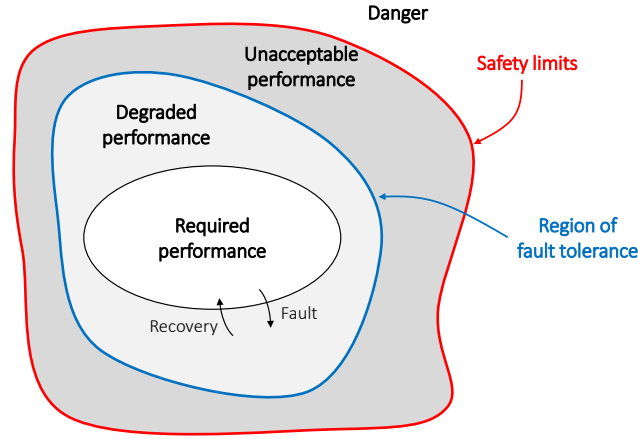


Figure 6.3.1 Safety versus fault tolerance. Adapted from [14].

Finally, *reliability* aims to evaluate the probability of remaining in the required or degraded performance domain. Although fault tolerance cannot change the reliability of a component, it improves the overall reliability of the system by ensuring that it remains operational after faults have occurred. The notion of reliability is of particular interest with regard to the SC-Light-UAS.2510 requirement, as it must be demonstrated that loss of control of the operation is not probable. Although the present thesis does not go into the details of reliability, its coverage in the context of multirotor UAV design is proposed in a separate work in collaboration with the High Reliability Aerospace Design Lab (HiRAD) of Concordia University [16]. In particular, the article [16] proposes a reliability analysis of different multirotor configurations. The reliability is calculated from the number of failures of the propulsion system determined as controllable, and the reliability of each individual component of the redundant propulsion system. It is shown that the octocopter has higher reliability than other configurations because it provides a good balance between fault tolerance and number of rotors (more rotors means potentially more faults). The coaxial hexacopter also shows promising reliability while maintaining a reduced weight compared to the octocopter design. For this configuration, the probability of failure of the UAV per flight hour is estimated to be 1.75×10^{-13} , using high-caliber equipment equivalent to transport aircraft category are used. If lower quality components are used (e.g. toy industry), the reliability drops to 1.68×10^{-4} failures per flight hour.

6.3.2 Existing research on fault-tolerant UAVs

Most research on fault tolerance for multirotor and fixed-wing vehicles focuses on the design of a controller under fault conditions to achieve fault tolerant control (FTC) [13, 17–29]. These FTC strategies rely on the fundamental property of reconfigurability. Indeed, the fault diagnosis that precedes the adjustment of the controller to the faulty situation requires a characterization of the fault against a known set of candidates [30, 31]. Without information about these candidate faults and how they affect the UAV, fault diagnosis cannot be achieved. Although reconfigurability is a fundamental prerequisite for FTC, few studies have considered its analysis for fixed-wing and multirotor aircraft.

Controllability of faulty aircraft

Reconfigurability can be assessed through the controllability and observability of the faulty aircraft. Observability depends mostly on the available measurements and the sensors and is not covered in the present manuscript. In the following, only the research work related to the controllability property will be examined.

Hassan and Taha [32] explore the controllability of a conventional airplane when it loses an elevator, aileron, rudder or engine. The authors apply the classical controllability theory to a linearized model of the aircraft. The classical controllability theory relies on the well-known Kalman rank condition [33]. Consider a linear time-invariant (LTI) system in the form

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \quad (6.1)$$

where $\mathbf{x} \in \mathbb{R}^n$ is the state vector, $\mathbf{u} \in \mathbb{R}^m$ is the control vector, $\mathbf{A} \in \mathbb{R}^{n \times n}$ is the state matrix and $\mathbf{B} \in \mathbb{R}^{n \times m}$ is the control matrix. A necessary and sufficient condition for the controllability of system 6.1 is that the controllability matrix defined by

$$\mathcal{C}(\mathbf{A}, \mathbf{B}) = [\mathbf{B} \quad \mathbf{A}\mathbf{B} \quad \mathbf{A}^2\mathbf{B} \quad \dots \quad \mathbf{A}^{n-1}\mathbf{B}] \quad (6.2)$$

is full rank, i.e. $\text{rank } \mathcal{C}(\mathbf{A}, \mathbf{B}) = n$. When assessing nonlinear systems, it is possible to linearize the equations and apply the Kalman rank condition to the controllability matrix of the linearized model. It should be noted that the controllability of the linearized model is only a sufficient condition for the controllability of the nonlinear system. That is, there exists a class of nonlinear systems that are linearly uncontrollable but nonlinearly controllable. To address this, Hassan and Taha further developed a method based on the Lie algebraic rank condition (LARC) for assessing the local accessibility² of nonlinear models of fixed-wing aircraft [35]. For application to multirotor UAVs, Tahavori and Hassan [36] proposed a

²For nonlinear systems, accessibility is a weaker form of controllability. A system has the accessibility property from a given point x_0 if the set of all attainable points from x_0 has a nonempty interior [34]. In addition, the system has the accessibility property if it has the accessibility property from every point in the n -dimensional space where $\mathbf{x}$ is defined. This typically implies that a trajectory may not be achieved with exactly the same reverse motion.

measure for the combined controllability and observability of nonlinear systems. It relies on the controllability and observability Gramians to calculate the smallest Hankel singular value, as suggested by Wu *et al.* [15]. This measure assesses how well the system remains controllable and observable in the presence of faults.

A major limitation of the Kalman rank condition, LARC, and Gramian approaches is their inability to handle constraints on the control vector $\mathbf{u}$. As shown in [32], this can lead to scenarios being incorrectly judged as controllable, e.g. by relying on negative thrust. This is an important issue for multirotor and fixed-wing designs where rotor thrust is unidirectional and the control surface deflection angles are bounded. In order to derive necessary and sufficient conditions for systems with control input constraints, the notion of *null controllability* has been introduced in the past [37–39]:

Definition 1 (Null controllability) *Consider a system constrained by $\mathbf{u} \in \Omega \subset \mathbb{R}^m$. The system is locally Ω -null controllable if there exists an open set $\mathcal{V} \subset \mathbb{R}^n$ containing the origin such that any $\mathbf{x}_0 \in \mathcal{V}$ can be controlled to $\mathbf{x}_1 = \mathbf{0}$ in finite time by some controller satisfying $\mathbf{u}(t) \in \Omega$ for all time $t \in [t_0, t_1]$. The system is globally Ω -null controllable if $\mathcal{V} = \mathbb{R}^n$.*

Lee and Markus [40] showed that the rank condition $\mathcal{C}(\mathbf{A}, \mathbf{B}) = n$ remains necessary and sufficient for the local Ω -null controllability of the linear system 6.1 only if $\mathbf{u} = \mathbf{0}$ is an interior point of the constraint set Ω . Consequently, because of the positive constraint applying to the thrust of multirotor UAVs ($\mathbf{u} \geq \mathbf{0}$), the rank condition is no longer sufficient. Necessary and sufficient conditions for the Ω -null controllability of linear systems with positive constraints were given by Brammer [38] and applied to multirotor UAVs around hover conditions by Du and Quan in [41]. Note that the notion of null controllability for multirotor UAVs physically translates into the ability to maintain hover flight. For this purpose, Du and Quan introduced an available control authority index (ACAI) to quantify the minimum control force that can be produced in all directions after compensating for the external disturbances (e.g., gravity). The ACAI calculation is divided into two main steps. The first step is to construct the attainable control set, which describes the reachable efforts generated by a system given its input limits. That is, it characterizes the UAV's capability to generate roll, pitch and yaw torque (L, M, N respectively) and thrust (T) with the available rotors. The attainable control set for a hexacopter is illustrated in Figure 6.3.2, where the four-dimensional polytope has been cut at $N = 0$ for better visualization.

The second step is to find the radius of the largest sphere that fits into the four-dimensional (T, L, M, N) polytope and is centered on the point of external disturbance. For multirotors in nominal hover conditions, the center is defined by the effect of gravity on the vehicle: $T = m_{uav}g$ and $L = M = N = 0$. The radius of the biggest enclosed sphere defines the ACAI. Physically, it measures the magnitude of the generable effort in the weakest direction. If the point of external disturbance is outside the attainable control set, then the ACAI is defined as negative. Finally, the ACAI adds to the rank condition of the controllability matrix

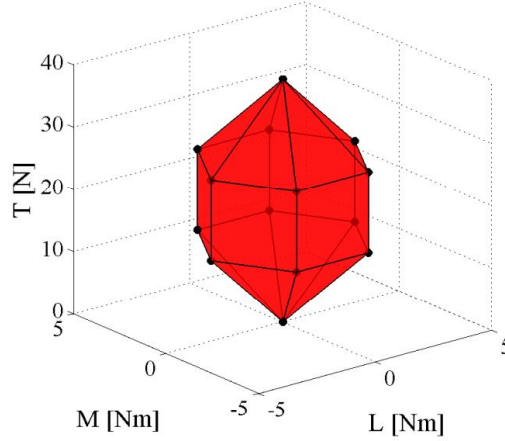


Figure 6.3.2 Attainable control set for a hexacopter in the absence of failures, cut at $N = 0$. Source: Schneider *et al.* [42].

to provide a necessary and sufficient condition for the linear controllability of multirotor UAVs in hovering conditions. The linear system 6.1 is locally Ω -null controllable if and only if the ACAI is strictly positive and the controllability matrix is full rank [41]. In addition, the larger the ACAI, the easier it is to control the UAV given the available rotors and external disturbances. By evaluating the ACAI for different faults, it is possible to identify the scenarios in which the multirotor UAV remains controllable and to select an appropriate fault-tolerant controller. The work of Du and Quan is very similar to that of Schneider *et al.* [42], which limits the quantification of control authority to a 2-dimensional measure (i.e., the radius of a circle) in the roll-pitch plane. Note, that the ACAI relates to the broader concept of degree of controllability (DoC). The DoC, although not having a unified definition at present, aims at quantifying the degree to which a system can be controlled [43, 44]. A time-dependent DoC, which enables the specification of a recovery time to return to a stable state (contrary to ACAI which is time-independant), is described in [45] and applied to multirotor UAVs in [46, 47]. However, the original publication [45] is written in Mandarin and the associated toolbox is no longer available. Research work conducted by Warren [48], published very recently before the present manuscript, successfully reimplemented the calculation procedure but is not yet open-source.

Finally, Lee *et al.* [49] proposed a method to analyze the local strong accessibility³ of input-constrained nonlinear multirotor systems. The method uses the sensitivity rank condition algorithm developed in [50]. However, it requires fully specifying the dynamics of the control inputs over time ($\mathbf{u}(t)$), which may not be known *a priori*.

³The local strong accessibility extends the concept of local accessibility by considering additional time constraints. For more details, see for instance [34].

Fault-tolerance and design optimization

All of the above methodologies apply to pre-defined UAV designs, where the configuration and characteristics of the propulsion system have already been determined. Recently, some authors have formulated optimization problems to identify a set of design parameters that optimizes a controllability metric in the presence of faults [47, 51, 52]. Kim *et al.* [51] find the optimal orientation for eight rotors of a VTOL UAV that maximizes a control authority measure similar to the ACAI (Figure 6.3.3). In particular, the optimized design maintains null controllability after two failure conditions which, for the initial design, were deemed uncontrollable. However, the thrust/torque characteristics of the rotors, such as maximum thrust, are considered constant.

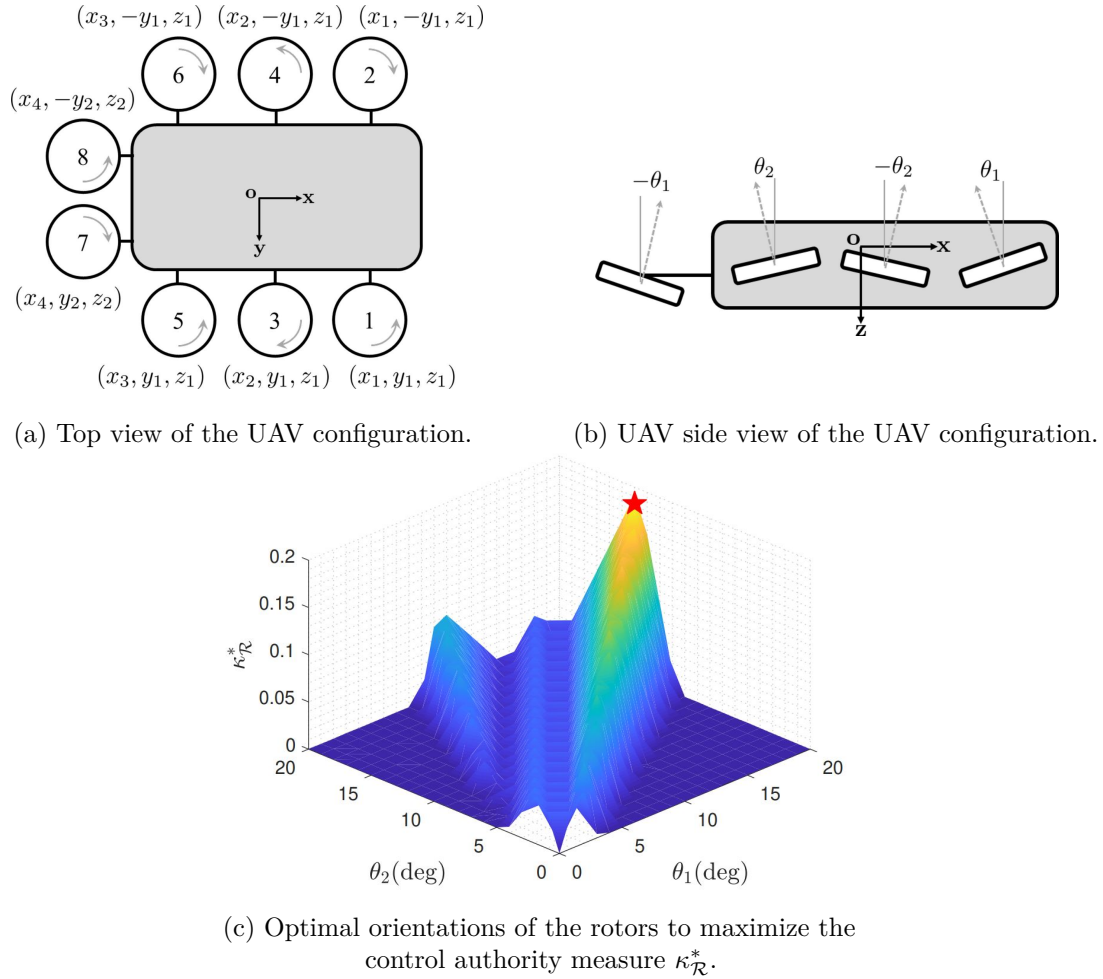
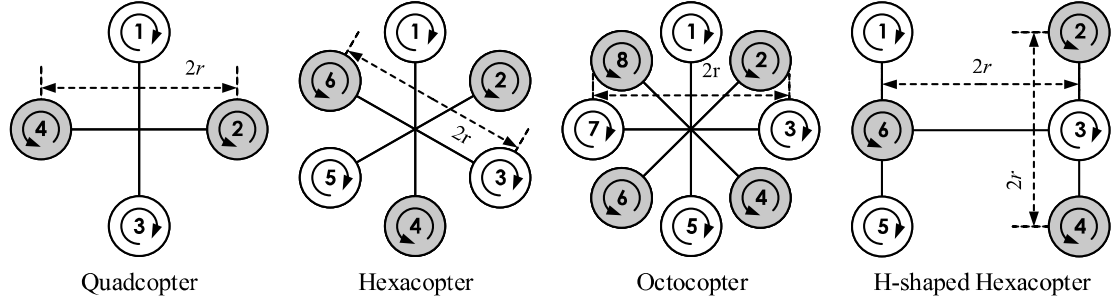


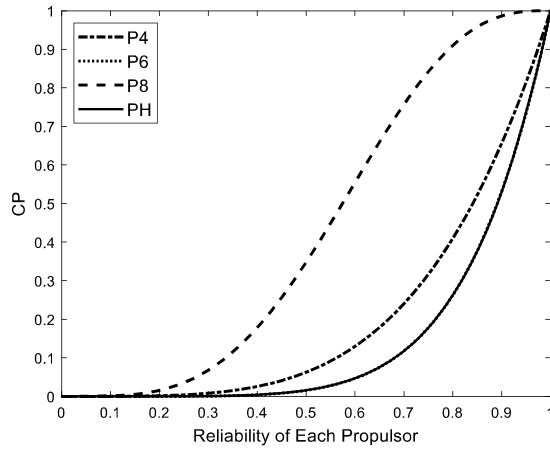
Figure 6.3.3 Case study from Kim *et al.* for control-centric design optimization. Source: [51].

Quan *et al.* [52] compared multirotor UAVs with various configurations to find the one with the highest controllable probability, that is, the overall reliability of the UAV given its fault tolerance and the failure rate of its components. The case study and the main results

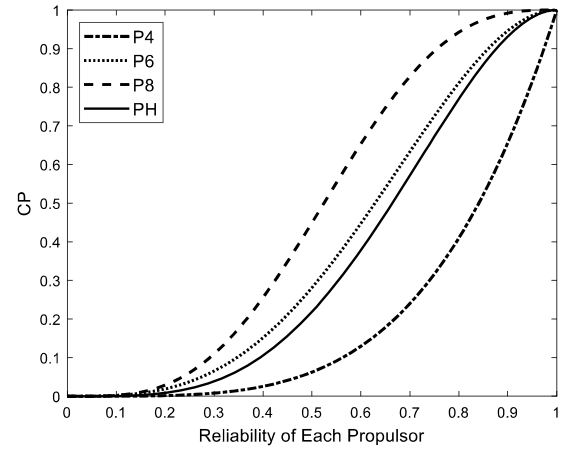
are shown in Figure 6.3.4. Regardless of the configuration, the physical parameters describing the rotors were fixed, so that no link to the sizing process was made.



(a) Multirotor configurations.



(b) Controllable probability of different configurations.



(c) Controllable probability when the yaw control is given up.

Figure 6.3.4 Case study from Quan *et al.* for comparing the controllable probability of different multirotor configurations, depending on the reliability of each rotor. Source: [52].

Notations: CP = controllable probability, P4 = quadcopter, P6 = hexacopter, P8 = octocopter, PH = H-shaped.

Finally, Du and Quan [47] optimized the propulsion system of a multirotor UAV based on the degree of controllability. The methodology relies on the exhaustive exploration and evaluation of a set of off-the-shelf components to select the candidate with the highest degree of controllability. The evaluation is carried out under nominal operating conditions and does not take failure cases into account.

6.4 Summary of the research gaps

The literature review revealed two major research gaps. First, there seems to be no methodology for assessing the controllability of fixed-wing and hybrid FW-VTOL UAVs in the presence of faults and under control input constraints. Particularly, the methodologies proposed for

multirotor UAVs cannot be applied to fixed-wing concepts without modification. Similarly to rotors, the control surfaces of fixed-wing and FW-VTOL UAVs, such as ailerons and elevators, are bounded by a maximum deflection angle. However, they differ in that the forces they generate are either unidirectional or bidirectional depending on the axis. Positive or negative aileron deflections always generate drag forces and yaw moments in the same direction, whereas pitch and roll moments will be either positive or negative. Ignoring this specificity can lead to non-physical conclusions, such as control surfaces generating thrust to ensure the controllability of the aircraft. In addition, unlike hovering multirotor UAVs, fixed-wing designs are subject to aerodynamic forces that may be introduced by, among other things, a control surface failure (e.g., jamming or run-away). As a result, the occurrence of a fault not only changes the attainable control set, but also the disturbances. Considering the additional aerodynamic forces, the control authority measure (e.g. ACAI) may be negative, meaning that the aircraft may be uncontrollable.

Secondly, the integration of fault-tolerance analysis with a sizing procedure for UAVs has not yet been achieved. Optimizations with controllability considerations are achieved either under nominal, fault-free operating conditions [47], or by considering constant properties for the rotors [51, 52]. Yet, in addition to the choice of UAV configuration, design optimization with controllability criteria calls for appropriate sizing of the control system (rotors and/or control surfaces).

Hence, the state-of-the-art lacks a consistent controllability analysis and fault-tolerant sizing for multirotor, fixed-wing and FW-VTOL UAVs. This is the purpose of the next Chapter 7.

Chapter 7

Controllability assessment and fault-tolerant sizing of UAVs

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7.1 Introduction

Chapter 6 has shown that the design of UAVs for many future applications must be driven by safety considerations in addition to the classical energy or mass considerations. In particular, the safety of UAVs depends mainly on their ability to maintain control objectives in the presence of faults. Therefore, the emergence of UAVs in a regulated context will depend on

the ability to investigate and compare different design concepts and optimize their sizing with respect to operational and safety requirements. Accordingly, the objective of the present chapter is to develop a new methodology combining a controllability assessment and a fault-tolerant sizing of UAVs to evaluate and explore various designs for safety-critical applications. The methodology is derived for the general case of hybrid FW-VTOL UAVs equipped with both rotors (with fixed thrust directions) and control surfaces (ailerons, elevators and rudders). Although control surfaces were not modeled in the sizing methodology of Chapter 3, their inclusion is essential for a meaningful controllability assessment. By treating the general case of hybrid FW-VTOL UAVs, the methodology can be easily adapted to any multirotor or fixed-wing UAV concept, as illustrated in the case studies of Section 7.7.

The aims of the present work are multiple. Firstly, a modification of the linearized state-space model of the UAV is proposed to reflect various faults in the effectors. Here, the term *effector* refers to both the control surfaces and the rotors. Faults range from a mild loss of effectiveness of an effector to the more critical run-away (or, lock-in-place) of a control surface, and are captured in the state and the control matrices. Secondly, a new control authority metric called the *extended available control authority index* (XACAI) is proposed to evaluate the linear controllability of the UAV around trimmed states. This index considers both the effectors' boundedness and the piecewise linear nature of efforts such as control surface drag. Thirdly, the control analysis is linked to the sizing of the effectors through a specific control reallocation problem. The proposed *sizing control allocation* allows for increasing the size of the effectors to maintain a desired control authority under fault conditions.

The chapter is organized as follows: Section 7.2 discusses the scope of the controllability assessment. A linear flight dynamics model and its adaptation in failure scenarios are introduced in Section 7.3. A new control authority index, the XACAI, is presented in Section 7.4. The control reallocation problem and its link with the effectors sizing is presented in Section 7.5. The overall methodology for controllability assessment and fault-tolerant sizing is described in Section 7.6. Finally, case studies covering multirotor, fixed-wing and hybrid FW-VTOL UAVs are proposed in Section 7.7. Section 7.8 concludes the chapter.

7.2 Scope of the controllability assessment

This section aims to discuss the controllability analysis of multirotor, fixed-wing and FW-VTOL UAVs and highlight the theoretical gap addressed by the introduction of the extended available control authority index (XACAI) in Section 7.4.

As pointed out in Chapter 6, the classical controllability theory relies on the controllability matrix rank check to assess linear time-invariant systems. While it serves as a necessary condition for linear controllability, it is inadequate for multirotor and fixed-wing designs. This

is due to its inability to account for the limitations imposed by the bounded and unidirectional nature of fixed-pitch propellers [53].

To overcome this limitation, Du *et al.* [41] expanded upon the conditions for null-controllability of linear systems with positive controllers given by Brammer [38]. They proposed the concept of ACAI [41], which quantifies the available control authority by measuring the remaining attainable control set, considering the bounded nature of rotor thrust and the effect of gravity on vehicle control. The controllability of the linear model is assessed by combining the controllability rank check and the ACAI, which serve as necessary and sufficient conditions [41]. This way, the ACAI effectively addresses the controllability challenges associated with multirotor designs subject to constrained control efforts. However, fixed-wing and FW-VTOL designs involve additional specificities in control efforts during operation and under specific failure conditions, such as control surface jamming and control surface drag, as outlined in Section 7.3 and 7.4 respectively. When a control surface jamming occurs, it can be considered as the introduction of a new disturbance while fixing the associated control variable. Furthermore, the surface drag results in symmetric efforts around a neutral position governed by even functions. Specifically, it is assumed that the surface drag is governed by an absolute function in the form $y = |x|$, as shown in Figure 7.2.1. This assumption is incompatible with the ACAI computation, which would incorrectly assume that the drag forces can become propulsive.

Therefore, a new approach is required to address these challenges. In this study, the XACAI is introduced to extend the ACAI calculations to specific efforts governed by the absolute function and to account for the additional disturbances introduced by the failures. The XACAI calculations are detailed in Section 7.4, while the controllability conditions and their applicability are summarized in Table 7.2.1.

This chapter only covers the null controllability (see Definition 1) of linear systems. Physically, null controllability demonstrates the ability to bring the UAV to hover or, in the case of fixed-wing concepts, to a complete stop. The development of alternative nonlinear approaches is worth exploring in future research, as the uncontrollability of the linear model does not necessarily imply uncontrollability in a nonlinear sense or in practical scenarios (see for instance [35]). In particular, this would allow to capture the quadratic evolution of the drag effort with the deflection angles of the control surfaces (Figure 7.2.1d).

7.3 Dynamics models

This section introduces the UAV dynamics state space model linearized and reformulated to represent rotor and control surface loss of effectiveness and lock-in-place failures. The dynamics model is illustrated with a FW-VTOL UAV concept.

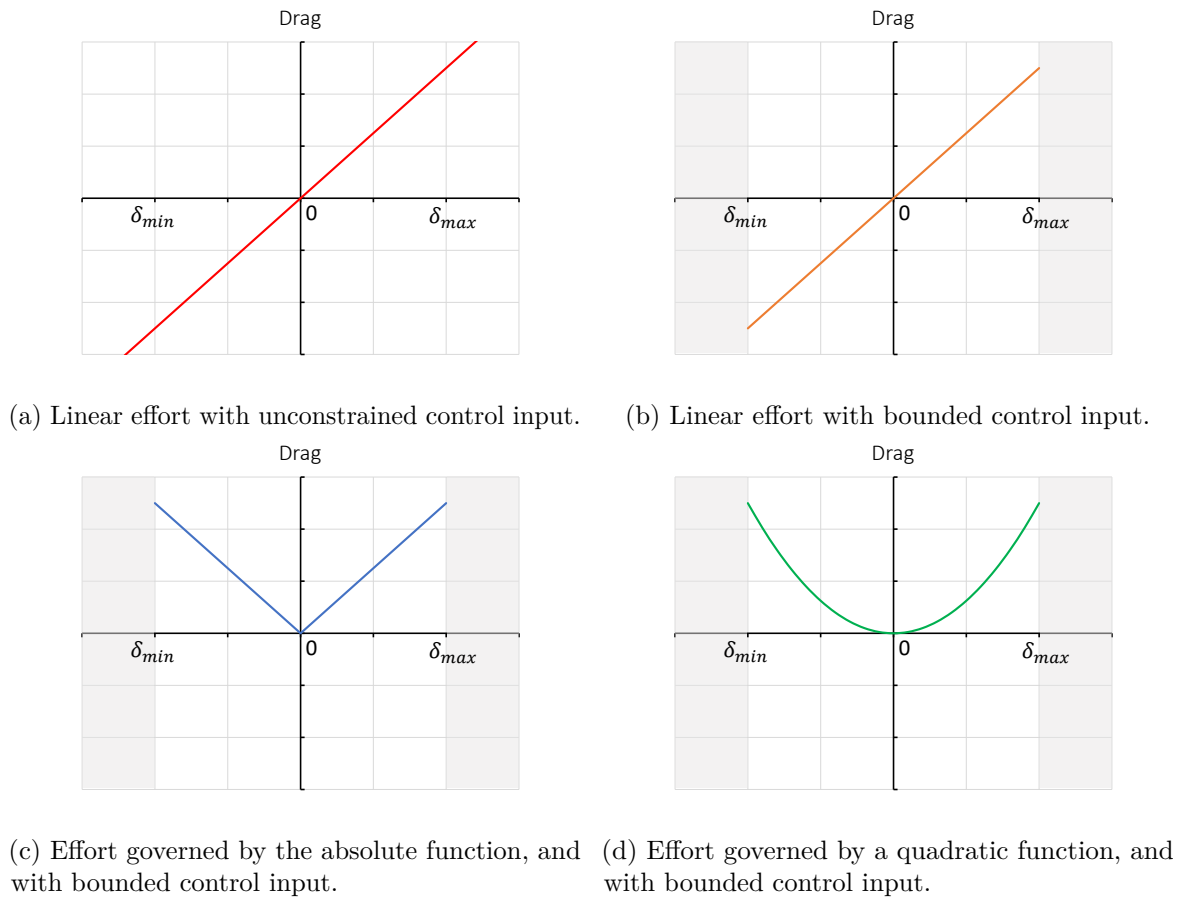


Figure 7.2.1 Assumptions about the drag generated by a control surface as a function of its angle of deflection δ . The efforts shown in (a) and (b) incorrectly assume that drag can become propulsive. The absolute function approximation shown in (c) is the main assumption for this chapter, although a quadratic function for the drag (d) would be even more representative.

Table 7.2.1 Controllability conditions of linear time-invariant models and their applicability.

Assumptions on control efforts	Necessary and sufficient conditions for linear controllability	Examples of applications
Linear efforts with unconstrained control inputs	Controllability matrix is full rank	• Multirotor UAVs with pitch-controlled propellers
Linear efforts with bounded control inputs	Controllability matrix is full rank and ACAI is strictly positive	• Multirotor UAVs with fixed propellers • Multirotor UAVs with pitch-controlled propellers
Efforts governed by the absolute function with bounded control inputs	Controllability matrix is full rank and XACAI is strictly positive	• Fixed-wing and FW-VTOL UAVs • Multirotor UAVs with fixed propellers • Multirotor UAVs with pitch-controlled propellers

7.3.1 Non-linear state-space model

The UAV model is obtained following a standard approach based on Newton's second law of motion. The rotation speed vector $\mathbf{\Omega} = [p \ q \ r]^T$ and the translation speed vector

$\mathbf{V} = [u_b \ v_b \ w_b]^T$ are expressed in the UAV body frame of reference. Deriving the dynamic equations in an inertial frame of reference provides the following dynamic model:

$$\begin{cases} m_{uav}(\dot{\mathbf{V}} + \boldsymbol{\Omega} \times \mathbf{V}) = \mathbf{F} \\ \mathbf{I}\dot{\boldsymbol{\Omega}} + \boldsymbol{\Omega} \times \mathbf{I}\boldsymbol{\Omega} = \mathbf{M} \end{cases} \quad (7.1)$$

where m_{uav} denotes the mass of the UAV and $\mathbf{I}$ denotes the inertia matrix. The external forces $\mathbf{F}$ [N] and external moments $\mathbf{M}$ [N.m] are due to the propulsive, aerodynamic and gravitational efforts acting on the UAV. The external efforts are either control or perturbation efforts. The propulsion system and the control surfaces generate the control efforts. The perturbation efforts come from the gravity effect and the aerodynamics of fixed surfaces (e.g., fuselage drag). The external forces and moments are divided accordingly:

$$\begin{bmatrix} \mathbf{F} \\ \mathbf{M} \end{bmatrix} = \begin{bmatrix} \mathbf{F}_u \\ \mathbf{M}_u \end{bmatrix} + \begin{bmatrix} \mathbf{F}_{\text{pert}} \\ \mathbf{M}_{\text{pert}} \end{bmatrix}, \quad (7.2)$$

where subscripts $\cdot_u$ and $\cdot_{\text{pert}}$ refer to the control efforts and perturbation efforts, respectively. The force and moment components of the control efforts are denoted as X, Y, Z and L, M, N , respectively, and are represented by the virtual control vector $\mathbf{u} \in \mathbb{R}^6$:

$$\mathbf{u} = [X \ Y \ Z \ L \ M \ N]^T \quad (7.3)$$

The virtual control vector components are functions of the control effector states and control effectiveness matrix $\mathbf{B}_f \in \mathbb{R}^{6 \times m}$, with m being the number of effectors, as follows:

$$\mathbf{u} = \mathbf{B}_f \boldsymbol{\delta}, \quad (7.4)$$

The control effectiveness matrix, which reflects the configuration and geometry of the UAV design (e.g., the position and size of the control surfaces), is further developed in Section 7.4. The control input vector $\boldsymbol{\delta} \in \mathbb{R}^m$ in Equation 7.4 contains the effectors' inputs such as the propeller thrust settings $\tau \in [0, 1]$ and the deflection angles [rad] of the elevators (η), ailerons (ρ), and rudders (ζ) [54]:

$$\boldsymbol{\delta} = [\rho_1, \dots, \rho_i, \eta_1, \dots, \eta_j, \zeta_1, \dots, \zeta_k, \tau_1, \dots, \tau_l]^T \quad (7.5)$$

The detailed scalar equations of motions, derived from the above equalities, can be found in 7.A. The motion variables and control notations introduced above are visually summarized in Figure 7.3.1, where $\boldsymbol{\Phi} = [\phi \ \theta \ \psi]^T$ represents the Euler angles (i.e., roll, pitch and yaw).

7.3.2 Linearized state-space model

The equations of motion are linearized using the small-disturbance theory applied to a reference flight condition. For example, the longitudinal speed becomes $u_b = u_{b_0} + \Delta u_b$,

given by [54]:

$$\mathbf{A} = \begin{bmatrix} \dot{X}_u & 0 & \dot{X}_w & 0 & 0 & 0 & 0 & -g \cos \theta_0 & 0 \\ 0 & \dot{Y}_v & 0 & \dot{Y}_p & 0 & \dot{Y}_r - u_{b_0} & g \cos \theta_0 & 0 & 0 \\ \dot{Z}_u & 0 & \dot{Z}_w & 0 & \dot{Z}_q + u_{b_0} & 0 & 0 & -g \sin \theta_0 & 0 \\ 0 & \dot{L}_v & 0 & \dot{L}_p & 0 & \dot{L}_r & 0 & 0 & 0 \\ \dot{M}_u & 0 & \dot{M}_w & 0 & \dot{M}_q & 0 & 0 & -g \dot{M}_w \sin \theta_0 & 0 \\ 0 & \dot{N}_v & 0 & \dot{N}_p & 0 & \dot{N}_r & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \quad (7.10)$$

The parameters $\dot{X}$, $\dot{Y}$, $\dot{Z}$, $\dot{L}$, $\dot{M}$ and $\dot{N}$ are aerodynamic stability derivatives [55]. Their expressions can be found in [54]. The detailed set of linearized scalar equations can be found in 7.A.

The deviation of the virtual control vector in Equation 7.7, $\Delta \mathbf{u}$, is further developed to include the contributions of the effectors:

$$\Delta \mathbf{u} = \mathbf{u} - \mathbf{u}_0 = \mathbf{B}_f \boldsymbol{\delta} - \mathbf{u}_0, \quad (7.11)$$

where $\mathbf{u}_0$ is obtained from the reference equilibrium condition and is representative of the perturbation efforts to be compensated. Finally, the state-space model is re-arranged as follows:

$$\Delta \dot{\mathbf{x}} = \mathbf{A} \Delta \mathbf{x} + \mathbf{B} \underbrace{(\mathbf{B}_f \boldsymbol{\delta} - \mathbf{u}_0)}_{\Delta \mathbf{u}} \quad (7.12)$$

7.3.3 Failure cases

This subsection introduces a new flight dynamics model to address failure cases. The considered effectors (control surfaces and propellers) have two distinct failure modes: loss of effectiveness and lock-in-place. Both cases are discussed below.

Loss of effectiveness

A loss of effectiveness is characterized by a reduction of the effectiveness of one or more effectors with respect to a nominal value. Practically, one or more effectors achieve a fraction of or no control command (i.e., complete failure). This failure mode is modeled by introducing a failure matrix, as described in [41]:

$$\mathbf{H} = \text{diag}(\eta_1, \dots, \eta_m), \quad (7.13)$$

where the parameter $\eta_i \in [0, 1]$ accounts for the loss of effectiveness of the i -th effector. A complete failure is characterized by $\eta_i = 0$. The control effectiveness matrix from Equation 7.4

becomes:

$$\mathbf{B}'_f = \mathbf{B}_f \mathbf{H}, \quad (7.14)$$

with superscript $'$ representing the failure case.

Lock-in-place

Following an actuator's failure, for example, through a run-away or jamming, a control surface may be stuck in a fixed position that is usually different from zero. When a lock-in-place failure occurs, the control authority over the jammed surface is lost. Additionally, unwanted aerodynamic efforts, which have to be balanced by the remaining effectors, are introduced. This leads to a revision of the state-space model.

The state-space model is linearized about the same initial conditions (e.g., steady-level flight) but with different reference inputs representing the new trimmed state. As for the loss of effectiveness, the contributions of the failed surfaces in the virtual control vector are set to zero (apply $\eta_i = 0$ for each jammed surface in Equation 7.13 to define $\mathbf{H}$). In addition, the efforts of the jammed surfaces are shifted to the perturbation efforts. In other words, the efforts of the jammed effectors are now part of the effort demand that must be satisfied by the remaining effectors. This translates to:

$$\mathbf{u}' = \mathbf{B}'_f \boldsymbol{\delta} \quad (7.15)$$

$$\mathbf{u}'_0 = \mathbf{u}_0 - \mathbf{B}_f (\mathbf{I}_m - \mathbf{H}) \boldsymbol{\delta}'_0, \quad (7.16)$$

where $\mathbf{I}_m \in \mathbb{R}^m$ is the identity matrix. The second term in Equation 7.16 represents the additional efforts to be compensated. The vector $\boldsymbol{\delta}'_0$ contains the control inputs derived from the reference equilibrium condition, which are revised to reflect the modified trimmed state. Its full determination is covered in Section 7.5. However, because of the form of Equation 7.16, only the positions of the jammed effectors are required to derive $\mathbf{u}'_0$. Finally, the state matrix is updated to reflect the new reference condition:

$$\mathbf{A}' = f(\mathbf{V}_0, \boldsymbol{\Omega}_0, \boldsymbol{\Phi}_0, \boldsymbol{\delta}'_0) = \mathbf{A} + \mathbf{A}_{\delta'_0}, \quad (7.17)$$

where $\mathbf{A}_{\delta'_0}$ represents the couplings introduced by the non-zero angles of the control surfaces in the trimmed state. For a steady, straight, and level reference flight, $\mathbf{A}_{\delta'_0}$ is expressed as follows:

$$\mathbf{A}_{\delta'_0} = \frac{2}{u_{b0}} \begin{bmatrix} \mathbf{B} \mathbf{B}_f \boldsymbol{\delta}'_0 & \mathbf{0}_{9 \times 8} \end{bmatrix} \quad (7.18)$$

The demonstration of Equation 7.18 is provided in 7.B.

7.4 Controllability assessment

This section synthesizes the calculation of the ACAI before proposing a new index encompassing multirotor, fixed-wing, and hybrid FW-VTOL designs.

7.4.1 Available Control Authority Index (ACAI)

The ACAI computation introduced by Du *et al.* [41] is based on a linear state-space model of multirotor drones in hovering conditions, such as:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}(\mathbf{u} - \mathbf{u}_0) \quad (7.19)$$

where $\mathbf{u} = [Z \ L \ M \ N]^T \in \mathbb{R}^4$ and $\mathbf{u}_0 = [m_{uav}g \ 0 \ 0 \ 0]^T \in \mathbb{R}^4$. Indeed, the rotors can only provide vertical thrust (Z axis) and 3-dimensional moments (L , M , N) to the vehicle. Apart from specific configurations where the rotors can be oriented horizontally (e.g., tilt rotor UAVs), multirotor effectors cannot act directly on the X and Y axes. The main component of the control authority is the previously defined virtual control vector $\mathbf{u}$, which results from the rotor thrust vector $\mathbf{f}$ as follows [41]:

$$\mathbf{u} = \mathbf{B}_f \mathbf{f}, \quad (7.20)$$

where $\mathbf{f} \in \mathbb{R}^m$ is the rotor thrust vector, and m is the number of rotors. The control effectiveness matrix $\mathbf{B}_f \in \mathbb{R}^{4 \times m}$ for multirotor UAVs is linked to the arrangement of the rotors:

$$\mathbf{B}_f = \begin{bmatrix} 1 & \cdots & 1 \\ -d_1 \sin \varphi_1 & \cdots & -d_m \sin \varphi_m \\ d_1 \cos \varphi_1 & \cdots & d_m \cos \varphi_m \\ k_\mu \omega_1 & \cdots & k_\mu \omega_m \end{bmatrix}, \quad (7.21)$$

where d_i is the distance of rotor i from the UAV center of mass, φ_i is its angular position in the $X - Y$ plane, $\omega_i \in \{-1, 1\}$ is its rotation direction (clockwise/counterclockwise), and k_μ is the ratio between the reactive torque and the thrust produced by the rotor.

Each rotor provides a purely positive and limited thrust, leading to the following constraint set for the rotor thrust vector [41]:

$$\mathcal{F} = \left\{ \mathbf{f} \mid \mathbf{f} = [f_1, \dots, f_m]^T, f_{max,i} \leq f_i \leq f_{max,i} \right\} \subset \mathbb{R}^m \quad (7.22)$$

where f_i is the thrust of i -th rotor, $f_{max,i}$ is the maximum thrust [N] of the i -th rotor, and $f_{min,i} = 0$. Equations 7.22 and 7.23 provide the virtual control vector constraint set [41]:

$$\Omega = \{ \mathbf{u} \mid \mathbf{u} = \mathbf{B}_f \mathbf{f}, \mathbf{f} \in \mathcal{F} \} \subset \mathbb{R}^4 \quad (7.23)$$

The ACAI quantifies the maximum effort achievable in all control directions (specifically, Z , L , M , and N), taking into account the impact of constrained rotor thrusts and the influence of a disturbance $\mathbf{u}_0$ (here representing gravity). More specifically, if $\mathbf{u}_0$ is contained within Ω , then the ACAI is the radius of the largest enclosed four-dimensional sphere centered at $\mathbf{u}_0$ in Ω . If $\mathbf{u}_0$ is not contained within Ω , then the ACAI is the radius of the largest enclosed sphere centered at $\mathbf{u}_0$ in the complementary set of Ω , denoted as Ω^c [41]:

$$\rho(\mathbf{u}_0, \partial\Omega) \triangleq \begin{cases} \min\{\|\mathbf{u}_0 - \mathbf{u}\| : \mathbf{u}_0 \in \Omega, \mathbf{u} \in \partial\Omega\} \\ -\min\{\|\mathbf{u}_0 - \mathbf{u}\| : \mathbf{u}_0 \in \Omega^c, \mathbf{u} \in \partial\Omega\} \end{cases} \quad (7.24)$$

where $\partial\Omega$ is the boundary of Ω .

As demonstrated in [56], $\partial\Omega$ can be described by a polytope of parallel hyper-plane segments. In an n -dimensional space, these hyper-plane segments have a dimension of $n - 1$. The ACAI is based on calculating the perpendicular distance between each parallel hyper-plane segment and projecting $\mathbf{u}_0$ on the axis of each of these perpendicular distances. Evaluating whether each projection of $\mathbf{u}_0$ lies between the hyper-plane segments or not determines if $\mathbf{u}_0$ is contained within Ω . In addition, the minimum among all distances between the projections of $\mathbf{u}_0$ and the hyper-plane segments provides the ACAI value.

The hyper-plane segments defining $\partial\Omega$ are obtained as follows. According to [56], each hyper-plane segment of $\partial\Omega$ is the image of an $n - 1$ dimensional boundary segment of the control hypercube $\mathcal{F}$, with n as the dimension of the virtual control vector (here, $n = 4$). We therefore define the following partitions of $\mathbf{f}$ and $\mathbf{B}_f$:

$$\mathbf{u} = \mathbf{B}_f \mathbf{f} = \begin{bmatrix} \mathbf{B}_{f1,k} & \mathbf{B}_{f2,k} \end{bmatrix} \begin{bmatrix} \mathbf{f}_{1,k} \\ \mathbf{f}_{2,k} \end{bmatrix} = \mathbf{B}_{f1,k} \mathbf{f}_{1,k} + \mathbf{B}_{f2,k} \mathbf{f}_{2,k}, \quad k = [1, \dots, p] \quad (7.25)$$

where $\mathbf{f}_{1,k} \in \mathbb{R}^{n-1}$, $\mathbf{f}_{2,k} \in \mathbb{R}^{m-(n-1)}$, and $\mathbf{B}_{f1,k}$, $\mathbf{B}_{f2,k}$ are the corresponding partitions of $\mathbf{B}_f$. The integer k indexes the sets of hyper-plane segments of $\partial\Omega$. $k = [1, \dots, p]$, where p is the number of combinations of $n - 1$ effectors out of m . The axis of the perpendicular distance between the k -th parallel hyper-plane segments is defined by a unit vector, $\boldsymbol{\xi}_k \in \mathbb{R}^n$, orthogonal to these segments. By definition, $\boldsymbol{\xi}_k$ belongs to the null space of the column space of $\mathbf{B}_{f1,k}$. Hence, $\boldsymbol{\xi}_k$ satisfies [41]:

$$\mathbf{B}_{f1,k}^T \boldsymbol{\xi}_k = 0 \text{ and } \|\boldsymbol{\xi}_k\| = 1 \quad (7.26)$$

Then, the k -th parallel hyper-plane segments can be characterized by fixing the values of $\mathbf{f}_{2,k}$ at their extrema and allowing the remaining $n - 1$ elements of $\mathbf{f}_{1,k}$ to vary between their limits [56]. It results that the perpendicular distance between the k -th set of parallel hyper-planes is [41]:

$$d = \left| \boldsymbol{\xi}_k^T \mathbf{B}_{f2,k} \right| |\Delta_k| \quad (7.27)$$

The vector Δ_k of the effector magnitudes is defined as:

$$\Delta_k = [\alpha_{k_1} \quad \cdots \quad \alpha_{k_{m-(n-1)}}] \quad (7.28)$$

where α_{k_i} represents the magnitude of the i -th effector of $\mathbf{f}_{2,k}$: $\alpha_{k_i} = f_{max,k_i} - f_{min,k_i}$. As noted in [41], the pairs of parallel hyper-planes are symmetric about the center $\mathbf{F}_c$ of Ω , such that the distance of each hyper-plane segment to $\mathbf{F}_c$ is equal to half the quantity provided in Equation 7.27. The center $\mathbf{F}_c$ is defined as follows¹:

$$\mathbf{F}_c = \mathbf{B}_f \left[\frac{f_{min,1} + f_{max,1}}{2} \quad \cdots \quad \frac{f_{min,m} + f_{max,m}}{2} \right]^T \quad (7.29)$$

Finally, to evaluate whether $\mathbf{u}_0$ resides between or outside the hyperplane segments, the distances are compared as follows:

$$d_k = \underbrace{\frac{1}{2} |\xi_k^T \mathbf{B}_{f2,k}| |\Delta_k|}_{\text{distance between } k\text{-th hyperplane-segment and } \mathbf{F}_c} - \underbrace{|\xi_k^T (\mathbf{F}_c - \mathbf{u}_0)|}_{\text{distance between } \mathbf{u}_0 \text{ and } \mathbf{F}_c} \quad (7.30)$$

The process can be visualized in Figure 7.4.1, considering $n = 2$ virtual control dimensions (L and M) and $m = 3$ rotors providing vertical thrust between $[0, f_{max}]$, for simplicity. A non-centered disturbance $\mathbf{u}_0$ is also considered for general understanding. The k -th pair of parallel segments (out of $p = 3$) in red is obtained by fixing $(f_1, f_3) = (f_{max}, 0)$ and $(f_1, f_3) = (0, f_{max})$, and varying f_2 between 0 and f_{max} .

The process is repeated for all sets $k = [1, \dots, p]$ of parallel hyper-plane segments. According to Equation 7.24, the ACAI is finally given by:

$$\rho(\mathbf{u}_0, \partial\Omega) = \min(d_1, \dots, d_p) \quad (7.31)$$

As demonstrated by Du *et al.* [41], a configuration is locally null-controllable in all directions of $\mathbf{u}$ (i.e., Z , L , M , and N) only if

$$\mathcal{C}(\mathbf{A}, \mathbf{B}) \text{ has full rank and } \rho(\mathbf{u}_0, \partial\Omega) > 0 \quad (7.32)$$

where $\mathcal{C}(\mathbf{A}, \mathbf{B})$ is the system's controllability matrix.

7.4.2 Hybrid FW-VTOL UAV Control Effectiveness Model

The definition of the control effectiveness matrix $\mathbf{B}_f$ is the starting point for the ACAI calculation. Its expression for multirotor configurations is provided in Equation 7.21. The present section aims to introduce the formulation of $\mathbf{B}_f$ for hybrid FW-VTOL UAVs with both control surfaces and rotors.

¹The original definition provided by Du *et al.* considers $f_{min} = 0$. The present definition aims for more general applicability.

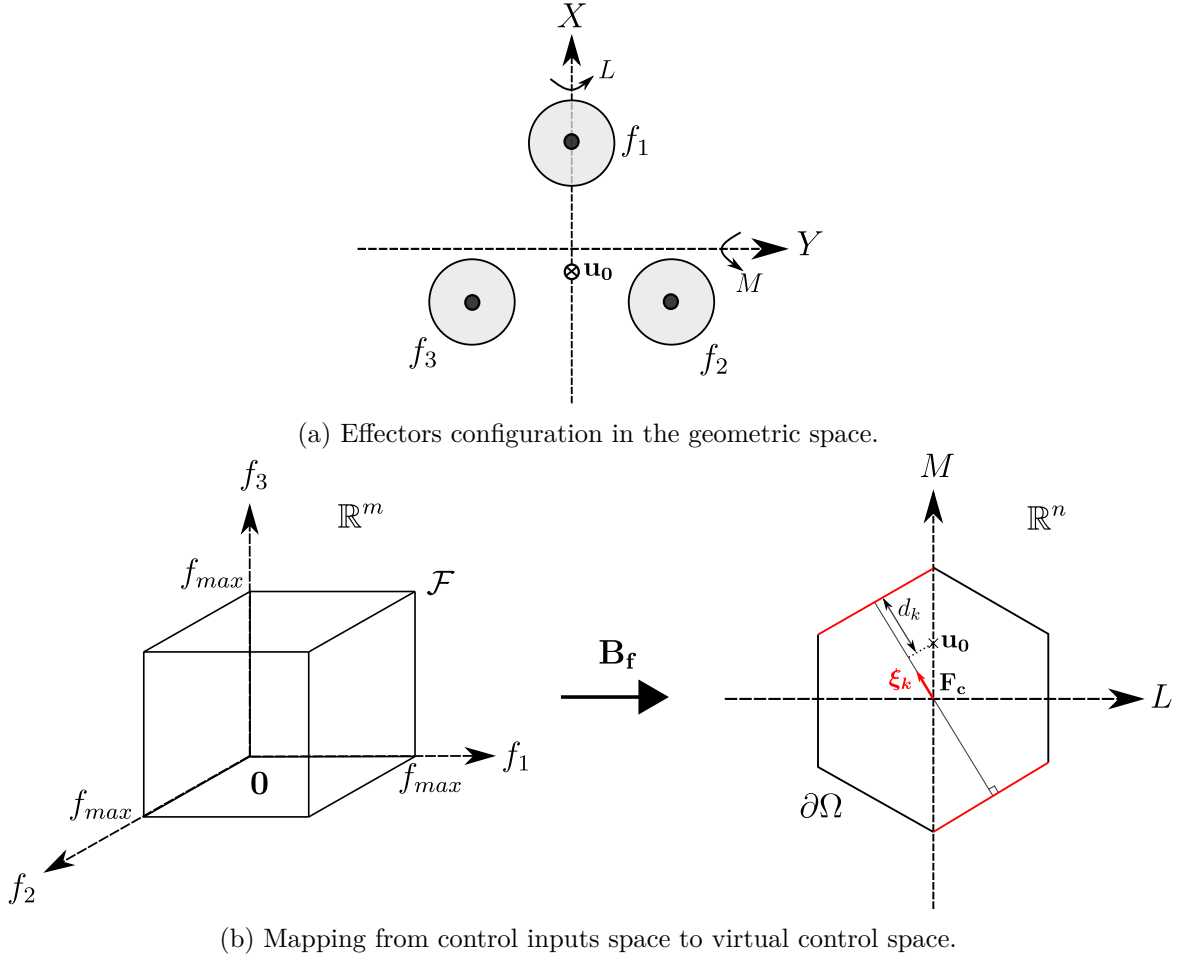


Figure 7.4.1 Visualization of the ACAI calculation process with a focus on the k -th pair of parallel segments. Here, $m = 3$ and $n = 2$ are considered for simplicity.

In the case of hybrid FW-VTOL UAVs, a level flight under steady conditions is assumed. This implies negligible angles of attack, bank angle, side-slip angle, and a constant air velocity along the flight direction. Control along the y -axis of the UAV is primarily achieved through fixed surfaces such as the wing and horizontal stabilizer. Moreover, there exists a coupling between the y -axis and the other axes, enabling control of the UAV's vertical motion by acting on these other axes. Therefore, a four-dimensional virtual control vector is considered, denoted as $\mathbf{u} = [X \ L \ M \ N]^T$. The effector control vector, represented as $\boldsymbol{\delta} = [\rho_1 \ \rho_2 \ \eta_1 \ \eta_2 \ \zeta \ \tau_p \ \tau_1 \ \tau_2 \ \tau_3 \ \tau_4]^T$, corresponds to the configuration illustrated in Figure 7.3.1. Note that for most fixed-wing UAVs, the number of elevators is reduced to one, since both sides are mechanically connected to a single actuator. However, for genericity, the case of two independent elevators is covered here. For consistency with the notations introduced in Section 7.3, the effector control vector $\boldsymbol{\delta}$ replaces the rotor thrust vector notation $\mathbf{f}$ used in the ACAI computation presented in Section 7.4.1.

Drag forces are essential aerodynamic components that introduce cross-coupling effects between control axes in the control surfaces. The drag forces must therefore be included in the assessment of control authority, specifically in the calculation of the control effectiveness matrix. These forces exhibit symmetric behavior governed by even functions, which can be approximated as absolute functions. As a result, the control effectiveness equations become highly nonlinear:

$$\begin{aligned} X &= \dot{X}_{\rho_1}|\rho_1| + \dot{X}_{\rho_2}|\rho_2| + \dot{X}_{\eta_1}|\eta_1| + \dot{X}_{\eta_2}|\eta_2| + \dot{X}_{\zeta}|\zeta| + \dot{X}_{\tau_p}\tau_p \\ L &= \dot{L}_{\rho_1}\rho_1 + \dot{L}_{\rho_2}\rho_2 + \dot{L}_{\eta_1}\eta_1 + \dot{L}_{\eta_2}\eta_2 + \dot{L}_{\zeta}\zeta + \dot{L}_{\tau_p}\tau_p + \dot{L}_{\tau_1}\tau_1 + \dot{L}_{\tau_2}\tau_2 + \dot{L}_{\tau_3}\tau_3 + \dot{L}_{\tau_4}\tau_4 \\ M &= \dot{M}_{\rho_1}\rho_1 + \dot{M}_{\rho_2}\rho_2 + \dot{M}_{\eta_1}\eta_1 + \dot{M}_{\eta_2}\eta_2 + \dot{M}_{\zeta}|\zeta| + \dot{M}_{\tau_1}\tau_1 + \dot{M}_{\tau_2}\tau_2 + \dot{M}_{\tau_3}\tau_3 + \dot{M}_{\tau_4}\tau_4 \\ N &= \dot{N}_{\rho_1}|\rho_1| + \dot{N}_{\rho_2}|\rho_2| + \dot{N}_{\eta_1}|\eta_1| + \dot{N}_{\eta_2}|\eta_2| + \dot{N}_{\zeta}\zeta + \dot{N}_{\tau_1}\tau_1 + \dot{N}_{\tau_2}\tau_2 + \dot{N}_{\tau_3}\tau_3 + \dot{N}_{\tau_4}\tau_4 \end{aligned} \quad (7.33)$$

where $\dot{X}_i, \dot{L}_i, \dot{M}_i, \dot{N}_i$ [N] are the control derivatives of the i -th effector on each axis. The control derivatives for the i -th propeller result from its maximum thrust f_{max} and positions x_{τ_i}, y_{τ_i} along the x-axis and y-axis, respectively:

$$\dot{L}_{\tau_i} = C_T \rho_{air} n_{max}^2 D^4 y_{\tau_i} = f_{max} y_{\tau_i} \quad (7.34)$$

$$\dot{M}_{\tau_i} = C_T \rho_{air} n_{max}^2 D^4 x_{\tau_i} = f_{max} x_{\tau_i} \quad (7.35)$$

$$\dot{N}_{\tau_i} = \omega_{\tau_i} C_Q \rho_{air} n_{max}^2 D^5 \quad (7.36)$$

where ρ_{air} [kg/m³] is the air density, n_{max} is the propeller speed for maximum thrust, D is the propeller diameter, and C_T [-] and C_Q [-] are the thrust and torque coefficients, respectively. The sense of rotation of the propeller is represented by ω_{τ_p} , which is equal to 1 for a clockwise rotation and -1 for a counterclockwise rotation. The control derivatives of the control surfaces reflect the aerodynamic effects due to their deflections. For example, on the x-axis:

$$\dot{X}_i = \frac{1}{2} \rho_{air} v_{air}^2 S_i C_{x,i} \quad (7.37)$$

where v_{air} [m/s] is the relative air velocity, S_i [m²] the i -th control surface area, and $C_{x,i}$ [-] is the aerodynamic coefficient specific to the i -th control surface geometry, profile, kinematics, and x-axis.

The presence of non-linearities, particularly the absolute functions in Equation 7.33, prevents the calculation of ACAI as presented in Section 7.4.1. To address this limitation, the following section introduces a new index to handle these non-linearities.

7.4.3 Extended Available Control Authority Index (XACAI)

The proposed XACAI extends the computation of ACAI to accommodate nonlinear control effectiveness relationships, including the one from Equation 7.33.

Step 1: Calculate the unit vectors ξ_k .

To begin, the control effectiveness equations are linearized to calculate the unit vectors, ξ_k , which are orthogonal to the hyper-plane segments forming the boundary of the attainable control set. This enables to define a control effectiveness matrix of the linearized efforts:

$$\mathbf{B}_{f,lin} = \begin{bmatrix} \dot{X}_{\rho_1} & \dot{X}_{\rho_2} & \dot{X}_{\eta_1} & \dot{X}_{\eta_2} & \dot{X}_{\zeta} & \dot{X}_{\tau_p} & 0 & 0 & 0 & 0 \\ \dot{L}_{\rho_1} & \dot{L}_{\rho_2} & \dot{L}_{\eta_1} & \dot{L}_{\eta_2} & \dot{L}_{\zeta} & \dot{L}_{\tau_p} & \dot{L}_{\tau_1} & \dot{L}_{\tau_2} & \dot{L}_{\tau_3} & \dot{L}_{\tau_4} \\ \dot{M}_{\rho_1} & \dot{M}_{\rho_2} & \dot{M}_{\eta_1} & \dot{M}_{\eta_2} & \dot{M}_{\zeta} & 0 & \dot{M}_{\tau_1} & \dot{M}_{\tau_2} & \dot{M}_{\tau_3} & \dot{M}_{\tau_4} \\ \dot{N}_{\rho_1} & \dot{N}_{\rho_2} & \dot{N}_{\eta_1} & \dot{N}_{\eta_2} & \dot{N}_{\zeta} & 0 & \dot{N}_{\tau_1} & \dot{N}_{\tau_2} & \dot{N}_{\tau_3} & \dot{N}_{\tau_4} \end{bmatrix} \quad (7.38)$$

We define each hyper-plane segment matrix, $\mathbf{B}_{f1,lin,k} \in \mathbb{R}^{n \times (n-1)}$, by isolating the column vectors of the effectors of the k-th hyper-plane segments in $\mathbf{B}_{f,lin}$, and ξ_k is obtained by applying $\mathbf{B}_{f1,lin,k}$ to Equation 7.26:

$$\mathbf{B}_{f1,lin,k}^T \xi_k = 0 \text{ and } \|\xi_k\| = 1 \quad (7.39)$$

Step 2: Partition the control effectiveness matrix between linear and absolute control inputs.

ACAI calculation based on a control effectiveness matrix applying to the linearized efforts, as defined in Equation 7.38, would incorrectly assume that the drag forces can become propulsive and lead to an incorrect controllability assessment. Therefore, the control effectiveness matrix $\mathbf{B}_f$ is divided into matrices relative to the linear and absolute control input vectors:

$$\mathbf{u} = \begin{bmatrix} \mathbf{B}_{f,l} & \mathbf{B}_{f,nl} \end{bmatrix} \begin{bmatrix} \delta \\ |\delta| \end{bmatrix} = \mathbf{B}_{f,l}\delta + \mathbf{B}_{f,nl}|\delta| \quad (7.40)$$

With $\mathbf{B}_{f,l}$, the linear-term matrix, expressed as follows:

$$\mathbf{B}_{f,l} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \dot{X}_{\tau_p} & 0 & 0 & 0 & 0 \\ \dot{L}_{\rho_1} & \dot{L}_{\rho_2} & \dot{L}_{\eta_1} & \dot{L}_{\eta_2} & \dot{L}_{\zeta} & \dot{L}_{\tau_p} & \dot{L}_{\tau_1} & \dot{L}_{\tau_2} & \dot{L}_{\tau_3} & \dot{L}_{\tau_4} \\ \dot{M}_{\rho_1} & \dot{M}_{\rho_2} & \dot{M}_{\eta_1} & \dot{M}_{\eta_2} & 0 & 0 & \dot{M}_{\tau_1} & \dot{M}_{\tau_2} & \dot{M}_{\tau_3} & \dot{M}_{\tau_4} \\ 0 & 0 & 0 & 0 & \dot{N}_{\zeta} & 0 & \dot{N}_{\tau_1} & \dot{N}_{\tau_2} & \dot{N}_{\tau_3} & \dot{N}_{\tau_4} \end{bmatrix} \quad (7.41)$$

And $\mathbf{B}_{f,nl}$ the nonlinear term matrix:

$$\mathbf{B}_{f,nl} = \begin{bmatrix} \dot{X}_{\rho_1} & \dot{X}_{\rho_2} & \dot{X}_{\eta_1} & \dot{X}_{\eta_2} & \dot{X}_{\zeta} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \dot{M}_{\zeta} & 0 & 0 & 0 & 0 & 0 \\ \dot{N}_{\rho_1} & \dot{N}_{\rho_2} & \dot{N}_{\eta_1} & \dot{N}_{\eta_2} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (7.42)$$

The subsequent partitions $[\mathbf{B}_{f1,l,k} \quad \mathbf{B}_{f2,l,k}]$ and $[\mathbf{B}_{f1,nl,k} \quad \mathbf{B}_{f2,nl,k}]$ are obtained by isolating the column vectors of the set of effectors of the k-th hyper-plane segments in $\mathbf{B}_{f,l}$ and $\mathbf{B}_{f,nl}$, respectively.

Step 3: Compute the distances d_k from the linear and absolute terms.

The vector $\Delta_{k,l}$ of the effector magnitude vector applied to $\mathbf{B}_{f2,l,k}$ is provided by Equation 7.28.

The effector magnitude vector relative to $\mathbf{B}_{f2,nl,k}$, $\Delta_{k,nl}$, is obtained by applying the absolute function to δ_i :

$$\Delta_{k,nl} = [\alpha_{k_1} \quad \cdots \quad \alpha_{k_{m-(n-1)}}] \quad (7.43)$$

$$\alpha_{k_i} = \max(|\delta_{max}|, |\delta_{min}|) \quad (7.44)$$

The minimum distance between the hyper-plane segment and $\mathbf{u}_0$ is then expressed as follows:

$$d_k = \frac{1}{2} \left(\left| \xi_k^T \mathbf{B}_{f2,l,k} \right| |\Delta_{k,l}| + \left| \xi_k^T \mathbf{B}_{f2,nl,k} \right| |\Delta_{k,nl}| \right) - \left| \xi_k^T (\mathbf{F}_c - \mathbf{u}_0) \right| \quad (7.45)$$

Finally, equations 7.31 and 7.32 are applied to assess the controllability of the UAV.

The controllability in failure conditions is assessed by applying the failure matrix $\mathbf{H}$ to the control effectiveness matrices defined above, and replacing $\mathbf{u}_0$ by $\mathbf{u}'_0$ as per Equation 7.16. In addition, the rank check of the controllability matrix is verified with the updated state matrix $\mathbf{A}'$ from Equation 7.17.

7.5 Control allocation and sizing factors

Failure case sizing aims to size the control effectors such that the UAV will have sufficient control authority following any failure assessed as controllable. To this end, this section introduces the control allocation problem and a method to derive failure case sizing factors linking the control analysis to the sizing process.

7.5.1 Control allocation problem formulation

Consider the system defined in Equation 7.12. Control allocation aims to determine a control input vector δ to achieve a desired virtual control effort $\mathbf{u}$:

$$\mathbf{B}_f \delta = \mathbf{u} \quad (7.46)$$

For redundant systems, Equation 7.46 is generally underdetermined, and an infinite number of vectors δ can be the solution, assuming the control allocation is feasible. A common way to deal with the non-uniqueness of solutions is to introduce a secondary objective, namely to minimize the energy cost of control allocation:

$$\begin{aligned} \min_{\delta} \quad & J = \|\delta\| \\ \text{s.t.} \quad & \mathbf{B}_f \delta = \mathbf{u}, \end{aligned} \quad (7.47)$$

where $||\cdot||$ is the l^2 -norm. This formulation admits an explicit solution given by the Moore-Penrose pseudo-inverse [57]:

$$\boldsymbol{\delta} = \mathbf{B}_f^+ \mathbf{u}, \text{ with } \mathbf{B}_f^+ = \mathbf{B}_f^T (\mathbf{B}_f \mathbf{B}_f^T)^{-1} \quad (7.48)$$

However, the above solution does not account for constraints on the effectors' inputs. Consequently, the solution might be physically unfeasible. In addition, the pseudo-inverse method does not handle non-linearities in the control effectivenesses, as introduced in Section 7.4.2. Among the alternative allocation methods, the optimization-based approaches allow for more complex problems to be solved, including constraints and non-linearities [57]. We propose to address the control allocation problem using the mixed optimization formulation, which consists of combining the allocation objective and energy objective as follows [58]:

$$\begin{aligned} \min_{\boldsymbol{\delta}} \quad & J = ||\mathbf{B}_f \boldsymbol{\delta} - \mathbf{u}||^2 + \lambda ||\boldsymbol{\delta}||^2 \\ \text{s.t.} \quad & \boldsymbol{\delta}_{min} \leq \boldsymbol{\delta} \leq \boldsymbol{\delta}_{max}, \end{aligned} \quad (7.49)$$

where $\lambda \in [0, +\infty]$ represents the priority of the allocation objective ($\lambda = 0$) over the energy objective ($\lambda = +\infty$) and is usually chosen close to zero.

Finally, the problem formulation is combined with Equation 7.39 to differentiate the linear efforts from those governed by the absolute function:

$$\begin{aligned} \min_{\boldsymbol{\delta}} \quad & J = ||\mathbf{B}_{f,l} \boldsymbol{\delta} + \mathbf{B}_{f,nl} |\boldsymbol{\delta}| - \mathbf{u}||^2 + \lambda ||\boldsymbol{\delta}||^2 \\ \text{s.t.} \quad & \boldsymbol{\delta}_{min} \leq \boldsymbol{\delta} \leq \boldsymbol{\delta}_{max}, \end{aligned} \quad (7.50)$$

The control reallocation problem consists of solving Equation 7.50 in the presence of faults, that is by applying the failure matrix $\mathbf{H}$ to the control effectiveness matrices $\mathbf{B}_{f,l}$ and $\mathbf{B}_{f,nl}$, and fixing the failed effectors.

7.5.2 Proposed sizing control allocation

The control allocation problem introduced above aims at finding the optimal control inputs to achieve a virtual effort at the UAV level, given a pre-defined design of the control effectors. However, there is no guarantee that an exact solution to the allocation objective ($\mathbf{B}_f \boldsymbol{\delta} = \mathbf{u}$) exists. Indeed, the set of attainable virtual efforts $\mathbf{u}$ is limited by the effectors' design, reflected by the control effectiveness matrix and the saturation constraints. In particular, the unfeasibility of control allocation in the presence of effector failures is a direct consequence of poorly sized controls. To address this issue, we propose a new formulation of the problem described in 7.49 that enables a fault-tolerant sizing of the effectors while solving the control reallocation problem.

Consider a predetermined sizing of the control effectors obtained by considering normal operations only. The initial sizing of the control surfaces, such as the rudder area and aileron position, is not within the scope of this thesis and is typically derived from effort

specifications (e.g., rudder hinge moment) or empirical data. The proposed methodology introduces sizing factors, which are multiplier factors applied to the specifications of the effectors, to incorporate failure scenarios within a resizing process. Specifically, for a given failure case, a sizing factor represents an increase in the effectiveness of an effector relative to its initial sizing:

$$\widetilde{\mathbf{B}}_{\mathbf{f}}' = \mathbf{B}_{\mathbf{f}}' \mathbf{K} = \mathbf{B}_{\mathbf{f}} \mathbf{H} \mathbf{K} \quad (7.51)$$

$$\mathbf{K} = \text{diag}(k_1, \dots, k_m) \in [1, +\infty]^m \quad (7.52)$$

where $\mathbf{B}_{\mathbf{f}}$ is the control effectiveness matrix obtained from initial sizing, $\widetilde{\mathbf{B}}_{\mathbf{f}}'$ is the revised control effectiveness matrix, $\mathbf{H}$ is the failure matrix representative of the failure case, k_i is the sizing factor applied to the i -th effector, and $\mathbf{K}$ is the sizing factor vector. The effectiveness of an effector is directly related to its sizing, as shown in Equation 7.37. For instance, doubling the surface area of an aileron leads to a doubling of its effectiveness ($k_i = 2$). Similarly, doubling the maximum thrust of a rotor results in a twofold increase in its effectiveness. The sizing factors must ensure that the allocation objective is achievable and that the resized system preserves a certain level of control authority. The preservation of control authority can be expressed by the following inequality:

$$\rho(\mathbf{u}'_0, \partial\widetilde{\Omega}') \geq \alpha \rho(\mathbf{u}_0, \partial\Omega), \quad (7.53)$$

where $\rho(\mathbf{u}'_0, \partial\widetilde{\Omega}')$ represents the control authority index, specifically the XACAI, of the fault-tolerant sizing in the presence of failures. The required control authority can be alleviated in cases of failure based on the probability of occurrence. In this context, the parameter $\alpha \in [0, 1]$ quantifies the acceptable reduction in control authority under failure compared to the initial sizing during normal operating conditions.

The sizing factor vector and the control authority requirement are added to the control allocation problem expressed by Equation 7.49, which leads to the following formulation:

$$\begin{aligned} \min_{\delta, \mathbf{K}} \quad & J = \underbrace{\|\mathbf{B}_{\mathbf{f}} \mathbf{H} \mathbf{K} \delta - \mathbf{u}\|^2}_{\text{allocation objective}} + \underbrace{\lambda \|\mathbf{K} \delta\|^2}_{\text{energy objective}} + \underbrace{\epsilon \|\mathbf{K} - \mathbf{I}_m\|^2}_{\text{sizing objective}} \\ \text{s.t.} \quad & \delta_{min} \leq \delta \leq \delta_{max}, \\ & \mathbf{K} \in [1, +\infty]^m, \\ & \rho(\mathbf{u}'_0, \partial\widetilde{\Omega}') \geq \alpha \rho(\mathbf{u}_0, \partial\Omega), \end{aligned} \quad (7.54)$$

where the last term in the function J represents the sizing minimization objective to minimize deviations from the initial sizing ($\mathbf{K} = \mathbf{I}_m$). Additional constraints can be imposed on the sizing factor vector, for example, to disable resizing of a specific effector or to restrict its oversizing. The parameters λ and ϵ are chosen to prioritize either allocation, energy, or sizing objectives. In practice, $\lambda \ll \epsilon \ll 1$. Finally, the allocation objective is to maintain the

equilibrium condition, that is $\mathbf{u} = \mathbf{u}'_0$.

The nonlinear least-square problem described above can be solved using various optimization methods. This study utilizes a trust region algorithm [59] implemented in the Python library SciPy [60]. To ensure the robustness of the results, a multi-start strategy was adopted.

Finally, resolving this problem for all failure cases provides not only the trimmed positions of the effectors but also the maximum sizing factors vector $\mathbf{K}_{\max} \in [1, +\infty]^m$

$$\mathbf{K}_{\max} = [\max(k_{1,1}, \dots, k_{1,p}), \dots, \max(k_{m,1}, \dots, k_{m,p})] \quad (7.55)$$

$$= [k_{1,max}, \dots, k_{m,max}], \quad (7.56)$$

where $k_{i,j}$ is the optimal sizing factor for effector $i \in [1, m]$ and failure case $j \in [1, p]$, obtained by solving Equation 7.54. $k_{i,max}$ is the maximum sizing factor for control effector i , considering all possible failures. Applying K_{max} to the initial effectors' sizing leads to a new design that is robust to failures.

7.6 Methodological framework

Figure 7.6.1 represents the overall methodology for a systematic controllability assessment and fault-tolerant sizing of UAVs under effector failures, using the extended design structure matrix (XDSM) notation [61]. Generic processes are depicted as green boxes, while parallelograms represent data inputs and outputs. Data connections are represented by vertical lines for inputs and horizontal lines for outputs. Thin black arrows indicate process connections. Stacked boxes indicate multiple executions corresponding to each failure case. The diagram shows a six-step procedure: concept definition, sizing optimization, XACAI calculation, control allocation, model revision for failure cases, and rank check of the controllability matrix. The chapter's scope is limited to the XACAI calculation, control allocation, model revision, and rank check.

The methodology starts by defining a concept of operation, which describes the characteristics of the unmanned aerial system from a user's perspective [62]. From the concept of operation, one or several candidate configurations of UAVs are proposed, along with design specifications and failure case definitions. Specifications include performance requirements and technology assumptions to assist in the design process. Also included in the specifications is a parametric state-space model representative of the UAV configuration. From the specifications, the UAV is optimally sized with the methodology presented in Chapter 3. By effectively sizing the UAV, this process provides actual values representative of the design and flight conditions for the state-space model. In parallel, the controllability assessment for failure cases is initiated. The XACAI presented in Section 7.4.3 is calculated for each failure case with non-restrictive effectiveness values for the effectors. This enables the identification of potentially controllable failure cases before the initiation of a fault-tolerant sizing. The

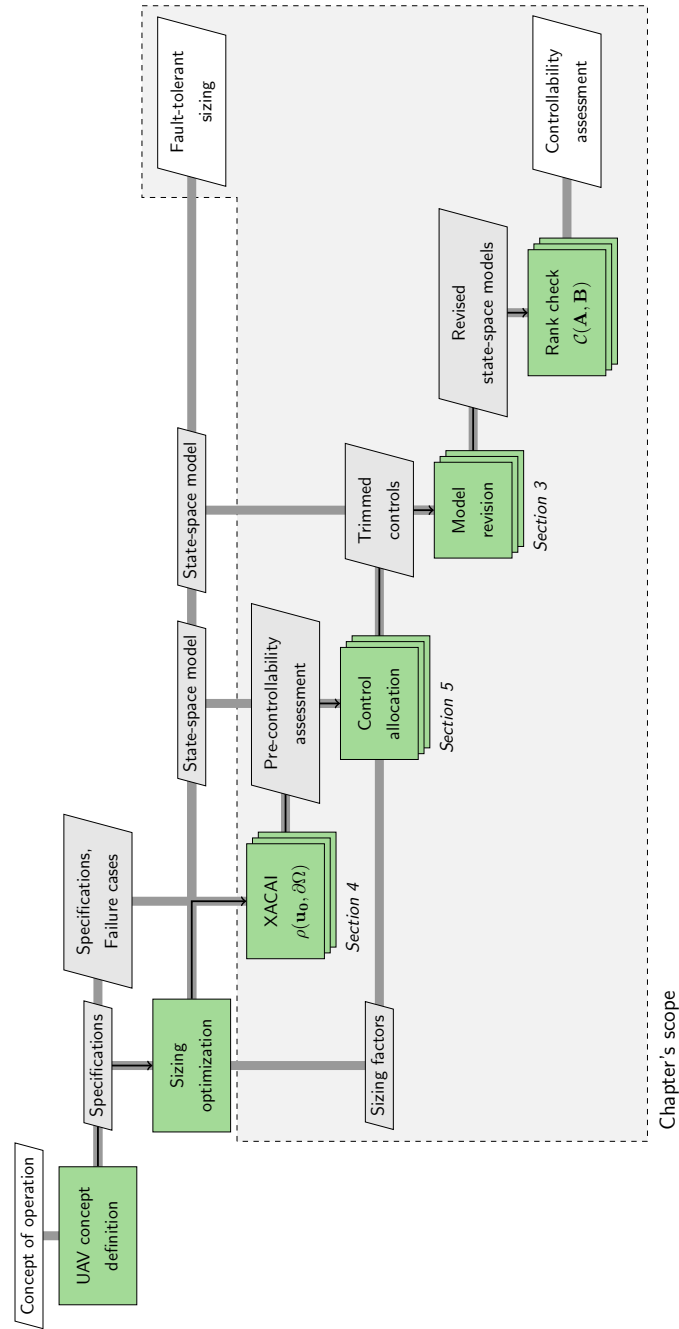


Figure 7.6.1 Diagram of the proposed methodology using the XDSM notation.

actual sizing of the effectors is achieved via the sizing control allocation presented in Section 7.5.2. The process takes the state-space model provided by the sizing optimization as inputs. The sizing factors required to ensure fault-tolerance are computed for each potentially controllable failure case. The resulting maximum sizing factors are then fed back into the sizing optimization so that the failure cases are included in the sizing process [16]. This

procedure is repeated until convergence to an optimal, fault-tolerant sizing of the UAV. Once the sizing has converged, the state-space model of the UAV is revised for each failure case. This process requires knowledge of the trimmed controls derived from the control allocation. Finally, the full rank condition is verified on each linearized state-space model. Together with the previously calculated XACAI, this concludes the overall fault-tolerance assessment.

The controllability assessment and fault-tolerant sizing methodology is implemented in a Python script available at the following GitHub repository: https://github.com/SizingLab/fault_tolerant_UAVs.

7.7 Case studies

The relevance of the methodology is illustrated through two case studies. Firstly, the controllability of a fixed-wing and a FW-VTOL UAV is assessed for various failures in the effectors, and sizing factors are determined for different requirements on the remaining control authority after failure. In particular, the benefits of the XACAI over the ACAI are demonstrated. Secondly, the impact of fault-tolerant sizing at the UAV level for multirotor configurations is illustrated by reintroducing the sizing factors into the sizing methodology of Chapter 3.

7.7.1 Relevance of the methodology for fixed-wing and FW-VTOL UAVs

The case study detailed in this section illustrates the application of the methodology proposed in Section 7.6 on two different concepts of UAVs to compare their fault-tolerance levels. Although the methodology may apply to any flight conditions by adjusting the linearized state-space model presented in Section 7.3, for the purpose of simplicity and conciseness, this case study focuses on failures occurring in cruise flight conditions. The controllability analysis is separated into two parts. In the first part, only the loss of effectiveness failures are considered. Any single failure or combination of two failures is covered. The study does not extend to triple failures, as their probability of occurrence is too remote. Then, the lock-in-place failures of the control surfaces are evaluated to demonstrate the contribution of the methodology. Any single or double failure of the control surfaces is covered, and only the worst cases regarding aerodynamic efforts (i.e., full deflection of the control surfaces) are assessed. Additionally, this section covers any combination of a control surface lock-in-place failure with a propeller loss of effectiveness. Finally, the maximum sizing factors obtained from the worst cases are provided to ensure consistency between the effectors' sizing and controllability analysis.

Concepts definition

The first concept consists of a fixed-wing UAV with two ailerons, two elevators, one rudder and a single rotor located at the nose-tip, as illustrated in Figure 7.7.1. Its design reflects

the E-flite Ultra Stick 25e, whose dynamics model, linearized around cruise flight conditions without failures, is provided in [63]. Expressly, this model assumes that the longitudinal and lateral dynamics are decoupled. The second concept consists of a hybrid FW-VTOL UAV



Figure 7.7.1 E-flite Ultra Stick 25e. Source: E-flite

with similar design specifications, except four VTOL rotors have been added. This concept is representative of the UAV depicted in Figure 7.3.1. On the y -axis, the VTOL rotors are centered on the right and left wings. On the x -axis, they are positioned on either side of the aerodynamic center, with a distance equal to twice the chord of the rectangular wing. The thrust-to-weight ratios for the propulsion and the VTOL rotors are based on market trends. Furthermore, the total masses of both concepts are assumed to be equal. The additional mass introduced by the VTOL propulsion is assumed to be balanced by a lower battery capacity. The VTOL rotors are typically inactive in nominal cruise flight conditions. However, their activation in the event of a failure adds a layer of redundancy to the flight controls. The specifications of both concepts are summarized in Table 7.7.1.

Controllability results

The results for loss of effectiveness failures are provided in Table 7.7.2. The controllability assessment indicates that the fixed-wing concept remains controllable after a complete loss of effectiveness of one aileron and/or one elevator. However, if both ailerons fail, the fixed-wing UAV is linearly uncontrollable. In fact, the XACAI is calculated as equal to zero, which reflects the absence of authority on the roll axis. In the absence of additional effectors for controlling the longitudinal axis, the fixed-wing UAV does not withstand the simultaneous failure of both elevators, as the controllability matrix is not full rank. Additionally, the

Table 7.7.1 Specifications of the UAVs for the case study. Adapted from [63].

Parameter	Fixed-wing	Hybrid FW-VTOL
Mass	1.959 kg	1.959 kg
Wing span	1.27 m	1.27 m
Wing chord (rectangular wing)	0.25 m	0.25 m
Wing surface area	0.32 m ²	0.32 m ²
Moment of inertia I_x	0.089 kg.m ²	0.089 kg.m ²
Moment of inertia I_y	0.144 kg.m ²	0.144 kg.m ²
Moment of inertia I_z	0.162 kg.m ²	0.162 kg.m ²
Max. deflection angles	±25 deg	±25 deg
Thrust-to-weight ratio (forward propulsion)	0.35	0.35
Thrust-to-weight ratio (VTOL)	N/A	1.25

fixed-wing concept is linearly uncontrollable if the rudder losses its effectiveness. The hybrid FW-VTOL remains controllable for any single or double failure thanks to its VTOL rotors.

Table 7.7.2 Results of the controllability assessment for loss of effectiveness failures.

Failures	Linear Controllability	
	Fixed-wing	Hybrid FW-VTOL
ρ_1 or ρ_2	Controllable	Controllable
η_1 or η_2	Controllable	Controllable
ζ	Uncontrollable	Controllable
$\rho_1 \& \rho_2$	Uncontrollable	Controllable
$\eta_1 \& \eta_2$	Uncontrollable	Controllable
$\rho_1 \& \eta_1$ or $\rho_1 \& \eta_2$ or $\rho_2 \& \eta_1$ or $\rho_2 \& \eta_2$	Controllable	Controllable
$\rho_1 \& \tau_1$ or $\rho_1 \& \tau_2$ or $\rho_1 \& \tau_3$ or $\rho_1 \& \tau_4$	N/A	Controllable
$\rho_2 \& \tau_1$ or $\rho_2 \& \tau_2$ or $\rho_2 \& \tau_3$ or $\rho_2 \& \tau_4$	N/A	Controllable

Note: ρ_i , η_i , and τ_i stand for aileron i , elevator i and rotor i , respectively.

ζ stands for the rudder. " ρ_1 or ρ_2 " represents the failure of either aileron 1 or aileron 2.

" $\rho_1 \& \rho_2$ " represents the failure of both aileron 1 and aileron 2.

The results for lock-in-place failures are provided in Table 7.7.3. The fixed-wing concept is not capable of withstanding any single lock-in-place failure, whereas the hybrid FW-VTOL concept maintains controllability in the presence of single or double lock-in-place failures, except for the rudder. To address a jammed rudder, the ailerons and VTOL rotors would need to be oversized by a factor greater than 3. However, this level of oversizing is impractical as it would significantly impact the UAV's initial design. Nevertheless, if the rudder were divided into two separate elements, the controllability in the event of a runaway of one of the two rudders would be preserved. Without considering triple failures, the controllability assessment clearly demonstrates the advantage of the hybrid FW-VTOL concept. Additionally, in the event of a jammed aileron or elevator, the activation of the VTOL rotors offers an advantageous control mechanism for maintaining balance in the roll

and pitch axes. This assumption relies on the prompt detection of the fault and the high responsiveness of the rotors.

Table 7.7.3 Results of the controllability assessment for lock-in-place failures.

Failures	Linear Controllability	
	Fixed-wing	Hybrid FW-VTOL
ρ_1 or ρ_2	Uncontrollable	Controllable
η_1 or η_2	Uncontrollable	Controllable
ζ	Uncontrollable	Uncontrollable*
$\rho_1 \& \rho_2$	Uncontrollable	Controllable
$\eta_1 \& \eta_2$	Uncontrollable	Controllable
$\rho_1 \& \eta_1$ or $\rho_1 \& \eta_2$ or $\rho_2 \& \eta_1$ or $\rho_2 \& \eta_2$	Uncontrollable	Controllable
$\rho_1 \& \tau_1$ or $\rho_1 \& \tau_2$ or $\rho_1 \& \tau_3$ or $\rho_1 \& \tau_4$	N/A	Controllable
$\rho_2 \& \tau_1$ or $\rho_2 \& \tau_2$ or $\rho_2 \& \tau_3$ or $\rho_2 \& \tau_4$	N/A	Controllable

* Controllability would require increasing the control effectiveness of the ailerons and VTOL rotors by a factor greater than 3.

Sizing factors

The controllability assessment has shown that the fixed-wing concept does not withstand any lock-in-place failure. This statement is valid regardless of the effectors' sizing. There exists no sizing factor that would result in a linearly controllable fixed-wing UAV. For the hybrid FW-VTOL concept, the sizing control allocation presented in Section 7.5 is applied to the failure scenarios assessed as controllable.

Several cases are analyzed, each corresponding to a distinct level of required control authority, as specified by Equation 7.53. The reference value for the XACAI is determined based on the initial sizing under normal operating conditions, without considering the involvement of the VTOL rotors. Subsequently, the sizing control allocation for failure cases includes the active VTOL rotors to ensure adequate control authority. The controllability results clearly demonstrated that relying solely on the control surfaces is insufficient to maintain controllability in the event of a lock-in-place failure. Table 7.7.4 presents a summary of the sizing control allocation cases, indicating the required control authority of the UAV in different failure cases relative to the reference XACAI value.

Figure 7.7.2 presents the maximum sizing factors obtained when the UAV is subject to single and double lock-in-place failures. Table 7.7.5 provides an extract of the detailed results for three specific failure scenarios. It should be noted that, given an aileron (or elevator) failure, an oversizing of the symmetrical aileron (or elevator) has been disabled to ensure design consistency. The results show that satisfaction of the allocation and control authority targets is achieved primarily by oversizing the VTOL rotors rather than the control surfaces. Low-speed flight conditions such as approach or landing generally drive the control surface sizing rather than the cruise flight scenario under study. Another reason for favouring the

Table 7.7.4 Summary of the sizing control allocation cases and associated required control authorities after failure.

Case	Requirement on control authority after failure	XACAI value
1	Control authority fully preserved*	1.76
2	25% deterioration of control authority*	1.32
3	50% deterioration of control authority*	0.88
4	75% deterioration of control authority*	0.44

* with respect to the initial sizing in nominal conditions.

oversizing of the VTOL rotors is that controllability in failure scenarios is mainly made possible by the presence of VTOL rotors. The control authority is more sensitive to the sizing of the latter than that of any other effectors. In the most restrictive cases regarding control authority, oversizing of the rudder is essential.

The simultaneous jamming of both ailerons is the most critical failure for the VTOL rotors sizing, although the simultaneous failure of both elevators results in sizing factors that are only slightly lower. When the control authority must be fully preserved (case 1), the concurrent failure of an aileron and a VTOL rotor is critical for sizing the rudder. Similar results are found when a 25% alleviation of the required control authority is allowed (case 2).

The sizing factors for the VTOL rotors highlight the importance of considering double failure scenarios during high-speed conditions in the sizing process, revising the conventional assumption that the takeoff scenario is the main driver for sizing the VTOL propulsion system. To conduct a more comprehensive comparison of the maximum thrusts during cruise flight and takeoff, it is necessary to consider the variation in the thrust coefficient that occurs between these two flight phases [64].

The results indicate that in the case of single failures, no significant oversizing of the control surfaces and VTOL rotors is necessary for this case study. Only the most critical scenario for control authority, referred to as case 1, requires a minor oversizing of the rudder by +3%.

Interestingly, further analysis has shown that if the rudder were split into two elements, the ailerons would also need to be oversized to meet the control authority requirement in the event of a jammed rudder. This oversizing would be +19% if a 50% reduction in authority is allowed (case 3), +90% for a 25% reduction in authority (case 2) and +162% if full authority must be maintained (case 1). When considering double failures, the most critical case is the simultaneous jamming of a rudder and an elevator. In addition to the oversizing of the VTOL rotors indicated in Figure 7.7.2, the ailerons would need to be oversized by +34%, +104% or +178% depending on the control authority requirement, i.e. for cases 3, 2 and 1 respectively.

Finally, a comparison between the XACAI and ACAI approaches highlights the methodological improvement. Figure 7.7.3 illustrates the maximum sizing factors for the most restrictive case of control authority, considering the inclusion (XACAI) and exclusion (ACAI)

Table 7.7.5 Three examples of sizing control allocation results for fully preserved control authority (case 1).

Failure scenario 1 out of 67										
	ρ_1	ρ_2	η_1	η_2	ζ	τ_p	τ_1	τ_2	τ_3	τ_4
Failures	$\times$	$\times$								
Control inputs δ'_0	25 deg	-25 deg	2.0 deg	2.0 deg	0.1 deg	0.31	0.83	0.01	0.03	0.69
Sizing factors	1.0	1.0	1.0	1.0	1.0	1.0	1.89	1.0	1.0	1.89

Failure scenario 2 out of 67										
	ρ_1	ρ_2	η_1	η_2	ζ	τ_p	τ_1	τ_2	τ_3	τ_4
Failures			$\times$	$\times$						
Control inputs δ'_0	3.4 deg	6.6 deg	25 deg	25 deg	-1.1 deg	0.35	0.87	0.93	0.15	0.09
Sizing factors	1.0	1.0	1.0	1.0	1.0	1.0	1.77	1.77	1.0	1.0

Failure scenario 3 out of 67										
	ρ_1	ρ_2	η_1	η_2	ζ	τ_p	τ_1	τ_2	τ_3	τ_4
Failures	$\times$							$\times$		
Control inputs δ'_0	-25 deg	-11.0 deg	-7.4 deg	-7.4 deg	3.8 deg	0.32	0.04	0	0.83	0.03
Sizing factors	1.0	1.0	1.0	1.0	1.27	1.0	1.0	1.0	1.05	1.0

Note: the thrust settings $\tau \in [0, 1]$ are unitless.

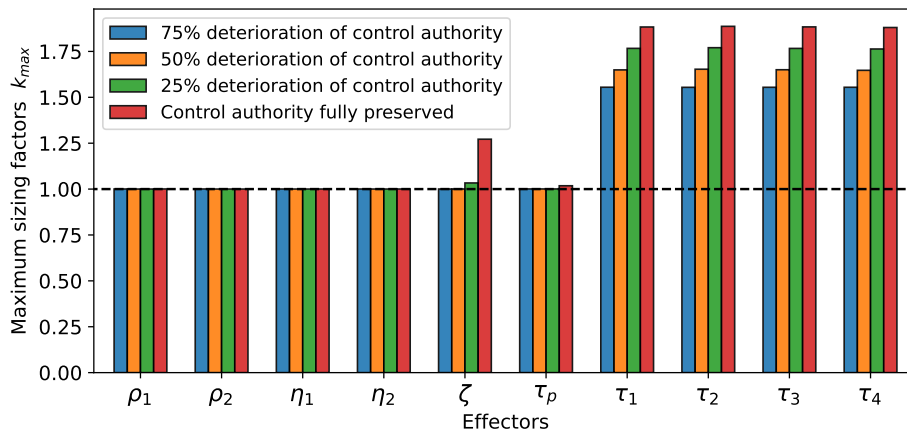


Figure 7.7.2 Maximum sizing factors obtained for the hybrid FW-VTOL concept subject to single and double lock-in-place failures, for different control authority requirements.

of control surface drag. Specifically, the drag forces were set to zero in the ACAI approach to prevent the control surfaces from generating thrust. The sizing of the VTOL rotors remains unchanged, as it is determined by the forces acting on the roll. However, the sizing of the rudder is influenced by the forces acting on the yaw, which include the rudder itself, drag forces from the deflection of other control surfaces, and to a lesser extent, the VTOL rotors. Therefore, accounting for the yaw moment generated by drag forces is crucial for accurately sizing the rudder. The ACAI approach, which neglects drag forces, fails to recognize the need for oversizing the rudder.

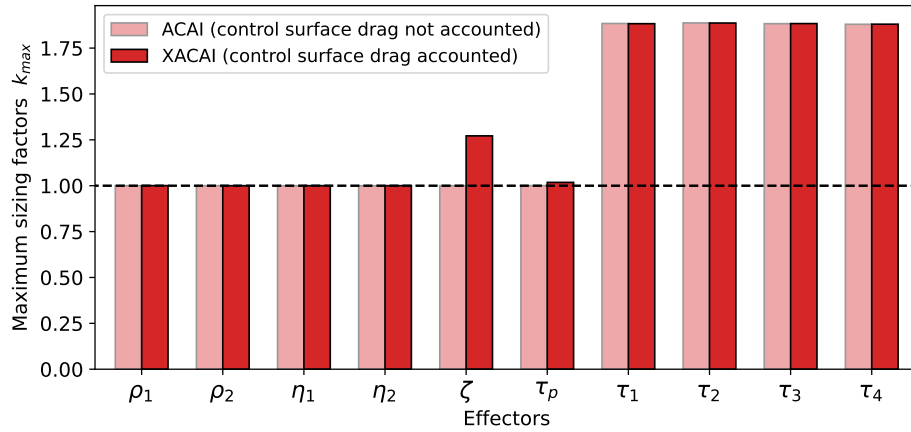


Figure 7.7.3 Comparison of the maximum sizing factors for the hybrid FW-VTOL concept under single and double lock-in-place failures, considering fully preserved control authority (case 1), using the ACAI and XACAI approaches.

7.7.2 Impact of fault-tolerant sizing at UAV level

This case study aims to illustrate the entire fault tolerant sizing process, i.e. the effect on the overall UAV sizing of modifying the size of the effectors to ensure sufficient control authority. The results are provided for various configurations of multirotor UAVs.

The concept of operation consists of the transportation of organs by drones within large cities. The mission involves transporting a payload of up to 20 kg, including an organ in an icebox (15 kg) and a transportation box attached to the UAV (5 kg), over a significant distance of 25 km. It is assumed that there is an emergency landing area halfway along the route. Therefore, a diversion distance of 12.5 km is considered to ensure that this area can be reached in the event of an inaccessible landing zone. The mission profile and the specifications used for the sizing process are provided in Figure 7.7.4 and Table 7.7.6, respectively. The mission is expected to occur beyond the visual line of sight (BVLOS) and over densely populated areas. This is a typical safety-critical operation that requires compliance with "SC-Light-UAS.2510 High-Risk". Accordingly, no loss of control should result from a single failure, and catastrophic conditions resulting from multiple failures

must be extremely improbable. Without going into the details of failure probability (i.e. the reliability assessment), fault tolerance to two simultaneous rotor failures is considered. Readers interested in the reliability assessment are referred to the full article [16].

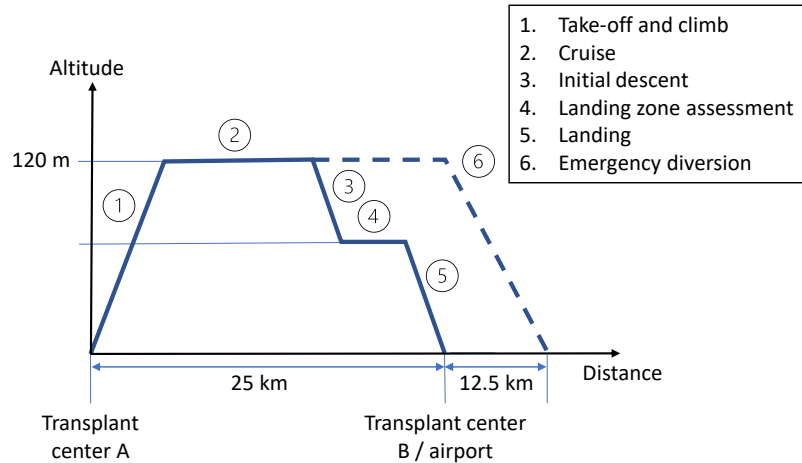


Figure 7.7.4 Profile of the safety-critical mission.

Table 7.7.6 Specifications of the safety-critical mission used for the sizing process.

Parameters	Values
Payload [kg]	20
Max. thrust-to-weight ratio [-]	1.6*
Climb height [m]	120
Climb speed [m/s]	3
Cruise distance [m]	25,000
Hover duration [min]	0.5**
Emergency diversion [m]	12,500

* Similar to the DJI Agras T20 thrust-to-weight ratio of 1.7

** Hover duration encompasses both descent and landing due to the absence of dedicated modeling for these phases.

Several multirotor UAV configurations, depicted in Figure 7.7.5, are sized with respect to the mission requirements and then evaluated. The sizing optimization is carried out using the methodology described in Chapter 3, with a mass minimization objective.

The controllability assessment based on the ACAI (the XACAI brings no additional information here) indicates that the quadcopter and hexacopter configurations cannot maintain control of all the control axes for some single failures [16, 41]. As shown in Table 7.7.7, the coaxial quadcopter can ensure control of all the control axes for any single failure but loses control for some double failures [16]. Consequently, the coaxial quadcopter is potentially compliant with the safety requirements of SC-Light-UAS.2510 Medium-Risk, as no single failure can have a catastrophic effect. The coaxial hexacopter and octocopter can maintain

control of all axes after any single and double failures [16]. Therefore, these designs are potentially compliant with the requirements of SC-Light-UAS.2510 High-Risk.

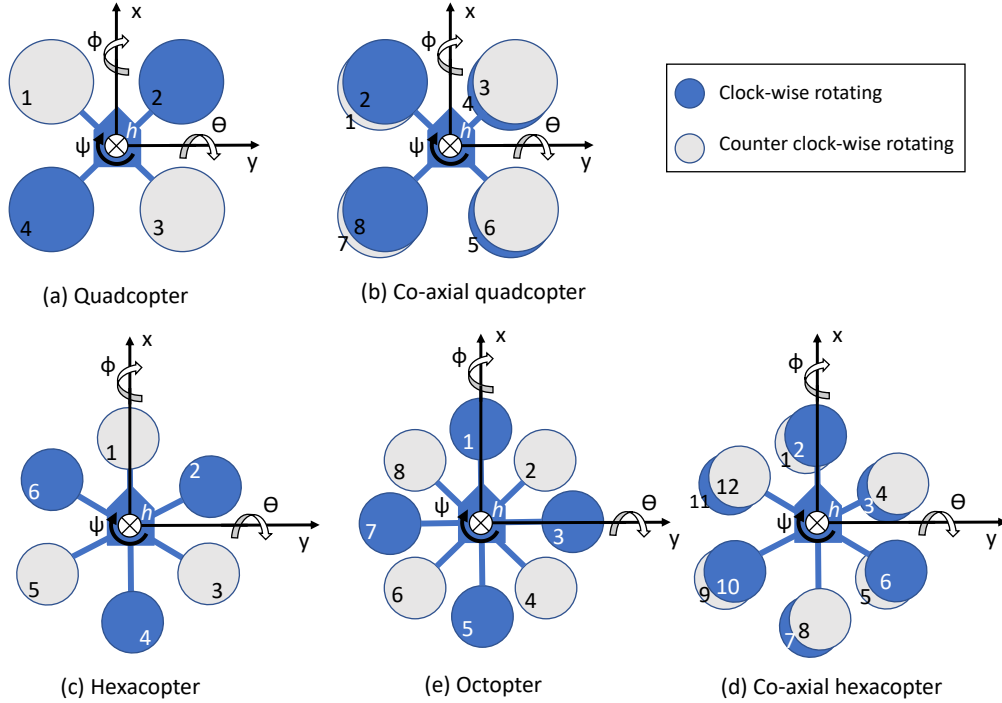


Figure 7.7.5 UAV configurations assessed in the case study.

Table 7.7.7 Results of the controllability assessment for the coaxial quadcopter.

	Controllable cases	Uncontrollable cases
Single rotor failures	$\tau_1, \tau_2, \tau_3, \tau_4, \tau_5, \tau_6, \tau_7, \tau_8$	None
Double rotor failures	$\tau_1 \& \tau_4, \tau_1 \& \tau_5, \tau_1 \& \tau_6, \tau_1 \& \tau_8, \tau_2 \& \tau_3, \tau_2 \& \tau_5, \tau_2 \& \tau_6, \tau_2 \& \tau_7, \tau_3 \& \tau_5, \tau_3 \& \tau_7, \tau_3 \& \tau_8, \tau_4 \& \tau_6, \tau_4 \& \tau_7, \tau_4 \& \tau_8, \tau_5 \& \tau_7, \tau_6 \& \tau_8$	$\tau_1 \& \tau_2, \tau_1 \& \tau_3, \tau_1 \& \tau_7, \tau_2 \& \tau_4, \tau_2 \& \tau_8, \tau_3 \& \tau_4, \tau_3 \& \tau_6, \tau_4 \& \tau_5, \tau_5 \& \tau_6, \tau_5 \& \tau_8, \tau_6 \& \tau_7, \tau_7 \& \tau_8$

For the three fault-tolerant configurations (coaxial quadcopter, coaxial hexacopter, and octocopter), the maximum sizing factors are obtained assuming the control authority must be fully preserved for all the failure cases assessed as controllable². Note that this is a rather strict requirement. The maximum sizing factors, which correspond to double failure scenarios, are provided in Table 7.7.8. Here, the scaling factors are reduced to single scalars applicable to all the rotors owing to the symmetry of the idealized case study designs.

Finally, the maximum sizing factors are reintroduced into the UAV sizing process. In particular, optimization constraints are added to ensure that the propulsion system is able to generate the emergency thrust $\tilde{f}_{max} = f_{max} \cdot K_{max}$. In a first approach, only one iteration

²A simplified approach based on the pseudo-inverse control allocation (Equation 7.48) was used to obtain the sizing factors. Interested readers are referred to the article [16].

Table 7.7.8 Maximum sizing factors obtained for the fault-tolerant configurations, considering fully preserved control authority.

Configuration	K_{max}
Coaxial quadcopter	200%
Coaxial hexacopter	186%
Octocopter	283%

(from sizing optimization to sizing factors and then back to sizing optimization) is performed. The mass breakdowns obtained for the coaxial quadcopter, coaxial hexacopter and octocopter, with and without failure case sizing, are presented in Figure 7.7.6. No feasible solution was found for the octocopter with the baseline mission. A downgraded mission with an emergency rerouting of 2.5 km instead of 12.5 km was applied to this concept for comparison. Despite this, the octocopter design did not converge with failure case sizing. The coaxial quadcopter and the coaxial hexacopter have comparable weights. The consideration of failure cases in the sizing resulted mainly in an increase of the motor mass, for example a 73% increase for the coaxial hexacopter. The other components are barely affected. The total increase in mass is 15% for the coaxial quadcopter and 11% for the coaxial hexacopter.

Overall, based on the results of the control analysis and the sizing, the coaxial hexacopter seems to offer the best compromise between fault tolerance and weight.

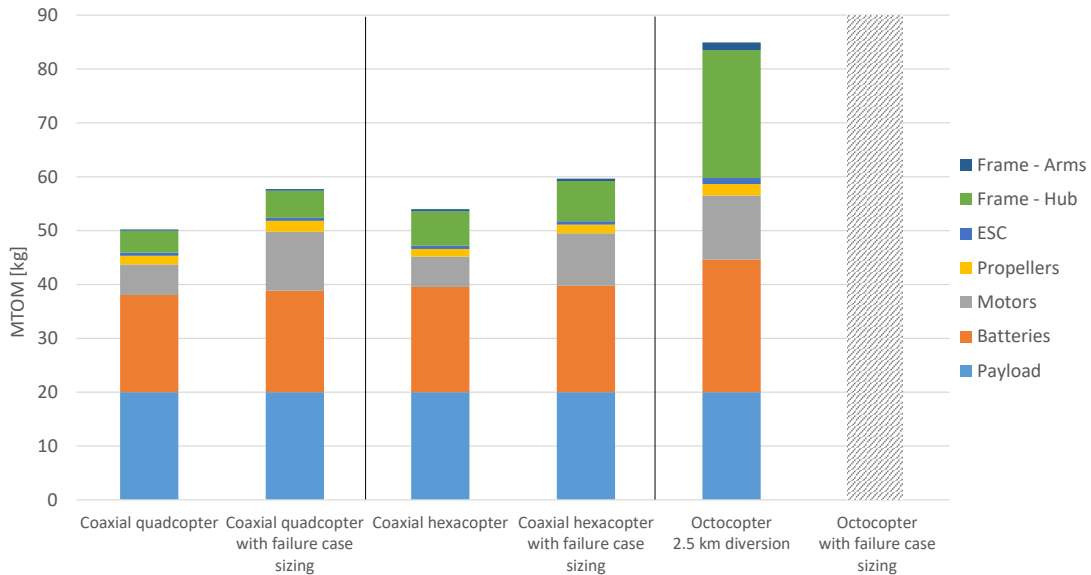


Figure 7.7.6 Comparison of different configurations of UAVs, with and without failure case sizing.

7.8 Conclusions and perspectives

Methodology and lessons learned

The present chapter introduces a new methodology for the controllability assessment and fault-tolerant sizing of UAVs subject to rotor and control surface failures. The contributions of this methodology are twofold. First, it enables the analysis of the null-controllability of linearized UAV models with constrained inputs and efforts governed by an absolute function. Second, it bridges the gap between controllability, failure scenarios, and the sizing of effectors. To this end, the following advancements were proposed:

- A linearized UAV model was derived and modified to account for failure cases specific to multirotor, fixed-wing and FW-VTOL configurations. In particular, lock-in-place failures (e.g., control surface jamming or run-away) were introduced into the state-space model and reflected in the controllability matrix.
- A new control authority index, the XACAI, was developed to address control surface drag and the introduction of disturbances due to lock-in-place failures. The UAV is linearly null-controllable only if the XACAI is strictly positive and the controllability matrix is full rank. Verifying the second condition requires calculating the UAV's trimmed state, which is obtained by solving a control allocation problem.
- A new control allocation problem was proposed to introduce a degree of freedom representative of the effectors sizing. As such, the control allocation method solves both the trimmed state calculation and the fault-tolerant sizing of the effectors.

The case studies demonstrate the effectiveness of the proposed methodology. First, the controllability of a fixed-wing UAV and a hybrid FW-VTOL concept was evaluated under different failure conditions. The study focused on a steady-level flight condition for simplicity and conciseness. The results highlight the higher fault tolerance of the hybrid FW-VTOL concept due to the presence of additional effectors (VTOL rotors) that can be used in case of failures. However, the study also reveals the need to oversize the VTOL rotors to ensure sufficient control authority in double failure scenarios, revising the conventional assumption that VTOL system sizing should be derived solely from take-off analysis. Furthermore, the comparison between the ACAI and XACAI approaches shows that using the ACAI method to support the sizing analysis leads to an underestimation of the required rudder oversizing. This is due to ACAI's inability to account for control surface drag. This discrepancy highlights the importance of the proposed XACAI approach. In a second case study, focusing on multirotor UAVs, the results of the control analysis (namely the sizing factors on the effectors) were integrated back into the sizing optimization developed in Chapter 3 to evaluate the implications at the UAV level. If the control authority has to be fully maintained in case of failures compared to the initial sizing under nominal conditions, the total mass increase of

the UAV exceeds 10%. Finally, the coaxial hexacopter configuration was found to be the most likely to comply with EASA regulations for safety-critical operations while maintaining an acceptable weight.

Perspectives

Further research should address the following limitations of the presented work. The methodology proposed in this study supports the evaluation of the intrinsic reconfigurability of UAVs through a linear null-controllability analysis. The controllability assessment is applied to a linearized system model and offers a sufficient condition for preserving or restoring control authority on the virtual control axes. It is important to note that this controllability assessment does not exclude the possibility of nonlinear controllability in the system, as discussed in Section 6.3.2. In addition, the methodology is time-independent and does not capture the dynamics of the effectors, which limits the accuracy of the assessment. The inclusion of constraints on control response times could significantly increase this accuracy. A promising step toward the integration of a time-dependent degree of controllability into the sizing methodology of Chapter 3 was recently proposed by Warren [48]. This has been made possible by the open source availability of the framework FAST-UAV developed in this thesis. Overall, the use of nonlinear controllability analysis and time-dependent control authority indices would help refine the fault-tolerant sizing of UAVs.

Appendix

Appendix 7.A Equations of motion

Scalar equations of motion

$$X + q_d S C_{x_0} - m_{uav} g \sin \theta = m_{uav} (\dot{u}_b + q w_b - r v_b) \quad (7.57)$$

$$Y + q_d S C_{y_0} + m_{uav} g \cos \theta \sin \phi = m_{uav} (\dot{v}_b + r u_b - p w_b) \quad (7.58)$$

$$Z + q_d S C_{z_0} + m_{uav} g \cos \theta \cos \phi = m_{uav} (\dot{w}_b + p v_b - q u_b) \quad (7.59)$$

$$L + q_d S C_{l_0} = I_x \dot{p} - I_{xz} \dot{r} + q r (I_z - I_y) - I_{xz} p q \quad (7.60)$$

$$M + q_d S C_{m_0} = I_y \dot{q} + r p (I_x - I_z) + I_{xz} (p^2 - r^2) \quad (7.61)$$

$$N + q_d S C_{n_0} = -I_{xz} \dot{p} + I_z \dot{r} + p q (I_y - I_x) + I_{xz} q r \quad (7.62)$$

q_d is the dynamic pressure [Pa], S is the UAV reference area [m²], C_{axis} are the aerodynamic coefficients of the fixed surfaces (e.g., wing and fuselage), and I are the moments of inertia [kg.m²]. ϕ , θ and ψ are the roll [rad], pitch [rad], and yaw [rad] angles. X , Y , Z , L , M , and N are the control efforts [N] originating from the propellers and the control surfaces.

Linearized equations of motion in steady, straight, and level flight

The linearized set of equations is obtained using the small-disturbance notation and neglecting the products of deviations (first-order approximation), according to [54].

$$\frac{\Delta X}{m_{uav}} + \dot{X}_u \Delta u_b + \dot{X}_w \Delta w_b - g \cos \theta_0 \Delta \theta = \Delta \dot{u}_b \quad (7.63)$$

$$\frac{\Delta Y}{m_{uav}} + \dot{Y}_v \Delta v_b + \dot{Y}_p \Delta p + (\dot{Y}_r - u_{b_0}) \Delta r + g \cos \theta_0 \Delta \phi = \Delta \dot{v}_b \quad (7.64)$$

$$\frac{\Delta Z}{m_{uav}} + \dot{Z}_u \Delta u_b + \dot{Z}_w \Delta w_b + (\dot{Z}_q + u_{b_0}) \Delta q - g \sin \theta_0 \Delta \theta = \dot{w}_b \quad (7.65)$$

$$\frac{\Delta L}{I_x} + \dot{L}_v \Delta v_b + \dot{L}_p \Delta p + \dot{L}_r \Delta r = I_x \dot{p} \quad (7.66)$$

$$\frac{\Delta M}{I_y} + \dot{M}_u \Delta u_b + \dot{M}_w \Delta w_b + \dot{M}_q \Delta q - g \dot{M}_w \sin \theta_0 = I_y \dot{q} \quad (7.67)$$

$$\frac{\Delta N}{I_z} + \dot{N}_v \Delta v_b + \dot{N}_p \Delta p + \dot{N}_r \Delta r = I_z \dot{r}, \quad (7.68)$$

where the parameters $\dot{X}$, $\dot{Y}$, $\dot{Z}$, $\dot{L}$, $\dot{M}$, $\dot{N}$ are aerodynamic stability coefficients obtained by a Taylor series expansion applied to the efforts' variation about the reference equilibrium condition [54]. For instance,

$$\dot{X}_u \approx \frac{\rho_{air} v_{air}^2 S C_{D_0}}{m_{uav} u_{b_0}}, \quad (7.69)$$

where C_{D_0} [-] is the parasitic drag coefficient of the airframe, ρ_{air} [kg.m⁻³] is the air density, and S [m²] is the reference area.

The insignificant aerodynamic derivatives in Equations 7.63 have been neglected.

Appendix 7.B State-space model for failure cases

For better understanding, the modification of the state-space model for failure cases is demonstrated in a first place on the x-axis, and focusing on an elevator. The scalar equation of motion on the x-axis, provided in 7.A, is re-written with a focus on the effort originating from the elevator, for clarity:

$$X_\eta + X_{others} = m_{uav}(\dot{u}_b + q w_b - r v_b), \quad (7.70)$$

where X_{others} are the efforts originating from the fixed surfaces, gravity, and control effectors, excluding the failed elevator. The effort from the elevator, X_η , is expressed as follows:

$$X_\eta = \frac{1}{2} \rho_{air} v_{air}^2 S_\eta C_{x,\eta} \eta \quad (7.71)$$

With S_η [m²] the elevator's area and $C_{x,\eta}$ [-] its aerodynamic coefficient specific to the x-axis, assumed to be constant. The above equation is linearized around straight and level flight conditions, that is $u_b = u_{b_0} + \Delta u_b$, $v_b = \Delta v_b$ and $w_b = \Delta w_b$. The air velocity is expressed from a first-order approximation as:

$$v_{air} = \sqrt{u_b^2 + v_b^2 + w_b^2} \quad (7.72)$$

$$= \sqrt{u_{b_0}^2 + 2u_{b_0}\Delta u_b + \Delta u_b^2 + \Delta v_b^2 + \Delta w_b^2} \quad (7.73)$$

$$\approx \sqrt{u_{b_0}^2 + 2u_{b_0}\Delta u_b}, \quad (7.74)$$

The elevator's effort on the x-axis becomes:

$$X_\eta = \frac{1}{2} \rho_{air} (u_{b_0}^2 + 2u_{b_0}\Delta u_b) S_\eta C_{x,\eta} (\eta_0 + \Delta \eta) \quad (7.75)$$

$$= \frac{1}{2} \rho_{air} u_{b_0}^2 S_\eta C_{x,\eta} \Delta \eta + \frac{1}{2} \rho_{air} u_{b_0}^2 S_\eta C_{x,\eta} \eta_0 + \rho_{air} u_{b_0} S_\eta C_{x,\eta} \eta_0 \Delta u_b \quad (7.76)$$

Introducing the elevator's control derivative, $\dot{X}_\eta = \frac{1}{2}\rho_{air}u_{b0}^2 S_\eta C_{x,\eta}$, one obtains:

$$X_\eta = \dot{X}_\eta \Delta\eta + \dot{X}_\eta \eta_0 + \frac{2}{u_{b0}} \dot{X}_\eta \eta_0 \Delta u_b \quad (7.77)$$

$$= \dot{X}_\eta \eta + \frac{2}{u_{b0}} \dot{X}_\eta \eta_0 \Delta u_b \quad (7.78)$$

Generalizing to all effectors' controls and efforts yields the following virtual control vector:

$$\mathbf{u} = \mathbf{B}_f \boldsymbol{\delta} + \frac{2}{u_{b0}} \mathbf{B}_f \boldsymbol{\delta}_0 \Delta u_b \quad (7.79)$$

The state-space model becomes

$$\Delta \dot{\mathbf{x}} = \mathbf{A} \Delta \mathbf{x} + \mathbf{B} \left(\mathbf{B}_f \boldsymbol{\delta} + \frac{2}{u_{b0}} \mathbf{B}_f \boldsymbol{\delta}_0 \Delta u_b - \mathbf{u}_0 \right) \quad (7.80)$$

$$= \left(\mathbf{A} + \frac{2}{u_{b0}} [\mathbf{B} \mathbf{B}_f \boldsymbol{\delta}_0 \quad \mathbf{0}_{9 \times 8}] \right) \Delta \mathbf{x} + \mathbf{B} (\mathbf{B}_f \boldsymbol{\delta} - \mathbf{u}_0) \quad (7.81)$$

Finally, taking into account the failures yields:

$$\Delta \dot{\mathbf{x}} = \underbrace{\left(\mathbf{A} + \frac{2}{u_{b0}} [\mathbf{B} \mathbf{B}_f \boldsymbol{\delta}'_0 \quad \mathbf{0}_{9 \times 8}] \right)}_{\mathbf{A}_{\delta'_0}} \Delta \mathbf{x} + \underbrace{\mathbf{B} [\mathbf{B}_f \mathbf{H} \boldsymbol{\delta}]}_{\mathbf{B}'_f} - \underbrace{(\mathbf{u}_0 - \mathbf{B}_f (\mathbf{I}_m - \mathbf{H}) \boldsymbol{\delta}'_0)}_{\mathbf{u}'_0} \quad (7.82)$$

Appendix 7.C Study case

State-space matrices of the Hybrid FW-VTOL concept.

The full-order state matrix, control matrix and control effectiveness matrix for the hybrid FW-VTOL UAV described in the case study (Section 7.7) are provided below. The system is linearized around cruise flight conditions, without failures, with a longitudinal speed $u_{b0} = 19m/s$ [63].

$$\mathbf{A}_0 = \begin{bmatrix} -0.38 & 0 & 0.60 & 0 & 0 & 0 & 0 & -9.80 & 0 \\ 0 & -0.64 & 0 & 0.46 & 0 & -18.21 & 9.50 & 0 & 0 \\ -0.98 & 0 & -10.65 & 0 & 16.74 & 0 & 0 & -0.21 & 0 \\ 0 & -2.02 & 0 & -12.47 & 0 & 4.05 & 0 & 0 & 0 \\ 0.18 & 0 & -5.39 & 0 & -16.55 & 0 & 0 & 0 & 0 \\ 0 & 1.30 & 0 & 0.86 & 0 & -3.09 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \quad (7.83)$$

$$\mathbf{B} = \begin{bmatrix} \mathbf{J}_f^{-1} \\ \mathbf{0}_{3 \times 6} \end{bmatrix}, \quad \mathbf{J}_f = \text{diag}(1.959, 1.959, 1.959, 0.089, 0.144, 0.162) \quad (7.84)$$

$$(7.85)$$

$$\mathbf{B}_f = \begin{bmatrix} 0. & 0. & -0.35 & -0.35 & 0. & 6.73 & 0. & 0. & 0. & 0. \\ 0. & 0. & 0. & 0. & 5.84 & 0. & 0. & 0. & 0. & 0. \\ 0. & 0. & -3.55 & -3.55 & 0. & 0. & 6.01 & 6.01 & 6.01 & 6.01 \\ -6.19 & 6.19 & 0. & 0. & 0.58 & 0.02 & 1.91 & -1.91 & -1.91 & 1.91 \\ 0. & 0. & -10.19 & -10.19 & 0. & 0. & 3.00 & 3.00 & -3.00 & -3.00 \\ 1.39 & -1.39 & 0. & 0. & -4.28 & 0. & 0.03 & -0.03 & 0.03 & -0.03 \end{bmatrix} \quad (7.86)$$

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Part IV

Environmental impacts assessment

The final part of the manuscript addresses the critical issue of environmental sustainability in the context of UAV design. Chapter 8 highlights the existing gap in methodologies for assessing the environmental impact of UAVs during the preliminary design phase. The lack of interfacing between life cycle assessment (LCA) and design tools calls for the development of an automated, integrated approach. Chapter 9 responds to this need by introducing an environmental module based on the LCA methodology that is seamlessly integrated with the sizing process. The proposed approach allows a comprehensive exploration of environmental impacts as early as the preliminary design phase, including sensitivity to technological assumptions and mission requirements. The results highlight the importance of battery production and renewal in the environmental footprint of UAVs. It is also shown that replacing the traditional mass minimization objective with an environmental objective results in slightly heavier but more environmentally friendly designs.

Author's contributions

Conference papers

F. Pollet, S. Delbecq, M. Budinger, J.-M. Moschetta, and T. Planès. Environmental Life Cycle Assessments for the Design Exploration of Electric UAVs. In *Aerospace Europe Conference, Joint 10th EUCASS 9th CEAS Conference*, Lausanne, Switzerland, July 2023. doi: 10.13009/EUCASS2023-548.

Chapter 8

State of the art of environmental assessment for UAVs

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8.1 Introduction and context

The market for UAVs in logistics applications has grown considerably over recent years. In 2022, more than 2,000 commercial drone deliveries occurred every day worldwide [1], and it is expected that 2,700 million parcels will be delivered in 2025 alone [2]. To support this growth, new projects of UAVs are being developed. US company Matternet was the first to receive FAA type certification for its *M2 drone* [3]. Amazon launched its *Prime Air* service in 2022, and began trials in several rural areas the same year [4]. Zipline has achieved more than 600,000 commercial deliveries with its fixed-wing UAV since 2016 [5]. To date, the development of UAVs has mainly been motivated by the expectation that these platforms will become cost-competitive with other transport modes [6]. However, the evolution of the drone market relies on two other cornerstones: regulation and public acceptance. In particular, the social acceptance of UAVs is strongly linked to environmental concerns, including, but not limited to, the impact on climate change. This point received the attention of EASA, which conducted a survey of European citizens showing that a vast majority of 74% would be in favor of the introduction of an eco-label for urban air operations, including drone deliveries [7].

While these novel aircraft concepts claim no direct emissions during flight owing to their electric propulsion, they are responsible for other environmental issues through the rest of the value chain, such as manufacturing and disposal. The development of UAVs must therefore be supported by a comprehensive evaluation of their environmental implications.

8.2 Life Cycle Assessment methodology

Life Cycle Assessment (LCA) has proven to be a valuable tool for quantifying the environmental impacts of products throughout their entire life cycle and across a range of indicators. The LCA methodology is standardized by the International Organization for Standardization (ISO), in particular ISO 14040 [8] and ISO 14044 [9]. It is defined as “a compilation and evaluation of the inputs, outputs and the potential environmental impacts of a product system throughout its life cycle” [8]. LCA proposes a specific approach to environmental analysis which includes the following features [10, 11]:

- **Life cycle approach:** from the extraction of raw materials, through processing and manufacturing, distribution, use and end-of-life, all stages of the product life cycle are taken into account.
- **Multi-criteria analysis:** LCA covers the main environmental concerns known to date (extraction of resources, impact of toxic substances on humans and ecosystems, etc.).
- **Quantitative assessment:** indicators are quantitative and derived with mathematical cause-effect relationships. The quantitative nature of LCA allows to compare different systems and scenarios.

While LCA has its advantages, it is not without its limitations. Typically, LCA does not address economic and social aspects. Certain environmental concerns, for example noise pollution from flying UAVs, are not considered. Finally, the standard LCA approach cannot say how sustainable a product is in absolute terms. The approach is essentially comparative, i.e. it is most often used to determine whether one product is better than another. Complementary tools may be used to support LCA results for more informed decision-making, including life cycle costing (LCC) and social life cycle assessment (S-LCA) [11].

As standardized by the ISO series, the LCA methodology operates in four main phases, as illustrated in Figure 8.2.1. The analysis based on these 4 stages is often iterative. For example, the results of a preliminary LCA can be used to further investigate specific assumptions and refine the results for a more detailed analysis.

Step 1: Goal and scope definition

An LCA starts by defining the aims of the study, such as who the study is intended for and what question it is attempting to answer. For instance, an LCA may serve to collect

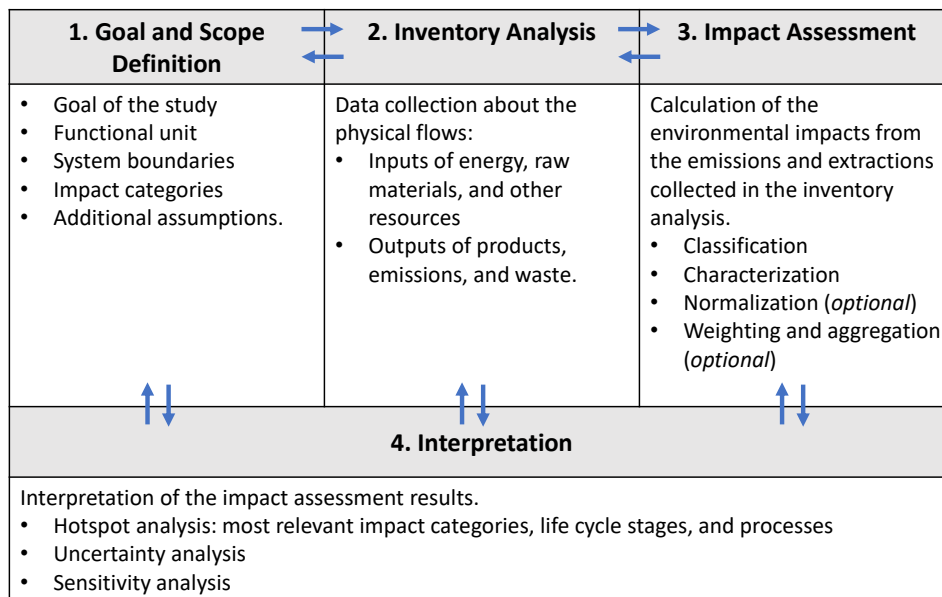


Figure 8.2.1 Phases of LCA, adapted from ISO 14040 [8]. The two-way arrows suggest an iterative process.

information on existing products and compare alternatives, or to support an eco-design approach for the development of new products. Next, when defining the scope of the study, a number of elements need to be considered and described. This involves defining the system and its intended function. A functional unit provides a quantitative characterization of the function. It allows several systems, or *scenarios*, to be compared on the same basis. For example, the functional unit “transport one kilogram of payload over one kilometer” can be used to compare different transport modes for commercial delivery. Additionally, the boundaries of the system need to be defined. This implies deciding which activities and processes are part of the life cycle and which are excluded. For example, the ground infrastructure that supports the operation of UAVs may or may not be included in the system. Where relevant, geographical and temporal boundaries can also be selected. Finally, the environmental impact categories to be assessed in Step 3 must be specified.

Step 2: Inventory analysis

The inventory analysis quantifies the various physical flows crossing the boundaries of the system, including resources, emissions, energy, waste and materials. The output of this step is the so-called Life Cycle Inventory (LCI), which lists the physical elementary flows (e.g., CO₂, crude oil in the ground, etc.) that are associated with the provision of the functional unit.

Different mathematical methods are available to compile the LCI, each having its advantages and limitations [12]. LCA software generally relies on matrix inversion methods to solve algebraic singularities, such as dependency loops between processes (the input of one process requires the output of another, and vice versa). The main limitation of this approach is that it requires significant computational resources and is therefore time-consuming [12].

Because of the large number of processes involved in most systems, the analysis often relies on generic data provided by databases, such as EcoInvent or ELCD. In these databases, the inventory data is pre-compiled for a number of existing products and services (e.g. production of a specific material or generation of electricity). However, the use of generic datasets may lead to a loss of accuracy in the results. On the one hand, it can be difficult to find a dataset that is properly applicable to a given study. Data may be provided averaged at a regional or global scale and may be valid for a specific temporal scope. On the other hand, some assumptions about the processes (e.g., technologies used in a manufacturing process) and difficulties in accessing accurate documentation can lead to poor data quality. A few databases assign a level of uncertainty to each process, making it possible to quantify the final uncertainty in the LCI.

Step 3: Impact assessment

The Life Cycle Impact Assessment (LCIA) translates the physical flows collected in the LCI into impacts on the environment.

First, the impact categories to be assessed must be defined. These categories are associated with indicators, which are quantifiable representations of the impacts, and characterization models. The characterization models are mathematical expressions linking the LCI results to the category indicators. In practice, predefined sets of impact categories, indicators and characterization models are found in existing LCIA methods available in LCA software. IMPACT World+ [13], ReCiPe2016 [14] and TRACI 2.0 [15] are among the most known methods. More recently, the European Commission has proposed the Environmental Footprint (EF) methods as a common framework for EU companies [16].

Once an LCIA method has been selected, the *classification* step assigns the LCI results to impact categories according to their potential effects. For example, the CO₂, CH₄ and N₂O emissions are classified within the climate change category. They may additionally be classified in other categories, such as acidification for CO₂.

Finally, for each impact category, the elementary flows are assessed according to their contributions. In this *characterization* step, each elementary flow E_i classified in a given category c is multiplied by the corresponding characterization factor CF_i provided by the characterization models. The impact score IS_c for the category c is the sum of all elementary

effects:

$$IS_c = \sum_i CF_i \cdot E_i \quad (8.1)$$

Here, we distinguish between two types of characterization factors, depending on the impact indicator they express. The *midpoint* characterization factors, associated with midpoint indicators, characterize impacts observed early in the environmental cause-effect chain. This includes, among others, ozone depletion, eutrophication, and water use. Midpoint indicators are easily measurable as they are directly observable on the field. However, their environmental significance is sometimes difficult to interpret. To overcome this limitation, *endpoint* indicators characterize impacts further downstream in the environmental mechanism, for example, impact on human health or ecosystem quality. Because they are more difficult to identify, endpoint models come with greater uncertainty. Yet, if considering environmental relevance as a measure of accuracy, both midpoint and endpoint indicators support the decision-making process to the same extent [11, 17]. Depending on the LCIA method, the characterization is proposed at midpoint level (e.g. Environmental Footprint), endpoint level, or both (e.g. IMPACT World+, ReCiPe2016). Figure 8.2.2 depicts the general framework of the IMPACT World+ method.

Two additional steps can be taken as part of the LCIA: normalization and weighting of the results. Although optional according to ISO standards, these steps support comparisons across impact categories, life cycle stages and processes, and facilitate communication of the results. For this reason, normalization and weighting are mandatory for studies conducted under the European EF framework. The purpose of the *normalization* step is to express the LCIA results on the same scale, relative to a reference system. Indeed, the characterization scores are expressed in units that vary between impact categories and are therefore hardly comparable. The dimensionless normalized score NS for a category c is obtained by multiplying the impact score by a normalization factor NF_c :

$$NS_c = IS_c \cdot NF_c \quad (8.2)$$

The normalization factors are derived according to the reference system. However, there are many ways to choose a reference system. For example, the EF method considers the emissions and resources used globally over one year per capita. Other normalization approaches focus on an estimate of the planet's carrying capacity [18] to ensure environmental sustainability. Here lies the main limitation of the normalization process: depending on the reference system used, the results obtained can be very different.

Last, *weighting* of the normalized results aims at reflecting the perceived relative importance of the impact categories. The weighted results can then be aggregated across impact

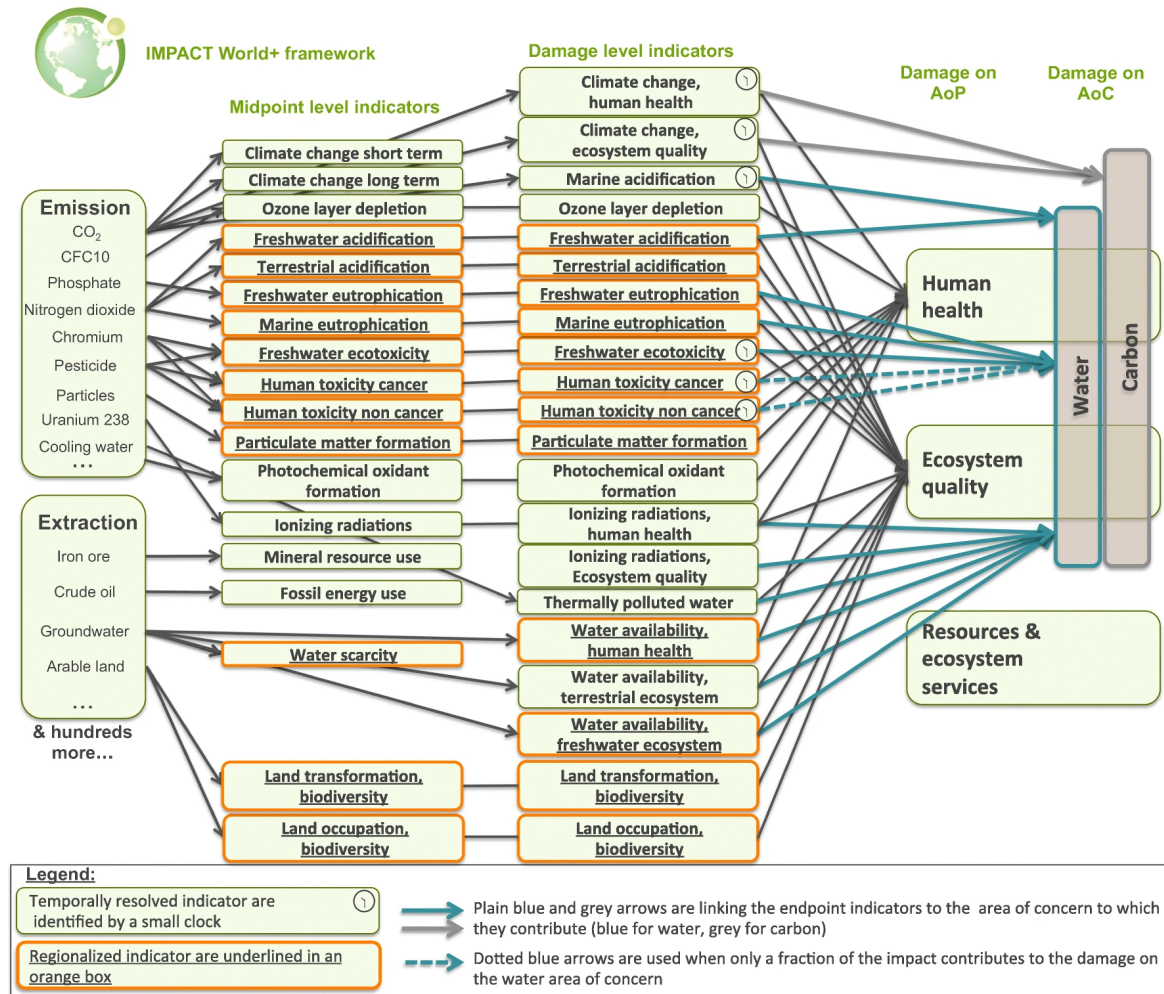


Figure 8.2.2 Example of the IMPACT World+ framework [13].

categories to produce a single overall score:

$$SS = \sum_c NS_c \cdot WF_c, \quad (8.3)$$

where SS is the single score, expressed for example in points per functional unit, and WF_c is the weighting factor applying to impact category c . It is important to note that weighting is based on subjective considerations. However, obtaining a single score facilitates a number of comparisons and the communication of the results.

Step 4: Interpretation

The interpretation is the final step in LCA. It aims to draw conclusions for the study while clarifying its limitations and providing recommendations. On the one hand, the results can be used to identify the most relevant impact categories, life cycle stages, and processes. Such

analyses provide valuable information for reducing the environmental impact of the system. On the other hand, to ensure the robustness of the results, it is necessary to verify the methodological choices and the data supporting the LCA. Besides checking the completeness of the methods and data, sensitivity studies can be carried out to ensure the reliability of the results. In particular, this involves exploring the effect of varying assumptions on the results through the evaluation of alternative scenarios. Finally, uncertainty studies on various LCA parameters can be conducted [19]. These uncertainties may be related to the inventory data, as conducted in [20], and/or the characterization factors of the environmental models. In the context of this thesis, these are added to the uncertainties relating to the design and performance of the modeled system.

8.3 LCA software tools for the eco-design of systems

Since the first commercial LCA software was released in 1989, *GaBi*, several dedicated LCA tools have been published. A non-exhaustive list of software for performing LCA studies includes *SimaPro*, *OpenLCA*, *Umberto*, *ASKOR* and *Ecodesign Studio*. These commercial software tools, which come with or without embedded LCI databases and LCIA methods, provide advanced capabilities to produce detailed LCA reports. Their main disadvantage is that they require manual entry of the data related to the product and the scenario, although developer versions allow for programming a COM interface to process external data automatically. For these reasons, commercial LCA software is primarily dedicated to LCA practitioners. On the opposite, some non-LCA-specific tools integrate life cycle data to support designers in the eco-design process. This includes computed-aided technologies (e.g. computer-aided design) software such as *3DEXPERIENCE* and *Grasshopper 3D* through the Bombyx project [21]. For earlier design stages, the use of simplified LCA approaches, also known as streamlined LCA, narrows the scope to a limited number of impact indicators, life cycle stages or processes. This allows for easier integration with existing engineering tools while maintaining a sufficient level of information to support eco-design decisions. This approach has been widely used to implement LCA in preliminary design methodologies [22–30]. For example, Debusschere *et al.* used estimates of the energy costs and CO_{2eq} emissions for the production of raw materials and electricity to optimize the design of a single-phase induction motor. Finally, advanced LCA software written in programming languages (e.g., Python) are of special interest to researchers or developers willing to implement specific functionalities. The open-source framework *Brightway* [31] has attracted particular interest in recent years due to its versatility and easy customization. However, very few studies to date have integrated the Brightway environment into design tools [32].

8.4 Applications to aerial vehicles

8.4.1 Environmental impacts of UAVs

Although the literature reviews show increasing interest in LCA in the aviation industry [33, 34], its application to UAVs is sparse. Stolaroff pointed out the need for a life cycle assessment to evaluate the potential impacts of UAVs for package delivery [35]. Some studies addressed this aspect with a focus on energy demand and CO₂ equivalent¹ (CO_{2e}) emissions related to the operating phase [36–38]. Goodchild and Toy [36] compared the CO_{2e} emissions of multirotor UAVs with those of conventional trucks to deliver parcels. It was established that a drone-based delivery system can emit less CO_{2e} than diesel trucks when the customer density is low and the distance is short. Chiang *et al.* [37] proposed an optimal routing model that would reduce the use of energy and CO_{2e} emissions when coordinating conventional vehicles and UAVs for deliveries. Kirschstein [38] further compared the energy demand and CO_{2e} emissions with electric trucks. The study shows that switching to an entirely drone-based system for deliveries in most situations is not desirable. However, none of the aforementioned studies has addressed the emissions related to the production stage of UAVs. Rodrigues *et al.* [39] and Stolaroff *et al.* [40] extended the scope to battery life cycle emissions, although other components (e.g. airframe) are not considered. More recently, Duriez *et al.* [29] focused on the CO₂ footprint of solar-powered High Altitude Long Endurance (HALE) drones when designed with different airframe materials. Emissions from the production of solar panels and batteries are included, but because this particular drone concept is fully powered by its solar panels, the use stage causes no emissions. Such a focus on the production stage is not applicable to other drone concepts that operate with electricity from the grid. Figliozzi [41] was the first to adopt a cradle-to-grave approach for comparing the CO_{2e} emissions of multirotor UAVs with ground vehicles. It is shown that UAV emissions per delivery can increase by up to 50% when the production stage is taken into account. Consequently, if customers can be grouped in a route, CO_{2e} emissions of UAVs are higher than those of tricycles and electric vans. Yet, the study from Figliozzi is limited to the impact on climate change. The production of batteries to power drones will likely put significant pressure on the supply of materials, resulting in various additional impacts. The environmental assessment should cover different impact categories to mitigate the risk of environmental burdens shifting. Recently, academic studies have adopted a multi-criteria approach to address this risk [42–45]. Yowtak *et al.* conducted a life cycle assessment on multiple impact categories such as global warming potential, photochemical oxidant formation potential, and respiratory effects (particulate matter formation) [45]. For grocery deliveries, the authors conclude that UAVs have less impact on climate change than conventional vehicles, but

¹The carbon dioxide equivalent (CO_{2e}) is a metric used to compare the radiative forcing of various greenhouse gases, expressed as a multiple of the effect that would produce the same mass of CO₂.

greater than battery electric vehicles. However, particulate matter formation can be much higher for UAVs than conventional vehicles, especially if the electricity mix is based on coal.

A critical element of environmental assessment is the adequacy of the data used in the life cycle inventory. As pointed out by Kirschstein [38], the results of an LCA are based on several assumptions restricting their credibility and their domain of validity. Most of the works mentioned above use an average value for estimating the energy consumption per flight distance [36, 37, 43, 45]. However, using an average value does not capture the effects of varying UAV designs and flight conditions. Some authors have proposed models to describe the energy demand as a function of the UAV's mass and the flight path [38–41]. The models are fitted with either real-world data of existing UAVs or generic values. In both cases, the energy consumption models do not take into account the individual characteristics of the components, which may vary from one design to another. Consequently, it does not allow for investigating the effects of future technologies. The constitution of a bill of materials for exploring the production and end-of-life impacts of UAVs is even more case-specific. It is systematically based on data from existing UAVs. Therefore, current research is not capable of exploring the environmental impacts of specific UAV designs that are not yet developed.

8.4.2 Integration of environmental assessment in the preliminary design of aerospace vehicles

In order to adequately model how a change in a system's design will affect the environmental footprint, some authors have developed streamlined LCA tools that interface with design methodologies [22–24, 28–30, 46, 47]. Johanning and Scholz [24] proposed a simplified LCA methodology for future integration into the conceptual design of aircraft. The inventory analysis and the impact assessment are performed with a few equations representing the main processes of the life cycle. Dallara *et al.* [23] developed a tool to evaluate a variety of aircraft production and operation scenarios. The tool takes a few key inputs such as aircraft empty weight and fuel consumption, and returns cradle-to-grave air emissions, energy consumption and water consumption, among others. The tool's capability is utilized to assess existing aircraft as well as conceptual designs. André [30, 46] proposed a procedural framework for sizing eVTOL concepts and assessing their greenhouse gas emissions, with a cradle-to-grave approach. Fabre *et al.* [28] developed and integrated an LCA module into an aircraft design framework, FAST-OAD [48]. It enables to assess the environmental impacts of next-generation aircraft, such as those using sustainable aviation fuels. More recently, Bellier [47] presented the first steps for integrating LCA into a multidisciplinary framework for space launcher design.

In all of the aforementioned studies, the integration of LCA into design methodologies is limited to post-processing modules. As such, the environmental impacts are reduced to outputs that do not influence the design process. Specifically, the LCA results are not being used to automatically guide the design process towards the most eco-efficient solution

or to constrain the design exploration space. A possible reason why this aspect has not been addressed in aircraft design is that the vast majority of the environmental impact of commercial aircraft is directly related to their fuel consumption [28]. However, UAVs are electric vehicles with short lifetimes, and the production phase accounts for a significant share of the environmental burdens [41, 42, 44, 45]. Recent work by Duriez *et al.* [29] has introduced a CO₂ metric into the conceptual design of a solar-powered UAV and performed a design optimization with the objective of minimizing the CO₂ footprint. In particular, the airframe material was used as a design variable to minimize CO₂ emissions. The results show that, for the solar-powered fixed-wing UAV, the lightest material gives the lowest CO₂ footprint, as the reduction in overall UAV mass allows the selection of smaller solar panels and batteries. Therefore, minimizing the CO₂ footprint was found to be equivalent to minimizing the mass of the UAV. However, as previously mentioned, the study does not account for the grid electricity that would be consumed if the drone were not equipped with solar panels. This case study contrasts with that of commercial aircraft, where the reduction in emissions is mainly achieved by minimizing energy consumption. Most UAV concepts are likely to fall somewhere in between these two opposing cases. For a given set of specifications, the UAV with the lowest environmental impact may not be the most energy-efficient nor the lightest, but rather a trade-off between the two optimums.

8.5 Summary of the research gaps

The literature review highlights a lack of methodology for assessing the environmental impact of UAVs from the preliminary design phase. On the one hand, the majority of studies are limited to climate change indicators and to a single stage of the life cycle. Secondly, LCA studies are essentially applied to pre-defined designs. No exploration of the design space is proposed. More generally, no research has been identified that fully integrates the LCA methodology with design tools for aerospace systems. As a result, adopting an eco-design approach for UAVs is costly and time-consuming. If design engineers are to consider environmental issues, a shift is needed from a manual, iterative approach to an integrated, automated one. In a sense, this involves extending multidisciplinary analysis and optimization work to include the environmental discipline. Such a development would make it possible to explore the sensitivity of environmental impacts to technological assumptions or mission requirements. In addition, environmental impact criteria could become an objective and/or constraint in a UAV design optimization process.

Chapter 9

Environmental Life Cycle Assessments for the Design Exploration of UAVs

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9.1 Introduction

As outlined in Chapter 8, current research is not capable of exploring the environmental impacts of UAV designs that are not yet developed, nor of providing the means to minimize their impacts. To do so, it is necessary to develop a UAV-specific LCA model and link it to a preliminary design framework.

The present chapter introduces a design exploration of multirotor UAVs with a focus on environmental impacts, using a life cycle and multi-criteria approach. For this purpose, a parametric model for performing the LCA of multirotor UAVs is introduced. The LCA model is linked to an LCI database providing essential processes for the production and operation stages of the vehicle. Additionally, the model parameters, such as component masses and electricity consumption, are assigned based on the outputs of the sizing tool developed in Chapter 3. This novelty allows for the assessment of environmental impacts right from the preliminary design phase. Notably, it enables designers to evaluate the sensitivities of these impacts to mission requirements and technology assumptions, as well as optimizing the vehicle sizing to minimize environmental burdens. The effectiveness of the proposed methodology is demonstrated through a case study involving a parcel delivery mission.

The chapter is organized as follows. First, the goals and scope of the life cycle assessment are defined in Section 9.2. The data collection and implementation of the environmental module are presented in Section 9.3. Results for a reference delivery mission are provided in Section 9.4, and a method for analyzing the direct and indirect contributions of each component to environmental impacts is proposed in Section 9.5. Sensitivity of the environmental impacts to mission requirements and battery technologies are discussed in Section 9.6 and Section 9.7, respectively. Section 9.8 addresses the sizing optimization problem with an environmental objective. Finally, an assessment of absolute sustainability is proposed in Section 9.9 before the conclusion in Section 9.10.

9.2 Goal and scope definitions

This section introduces the LCA scope definition. First, the goal of the study and the functional unit are detailed. Then, the UAV modeling and the associated technological assumptions are presented. Finally, further information on the LCA is provided, including the process tree, system boundaries, and the environmental impacts considered for analysis.

9.2.1 Goal of the study

The goal of this study is to evaluate the environmental impacts associated with the entire life cycle of a multirotor UAV designed for a typical delivery mission. Specifically, the study focuses on identifying the main contributors to environmental burdens (Sections 9.4 and 9.5), assessing the sensitivity of the drone environmental performance to mission requirements

(Section 9.6) and technological choices (Section 9.7), and exploring opportunities for impact minimization through sizing and optimization (Section 9.8). Therefore, the study aims to provide a quantitative basis for decision-making on how to improve the sustainability of multirotor UAVs.

9.2.2 Functional unit

In the context of LCA, the functional unit represents a quantified description of the system and its performance. It provides a reference unit for the reporting of environmental impacts. In this study, the functional unit is intended to represent a typical UAV delivery application. The reference application, from which the functional unit is derived, is defined as follows:

Deliver a 2 kg payload within a 7.5 km radius, at the rate of 5 deliveries per day, 260 days per year, for a period of 2 years.

However, when comparison with alternative scenarios is required, the results will be expressed per kilogram.kilometer (kg.km), representing the transportation of 1 kg of payload over 1 km.

Specifically, a mission starts with a take-off before initiating a vertical climb at a rate of 3 m/s to an altitude of 120 meters, which is the maximum flight height generally authorized by EASA [49]. It is followed by a level flight over a distance of 7.5 km to the delivery area. This is slightly higher than Matternet's five-kilometer delivery route launched in 2022 [50], but below the announced maximum radius of 10 km (20 km total distance) for their M2 quadcopter [51, 52], which received FAA type certification in 2022 [3]. The choice of cruising speed typically depends on optimum energy consumption, which is linked to parameters such as the drone mass and aerodynamic performance, as shown in Chapter 3. However, more stringent considerations could be considered for deliveries, such as safety, cost, and delivery time [53]. In the present work, the cruising speed is assumed to be fixed at 15 m/s. The cruise phase is followed by a hover phase to deliver the parcel. A hover time of 5 minutes is chosen, including a safety margin. Finally, the 7.5 km return flight is performed empty, without the parcel. It is assumed that the drone makes an average of 5 deliveries daily, throughout the 260 working days of a year. The 2-year lifespan of the UAV is in line with the arrival of the next generation of UAVs every two years, leading to the retirement of older products [54]. In total, the delivery rate represents a lifetime of 2,600 missions, also referred to as cycles in the following. Table 9.2.1 summarises the main choices for the functional unit. Given the wide variability of values found in the literature [41, 42, 45, 53], ranges of variation used for subsequent sensitivity analyses are also presented.

9.2.3 System modeling and assumptions

The system consists of a hexacopter UAV with coaxial rotors. This choice is consistent with the results of the controllability assessment of safety-critical UAVs presented in Chapter 7.

Table 9.2.1 Assumptions and sensitivity analysis ranges for the functional unit definition

Functional unit parameter	Reference value	Range of variation
Payload mass	2 kg	[0.5 - 5] kg
Delivery radius	7.5 km	[3 - 12.5] km
Cruise speed	15 m/s	-
Cruise altitude	120 m	-
Lifespan	2,600 cycles	[0 - 10,000] cycles

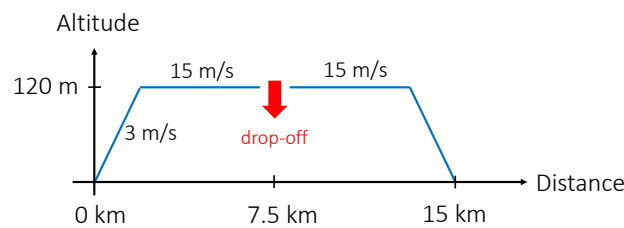


Figure 9.2.1 Flight profile for the reference mission.

Indeed, the coaxial hexacopter concept was shown to be robust to double rotor failures while maintaining a reasonable mass compared to other concepts of multirotor UAVs (e.g. octocopters). This is of major importance as the delivery mission may occur in densely populated areas.

The sizing of the UAV and its components is obtained using the methodology presented in Chapter 3. Specifically, the UAV is optimized with respect to mission requirements and technology assumptions (e.g. battery energy density) while minimizing its mass. Here, the mission requirements correspond to the operational mission defined by the functional unit. This is an optimistic case, as in practice UAVs are oversized compared to the actual missions they perform. Appendix 9.A lists the technology assumptions used for fitting the models representing the main components. In addition to the airframe and propulsion system components, an additional 1 kg of mass is included to model the payload attachment and release mechanism but is not considered in the life cycle inventory. The reference lithium-ion battery features NMC chemistry for the cathode — a mix of lithium, nickel, manganese and cobalt. It is expected to last 400 cycles before replacement¹, and its combined charge-discharge efficiency caused by internal inefficiencies is assumed to be 85%.

9.2.4 Sensitivity scenarios

The reference scenario, described by the reference values in Table 9.2.1 and the above system definition, is complemented by alternative scenarios for the purposes of sensitivity analyses and robustness of the results. First, the robustness of the results to the assumptions underlying the

¹DJI Matrice 30 Series TB30 Intelligent Flight Battery.

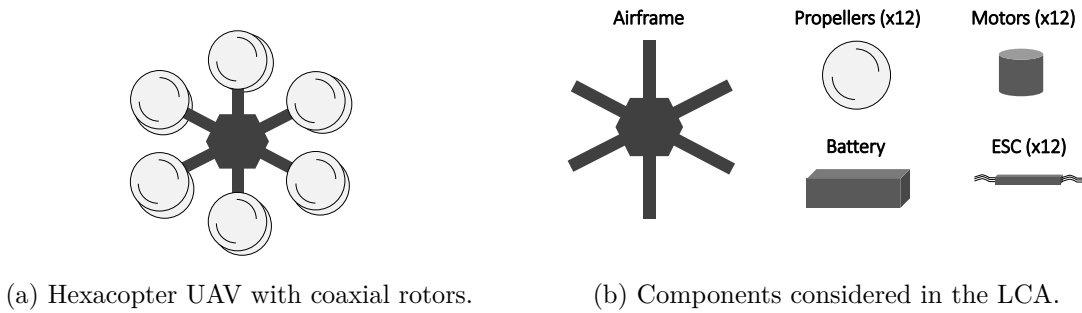


Figure 9.2.2 Conceptual view of the hexacopter UAV and its components.

functional unit is studied. Both UAV lifespan and mission requirements are varied to assess their influence on environmental performance. Specifically, for the mission requirements, the payload mass and delivery radius are varied in accordance with Table 9.2.1. By applying the optimal sizing methodology presented in Chapter 3, the variation of these parameters induces different design alternatives, thereby influencing the environmental impacts. The results are provided in Section 9.6. Then, two additional scenarios are introduced to allow a comparison of alternative battery chemistries with varying energy densities and lifetimes. The results, presented in Section 9.7, allow identifying the most promising areas for future technological developments. Finally, the influence of the optimization objective used in the sizing optimization of the UAV is assessed. The mass minimization objective is compared with an in-flight energy objective, and the introduction of an environmental function makes it possible to obtain the sizing that has the least impact on the environment. A detailed comparison of the three optimization objectives is presented in Section 9.8.

9.2.5 Process tree

In LCA, a process tree is a hierarchical representation of the processes and flows required to fulfill the functional unit being assessed. Figure 9.2.3 provides an overview of the materials and energy required to fulfill the delivery application. The foreground and background processes are distinguished. Foreground processes refer to the processes within the scope of the UAV design phase (e.g. the components' masses). On the other hand, background processes include processes that are not under the direct control of the designer and that precede or follow the sizing process. For example, the propeller manufacturing process is beyond the designer's control. The background processes are defined by generic data obtained directly from the EcoInvent 3.9 database [55]. Detailed information on the EcoInvent processes used in this study is provided in Section 9.3. The achievement of the functional unit requires a UAV and electricity from the grid to ensure its operation. The UAV is composed of several components, each of which has a mass determined by the sizing methodology presented in Chapter 3. In addition, the cumulative mass of batteries required to fulfill the

functional unit depends on their replacement rate, which is defined by their specific lifetime and the number of mission executions. All these quantities, known as reference flows, differ according to the scenarios previously introduced. Indeed, the flows quantifying the amount of components and electricity depend on the assumptions underlying the delivery mission, the UAV lifespan and the battery chemistry used in the design. The parameters used for building the process tree are described in Table 9.2.2.

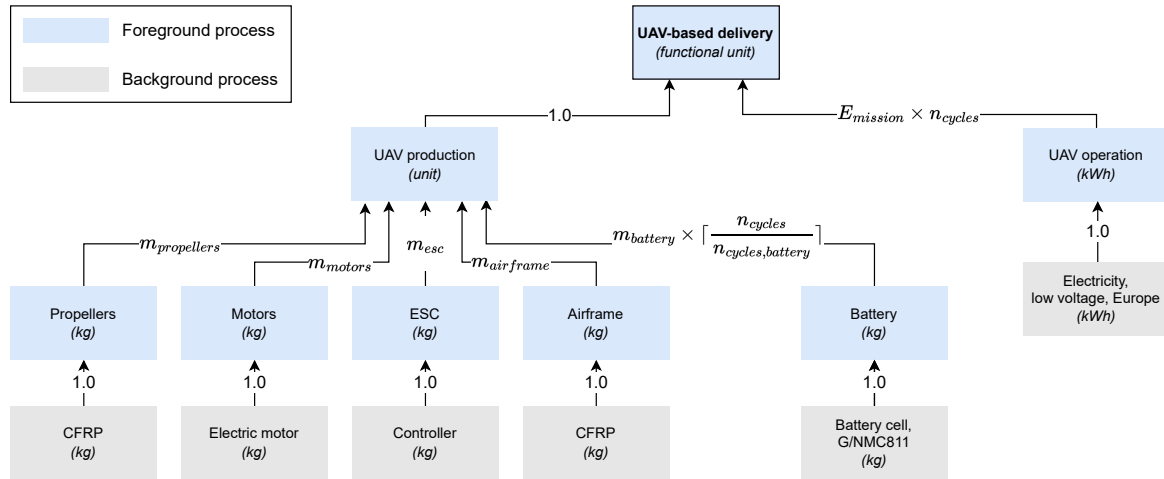


Figure 9.2.3 Process tree for the delivery application, over the entire lifecycle of the UAV.

Table 9.2.2 Parameters for modeling the reference flows

Parameter	Unit	Description
$m_{propellers}$	kg	Mass of propellers
m_{motors}	kg	Mass of motors
m_{esc}	kg	Mass of electronic speed controllers
$m_{battery}$	kg	Mass of battery
$m_{airframe}$	kg	Mass of airframe
$E_{mission}$	kWh	Energy from electrical grid consumed over a single mission
n_{cycles}	-	Number of times the mission is performed
$n_{cycles,battery}$	-	Number of charge-discharge cycles before replacing battery

9.2.6 System boundaries

The scope of the study encompasses the production and operation (or, use) stages of the UAV. The production stage includes processes such as materials extraction and processing, components manufacturing, and transport of the components to the assembly facility. Additionally, the study considers the electricity consumed during the use stage, specifically the energy required for the flights. However, certain aspects are excluded from the assessment due to limited data availability and challenges with allocation rules. These exclusions include

the assembly of the components, the delivery or transport of the UAV to the operating area, and the end-of-life treatment of the components. Furthermore, the study does not include ground infrastructure and ancillary systems such as battery chargers, nor the parcel and its release mechanism.

9.2.7 Environmental impact categories

The impact assessment relies on the EU Product Environmental Footprint (PEF) method [16] which covers 16 environmental impacts, including but not limited to climate change, resource depletion, and human toxicity. The PEF provides a set of characterization factors for each impact, allowing the Life Cycle Inventory, described in the following section, to be mapped to each specific environmental impact. Following EF's recommendations, the impacts are then normalized against reference values to better reflect their relative importance. The reference values used in the PEF consider the average environmental pressures of one person over one year [56]. Finally, the results are weighted and aggregated to obtain a single score [57]. This facilitates comparison of the results, for example when assessing different scenarios.

9.3 Data collection and implementation

9.3.1 Life Cycle Inventory

In this study, data collection for processes supporting the LCI is based exclusively on the EcoInvent database version 3.9 [55], although the lack of specific data for UAVs requires some simplifying assumptions. The main choices that were made for building the LCI are provided here. The propellers are assumed to be made entirely of Carbon Fiber Reinforced Plastic (CFRP), similar to the carbon propellers manufactured by APC². The composition of the brushless motors used in the UAV is assumed to be equivalent to that of electric scooter motors. Similarly, the electronic speed controller (ESC) is represented by an electric scooter controller. The airframe, which consists essentially of the body and the arms supporting the rotors, is made entirely of CFRP. The battery technology chosen for the reference scenario is based on a G/NMC811 chemistry, with a graphite anode and a lithium-based cathode containing nickel, manganese and cobalt in an 8:1:1 ratio. Additional battery chemistries are introduced in Section 9.7. The manufacturing processes for all these components include the extraction, transport and transformation of raw materials as well as the delivery of the manufactured component. The average global production (denoted by *GLO*) is chosen for the production of the components. This ensures general applicability and best reflects the current market. However, future research could explore the possibility of providing region-specific results. Finally, operations at the European level are considered. The battery of the UAV is therefore powered by the European electricity mix. Table 9.3.1 provides the EcoInvent

²APC Propellers: <https://www.apcprop.com/>

processes corresponding to these assumptions. From the process tree shown in Figure 9.2.3 and the EcoInvent processes listed in Table 9.3.1, it is possible to build the LCI and to determine the quantities of emissions to and extractions from the environment for each scenario. These cumulative exchanges are essential for evaluating the damages across the 16 impact categories using the characterization factors provided in the PEF method.

Table 9.3.1 EcoInvent 3.9 processes for the LCI.

Component	Unit	EcoInvent processes
Propeller	kg	Market for carbon fibre reinforced plastic, injection moulded [GLO]
Motor	kg	Market for electric motor, for electric scooter [GLO]
ESC	kg	Market for controller, for electric scooter [GLO]
Battery	kg	Market for battery cell, Li-ion, NMC811 [GLO]
Airframe	kg	Market for carbon fibre reinforced plastic, injection moulded [GLO]
Energy consumption	kWh	Market group for electricity, low voltage [EU]

Note: Terminology [LOC] refers to the geographic location: Europe without Switzerland (EU) and Global (GLO).

9.3.2 Implementation

For the purpose of this study, a specific environmental module was developed and linked to the design optimization framework introduced in Chapter 3. By efficiently sizing the UAV from a set of specifications, the process yields key component characteristics, such as mass, along with in-flight energy consumption. These outputs, which are essential for conducting the LCA, link the design process to the environmental module. Figure 9.3.1 represents the integration of the environmental module with the sizing optimization, using the XDSM notation [58]. The sizing framework and the environmental module are depicted as green boxes, while parallelograms represent data inputs and outputs. Data connections are represented by vertical lines for inputs and horizontal lines for outputs. The environmental module consists of a 3-step procedure, as shown in Figure 9.3.2.

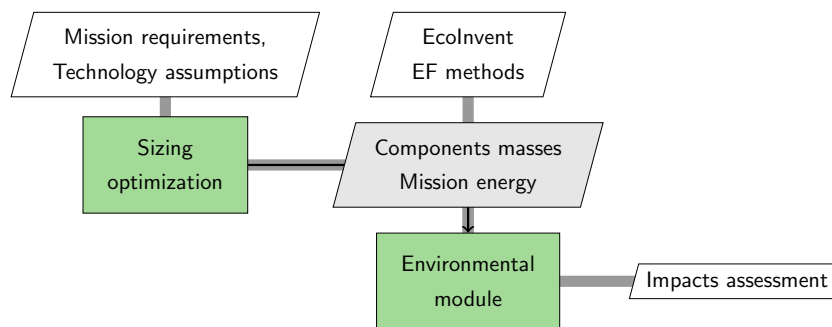


Figure 9.3.1 Overview of the methodology using the XDSM notation.

The first step consists of building a parametric model for the life cycle inventory, as represented by the process tree in Figure 9.2.3. For this purpose, the background processes are imported from the EcoInvent database, and the parameters for modeling the reference flows are created. The functional unit is defined as a parameterized combination of the foreground processes, which are themselves expressed by the background processes. In this approach, the inventory can be expressed in the form of a symbolic expression referring to the background processes:

$$\begin{aligned} \text{Inventory} &= f(P_i, B_j) \\ &= m_{propellers} \times B_{propeller} + m_{motors} \times B_{motor} + m_{esc} \times B_{controller} \\ &\quad + m_{airframe} \times B_{airframe} + m_{battery} \times \left[\frac{n_{cycles}}{n_{cycles,battery}} \right] \times B_{battery} \\ &\quad + E_{mission} \times n_{cycles} \times B_{electricity} \end{aligned} \quad (9.1)$$

where P_i are the parameters from Table 9.2.2, and B_j are the background processes introduced in Table 9.3.1. The parametric inventory is created once and for all, and is common to all life cycle assessments carried out in this study.

In the second step, the parameters' values are assigned from the outputs of the sizing optimization, which differ depending on the mission requirements and technology assumptions. Additionally, LCA-specific variables (e.g. UAV lifespan) are introduced. This second step is crucial in linking the LCA to the design framework.

Finally, the parametric inventory and the assigned parameters are fed into the sub-module for impact assessment. The impacts are calculated in two stages, as proposed by Jolivet [59]. During instantiation, i.e. the first time the sub-module is called, the impacts of each background process are computed. This initial calculation, which requires solving a linear system of equations (e.g. with matrix inversion), is time-consuming due to the extensive data associated with the background processes [60]. The impacts are then substituted into the parametric inventory, allowing the creation of algebraic formulas for each impact that involve only the parameters P_i [59]:

$$\text{Impact}_k = f_k(P_i) \quad (9.2)$$

Therefore, the subsequent evaluations of the impacts are processed in a very short time. This is important when evaluating a large number of scenarios (e.g. for sensitivity analyses) and when performing optimizations where numerous iterations and function evaluations are required to converge on the optimal solution, as presented in Section 9.8. Lastly, the results for each impact category are normalized and weighted on the basis of the factors introduced by the PEF [16]. As recommended for PEF studies, a single score aggregating all the impacts is also returned.

The environmental module is fully integrated with the design framework developed in Chapter 3. It leverages the capabilities of *Brightway* for performing LCA in the Python

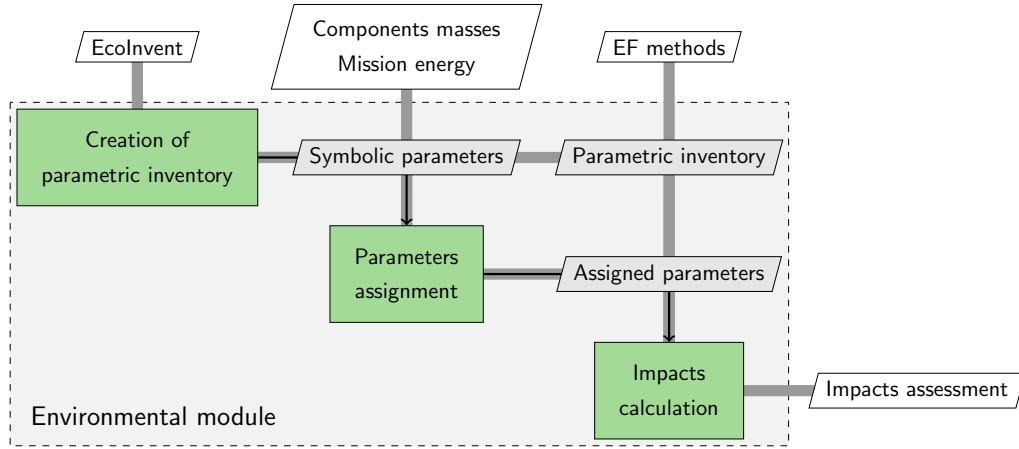


Figure 9.3.2 Representation of the environmental module using the XDSM notation.

programming language [61]. The implementation of parameterized models for the LCA is enabled by the *lca_algebraic* library [59]. This layer on top of *Brightway* brings symbolic calculus to LCA, which is essential for building the parametric inventory and supporting fast processing speeds of the impact assessment. The initialization of the environmental module, which involves the calculation of the background processes, is performed in 176.2 seconds on a 1.8 GHz quad-core processor (Intel Core i7-10610U). At runtime, the calculation for all 16 impact categories takes an average of 0.095 seconds.

9.4 Results for the reference scenario

The main results of the environmental study for the reference scenario are presented here. The key design parameters of the UAV, obtained from the specifications through the sizing optimization, are provided in Table 9.4.1. The single score for the design thus obtained is equal to 0.095 points over the entire life cycle, or 2.4×10^{-6} points/kg.km.

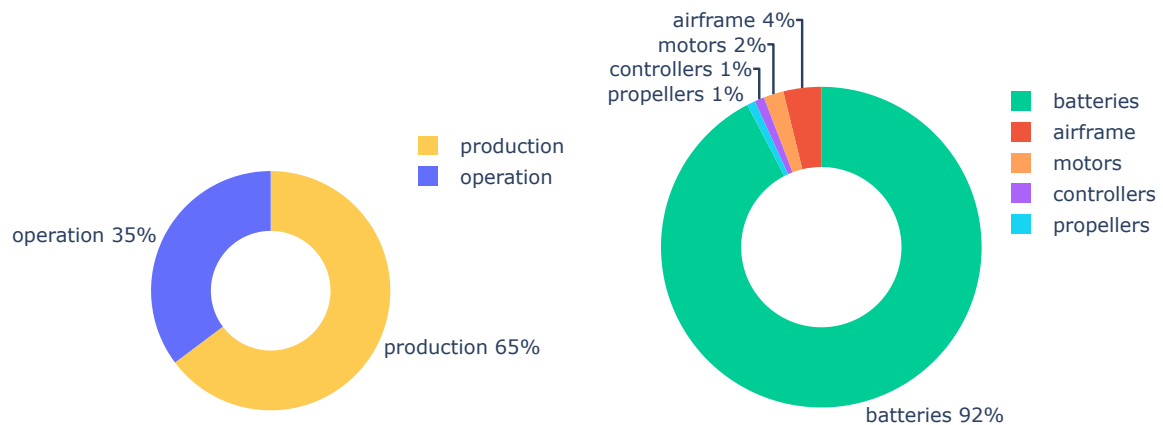
Figure 9.4.1 provides more information on the contributions to this single score. The most critical stage is the production of the components, which accounts for 65% of the total environmental footprint. For this stage, battery production represents more than 90% of the impacts. This is due to the battery being replaced every 400 cycles, i.e. 6 times during the 2 years of use. The contributions of the airframe, propellers, motors and electronic speed controllers are negligible, particularly due to their low mass. Early replacement of these components, for example in the event of a malfunction, would not significantly affect the results.

As shown in Figure 9.4.2, the most relevant impact categories are resource use — both metals and minerals, and non-renewable energy — and climate change. These categories account for 37%, 13%, and 14% of the total impacts, respectively. The operation stage

Table 9.4.1 Results of the sizing optimization with a mass minimization objective for the coaxial hexacopter, for the reference scenario.

UAV total mass *	5.62 kg	Battery mass	1.69 kg
Motors mass	0.42 kg	Airframe mass	0.35 kg
Propellers mass	0.08 kg	Arms length	0.21 m
Propellers diameter	0.21 m	Mission energy	257 Wh
ESC mass	0.08 kg	Mission duration	23 min

* including payload mass.



(a) Production and operation stages.

(b) Components production and renewal.

Figure 9.4.1 Contributions to the single overall score for the reference scenario.

is particularly important regarding climate change and non-renewable energy use, as it represents more than half of the impacts in these categories (Figure 9.4.3). The consequences are twofold. Firstly, the transition to a low-carbon electricity mix is essential for operating UAVs. Secondly, reducing energy consumption is a key factor for mitigating the impact on climate change, provided that it is not accompanied by a transfer of the impact to the production stage. Finally, reducing the depletion of metals and minerals requires batteries that consume fewer critical compounds, last longer so they have to be replaced less frequently, or can be easily recycled. Again, particular attention must be paid to the transfer of impacts between the production and operation stages due to changes in battery technology. This point is covered in Section 9.7.

As pointed out in Chapter 8, uncertainty inherited in LCA steps is a key factor influencing the reliability of the findings. We provide here an analysis of the uncertainty resulting from variability in the inventory data. For this purpose, a Monte Carlo experiment with 1,000 samples was carried out on the data provided by the EcoInvent 3.9 database. Uncertainty on the parameters describing the UAV design and uncertainty on the characterization factors are not considered. Future work should include these aspects, for instance by combining

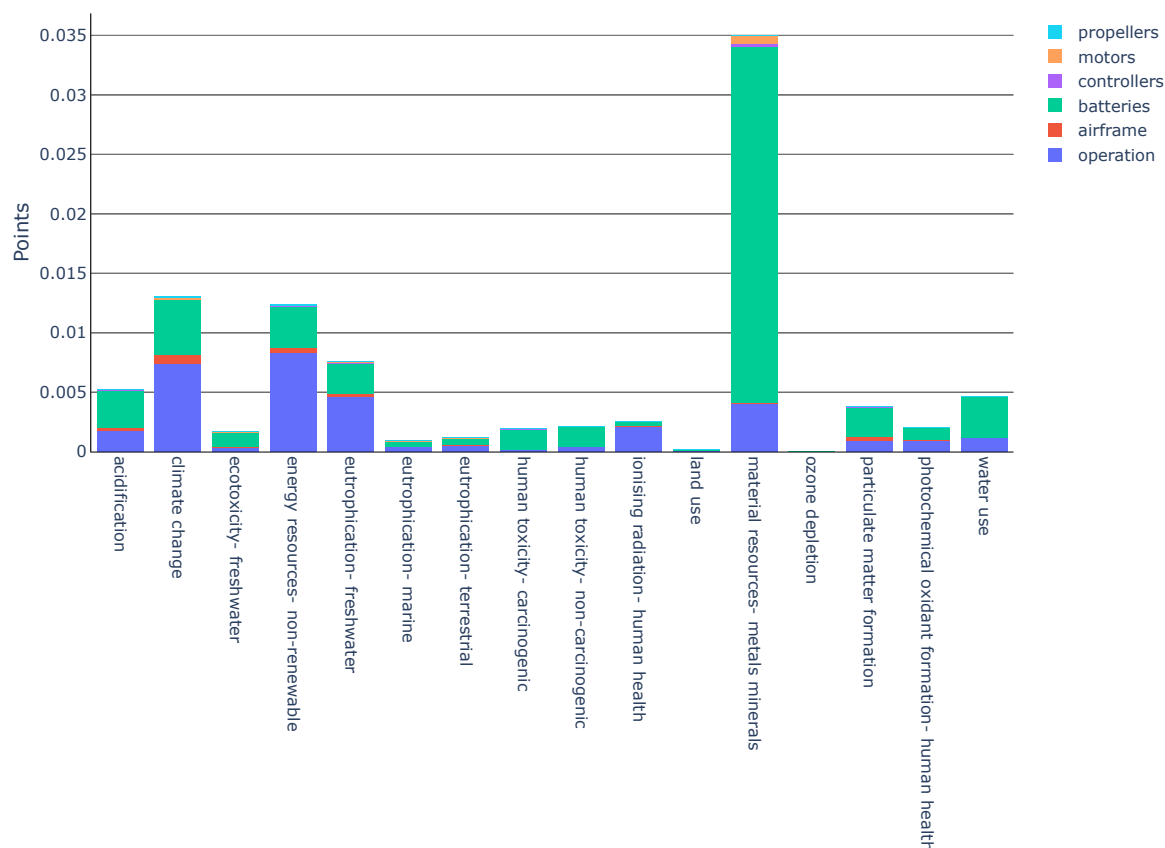


Figure 9.4.2 Normalized and weighted scores for the reference scenario.

the design uncertainties obtained in Chapter 5 with the LCA study. The results of the uncertainty analysis for each of the 16 characterized impacts are presented in Figure 9.4.4 and Table 9.4.2.

A lack of consistency between the static value used in the deterministic approach and the statistical mean of the uncertainty analysis is observed for some impact categories. This is due in particular to static calculations using the median as the most representative value of a distribution [55]. For the most common distributions found in the EcoInvent database, typically lognormal distributions, the mean value differs from the median value. Consequently, the mean obtained from a Monte Carlo experiment involving a large number of lognormal distributions is biased towards higher values compared to the static calculation result. It should also be noted that EcoInvent v3.9 was reported to have issues with lognormal distributions. As this may affect the results of the uncertainty analysis, future work should migrate to a corrected version of EcoInvent.

Finally, a very high coefficient of variation is observed for three impact categories, namely freshwater ecotoxicity and human toxicity — both carcinogenic and non-carcinogenic.

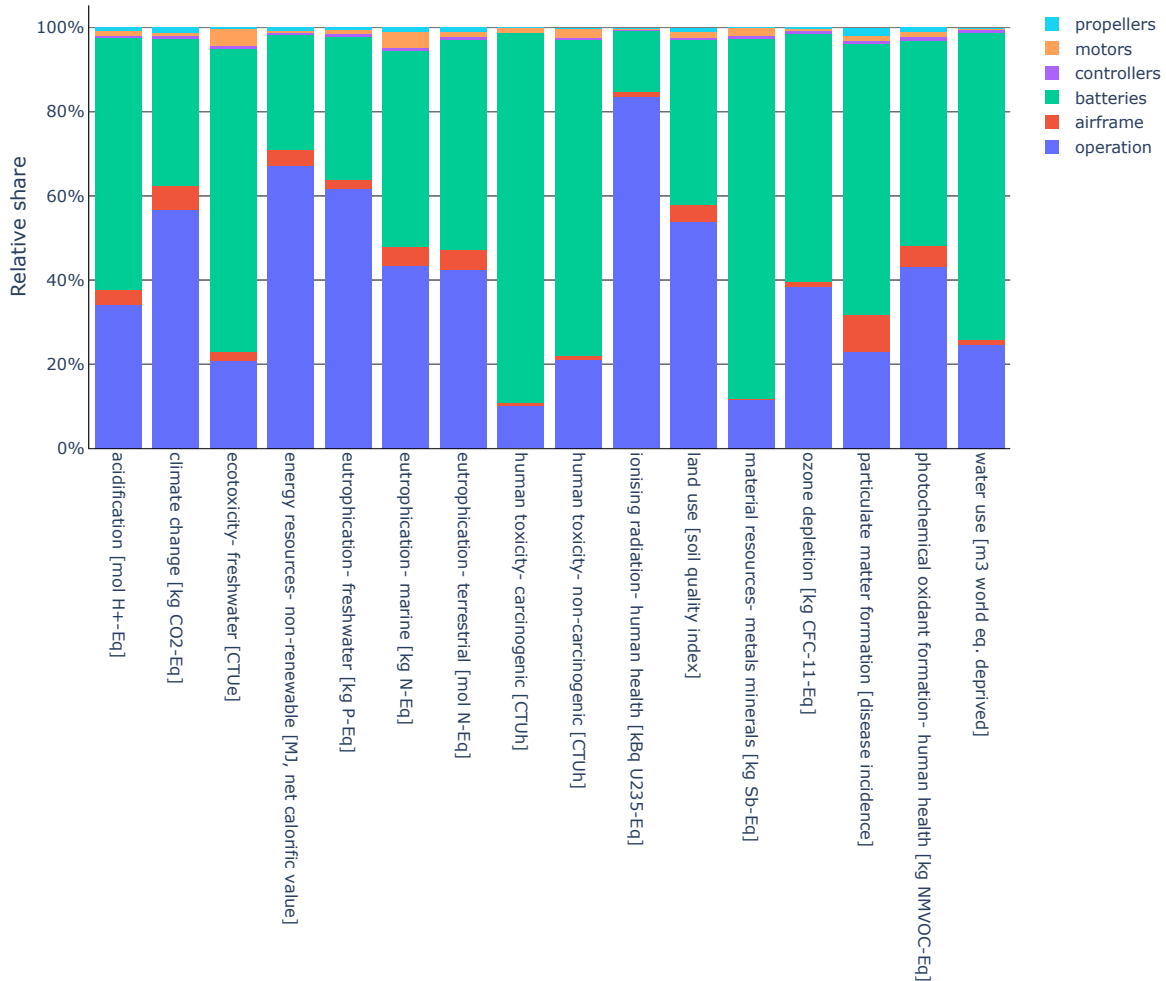


Figure 9.4.3 Relative contributions to each impact category for the reference scenario.

Interestingly, these impact categories are also the ones with the lowest weighting factors in the PEF method: 1.92%, 2.13% and 1.84%, respectively. This finding is consistent with the development of the PEF weighting approach, in which impact categories with more reliable results are given greater weight than the impact categories with less robust results [57]. Here again, switching to a corrected version of EcoInvent may reduce the level of uncertainty.

9.5 Hotspot analysis: contributions of components masses, efficiencies and productions to the environmental score.

The results presented in Section 9.4 indicate that the operation stage is responsible for a significant share of the environmental burdens. Minimization of the impacts occurring at this stage can be achieved in three distinct ways, each involving different stakeholders. A

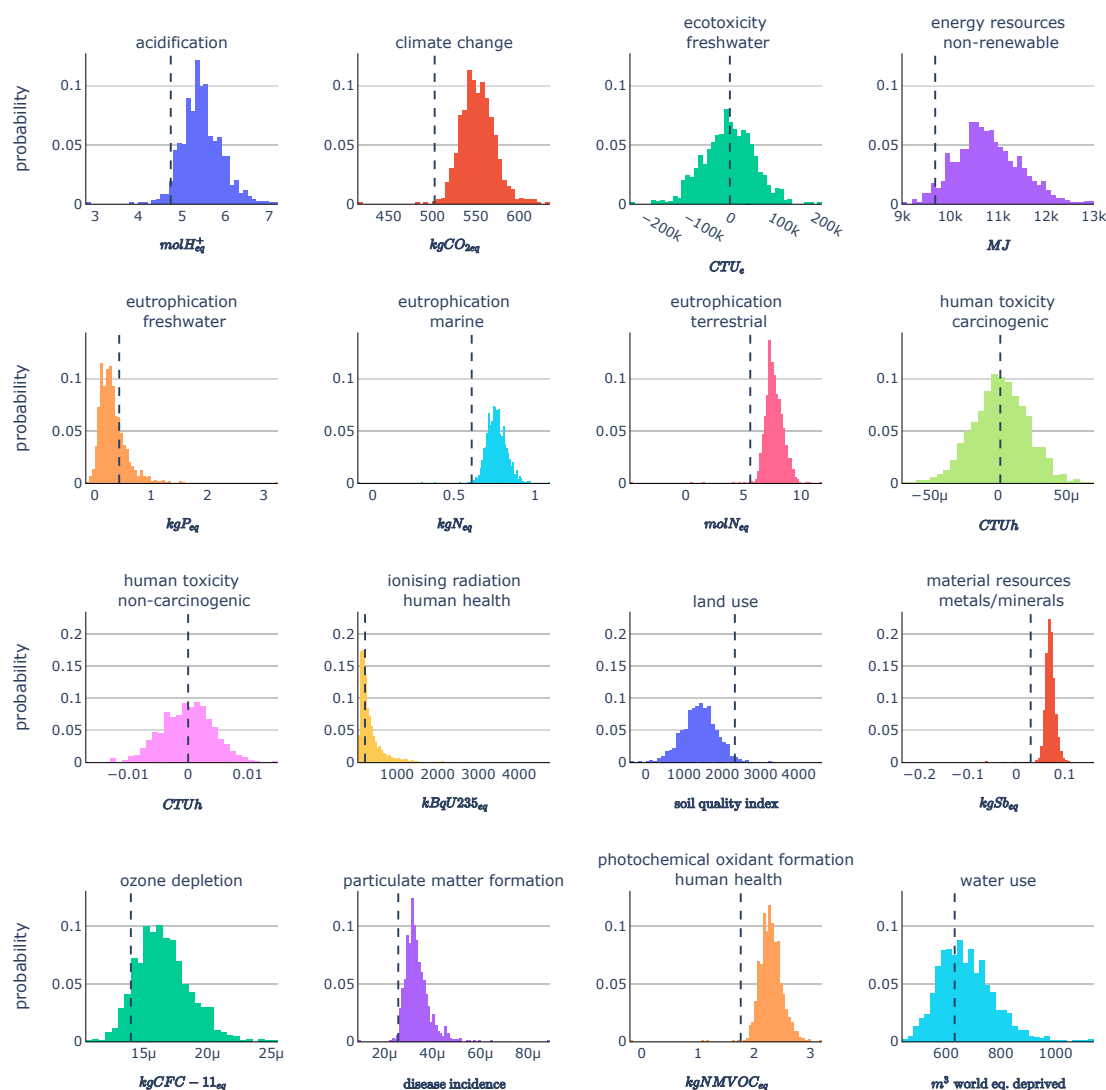


Figure 9.4.4 Uncertainty analysis obtained with 1,000 Monte Carlo simulations on the LCI data, for the reference scenario. Dashed vertical lines represent the static values used in the deterministic analysis.

cleaner electricity mix (e.g., with lower carbon intensity) can be used to power the flights. Alternatively, the flight path can be optimized to save energy. Finally, the design of the UAV can be improved to reduce energy consumption. Of the three areas for improvement, only the last falls within the designer's remit. However, identifying the most relevant improvements for the design is not an easy task. Energy consumption during flight depends on several parameters, including the mass of the components and the efficiency of the propulsion chain. We propose here a method for attributing in-flight energy consumption to the respective masses and efficiencies of the UAV components. By combining this analysis with the

Table 9.4.2 Uncertainty analysis obtained with 1,000 Monte Carlo simulations on the LCI data, for the reference scenario.

Impact category	Unit	Static value	Mean μ	Standard deviation σ	Coefficient of variation σ/μ
Acidification	molH ⁺ _{eq}	4.7	5.5	4.5×10^{-1}	8%
Climate change	kgCO _{2eq}	5.0×10^2	5.5×10^2	2.0×10^1	4%
Ecotoxicity: freshwater	CTU _e	3.8×10^3	1.1×10^3	5.9×10^4	5515%
Energy resources: non-renewable	MJ	9.7×10^3	1.1×10^4	6.5×10^2	6%
Eutrophication: freshwater	kgP _{eq}	4.3×10^{-1}	3.2×10^{-1}	2.5×10^{-1}	79%
Eutrophication: marine	kgN _{eq}	6.1×10^{-1}	7.6×10^{-1}	6.9×10^{-2}	9%
Eutrophication: terrestrial	molN _{eq}	5.6	7.7	8.9×10^{-1}	12%
Human toxicity: carcinogenic	CTU _h	1.5×10^{-6}	2.0×10^{-6}	2.0×10^{-5}	1000%
Human toxicity: non-carcinogenic	CTU _h	2.7×10^{-5}	-1.3×10^{-4}	4.4×10^{-3}	-3275%
Ionising radiation: human health	kBq U235 _{eq}	2.1×10^2	3.6×10^2	3.8×10^2	105%
Land use	soil quality index	2.3×10^3	1.4×10^3	4.8×10^2	34%
Material resources: metals/minerals	kgSb _{eq}	3.0×10^{-2}	6.9×10^{-2}	1.6×10^{-2}	23%
Ozone depletion	kgCFC-11 _{eq}	1.4×10^{-5}	1.7×10^{-5}	2.0×10^{-6}	12%
Particulate matter formation	disease incidence	2.6×10^{-5}	3.4×10^{-5}	5.0×10^{-6}	15%
Photochemical oxidant formation	kgNMVOC _{eq}	1.8	2.3	2.1×10^{-1}	9%
Water use	m ³ world eq. deprived	6.3×10^2	6.7×10^2	1.0×10^2	15%

production-related impacts, the aim is to identify the pathways through which a component generates environmental damages.

9.5.1 Hotspot Analysis Method

An analytical approach is presented here to attribute the environmental impact of a multirotor UAV to the masses, efficiencies and production of its components. Although the proposed method applies to multirotor UAVs and, more generally, to rotorcraft, it can easily be extended to other systems using the appropriate governing laws of physics.

The environmental impacts of a UAV are related to its in-flight energy consumption and the production of its components:

$$EI = \sum_i c_{m,i} m_i + c_e E_{operation}, \quad (9.3)$$

where EI is the environmental impact (e.g., climate change), m_i are the components' masses and $E_{operation}$ is the energy consumed during the operation stage. The characterization factors c_i represent the impact intensity of each element. The energy consumption is directly related to the power requirements for the flight mission, its duration and the number of times the mission will be completed during the UAV lifetime:

$$E_{operation} = n_{cycles} \cdot E_{mission} = n_{cycles} \cdot \int_0^{t_{mission}} P(t) dt. \quad (9.4)$$

For simplicity, the power is assumed to be time-independent in the following analysis and is derived considering forward flight conditions.

Equations of power

For multirotor UAVs, the propulsive power, which excludes losses in the propulsion chain,

can be determined by applying the principle of conservation of energy [62]:

$$P_{propulsive} = F(v_i + V_\infty \sin \alpha) = Fv_i + DV_\infty, \quad (9.5)$$

where α is the angle of attack of the propeller disk and $D = \frac{1}{2}\rho SV_\infty^2 C_D$ is the aerodynamic drag. In forward flight, Glauert's high speed approximation gives a simple expression of the thrust F as a function of the air density ρ , the propellers' disk area A_{pro} , the free-stream velocity V_∞ and the induced velocity v_i [62]:

$$F \approx 2\rho A_{pro} v_i V_\infty \quad (9.6)$$

Replacing the disk area by the number of propellers N_{pro} and their diameter d_{pro} gives

$$v_i = \frac{2F}{\pi \rho d_{pro}^2 N_{pro} V_\infty} \quad (9.7)$$

Then, projection of the forces acting on the UAV yields $F^2 = (m_{uav}g)^2 + D^2$. Finally, the propulsive power is expressed as

$$P_{propulsive} = \frac{2g^2}{\pi \rho V_\infty d_{pro}^2 N_{pro}} m_{uav}^2 + \frac{1}{2} \rho C_D S V_\infty^3 \left(1 + \frac{C_D S}{\pi d_{pro}^2 N_{pro}} \right) = A m_{uav}^2 + B_1 C_D + B_2 C_D^2 \quad (9.8)$$

Where A , B_1 and B_2 are constants depending on the operating conditions (e.g., airspeed) and the drone geometry (propeller diameter and airframe area). The propulsive power is divided by the total efficiency η of the propulsion chain to obtain the actual power consumed from the power grid:

$$P = \frac{P_{propulsive}}{\eta} \quad (9.9)$$

Replacing with the individual efficiencies of the propulsion chain η_i and the components' masses m_i yields:

$$m_{uav} = m_{pay} + \sum_i m_i, \quad \eta = \prod_i \eta_i, \quad (9.10)$$

$$P(m_i, \eta_i, C_D) = \frac{1}{\prod_i \eta_i} \left[A \left(m_{pay} + \sum_i m_i \right)^2 + B_1 C_D + B_2 C_D^2 \right] \quad (9.11)$$

Where m_{pay} is the payload mass.

Approximation with Taylor polynomials

As expressed by Equation 9.11, power consumption can be attributed to three main factors: the masses of the components (m_i), the efficiencies of the components within the propulsion chain (η_i), and the aerodynamic drag, assumed to be essentially related to the airframe

drag coefficient (C_D). To further capture the contribution of each factor (mass, efficiency, drag) and each component (airframe, propeller, motor, controllers, battery) to the power consumption, the power expression is approximated by a Taylor polynomial around a reference point $P_0(m_{i,0}, \eta_{i,0}, C_{D,0})$:

$$\begin{aligned}
 P(m_i, \eta_i, C_D) = & P_0 \Rightarrow \text{Order 0} \\
 & + \sum_i \frac{\partial P_0}{\partial m_i} (m_i - m_{i,0}) + \sum_i \frac{\partial P_0}{\partial \eta_i} (\eta_i - \eta_{i,0}) + \frac{\partial P_0}{\partial C_D} (C_D - C_{D,0}) \Rightarrow \text{Order 1} \\
 & + \frac{1}{2} \left(\sum_i \frac{\partial^2 P_0}{\partial m_i^2} (m_i - m_{i,0})^2 + \sum_i \frac{\partial^2 P_0}{\partial \eta_i^2} (\eta_i - \eta_{i,0})^2 + \frac{\partial^2 P_0}{\partial C_D^2} (C_D - C_{D,0})^2 \right. \\
 & + \sum_{i \neq j} \frac{\partial^2 P_0}{\partial m_i \partial m_j} (m_i - m_{i,0})(m_j - m_{j,0}) + \sum_{i \neq j} \frac{\partial^2 P_0}{\partial \eta_i \partial \eta_j} (\eta_i - \eta_{i,0})(\eta_j - \eta_{j,0}) \\
 & + \sum_{i,j} \frac{\partial^2 P_0}{\partial m_i \partial \eta_j} (m_i - m_{i,0})(\eta_j - \eta_{j,0}) + \sum_i \frac{\partial^2 P_0}{\partial m_i \partial C_D} (m_i - m_{i,0})(C_D - C_{D,0}) \\
 & \left. + \sum_i \frac{\partial^2 P_0}{\partial \eta_i \partial C_D} (\eta_i - \eta_{i,0})(C_D - C_{D,0}) \right) \Rightarrow \text{Order 2} \\
 & + \text{higher order terms}
 \end{aligned} \tag{9.12}$$

The order of the Taylor polynomial is chosen to ensure sufficient accuracy in the approximation. In this study, a 2nd order approximation yields a 26% error and is deemed acceptable. The reference point P_0 in Equation 9.12 is chosen to be an ideal design, that is:

$$\begin{cases} m_{1,0} = \dots = m_{n,0} = 0 \\ \eta_{1,0} = \dots = \eta_{n,0} = 1 \\ C_{D,0} = 0 \end{cases} \tag{9.13}$$

And $P_0 = A m_{pay}^2 = \frac{2g^2}{\pi \rho V_\infty d_{pro}^2 N_{pro}} m_{pay}^2$.

Finally, a specific allocation rule is defined to attribute the 2nd order cross-terms half to their respective contributors. The remaining term P_0 , which represents the minimum theoretical power to lift a payload of mass m_{pay} with a given propeller disk area, is left unallocated. In this study, its contribution to the final power consumption was found equal to 4.1%. That is, the vast majority of the power demand is due to the non-ideal nature of the components. The contributions of each component to the final power consumption are summarized in Table 9.5.1. These contributions are introduced in Equation 9.4 and, ultimately, in Equation 9.3 to allow comparison of the different factors affecting the environmental impacts.

Future work could further develop this hotspot analysis method to identify the tipping points at which the integration of a technology could be considered beneficial for the

Table 9.5.1 Contributions to power consumption and environmental impacts for the i^{th} component.

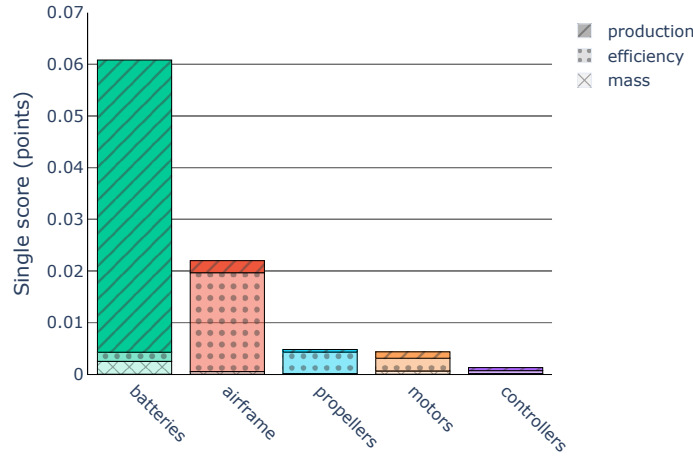
Factor	Power consumption	Environmental impact
Mass	$P_{mass,i} = Am_i \left[m_{pay} + m_{uav} - \frac{1}{2}m_{pay} \sum_j (\eta_j - 1) \right]$	$EI_{mass,i} = c_e P_{mass,i} n_{cycles} t_{mission}$
Efficiency	$P_{eff,i} = \frac{Am_{pay}^2}{2} (1 - \eta_i) \left[2 - \eta_i + \sum_j (1 - \eta_j) + \frac{m_{uav}}{m_{pay}} \right] + \frac{B_1}{2} (1 - \eta_i)$	$EI_{eff,i} = c_e P_{eff,i} n_{cycles} t_{mission}$
Aerodynamics	$P_{aero} = B_1 C_D + B_2 C_D^2 - \frac{B_1}{2} C_D \sum_j (\eta_j - 1)$	$EI_{aero,i} = c_e P_{aero,i} n_{cycles} t_{mission}$
Production	—	$EI_{prod,i} = c_{m,i} m_i$

environment (e.g., lighter but less efficient motor). This will require comparing the equations relating to the mass, efficiency and production of the components through sensitivity studies.

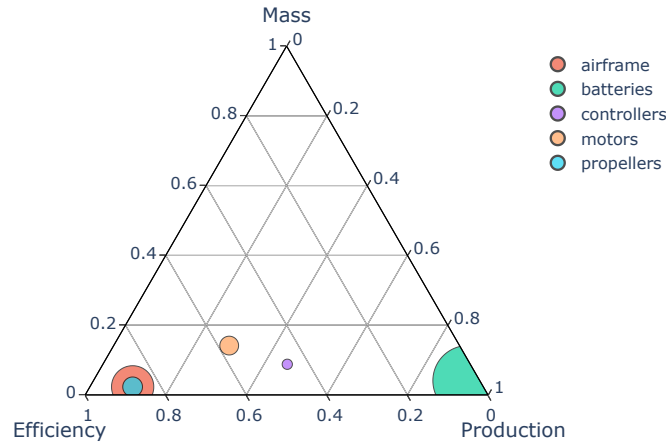
9.5.2 Results

The results obtained by applying the hotspot analysis method to the reference scenario are presented in Figure 9.5.1. For the sake of clarity, the aerodynamic contribution of the airframe is reported in the *efficiency* category. The ternary plot in Figure 9.5.1b depicts the ratios of the three factors — mass, efficiency and production — as positions in an equilateral triangle. A vertex in the triangle corresponds to the exclusive contribution of the associated factor. For example, a data point close to the vertex representing the efficiency indicates a high relative importance of the component efficiency in the environmental impact. The size of the data points represents the total contribution of a component to the overall single score, as reflected in the bar chart in Figure 9.5.1a.

The environmental impact of batteries is mainly due to their production and replacement. It is therefore expected that batteries with longer lifetimes and cleaner chemistries, even when heavier, will lead to a reduction in the impact. In contrast, the airframe contributes to environmental damages mainly through aerodynamic losses. The airframe mass is less critical for this particular case study, as the UAV mass is mostly attributed to the battery and payload. When the in-flight energy consumption is attributed to the components' imperfections, the airframe alone accounts for 23% of the overall environmental score. Similarly, the inefficiency of the propellers (35% losses in forward flight) leads to increased energy consumption, resulting in higher environmental impact. Although the proposed method does not capture non-linear effects such as the 'snowball' effect of mass variation, the weight of the airframe and propellers is not a critical factor. Design efforts for these parts should focus on improvements in aerodynamics and efficiency. Finally, the impacts of the controllers and the electric motors are equally shared between the production stage and the energy losses during the operation stage as a result of their inefficiency.



(a) Absolute contributions of the components.



(b) Relative contributions of mass, efficiency and production factors for each component.

Figure 9.5.1 Contributions of components masses, efficiencies and productions to the overall single score, for the reference scenario.

9.6 Sensitivity to UAV lifespan and mission requirements

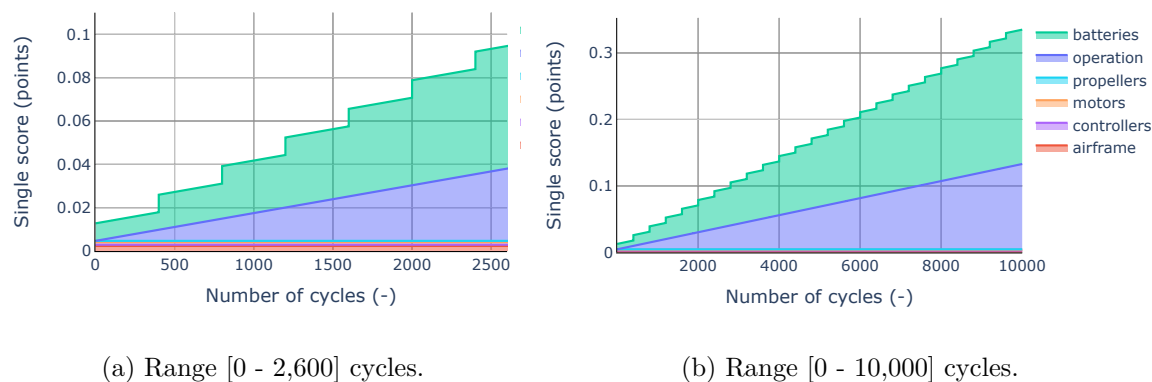
Alternative scenarios are assessed to investigate the effects of a change in the functional unit assumptions using the integrated design methodology (Figure 9.3.1). Specifically, the payload mass, delivery radius and lifespan of the UAV are varied. The application is now defined as:

Deliver a x kg payload within a y km radius, at the rate of z_1 deliveries per day, z_2 days per year, for a period of z_3 years.

Sensitivity to UAV lifespan

First, the lifespan of the UAV, $z = z_1 z_2 z_3$, is varied. In practice, this means using the UAV more intensively, flying more missions per day or using the UAV for a longer period of time before replacing it. Figure 9.6.1 provides the evolution of the single overall score with the

total number of cycles, or missions, in the functional unit. Battery replacement every 400 cycles is responsible for the steps observed in the plots. The proportion of impacts caused by the operation and production stages varies only slightly. For 1000 cycles, 38% of the total environmental footprint is attributed to operation, and 62% to production. At 10,000 cycles these figures are 31% and 69% respectively. This result contrasts with life cycle analyses of conventional kerosene-fueled aircraft where the manufacturing stage quickly becomes negligible compared to the kerosene life cycle [28]. However, if attributing the impacts of the batteries to the operating stage rather than the production stage, as André and Hajek propose [30, 46], similar results to those for conventional aircraft are obtained. With this approach, the system comprising the batteries and the energy supply (i.e. electricity from the grid) accounts for up to 98% of the total environmental footprint when the number of cycles increases to 10,000. This is in the same order of magnitude as the production and combustion of jet fuel for conventional aircraft.



(a) Range [0 - 2,600] cycles.

(b) Range [0 - 10,000] cycles.

Figure 9.6.1 Variation of the single overall score with the number of cycles.

Table 9.6.1 provides the single scores for different numbers of cycles. The results are expressed for the entire life cycle, per cycle and per kilogram of payload transported per kilometer (kg.km). The results show little change in the score per kg.km with usage intensity.

Table 9.6.1 Single overall score for different numbers of cycles.

Number of cycles	Single score (points)			
	Total	Per cycle	Per kg.km	Relative change per kg.km
1,000	0.042	4.18×10^{-5}	2.78×10^{-6}	+ 15 %
2,600 (reference)	0.095	3.64×10^{-5}	2.43×10^{-6}	—
10,000	0.335	3.35×10^{-5}	2.23×10^{-6}	- 8 %

Finally, the variations in the contributions of components masses, efficiencies and productions as a function of the UAV lifetime are illustrated in Figure 9.6.2. With the exception of the batteries, which must be replaced frequently, the production stage becomes marginal

when the intensity of use increases.

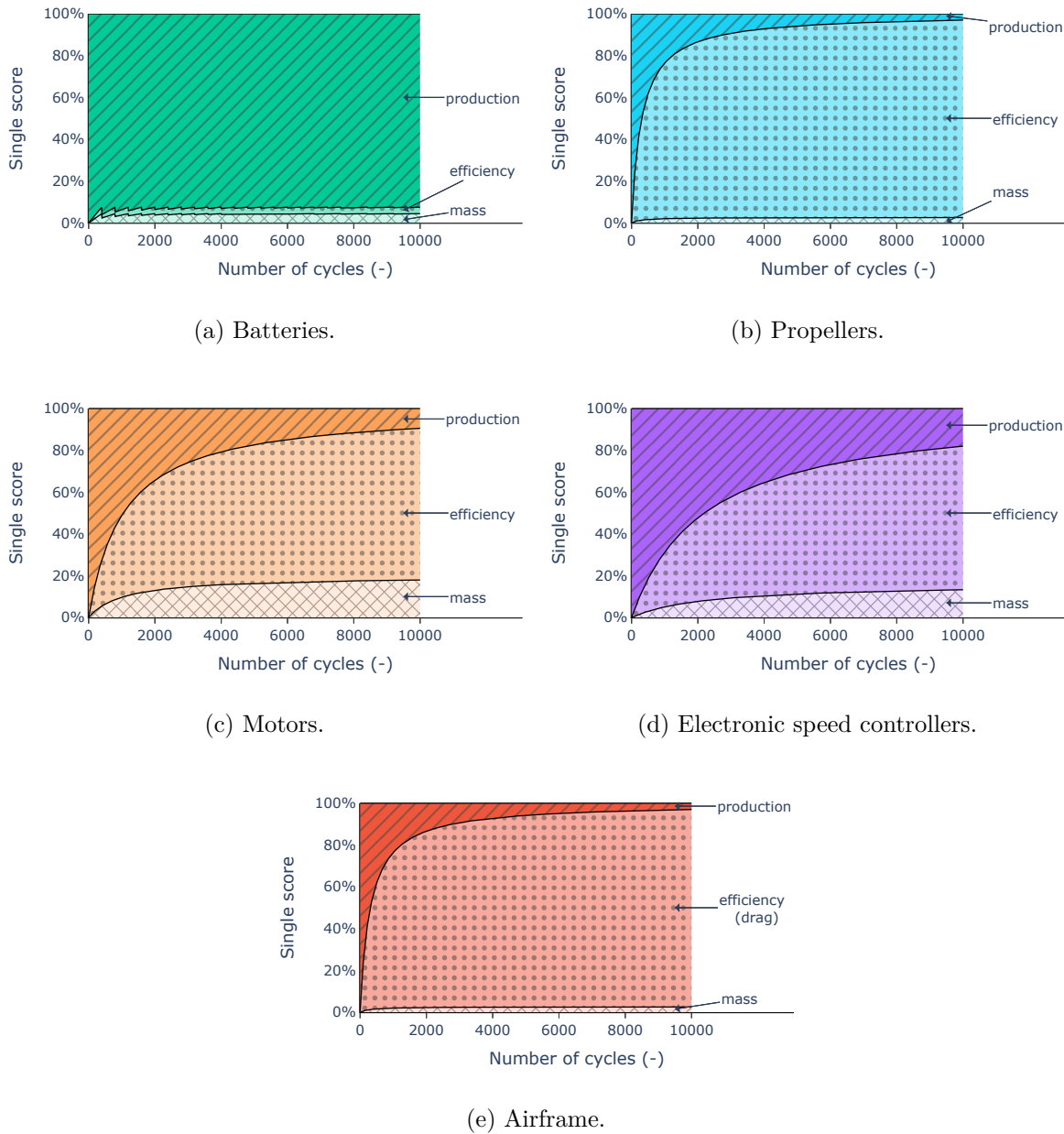


Figure 9.6.2 variations in the contributions of components masses, efficiencies and productions with the UAV lifespan.

Sensitivity to mission requirements

Secondly, the main design-related parameters of the functional unit are varied. In this study, the two parameters of interest are the payload mass and the delivery distance. These

parameters define the delivery mission, which in turn affects the sizing of the UAV through design optimization. Indeed, as stated in Section 9.2.3, the UAV is minimally sized for achieving the mission specified by the functional unit. According to Table 9.2.1, the payload mass is varied in the range [0.5 - 5] kg, and the delivery distance in the range [3 - 12.5] km. The payload attachment and release mechanism is assumed to be unchanged regardless of the payload mass. Figure 9.6.3 shows the results of the design process in terms of both UAV mass and environmental single score, for a fixed number of 2,600 cycles. The environmental score is expressed in points/kg.km to allow comparison on a common basis.

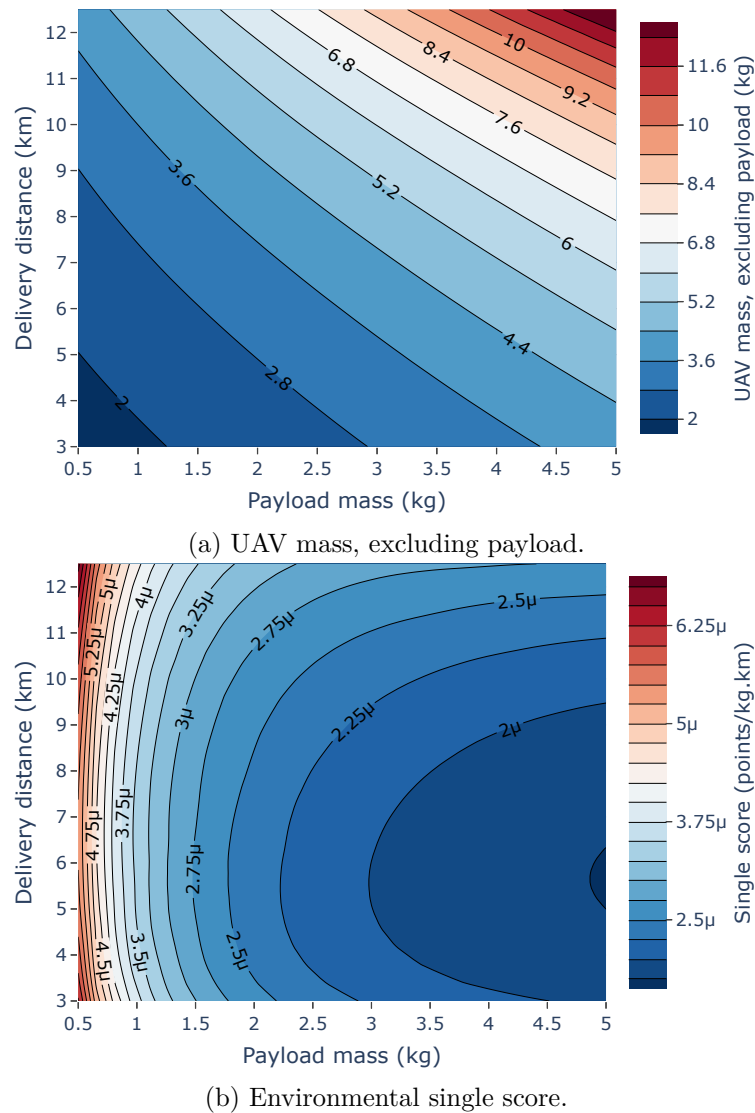


Figure 9.6.3 Results of the design optimization for various payload masses and delivery distances.

It is observed that UAVs designed for carrying heavy loads on short distances yield lower impact than their counterparts. This design area puts the emphasis on power-related criteria

(e.g. to achieve take-off with a heavy payload) rather than endurance. Since battery sizing is primarily influenced by endurance criteria, this domain allows the battery to be lighter, thereby reducing its environmental impact. The increase in mass of the propellers, motors, speed controller and airframe, which are primarily sized from power requirements, does not outweigh the savings offered by the reduced battery mass. Indeed, the single overall score is more sensitive to the battery mass than that of the other components. This mechanism also explains why the environmental score is almost independent of the total mass of the UAV. In fact, a long endurance, low carrying capacity UAV may have the same mass as a UAV carrying heavy loads over short distances. For the same total mass, the different mass distribution of the components results in different environmental impacts. The inverse effects observed for very short distances, that is, an increase of environmental impacts for distances below 5 km, coincide with a tipping point for the sensitivity of battery sizing to the mission distance. For these distances, the take-off and the payload drop-off phases, whose duration is incompressible, are no longer negligible compared to the cruising phase. As a result, a reduction in the delivery distance no longer makes it possible to reduce in the same proportions the energy consumed over the mission, and ultimately the size of the battery and its impact.

9.7 Comparative analysis of different battery chemistries

The findings in Section 9.4 demonstrate that the environmental impacts of the UAV are primarily attributed to batteries, specifically due to the depletion of metals and minerals. Therefore, the consequences of modifying the battery chemistry employed in the UAV design are investigated. To this end, two alternative battery chemistries are introduced alongside the reference battery, which utilizes G/NMC811 chemistry. The first alternative battery uses lithium ferro-phosphate (LFP, or LiFE) as the cathode material and graphite as the anode. Despite LFP batteries having lower energy density compared to NMC batteries, they provide a significantly longer cycle life, reaching several thousand cycles. The present study considers a conservative value of 1,500 cycles [63], along with an energy density of 119 Wh/kg [64]. The second alternative battery replaces the conventional graphite anode with a silicon nanowire anode (referred to as "Si"), as proposed by Amprius [65, 66]. Although the anode can be paired with various cathode materials, an NMC811 chemistry is selected in this study. This technology allows for achieving high energy densities at the expense of reduced battery lifespan. For the specific application under study, an energy density of 360 Wh/kg is assumed, corresponding to a C-rate of 3C for balance between power and energy [65]. Although Amprius claims a cycle life of 325 cycles for a C/5 battery, which corresponds to very high endurance applications [65], an increased charge-discharge rate is likely to reduce this specification [67]. Consequently, a conservative value of 150 cycles is assumed for the Si/NMC chemistry. Table 9.7.1 provides a summary of the characteristics of the three battery

chemistries. Detailed information on the life cycle inventory, more specifically the process tree and the EcoInvent processes related to the batteries, is provided in Appendix 9.B.

Table 9.7.1 Assumptions for the three battery chemistries used in this study.

Battery chemistry	Energy density (Wh/kg)	Proposed lifespan (cycles)
G/NMC (reference)	191 [68]	400 [69]
G/LFP	119 [64]	1500 [63]
Si/NMC	360 [65]	150 [65]

Table 9.7.2 shows the results for the UAV designs obtained with the three battery chemistries. The single overall score is provided for the transport of a 2 kg payload over a distance of 7.5 km during 2,600 cycles. The differences observed in the total mass of the UAVs are due to the lighter (or heavier) batteries, which in turn lead to the oversizing (or undersizing) of the other components to accommodate the extra mass.

Table 9.7.2 Sizing results and environmental scores for the UAV designs obtained with the three battery chemistries.

Battery chemistry	UAV design		Environmental single score			
	Total mass* (kg)	Battery mass (kg)	Points**	Relative variation	Share production	Share operation
G/NMC (reference)	5.62	1.69	0.095	—	65 %	35 %
G/LFP	9.73	4.40	0.093	- 2 %	41 %	59 %
Si/NMC	4.48	0.80	0.156	+ 65 %	83 %	17 %

* including payload mass.

** over the entire life cycle.

On the one hand, the design based on the Si/NMC chemistry requires a significantly lower battery mass to fulfill the delivery mission. However, despite the mass reduction, the Si/NMC battery does not decrease any of the environmental impacts compared to the G/NMC reference battery, as shown in Figure 9.7.1. Specifically, the single overall score is increased by +65% compared to the reference chemistry. This is primarily due to the additional processing required to produce the silicon nanowire used in the anode, as detailed by Li *et al.* [70], and the very short lifespan of the battery, resulting in frequent replacement. Indeed, 18 batteries must be produced to fulfill the application over the two-year period. The impact of this can be seen in the proportion of the single score attributed to the production stage, which accounts for 83% in the Si/NMC-based design. In other words, although the Si/NMC chemistry offers energy savings due to its lighter weight, these savings are outweighed by the significant battery production required for the continuous operation of the UAV. This is illustrated in Figure 9.7.2, where the contribution of battery production to environmental impact far exceeds the contributions of any other factor.

On the other hand, the design involving G/LFP chemistry is considerably heavier than its counterparts due to its lower energy density. However, the increase in mass is offset by

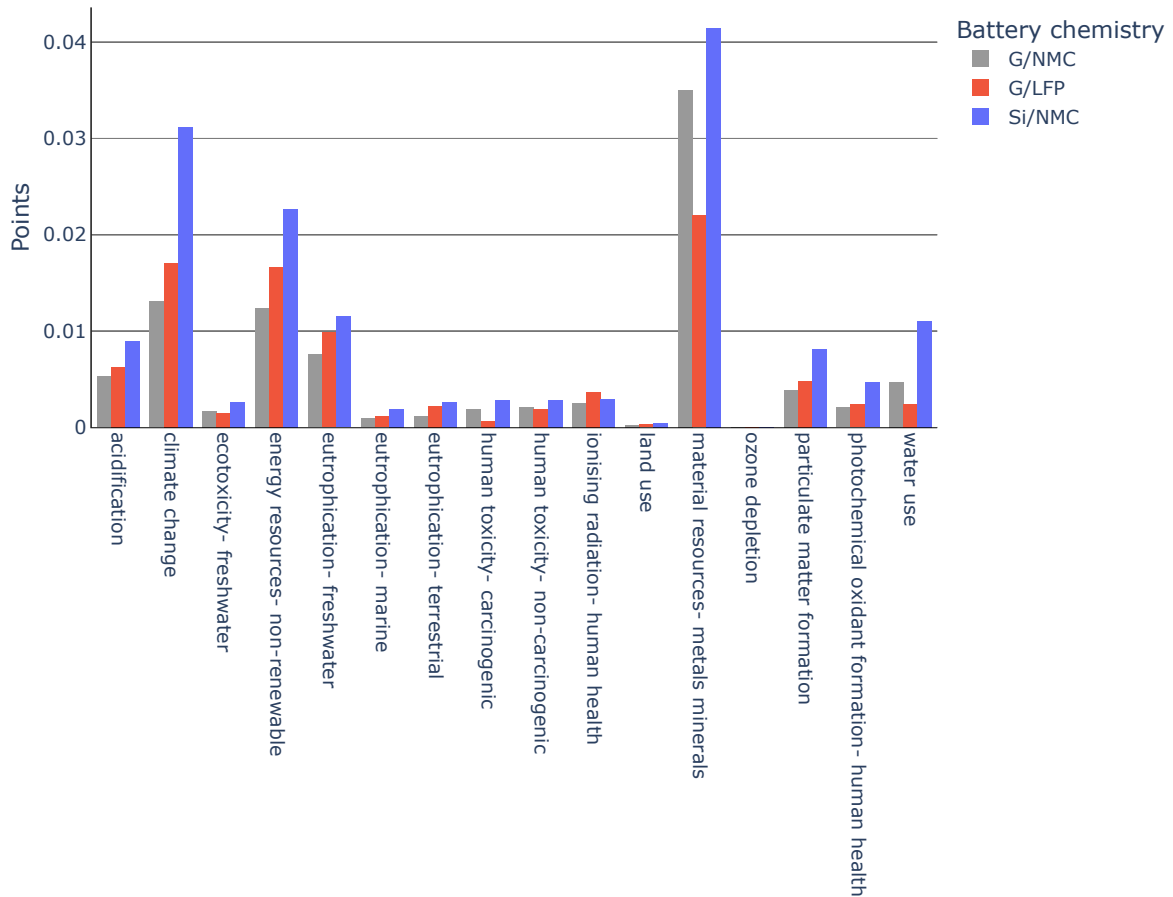


Figure 9.7.1 Normalized and weighted results for the UAV designs obtained with the three battery chemistries.

the long lifetime of the battery, which is almost four times that of the G/NMC chemistry and ten times that of the Si/NMC chemistry. By increasing the weight of the UAV while reducing the number of replacements, the G/LFP battery shifts the environmental burdens to the operation stage. With this technology, the airframe aerodynamics and the efficiency of the propulsion chain become critical, as depicted in Figure 9.7.3. Interestingly, the airframe is now the main contributor to the environmental score, particularly due to its larger surface area (+ 44% compared to the reference design), which increases aerodynamic drag. Alongside improving the aerodynamics of the airframe, the use of a low-carbon electricity mix would help to reduce the impacts on climate change and the consumption of non-renewable energy resources. Overall, despite a different distribution of impacts, G/NMC- and G/LFP-based designs have similar environmental single scores.

Interestingly, the usage intensity does not significantly affect the comparisons, as shown in Figure 9.7.4, which illustrates the change in the single score with the total number of cycles for the application.

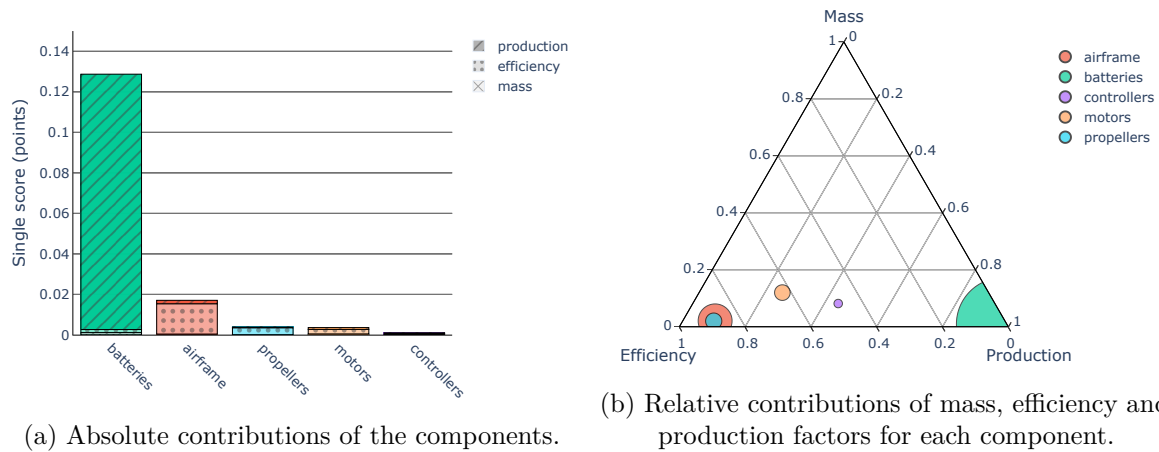


Figure 9.7.2 Contributions of components masses, efficiencies and productions to the overall single score, for the Si/NMC-based design.

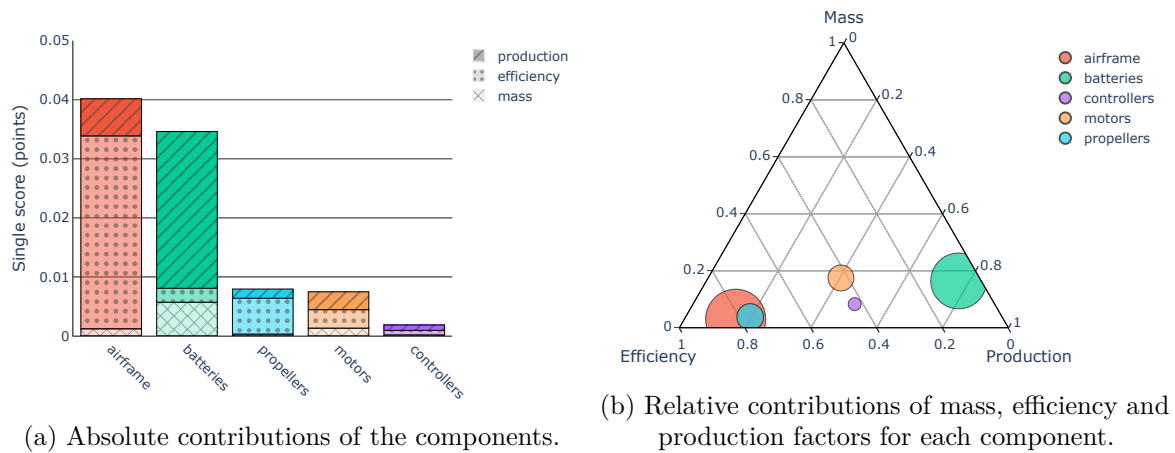


Figure 9.7.3 Contributions of components masses, efficiencies and productions to the overall single score, for the G/LFP-based design.

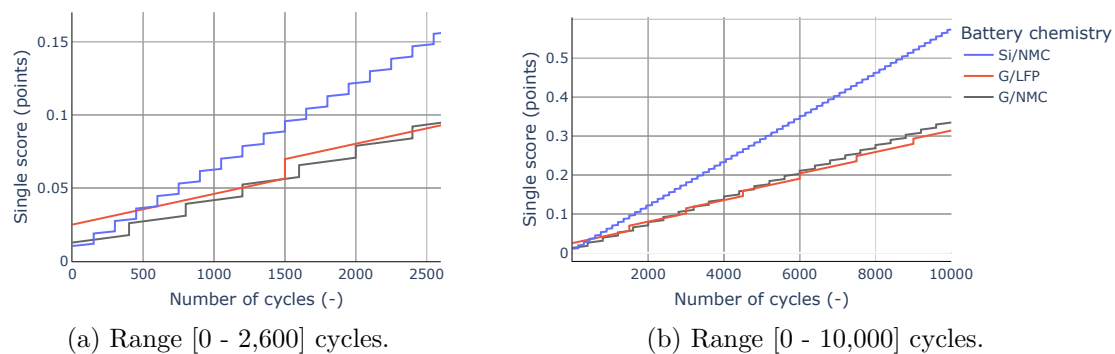


Figure 9.7.4 Variations of the single overall scores with the number of cycles, for the UAV designs obtained with the three battery chemistries.

9.8 Sizing optimization for impacts minimization

Sections 9.6 and 9.7 have explored how changes in the design specifications, namely the mission requirements and the technology assumptions, affect the environmental performance of multirotor UAVs. Typically, given a set of specifications, the UAV and its components are sized using the optimization framework presented in Chapter 3. So far, the sizing optimization was solved with a mass minimization objective, and only then the environmental impacts of the UAV were returned. This section investigates alternative optimization objectives and their influence on both the design and the environmental impacts. To this end, the environmental module is further integrated into the optimization framework, allowing the environmental score to be selected as an optimization objective instead of the mass or in-flight energy consumption (Figure 9.8.1). Here, the environmental score refers to the single overall score. Because of the weighting process underlying the single overall score, the environmental optimization can be perceived as a multi-objective optimization solved with a weighted sum approach. That is, the optimization function is a weighted sum of the 16 environmental impact categories, the multiplying factors being provided by the PEF method [57].

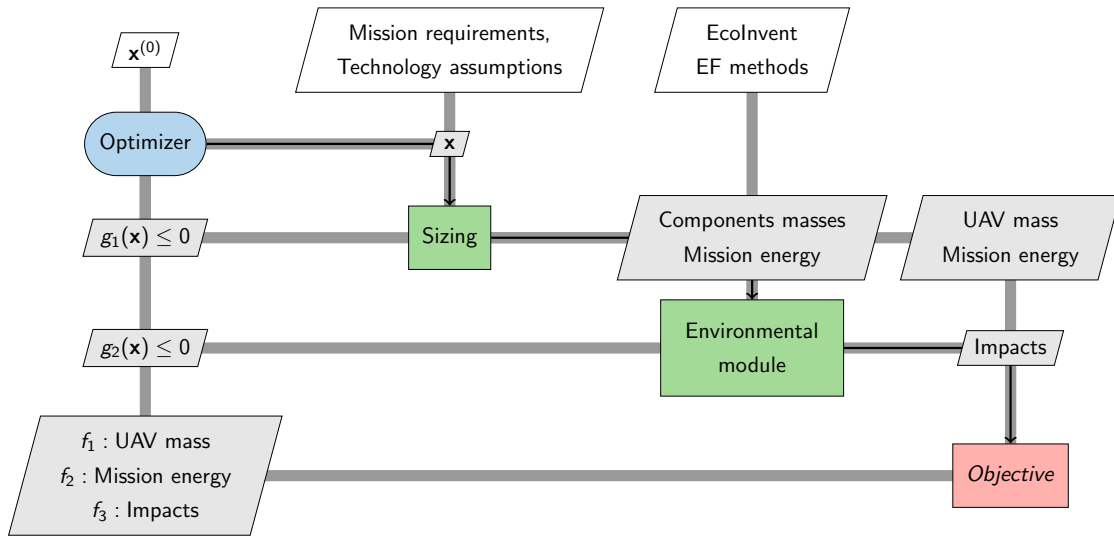


Figure 9.8.1 XDSM showing the integration of the environmental module within the optimization problem.

9.8.1 Efficient implementation

As presented in Chapter 2, multi-disciplinary optimization typically involves an iterative process to converge to the best design candidate. The high number of iterations can lead to significant computation times if the functions to be evaluated are numerically expensive. For this reason, LCA-oriented optimization may prove to be impossible in practice if many

processes are involved in the LCA model. For example, evaluation of the background processes involved in the multirotor UAV model takes hundreds of seconds (Section 9.3). If evolutionary algorithms are used to solve the optimization problem, the computation time can easily reach several hours due to the large number of candidate solutions to be evaluated. On the other hand, gradient-based optimizers rely on the derivatives of the objective function, which have to be estimated numerically if their analytical expressions are not provided. Here again, the iterative evaluation of environmental impacts and their derivatives can lead to unacceptable calculation times in the context of early design studies.

The novelty of the proposed implementation lies in the use of algebraic functions for the fast calculation of the LCA, as described in Section 9.3. The background processes are evaluated in an initialization step prior to the iterative MDO resolution. From the static values of the background processes, algebraic formulas for each impact, which are only functions of the LCA parameters, are compiled [59]. The functions are evaluated in less than a second at each iteration of the optimization, allowing fast convergence to the optimal solution. This approach is further advanced for when gradient-based algorithms are used to solve the optimization. Instead of numerically estimating the impacts' derivatives with respect to the LCA parameters, for instance, with a finite difference scheme, symbolic expressions of the gradients are compiled with the *SymPy* library [71]. This improves the speed and accuracy of the derivative calculation. A visualization of the overall process for fast impact calculation is proposed in Figure 9.8.2.

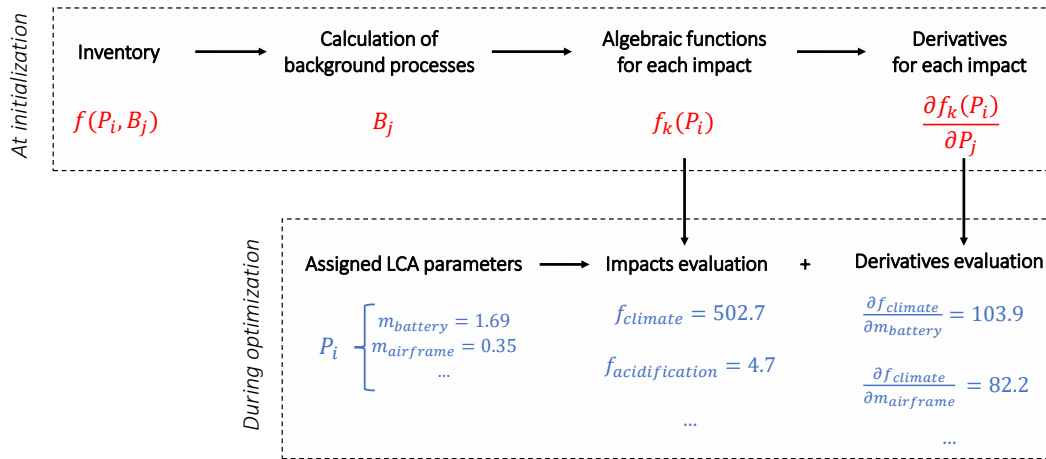


Figure 9.8.2 Visualization of the efficient implementation for fast impacts calculation. Variables in red represent symbolic functions or parameters. Variables in blue represent numerical values.

Thanks to the efficient implementation, the optimization problem converges to the solution with minimum environmental score in 40.58 seconds on a 1.8 GHz quad-core processor using the *SLSQP* algorithm, with a tolerance of 10^{-4} points on the objective value.

9.8.2 Results

The sizing optimization is solved for three different objectives, namely to minimize the UAV mass, the in-flight energy consumption, and the environmental single score. The results, summarized in Table 9.8.1 and Figure 9.8.3, show that the three objectives lead to distinct designs and environmental impacts.

Table 9.8.1 Optimization results obtained when minimizing the mass, the energy consumption, and the environmental impact, respectively.

Optimization objective	UAV mass* (kg)	UAV width & length (cm)	Energy per mission (Wh)	Propulsion efficiency
Mass (reference)	5.62	63×63	257	49%
Energy	6.24	88×88	223	53%
Environmental impact	5.98	81×81	225	52%

* Including payload mass.

Table 9.8.1 (continued).

Optimization objective	Environmental single score	
	Points**	Relative variation
Mass (reference)	0.095	—
Energy	0.087	- 8%
Environmental impact	0.086	- 9%

** Over the entire life cycle.

Specifically, the environmental objective allows the overall single score to be reduced by 9% compared to the reference design, obtained with a mass minimization objective. Interestingly, the design with the lowest energy consumption is not the lightest. This results from two opposite trends occurring in the sizing process. On the one hand, propulsive power decreases with larger propellers by means of the momentum theory. On the other hand, the selection of larger propellers requires bigger motors and longer arms for the multirotor UAV. Although the energy savings offered by the reduced propulsive power result in a smaller battery, the total mass of the UAV increases because of the larger propellers. This trend can be observed in Figure 9.8.4, which shows the evolution of the components' characteristics across the different optimization objectives. Overall, the heavier UAV benefits from the energy savings offered by a more efficient propulsion.

Optimization of the UAV sizing with an environmental objective leads to a compromise between energy minimization and mass minimization, as shown by the Pareto front in

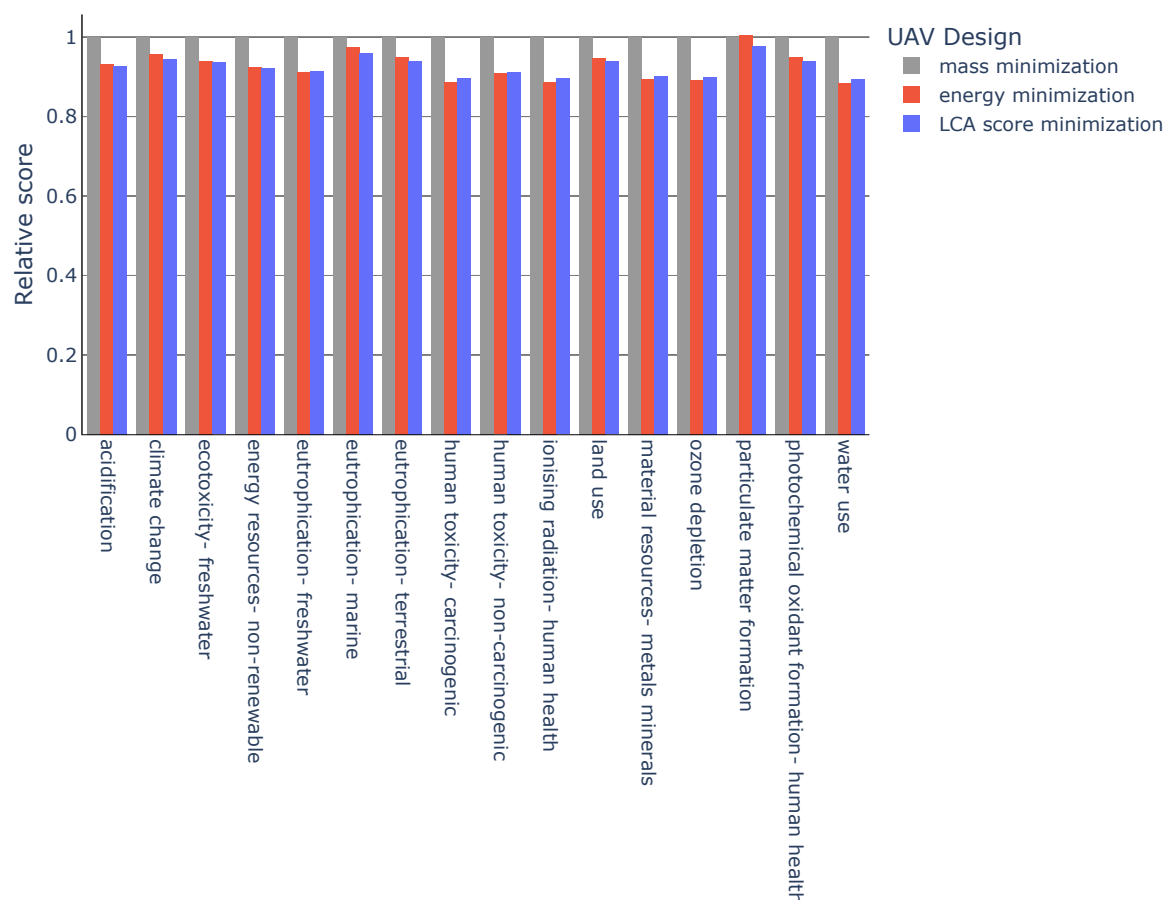


Figure 9.8.3 Comparison of the environmental scores in each impact category, with respect to the reference design obtained with a mass minimization objective.

Figure 9.8.5. As the design approaches the point of minimum energy consumption, the impacts associated with the batteries and electricity production are reduced. However, as the design moves further away from the point of minimum mass, the other components, i.e. the motors, propellers and airframe, see their impacts increase. These two trends result in a global optimum for the environmental score. This optimum depends on the design specifications, i.e. the mission requirements and technology assumptions. For example, for the Si/NMC-based design studied in Section 9.7, the environmental impacts are largely related to battery production. Minimizing the environmental score therefore leads to a reduction in the size of the battery, which is a direct consequence of energy consumption, so that the environmental and energy optimums are one and the same (Figure 9.8.6b). Finally, the gains offered by the environmental optimization are more significant for the Si/NMC-based design (-16% w.r.t. a mass minimization objective) than for the G/LFP one (-2%).

It is also worth noting that the environmental score is more sensitive to design variations near the point of minimum mass than near the point of minimum energy consumption. Future

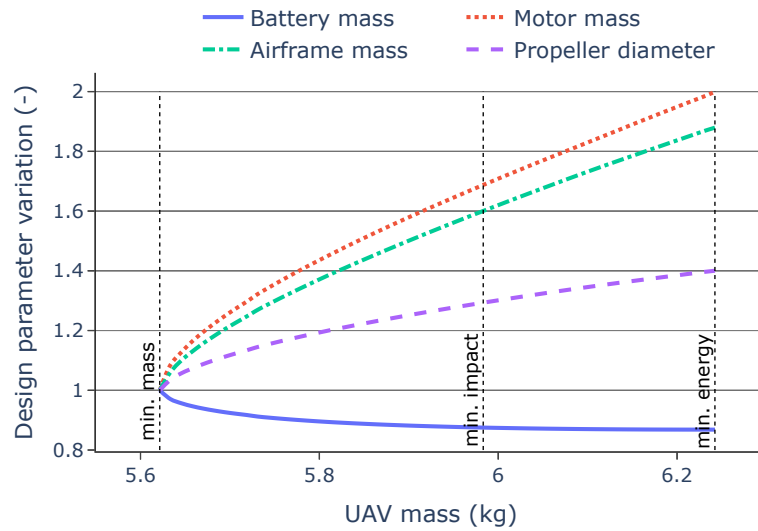


Figure 9.8.4 Variations of the design parameters with respect to the reference design obtained with a mass minimization objective.

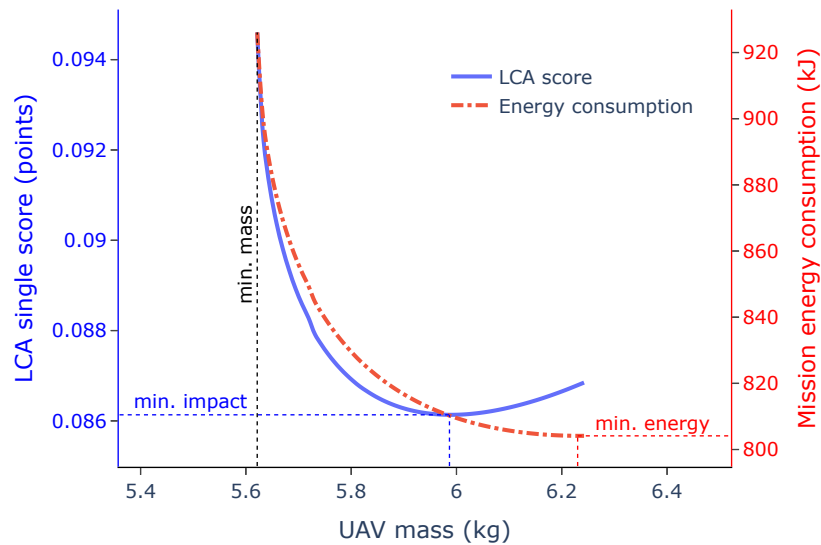


Figure 9.8.5 Pareto front for the sizing optimization, showing the points of minimum mass, minimum energy consumption, and minimum environmental score.

work will have to evaluate in more detail the robustness of the environmental optimum to design uncertainties, by establishing a link with the research conducted in Chapter 5.

Finally, two additional optimizations were performed to minimize climate change alone and material resource depletion alone. The results, shown in Table 9.8.2 for the reference G/NMC-based design, provide evidence of the robustness of the optimization to the weighting of the impact categories.

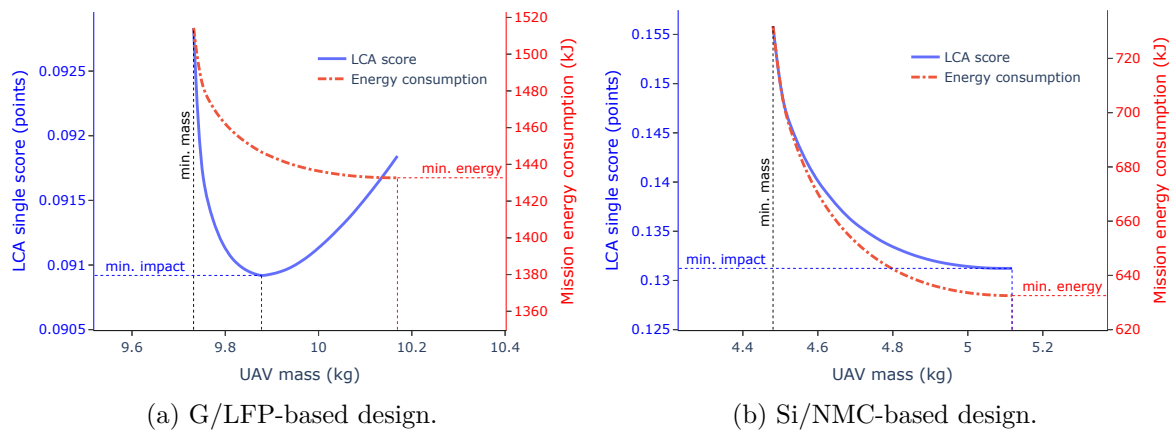


Figure 9.8.6 Pareto fronts for the alternative battery chemistries used in the design.

Table 9.8.2 Optimization results obtained when minimizing the environmental single score, the climate change, and the material resources depletion, respectively.

Optimization objective	UAV mass (kg)	Single score (Points*)	Climate change (kgCO _{2eq} *)	Material resources (kgSb _{eq} *)
Single score	5.98	8.61×10^{-3}	4.71×10^2	2.64×10^{-1}
Climate change	5.87	8.65×10^{-3}	4.69×10^2	2.66×10^{-1}
Material resources	6.08	8.63×10^{-3}	4.76×10^2	2.63×10^{-1}

* Over the entire life cycle.

9.9 Towards an absolute environmental sustainability assessment

The results presented so far only support a relative comparison of different UAV designs in terms of eco-efficiency. As such, they cannot be an indication of how sustainable multirotor UAVs and their applications may be. This requires a comparison of the environmental impacts against sustainability references, typically derived from the concept of planetary boundaries [72, 73]. Here we present some preliminary results of an *absolute sustainability* assessment [74] for different uses of UAV delivery at individual scale.

As described in Chapter 8, the factors used for normalizing the impacts in the original PEF method describe a reference system consisting of all the impacts on a global scale in 2010. For the purpose of absolute sustainability assessment, the normalization factors are adapted so that they now define sustainability thresholds over the 16 impact categories, as proposed by Sala *et al.* [75]. Specifically, these factors link the results of the impact assessment to the concept of planetary boundaries which represent the Earth carrying capacity and, as such, a *safe operating space* for human development [72, 73]. For example, for climate change, an atmospheric CO₂ concentration of 350 ppm and an increase in radiative forcing of +1.0 W.m² relative to preindustrial levels is estimated by Steffen *et al.* [73] as a limit for the safe operating space. This translates into 985 kg CO_{2eq} per person per year if shared equally

among the world's population, according to [18, 75]. By expressing the results of the LCA in the metrics of the planetary boundaries, it becomes possible to quantify the share of safe operating space taken up by a given usage or activity. Here we propose to assess the environmental sustainability of last-mile delivery of small parcels using multirotor UAVs.

To this end, three scenarios are defined. The functional unit is expressed as an annual personal usage, in kilogram.kilometers of parcels delivered over a year for one person. The first scenario involves the transportation of 180 kg.km of payload per year. For example, this corresponds to a monthly delivery of a 2 kg parcel within a radius of 7.5 km. The second scenario increases the frequency to a weekly delivery of a 2 kg parcel, i.e. 780 kg.km/year. Finally, the third scenario considers a usage intensity of 4,500 kg.km/year. This is equivalent to delivering all the food groceries for an average French consumer in 2022 [76, 77], assuming a distance of 7.5 km between the warehouse and the customer's house. For all three scenarios, it is assumed that the delivery missions are performed with the reference UAV design introduced in Sections 9.2 and 9.4, under nominal operating conditions. The impacts associated with the transportation of one kilogram.kilometer of payload with the reference UAV are summarized in Table 9.9.1. Also provided in Table 9.9.1 are the planetary boundaries, expressed per person per year assuming an egalitarian distribution [78].

Table 9.9.1 Estimated impacts for the transportation of one kg of payload over one km, and comparison with planetary boundaries [75].

Impact category	Unit	Planetary boundary [75] per person.year	Estimated impact Value per kg.km
Acidification	molH ⁺ _{eq}	1.45×10^2	1.22×10^{-4}
Climate change	kgCO _{2eq}	9.85×10^2	1.29×10^{-2}
Ecotoxicity: freshwater	CTU _e	1.90×10^4	9.63×10^{-2}
Energy resources: non-renewable	MJ	3.24×10^4	2.48×10^{-1}
Eutrophication: freshwater	kgP _{eq}	8.40×10^{-1}	1.11×10^{-5}
Eutrophication: marine	kgN _{eq}	2.90×10^1	1.56×10^{-5}
Eutrophication: terrestrial	molN _{eq}	8.87×10^2	1.43×10^{-4}
Human toxicity: carcinogenic	CTU _h	1.39×10^{-4}	3.93×10^{-11}
Human toxicity: non-carcinogenic	CTU _h	5.93×10^{-4}	6.87×10^{-10}
Ionising radiation: human health	kB _q U235 _{eq}	7.62×10^4	5.49×10^{-3}
Land use	soil quality index	1.26×10^6	5.97×10^{-2}
Material resources: metals/minerals	kgSb _{eq}	3.17×10^{-2}	7.57×10^{-7}
Ozone depletion	kgCFC-11 _{eq}	7.80×10^{-2}	3.59×10^{-10}
Particulate matter formation	disease incidence	7.49×10^{-5}	6.58×10^{-10}
Photochemical oxidant formation	kgNMVOC _{eq}	5.88×10^1	4.51×10^{-5}
Water use	m ³ world eq. deprived	2.63×10^4	1.62×10^{-2}

In order to support the sustainability assessment in absolute terms, a share of the safe operating space (SoSOS) must be assigned to the transportation of small parcels. The usage will be deemed sustainable if its impacts do not exceed the allocated SoSOS. The choice of an allocation principle is the result of a political decision on which there is no global consensus. The present study is based on the status quo sharing principle (or, "grandfathering") in which the SoSOS allocated to an activity or an industrial sector corresponds to its current

contribution to environmental impacts [79]. However, it should be emphasized that the choice of another sharing principle would greatly influence the results of the sustainability assessment [80]. A SoSOS of 0.1% is allocated to the last-mile transport of small parcels. This figure is an estimate for the current share of $\text{CO}_{2\text{eq}}$ emissions from this category of freight in France. Specifically, last mile delivery accounted for 0.94% of France's total $\text{CO}_{2\text{eq}}$ emissions in 2020 [81–83]. It is further assumed that around 10% of the goods and materials delivered in the last kilometers of the logistics chain are small items that could be transported by multirotor UAVs, leading to 0.1% of France's total impact on climate change. Finally, in the absence of additional data, this allocation is applied to all other environmental impact categories. Future work should differentiate the allocations by impact category.

The results for each scenario, expressed in percentages of the safe operating space at an individual level, are presented in Figures 9.9.1 and 9.9.2. When looking at climate change, none of the scenarios can be considered absolutely sustainable, as the emissions exceed the allocated SoSOS for last-mile delivery of small parcels indicated by the red circle.

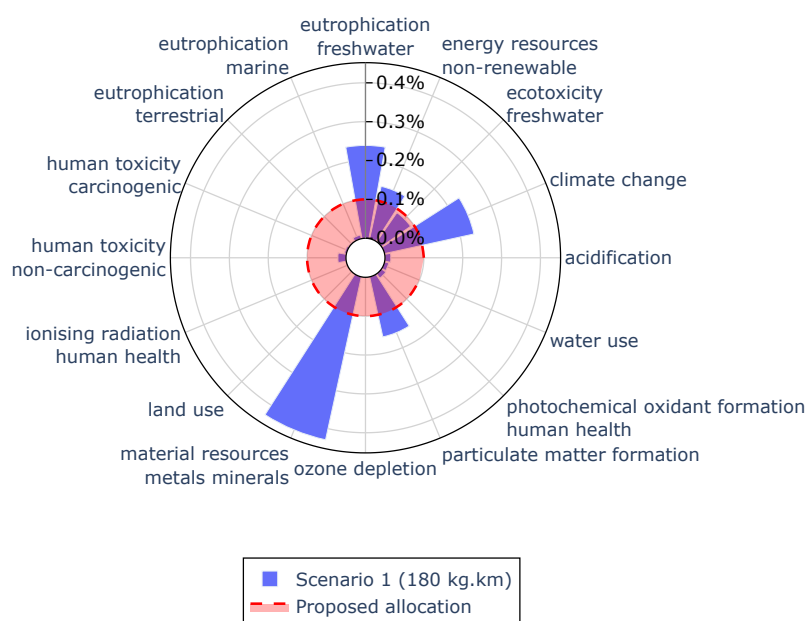


Figure 9.9.1 Percentage share of the safe operating space taken up by scenario 1 (in blue), expressed across the 16 impact categories of the PEF. The red circle represents the proposed allocation to last-mile delivery of small parcels.

If the drone delivery sector is to achieve sustainability objectives, a number of measures need to be taken, which can be divided into three categories:

1. Improving the environmental efficiency of UAVs and their operations. The multirotor UAV evaluated in this study emits $12.9 \text{ gCO}_{2\text{eq}}/\text{kg.km}$. This figure is ten times higher than that of diesel vans in the 2-2.5 tonne category (0.2 to $1.8 \text{ gCO}_{2\text{eq}}/\text{kg.km}$ [84]),

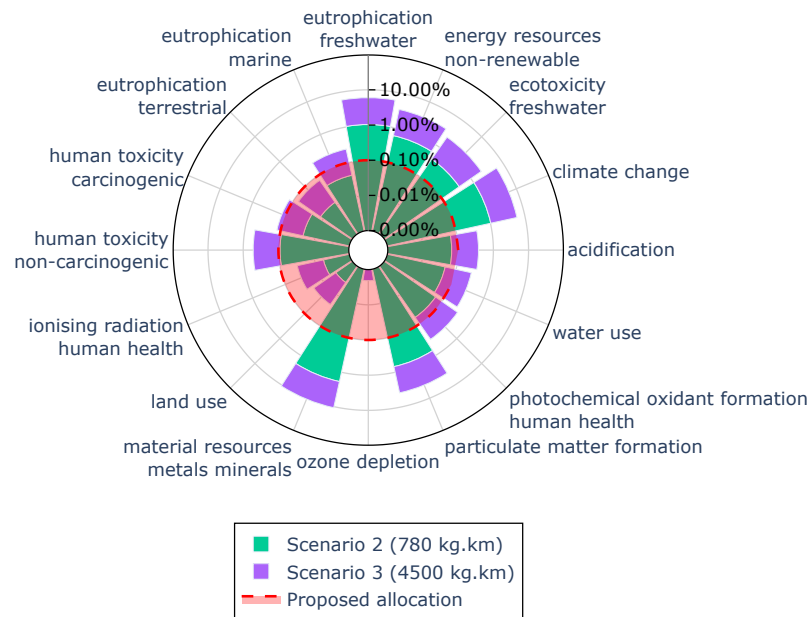


Figure 9.9.2 Percentage shares of the safe operating space taken up by scenario 2 (in green) and scenario 3 (in purple), expressed across the 16 impact categories of the PEF. The red circle represents the proposed allocation to last-mile delivery of small parcels. The scale is logarithmic.

which are now widely used for last-mile deliveries. UAV concepts equipped with a fixed wing, such as hybrid FW-VTOL UAVs, are likely to reduce environmental burdens due to their lower energy consumption. A cleaner electricity mix would also help mitigate environmental impacts. However, the high material consumption of electric UAVs designed with current technologies is unlikely to improve the environmental score sufficiently.

2. On the demand side, limit the use of electric drones for delivery applications. For example, by restricting their use to medical services.
3. Increasing the SoSOS assigned to the last mile delivery of small parcels by applying alternative allocation principles [80]. This political choice would involve reducing the share allocated to other areas of activity.

Figure 9.9.3 provides a sensitivity analysis for these three parameters with a focus on climate change.

9.10 Conclusions and perspectives

Methodology and lessons learned

This chapter presents the application of life cycle assessments to the early design of multirotor

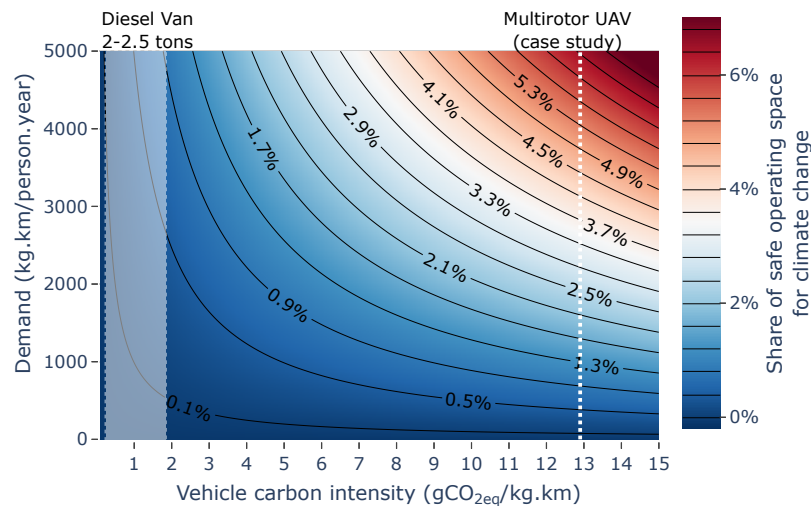


Figure 9.9.3 Sensitivity analysis on the environmental efficiency, usage intensity and allocated SoSOS for the climate change indicator.

UAVs. In doing so, it provides insights for making the right design choices to mitigate the environmental impacts of UAVs. More specifically, the contributions of the present work are as follows:

- A new environmental module based on the LCA methodology was developed and integrated into an existing framework for the preliminary design of UAVs. The module uses the capabilities of the *Brightway* and *lca_algebraic* Python libraries to build a parametric LCA model for UAVs and relies on EcoInvent to collect inventory data. By directly linking the LCA to the outputs of the sizing process through the model parameters, the environmental analysis efficiently calculates impacts for various design alternatives driven by mission requirements and technology assumptions.
- Application of the methodology to a case study shows that battery production and renewal contribute significantly to the environmental impact of UAVs. For this reason, UAVs designed to cover short distances have a better environmental score per kg.km of payload carried, as the size of the battery increases rapidly with range.
- It is also shown that using batteries with a higher energy density does not necessarily reduce the impacts. In fact, battery lifespan is a key parameter and the frequent renewal of short-life batteries, such as those with Si/NMC chemistry, can largely outweigh the gains in mass and in-flight energy consumption. The use of heavy but long-lasting batteries (e.g., LFP batteries) transfers the environmental impacts to the operation stage. This makes the aerodynamics of the airframe a priority to reduce energy consumption.

- A more advanced integration of the environmental module into the design framework was proposed, which allows to size the components in order to minimize the life-cycle environmental footprint. The environmental optimum was found to be a compromise between minimizing the UAV mass and minimizing the flight energy consumption, which are two distinct solutions. For the case studies discussed in this chapter, environmental optimization resulted in reductions ranging from 2% to 16% of the environmental impact. The savings are most significant for UAV designs that require frequent battery replacement, namely UAVs with Si/NMC batteries.
- Finally, an assessment of the absolute sustainability of UAVs for last-mile delivery was proposed. The transport of goods by multirotor UAVs, even if only occasionally used, is found to be unsustainable under a status quo sharing principle. In particular, a tenfold reduction in carbon intensity must be achieved for multirotor UAVs to be environmentally competitive with conventional ground vehicles. This gap could be addressed by using fixed-wing UAVs instead of multirotor drones.

Perspectives

Further research should address the limitations and perspectives of the present work. From a methodological perspective, the uncertainties underlying the LCA results need to be covered more thoroughly, by including the uncertainties related to the design and to the characterization factors employed in the LCA. In particular, the robustness of the environmental optimum in the sizing process should be investigated. From an application perspective, the LCA model of multirotor UAVs can be further detailed by including additional components and life cycle stages. Accounting for the disposal and recycling of the components, especially the batteries, could have a significant influence on the results. In addition, the assessment needs to be extended to fixed-wing and hybrid FW-VTOL UAVs. Combined with existing studies on the environmental impact of ground vehicles, this could provide valuable information for comparing and selecting the most appropriate vehicle for a given application. Addressing these limitations holds promise for advancing aerospace systems towards more environmentally conscious designs.

Appendix

Appendix 9.A Reference technologies and assumptions for the sizing optimization

Table 9.A.1 Reference technologies and assumptions for the components' models used in the sizing optimization.

Component	Reference technology / Assumptions
Propeller	APC Propellers for Multi-Rotors
Motor	AXI 5325/16 GOLD LINE
Electronic Speed Controller	Turnigy K-Force 70HV
Battery	DJI TB30 Battery with G/NMC811 chemistry Energy density $\rho = 191$ Wh/kg, 400 cycles lifetime, 85% charge-discharge efficiency
Airframe	Carbon fibre-reinforced plastic Density $\rho = 1700$ kg/m ³ , Allowable stress $\sigma = 280$ MPa, Drag coefficient $C_D = 1.0$

Appendix 9.B Process tree and EcoInvent processes for the batteries production

The production processes for the G/NMC and G/LFP batteries were taken from EcoInvent 3.9. In the case of the Si/NMC chemistry, the activity has been reconstructed using the data provided by Li *et al.* [70]. More specifically, the EcoInvent activity for the G/NMC battery is used as the basis, with the graphite component in the anode replaced by silicon nanowire.

Table 9.B.1 EcoInvent 3.9 processes for the different battery chemistries.

Battery chemistry	Unit	EcoInvent processes
G/NMC	kg	Market for battery cell, Li-ion, NMC811 [GLO]
G/LFP	kg	Market for battery cell, Li-ion, LFP [GLO]
Si/NMC	kg	Market for battery cell, Li-ion, NMC811 [GLO]
	kg	Market for silicon, solar grade [GLO]
	kWh	Market group for electricity, medium voltage [CN]

Note: [CN] refers to China geographical location.

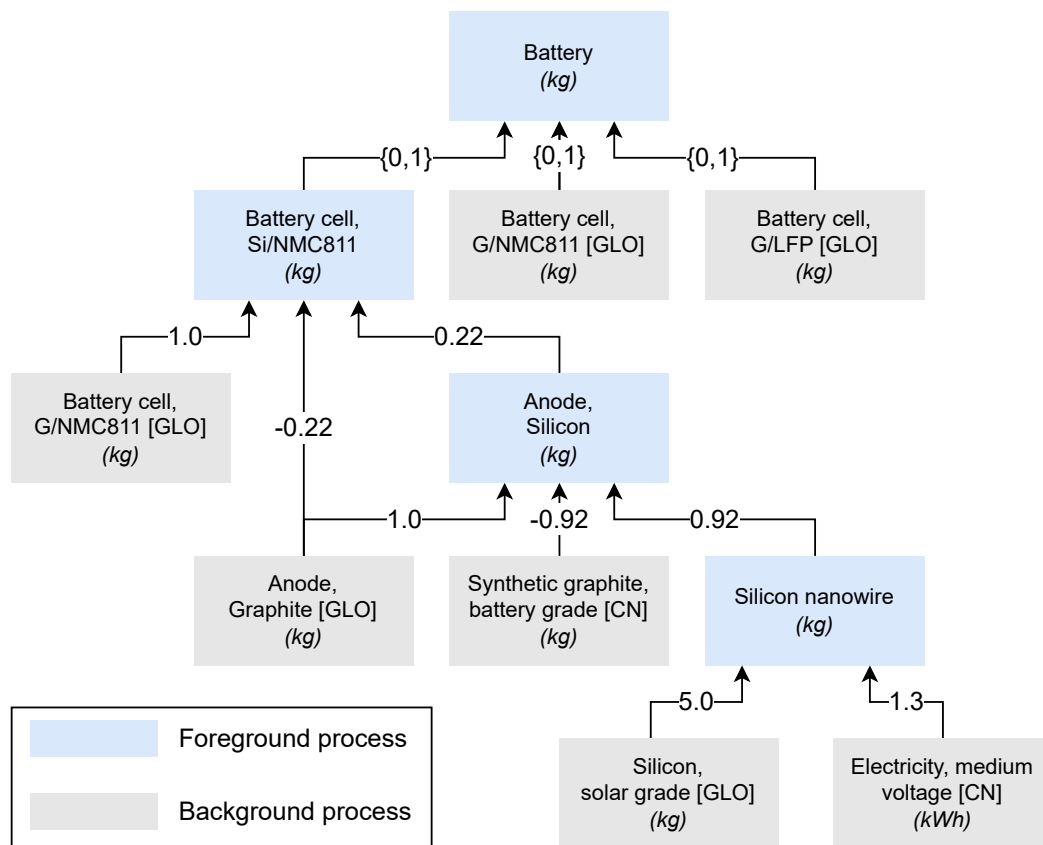


Figure 9.B.1 Process tree for the batteries production. $\{0, 1\}$ represents a boolean option.

Appendix 9.C Normalization and weighting factors

Table 9.C.1 Normalization and weighting factors for the Environmental Footprint methods [56, 57].

Impact categories	Normalization factors	Weighting factors
Acidification	5.56×10^1 mol H ⁺ eq.	6.20 %
Climate change	8.10×10^3 kg CO ₂ eq.	21.06 %
Ecotoxicity, freshwater	4.27×10^4 CTU eq.	1.92 %
Eutrophication, freshwater	1.61 kg P eq.	2.80 %
Eutrophication, marine	1.95×10^1 kg N eq.	2.96 %
Eutrophication, terrestrial	1.77×10^2 mol N eq.	3.71 %
Human toxicity, cancer	1.69×10^{-5} CTUh	2.13 %
Human toxicity, non-cancer	2.30×10^{-4} CTUh	1.84 %
Ionising radiation, human health	4.22×10^3 kBq U-235 eq.	5.01 %
Land use	8.19×10^5 pts	7.94 %
Ozone depletion	5.36×10^{-2} kg CFC-11 eq.	6.31 %
Particulate matter	5.95×10^{-4} disease incidences	8.96 %
Photochemical ozone formation - human health	4.06×10^1 kg NMVOC eq.	4.78 %
Resource use, fossils	6.50×10^4 MJ	8.32 %
Resource use, minerals and metals	6.36×10^{-2} kg Sb eq.	7.55 %
Water use	1.15×10^4 m ³ eq. deprived	8.51 %

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Chapter 10

Conclusions

10.1 Summary

The thesis provides a comprehensive exploration of UAV design from multiple perspectives, addressing critical gaps in sizing, uncertainty management, fault-tolerant design, and environmental impact assessment. The schematic summary in Figure 10.1.1 summarizes the interrelated research areas, each contributing to the goal of advancing UAV design, safety, and sustainability.

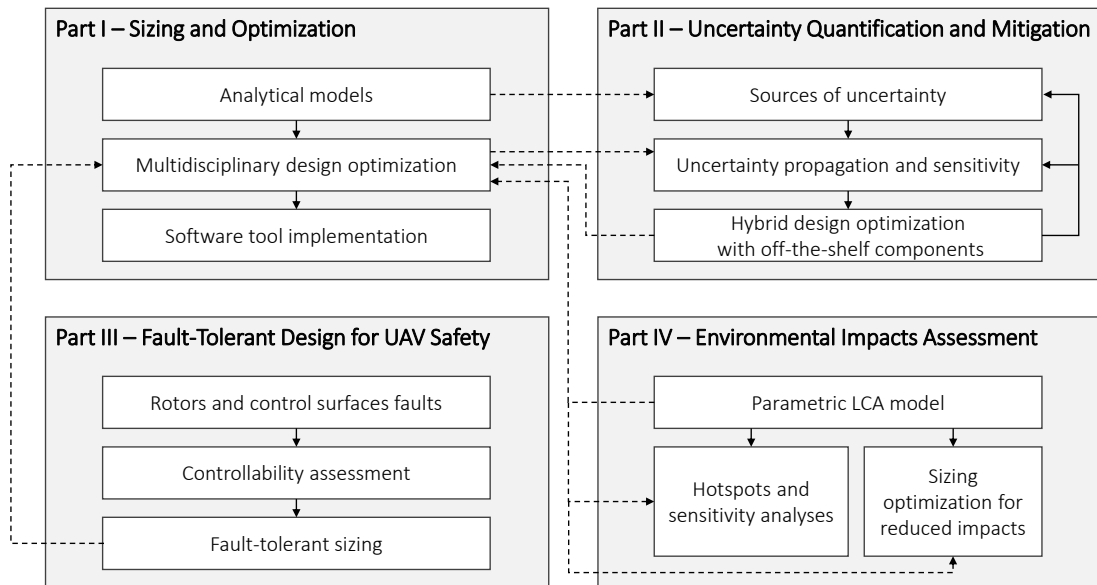


Figure 10.1.1 Schematic summary of the thesis.

The thesis introduction in Chapter 1 sets the stage by presenting the background and motivation for the research.

Part I addresses the sizing and optimization of UAVs. Chapter 2 reviews the existing methodologies for sizing UAVs, highlighting limitations and gaps in current approaches. It identifies the need for a unified, open-source tool for sizing multirotor, fixed-wing, and FW-VTOL UAVs. The third chapter presents a comprehensive and computationally efficient approach for the optimal sizing of UAVs. The methodology is based on analytical models describing the different disciplines and components applicable to multirotor, fixed-wing, and FW-VTOL concepts, and relies on a well-defined MDO formulation. The implementation of the methodology in a software tool demonstrates its efficiency and versatility for preliminary sizing.

Part II focuses on uncertainty management in the context of preliminary design. Chapter 4 reviews the literature on uncertainty analysis in the design process, highlighting the lack of methods addressing uncertainties associated with the availability of off-the-shelf components. It sets the stage for developing a new approach to uncertainty quantification and mitigation. Chapter 5 introduces a methodology for addressing uncertainties associated with both the design models introduced Chapter 3 and the availability of off-the-shelf components. It characterizes and compares input and model uncertainties with an application to a multirotor UAV. The chapter proposes a new approach to design optimization that allows the substitution of analytical models by a direct selection of off-the-shelf components from a database, thereby reducing design uncertainties.

Part III covers the fault-tolerant design of UAVs to improve safety. Chapter 6 reviews the literature and regulatory framework for safety-critical UAVs and identifies gaps in controllability assessments for fixed-wing and hybrid FW-VTOL UAVs. It emphasizes the need for methodologies that integrate fault tolerance analysis with design procedures. Chapter 7 presents a novel methodology for controllability assessment and fault-tolerant sizing of UAVs that addresses faults in both rotors and control surfaces. Specifically, a new control authority index, the XACAI, is developed, and the introduction of sizing factors in the control re-allocation following fault conditions allows the control analysis to be linked to the sizing methodology of Chapter 3. The approach is applied to various UAV concepts and its effectiveness in ensuring safety under fault conditions is demonstrated.

Finally, Part IV focuses on the environmental sustainability of UAVs. Chapter 8 introduces the life cycle assessment methodology, and reviews its current applications to aerial vehicles. In particular, the chapter identifies a gap in the integration of LCA with design tools for aerospace systems. Chapter 9 presents the development of an environmental module for the preliminary design of UAVs. The module, which consists of a parametric LCA model, is integrated into the MDO framework developed in Chapter 3. It allows a range of studies to identify the key drivers of environmental impacts and then optimize the design to minimize those impacts. The first steps towards an absolute sustainability assessment are also provided.

10.2 Contributions

The work presented in this manuscript aims to answer the specific research questions and the consequent objectives presented in Chapter 1. The main contributions can be outlined by recalling the initial questions and objectives to which they relate.

Research questions:

RQ1. *What methodologies can be employed for the preliminary design of UAVs?*

A comprehensive methodology for the design of UAVs must leverage MDO principles to efficiently solve the disciplinary couplings specific to UAV systems and the optimization problem. In the thesis, the NVH formulation is implemented to solve the couplings, and the complexity of the optimization problem is reduced by applying the First Monotonicity Principle and introducing sizing coefficients to explicitly handle some of the optimization constraints. In addition, the work of the gradient-based optimization algorithms is facilitated by the use of analytical models expressed in the continuous mathematical domain.

RQ2. *How can the sizing be optimized for various UAV configurations and applications?*

The design framework must include a range of models to describe the many disciplines with a component-level approach, so that models can be added, removed, or swapped to match the desired concept or configuration of UAV. This capability is facilitated by implementing the methodology in a numerical platform that allows the models to be combined and organized prior to performing optimizations on these models. In addition, the use of optimization variables whose bounds are independent of the application, as defined by the NVH formulation and the introduction of sizing coefficients, allow the design problem to be solved for a wide range of specifications without requiring adjustments.

RQ3. *What are the primary sources of uncertainties in the preliminary design of UAVs, and how do they impact the results of the sizing process?*

The two main sources of uncertainty in preliminary design are associated with the estimation models used to derive the components parameters and the inputs to these models, namely the design variables. In particular, for systems featuring off-the-shelf components, the value of the design variables selected during the sizing process may not be representative of the actual possibilities offered by the component marketplace. This is because the resolution of the design problem in a continuous design space is inconsistent with the finite number of off-the-shelf components available in the market. However, for UAVs, most of the uncertainty about the system performance is caused by inaccuracies in the estimation models (e.g., model fitting and validity domain) rather than the limited availability of components.

RQ4. *What strategies can be developed to mitigate these uncertainties?*

During the design process, replacing analytical models with a direct selection of components from a database eliminates the uncertainty associated with both estimation models and component availability. However, this requires the use of alternative optimization strategies. One strategy is to reformulate the optimization problem to accommodate mixed continuous and discrete variables. In the thesis, this approach was rejected because it is inconsistent with the efficient problem formulation developed previously. In particular, it would have compromised the advantages of using continuous normalized variables to address a wide range of specifications. Instead, the problem formulation (involving only continuous variables) was left unchanged, and the discrete selection process was integrated within the sizing procedure. That is, the continuous variables defining the components are mapped to the nearest available component. An evolutionary strategy algorithm, CMA-ES, is then used to solve the optimization problem.

RQ5. *What should be considered when evaluating the fault tolerance of different UAV configurations, and what methods can be applied?*

First, the evaluation of the fault tolerance of various UAV configurations must consider the unidirectional nature of rotor thrust, as well as the invariantly positive nature of the drag efforts produced by the control surfaces. Without this, it would be incorrectly assumed that fixed pitch propellers can provide negative thrust and that drag can become propulsive, resulting in an erroneous assessment. Secondly, the evaluation of the null controllability in the presence of faults, which supports the fault tolerance analysis, requires the calculation of a control authority index that captures the aforementioned specificities of the effectors. For this purpose, the XACAI has been developed in this thesis.

RQ6. *What metrics and methods can be used to establish a link between control analysis and UAV sizing?*

One way of sizing the UAV, and more specifically the rotors and control surfaces, to ensure sufficient control authority following any failure assessed as controllable is to introduce the so-called sizing factors. For this purpose, the control allocation problem must be redefined to include an additional degree of freedom to oversize the effectors if the control authority under failure conditions is deemed insufficient. Solving the "sizing control allocation" for each expected fault scenario provides the value of the sizing factors that are subsequently introduced into the sizing process to increase the control authority of the UAV design under degraded conditions.

RQ7. *How can the environmental footprint of UAVs be systematically assessed in the design process?*

The systematic assessment of environmental impacts during the design process of UAVs requires the development of an environmental module based on the LCA methodology. Specifically, the LCA calculations must be parameterized to allow automatic linkage between the sizing results and the environmental assessment. To speed up the calculations, the parametric LCA model must distinguish between the "background" processes that are independent of the UAV design (e.g., how the electricity is produced) and the "foreground" processes that are directly affected by changes in the design (e.g., the size of the battery).

RQ8. *What are the main contributors to the environmental impacts of electric UAVs, and how can these impacts be minimized?*

The main contributors to environmental impacts are the production of the batteries and the electricity consumed to power the drone. The impacts associated with batteries can be reduced by selecting battery types that offer a longer life expectancy. However, battery chemistries with longer lifetimes tend to be heavier, thereby transferring the impacts to the electricity consumption. Therefore, selecting a battery type for reduced impact is not a simple task. In addition, the aerodynamics of the airframe play an important role in minimizing the impacts associated with energy consumption. For a given configuration and set of technologies, properly sizing a UAV can help reduce its environmental footprint. Replacing the traditional mass minimization objective with an environmental objective results in slightly heavier but more environmentally friendly designs.

Objectives:

- O1.** The first objective was to develop a methodology for sizing and optimizing different UAV configurations across a range of applications. To this end, a methodology has been developed that covers multirotor, fixed-wing and FW-VTOL UAV concepts. The methodology adapts to the designer's needs by using either analytical estimation models (Chapter 3), databases of existing components (Chapter 5), or both to efficiently converge to the optimal candidate. In addition, the formulation of the optimization problem ensures the robustness of the design process with respect to the specifications and its fast convergence. The methodology was implemented in an open-source software tool, providing a framework for future collaborative work.
- O2.** The second objective was to characterize, quantify, and effectively mitigate uncertainties in UAV design. To this end, the sources of uncertainty in UAV design have been characterized and quantified in Part II of the manuscript, focusing on uncertainties related to scaling laws and regression models, and the discrete availability of off-the-shelf components. Specific techniques (e.g., Monte Carlo and Sobol' indices) were used to propagate uncertainty and rank sources of uncertainty by importance in terms of

UAV performance and dimensions. A "hybrid" design methodology was introduced to successfully mitigate uncertainties in multidisciplinary design optimization by allowing the optimization of systems with both custom and off-the-shelf components.

- O3.** The third objective was to establish a methodology for the controllability assessment and fault-tolerant sizing of UAVs. This has been achieved through the development of a new control authority index, the XACAI, which successfully accounts for the constraints posed by the physical behavior of the control surfaces and the rotors. In addition, a modified control allocation procedure is applied to determine sizing factors for the effectors, ensuring their proper sizing to provide the required control authority under fault conditions.
- O4.** Finally, the fourth objective was to develop and integrate an environmental assessment model into a preliminary design framework to improve the sustainability of UAVs. To this end, an efficient parametric LCA model was developed and integrated into the previously defined design methodology. This approach enables the exploration of the sensitivity of environmental impacts to technological assumptions, mission requirements, and sizing objectives, thereby contributing to the improvement of UAV sustainability.

10.3 Limitations and propositions for future work

The research conducted in this thesis contributes to the field of UAV design in several ways, but also acknowledges certain limitations that suggest directions for future research.

The software tool developed for the optimal sizing of UAVs provides a basis for various design studies and applications, but could be advanced with further discipline-specific models. For example, structural models for the landing gear and payload attachment could be developed, along with the integration of models for sizing the control surfaces of fixed-wing designs. In addition, airframe aerodynamics for multirotor UAVs could be further investigated, for example, by performing computational fluid dynamics (CFD) analyses of different airframe shapes and sizes. The modeling of alternative energy sources, such as hydrogen fuel cells and solar panels, would further extend the design domain covered by the methodology.

The uncertainty analysis would mainly benefit from an extension to fixed-wing and FW-VTOL designs as well as the coverage of dynamic flight scenarios. This requires the quantification of uncertainties in the airframe aerodynamic models and the propeller performance model in non-axial conditions. In addition, the selection of off-the-shelf components in the hybrid MDO methodology could be refined by weighting the design variables when searching for the nearest product in a database, thus improving the optimization performance. More advanced research could also explore the area of uncertainty-based MDO to develop reliable and/or robust designs in spite of uncertainties. Finally, future work could focus on tolerancing to bridge the gap between preliminary and detailed design, for example by

providing a specification with tolerances for each component or subsystem, thereby ensuring a contained level of uncertainty in the final integration.

The methodology developed in Part III of the thesis provides valuable insights into UAV controllability in the presence of faults, but has its limitations. The linear null-controllability analysis does not exclude the possibility of nonlinear controllability. Failure scenarios assessed as uncontrollable should be further investigated using nonlinear approaches. In addition, the time-independent nature of the methodology may limit the applicability of the findings. Future work should explore time-dependent control authority indices to refine the fault tolerant sizing.

Finally, the developments and findings for the environmental assessment of UAVs should be extended to new applications, configurations, and energy sources (e.g., hydrogen fuel cells and internal combustion engines). Regarding uncertainties, a more detailed analysis should include the uncertainties in the characterization factors and the design parameters, in addition to the quality of the inventory data. An interesting approach would be to shift the focus of the case studies from minimizing environmental impact to restricting the design domain based on environmental constraints imposed by regulations and policies. This would be more relevant to industrial practice, where UAVs are more likely to be designed to minimize production costs given a set of requirements and constraints. Lastly, a prospective approach could be used to assess the impacts of future drone technologies (e.g., new battery chemistries) along with evolving means of energy production and industrial processes for manufacturing the components. Coupling these considerations with drone market forecasts would provide deeper insight into the absolute sustainability of UAVs.

More broadly, the application of the developed methodologies to a wider range of case studies holds promise for advancing the design of UAVs toward more efficient, safe, and sustainable developments.

